

**Hegde's
PocketGuide to
Treatment in
Speech-Language
Pathology**

Fourth Edition

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Preface to the Fourth Edition

The fourth edition of this PocketGuide to treatment procedures in speech-language pathology has been updated and expanded to offer even more comprehensive coverage of treatment procedures than did the third edition. New entries reflect changes in the nomenclature of disorders and treatment approaches. Several weaker and unfounded treatment procedures or approaches have been eliminated. As before, whether described treatment procedures are supported by experimental evidence or not is briefly noted to assist a critical evaluation of procedures. All entries have been further streamlined. Entries have been edited with a view to make them more succinct and direct. References to major entries have been updated.

Two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, have been simultaneously revised to update and expand information on disorders and assessment, respectively. As a set, the three PocketGuides offer comprehensive information on the characteristics of communication disorders, their epidemiology, etiology, and brief overview of major theories; assessment approaches and procedures; and treatment approaches and techniques. The three guides serve a dual function: First, they are detailed enough for the student clinicians as well as the more established practicing clinicians. Second, the guides are succinct enough to provide an overview of the entire range of knowledge in speech-language pathology.

This fourth edition of the PocketGuide to treatment procedures is designed for clinical practitioners and students in communicative disorders. The PocketGuide combines the most desirable features of a specialized dictionary of terms, clinical resource book, and textbooks on treatment of communicative disorders. It is meant to be a quick reference book like a dictionary because the entries are alphabetized; but it

offers more than a dictionary because it specifies treatment procedures in a “do this” format. The PocketGuide is like a resource book in that its main objective is to describe practical treatment procedure, but it offers more than a resource book by clearly specifying the steps involved in treating clients. The PocketGuide is like standard textbooks that describe treatment procedures; but it organizes the information in a manner conducive to more ready use and easier access.

How the PocketGuide Is Organized

All main entries for treatment of communication disorders are printed in bold and blue color. Each cross-referenced entry is underlined. Each main disorder of communication is entered in its alphabetical order. Subcategories or types of a given disorder are described under the main entry (e.g., Broca's Aphasia under Aphasia, Ataxic Dysarthria under Dysarthria, Neurogenic Stuttering under Fluency Disorders).

Specific techniques, most of them with general applicability across disorders (e.g., Modeling, Biofeedback, or Turn-Taking) also are alphabetized. Techniques that apply across disorders are described at their main alphabetical entry (e.g., Modeling under M). When appropriate, the reader also is referred to the disorders for which the techniques are especially appropriate.

For most disorders, a general and composite treatment procedure is first described. For example, there is a general treatment program each described for such disorders as aphasia, speech sound disorders, stuttering, and language disorders in children. Following this description of a comprehensive treatment procedure, specific techniques or published treatment programs are described (e.g., social approaches to treating aphasia, pragmatic language intervention for children with language disorders, syllable prolongation in stuttering). Organization of entries varies somewhat for different disorders, but they follow a general format that begins with a brief description of the disorder, general guidelines on treatment, ethnocultural considerations in treatment, comprehensive treatment procedures in many cases, ending with specific treatment programs or procedures when available. Major and current references are given at the end of each main entry and at the end of specific treatment technique or program.

Many treatment concepts and procedures are cross-referenced. All cross-referenced entries are underlined. Therefore, the reader who comes across an underlined term can look up that term in its alphabetical order.

How to Use This PocketGuide

There are two methods for the clinician to use this guide. In the first method, the clinician looks up treatment procedures by disorders in their alphabetical order. *Major Entries* (the table contents) will quickly refer the reader to specific communication disorders described in the guide. Clinicians can quickly gain access to treatment procedures for aphasia to voice disorders described in their alphabetical order. Under each of the main entries for major disorders, the clinician may look up subentries or specific types of disorders. For example, under Dysarthrias, the clinician will find the various types of dysarthria, also entered alphabetically, from ataxic dysarthria to unilateral upper motor neuron dysarthria. Similarly, following the main entry for Aphasia, the different varieties of aphasia are described in their alphabetical order.

In the second method, the clinician looks up a treatment procedure by its name. For example, the clinician can look up such specific treatment techniques as the following in their alphabetical order: activity-based language intervention, air-flow management in stuttering, conversational repair strategies, delayed auditory feedback, event structure, functional equivalence training, mand model, melodic intonation therapy, and so forth. In many cases, the reader who finds a specific treatment technique in the general alphabetized order is referred to the specific disorder for which the technique is relevant.

A Caveat

Serious attempts have been made to include most treatment techniques described in the literature. However, the author is aware that not all techniques have been included. Some have been excluded because of their transparent lack of logic, appropriateness, or even expectation of desirable effects. A few are described briefly with the cautionary note that there is no evidence to support its practice. Most importantly, in

any task such as this that requires encyclopedic review of literature, omission of a procedure that deserves inclusion is an acknowledged and unintended limitation. The reader may be more often correct in assuming that a technique was omitted inadvertently than to assume that it was considered and rejected.

The author did not set for himself the impossible goal of including all treatment techniques. The practical goal was shaped more by such qualifiers of treatment techniques as *most*, the *major*, the *generally effective*, the *most widely practiced*, and so forth. Such qualifiers necessarily involve judgment with which clinicians will disagree. If some techniques included do not meet these qualifiers, that is fine; the author would rather err in that direction. On the other hand, errors of omission are correctable through revisions. Therefore, the author is open to suggestions from clinicians and researchers.

Although most treatment techniques in communicative disorder are in need of treatment effectiveness or efficacy data, those that are especially deficient are noted in their description or definition. Those treatment techniques that have especially strong supportive evidence also are noted. In most cases, unfortunately, information on effects and efficacy is unavailable or ambiguous. This guide is not a means of evaluating treatment techniques; such evaluation is solely the responsibility of the clinician who selects treatment techniques. To help the clinician make such evaluations, procedures and experimental designs that are used in treatment efficacy research are included in this guide. Also included are suggested *Treatment Selection Criteria*.

About the Author

M.N. Hegde, PhD, is Professor Emeritus of Speech-Language Pathology in the Department of Communicative Disorders at California State University, Fresno. A highly regarded author in speech-language pathology, his books include leading texts in academic courses and valuable resources for clinicians. His books have been used in worldwide in speech-language pathology programs.

He holds a master's degree in experimental psychology from the University of Mysore, India, a post master's diploma in Medical (Clinical) Psychology from Bangalore University, India, and a doctoral degree in Speech-Language Pathology from Southern Illinois University at Carbondale.

Dr. Hegde is a specialist in fluency disorders, language disorders, research methods, and treatment procedures in communicative Disorders. He has made numerous presentations to national and international audiences on various basic and applied topics in communicative disorders and experimental and applied behavior analysis. He also has served on the editorial boards of scientific and professional journals and continues to serve as an editorial consultant to *Journal of Fluency Disorders* and the *American Journal of Speech-Language Pathology*.

Dr. Hegde is a recipient of various honors including the Outstanding Professor Award from California State University-Fresno, CSU-Fresno Provost's Recognition for Outstanding Scholarship and Publication, Distinguished Alumnus Award from the Southern Illinois University Department of Communication Sciences and Disorders, and Outstanding Professional Achievement Award from District 5 of California Speech-Language-Hearing Association. Dr. Hegde is a Fellow of the American Speech-Language-Hearing Association.

Acknowledgments

I am pleased to note that this new edition is being published by Plural, whose predecessor, Singular, was the original publisher of the first edition. I would like to thank Valerie Johns, Executive Editor, Nicole Hodges, Assistant Editor, Linda Shapiro, Production Coordinator, Jessica Bristow, Production Assistant, and Angie Singh, President and CEO of Plural for their excellent support throughout the preparation of the new edition of this PocketGuide.

A

ABA Design. A single-subject treatment research design used to evaluate treatment effects; a target behavior is first baserated (A), taught with the procedure to be evaluated (B), and then reduced (A) by withdrawing treatment to show that the teaching was effective; if the treatment procedure you wish to select has been evaluated with this design, examine whether the investigator used the following procedure to establish its effectiveness:

- Baserated the target behavior to be taught
- Applied the new treatment to be evaluated
- When the target behavior increased, withdrew treatment
- Charted the outcomes to show that the results for the baserate and withdrawal conditions were similar but those for the treatment condition were different.

ABAB Design. A single-subject treatment research design used to evaluate treatment efficacy; a target behavior is first baserated (A), taught by applying the treatment program (B), reduced by withdrawing or reversing the treatment (A), and then taught again by reapplying the treatment (B) to show that the teaching was effective; the design has two versions: Reversal and Withdrawal; if the treatment procedure you wish to select has been evaluated with this design, examine whether the investigator used the following procedure to establish its effectiveness:

- Baserated the behavior to be taught
- Applied the new treatment to be evaluated for the target behavior
- Briefly, applied the treatment to another behavior or simply withdrew treatment
- Again treated the target behavior
- Charted the outcomes to show that the two no-treatment conditions were convincingly different from the two treatment conditions.

ABAB Reversal Design. A single-subject treatment design for evaluating treatment effects; a desirable behavior is baserated (first A), taught (first B), reduced by teaching its counterpart (second A), and then taught again (second B) to show that the teaching was effective; if the treatment

procedure you wish to select has been evaluated with this design, examine whether the investigator used the following procedure to establish its effectiveness:

- Baselerated the behavior to be taught
- Applied the new treatment to be evaluated for the target behavior
- Briefly, applied the treatment to an incompatible behavior
- Again treated the target behavior
- Charted the outcomes to show contrasting rates of behaviors under the baserate and experimental conditions.

ABAB Withdrawal Design. A single-subject research design for evaluating treatment effects; a desirable behavior is baselerated (A), taught (B), reduced by withdrawing the treatment (A), and then taught again (B) to show that teaching was effective; if the treatment procedure you wish to select has been evaluated with this design, examine whether the investigator used the following procedure to establish its effectiveness:

- Baselerated the target behavior to be taught
- Applied the new treatment to be evaluated
- When the behavior increased, withdrew treatment
- Reapplied the treatment to the target behavior
- Charted the outcomes to show that the behavior varied according to the treatment and withdrawal operations

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Agraphia. To treat lost or impaired writing skills associated with cerebral pathology or injury that may also be associated with reading problems (Alexia), see Treatment of Aphasia: Writing Problems; note that treatment for agraphia may have different parameters than treatment of writing problems in children who simply have not mastered the writing skills; see the two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, for description of different types and assessment procedures.

Airflow Management. A stuttering treatment target within the comprehensive fluency shaping procedure; includes inhalation of air, slight exhalation before initiating phonation, and sustained airflow throughout an utterance; for procedures see Fluency Disorders (Stuttering; Treatment of Stuttering: Specific Techniques or Programs).

Alaryngeal Speech. To teach speech without a biological larynx—a mode of communication for persons whose larynges have been surgically removed—see Laryngectomy.

Alerting Stimuli. Various means of drawing the individual's attention to the imminent treatment stimuli; needed whenever the individual's attention is likely to wander; include such statements as "Get ready! Here comes the picture!" or "Look at me, I am about to show you how," or such nonverbal cues as touching the individual's hand just before presenting a stimulus; important in treating individuals with autism spectrum, aphasia, dementia, right hemisphere syndrome, and children with attention deficit disorders.

Alexia. Treating reading problems of adults who have neurological impairments (e.g., strokes, neurodegenerative diseases); does not refer to teaching children who have not mastered grade-appropriate reading skills, called dyslexia, which is often due to inadequate instruction or learning disabilities; may be associated with writing problems (Agraphia) in some, isolated in others; for treatment of alexia in individuals with neurological communication disorders, see Treatment of Aphasia: Reading Problems; see the two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Alphabet Board. A means of teaching basic communication skills to individuals who have limited verbal language skills; also may be used to reduce the speech rate to improve intelligibility in individuals with hypokinetic dysarthria; it contains a communication board with the alphabet printed

on it; may also contain a few words and sentences; the individual simultaneously speaks (to the extent he or she can) and points to the printed first letter of each spoken word; the “listener” reads what is pointed out and thus understands the message; helps slow down the rate of speech in individuals whose speech rate is excessive (e.g., some individuals with Dysarthria).

Alternative Communication. To teach methods of non-oral, nonvocal communication that serve as *alternatives* to oral speech and language, see Augmentative and Alternative Communication; only in a few extreme cases are the methods totally *alternative*; most non-oral, nonvocal means of communication augment oral and vocal communication, regardless of how limited the vocal and verbal skills might be.

Alzheimer’s Disease. Intervention for individuals with Alzheimer’s disease is the same as that for dementia; intervention may be beneficial in slowing down deterioration; direct intervention to sustain the skills as long as possible and family and caregiver intervention to help them interact effectively with the individual are the two main components of intervention; see Dementia for management details.

American Indian Hand Talk (AMER-IND). A system of nonverbal communication used by native Americans to communicate with members of other tribes with different languages; a manual interlanguage; the signs represent ideas and many are pictographic; gestures may be produced in series to express more complex ideas, called agglutination; many signs are one-handed; used in teaching Augmentative Communication, Gestural (Unaided).

American Sign Language (ASL or AMESLAN). A highly developed manual (gestural) language used mostly by deaf persons in the United States; a communication target for certain nonverbal or minimally verbal persons; each sign or gesture may represent a letter of the English alphabet, a word, or a phrase; signs provide phonemic,

morphologic, and syntactic information; used in teaching Augmentative Communication, Gestural (Unaided).

Amyotrophic Lateral Sclerosis (ALS). To treat motor speech disorders associated with this progressive neurological disease in which the upper and lower motor neurons degenerate, see Dysarthrias.

Anomia. Treatment of naming difficulties (anomia) is essential in many individuals with neurological diseases or disorders who exhibit word finding problems; people with traumatic brain injury, dementia, and especially those with aphasia need treatment for their naming problems; see Aphasia for treatment strategies.

Antecedents. Important elements of behavioral treatment of communication disorders; events that occur before responses; stimuli or events the clinician presents in treatment; to make treatment stimuli effective, select them from the individual's natural environment whenever possible or use common stimuli; select stimuli that are ethnoculturally appropriate for the individual; antecedents may be:

- Common objects or objects from the individual's home environment (e.g., a child's favorite toy or book)
- Pictures that are colorful, unambiguous, and ethnoculturally appropriate
- Re-created or enacted events to show actions and scripts
- Instructions, demonstrations, modeling, prompting, manual guidance, and other special stimuli

Aphasia. Treatment of aphasia—a language disorder caused by recent brain injury—involves multiple treatment targets; initially and in a hospital setting, the treatment may be managed by a team of multiple professionals; eventually, most individuals with aphasia may receive communication treatment as outlined; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology and ethnocultural considerations, neuropathology,

and aphasic symptomatology; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for assessment procedures.

Treatment of Aphasia: General Guidelines

- Note that there is both controlled and uncontrolled evidence to suggest that aphasia treatment is effective and that all individuals are candidates for treatment
- Conduct a detailed assessment; see the cited sources and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*
- Reduce the effects of the residual deficits on the personal, emotional, social, family, and occupational aspects of the individual's life
- Teach compensatory strategies (e.g., signing, gestures)
- Counsel family members to help them cope with the residual deficits
- Give a realistic prognosis that modifies the individuals' and the family members' expectations
- Develop a variety of task- and individual-specific treatment procedures as illustrated in this outline
- Choose functional communication targets rather than grammatical correctness
- Sequence target behaviors in treatment; move from simple to complex tasks
- Offer an intensive treatment program; the greater the frequency of weekly sessions, the higher the progress
- Use such extra stimuli as instructions, prompts, modeling, pictures, and objects in initial stages of treatment; fade such extra stimuli used in treatment
- Use only natural stimuli (e.g., only a question, not a prompt) to evoke speech in later stages of treatment
- Program natural consequences for functional communication targets (e.g., smile and approval to reinforce verbal expressions; real objects to reinforce requests for objects)
- Provide immediate, response-contingent reinforcement for correct or effective responses and give equally immediate corrective feedback for incorrect responses

- Teach and reinforce self-monitoring skills to reduce errors and to sustain treatment gains in the natural environment
- Train family members to evoke, prompt, reinforce, and maintain communicative behaviors
- Offer group treatment sessions to reinforce verbal skills in the context of social communication and social integration
- Judge when it is not useful or ethical to continue the treatment

Treatment of Aphasia: Ethnocultural Guidelines

- Consider the ethnocultural, linguistic, and economic background of the individual in planning treatment
- Gain an understanding of the individual's family and its economic resources to pay for extended treatment, afford regular transportation, ability and willingness to keep regular appointments
- Help find public and private resources that support the individual's continued treatment and rehabilitation
- Assess the family members' educational level, emphasis on communication skills, and their willingness and time available for helping the individual
- Understand the individual's family constellation and communication patterns (e.g., living in an extended family; the individual's role in educating and raising grandchildren)
- Evaluate the individual's linguistic background and especially if the individual speaks a different dialect or form of standard English (e.g., African American English or Spanish-influenced English); premorbid literacy level and the current need for literacy skills (e.g., Does the individual need treatment for reading and writing or will functional communication suffice?)
- Assess communication needs of a bilingual individual in both languages or, at the least, in the dominant language
- Select treatment stimuli that are available in the individual's home, and, if appropriate, work environment

- Carefully describe the treatment procedures and note the effects they produce or fail to produce; modify the treatment procedure in light of the individual's performance and ethnocultural background

Treatment of Aphasia: Auditory Comprehension. Auditory comprehension is the least researched of the aphasia treatment procedures; there is no controlled evidence to support a time-consuming auditory comprehension treatment for individuals with aphasia; typical and repeated trials in which the clients are asked to point to objects or words may not produce beneficial effects on comprehending conversational speech; there is no evidence to suggest that improved comprehension (if that is demonstrated) results in improved production; dealing with evident comprehension problems in the context of teaching functional and social communication skills may be the best strategy; evidence suggests that when production skills improve, comprehension skills also improve with no additional and direct treatment for comprehension; a few general guidelines and management suggestions may be considered for individual clients with significant speech comprehension deficits; these suggestions may be useful for most individuals with aphasia.

Promoting Auditory Comprehension: General Guidelines

- Select picturable verbs and other words that give a clue to auditory comprehension
- Select unambiguous stimulus pictures to be used in treatment
- Use shorter and simpler sentences
- Use active sentences and avoid passive and indirect expressions
- Build personally relevant information into treatment tasks
- Speak at a slower rate; pause frequently; and give additional stress on key terms
- Conduct treatment in quieter environment with little or no distraction

- Give redundant messages and instructions; repeat them
- Speak in connected speech that gives context, rather than isolated words or sentences that do not
- Give the individual limited response choices; do not confuse by demanding multiple responses at the same time
- Pair auditory treatment stimuli with appropriate visual stimuli; use objects whenever possible; if not, use realistic, colorful, unambiguous pictures
- Make your face visible to the individual as you speak
- Draw the individual's attention before presenting treatment trials if necessary (give Alerting Stimuli; e.g., "Look at my face," "Here comes the picture," "Listen! I am going to ask you to do something").

Sequence of Auditory Comprehension Treatment

Comprehension of Words. Ask the individual to point to the items named; positively reinforce correct responses; repeat the trials for stimuli to which the individual gave a correct response; ask the individual to name:

- Various body parts
- Everyday objects and pictures you display in front of the individual
- Actions depicted in various individual pictures or pictures in story books

Comprehension of Spoken Sentences. Accept an appropriate verbal or nonverbal (gestural) response that suggests good comprehension; reinforce positively. Target comprehension of:

- Simpler sentences before more complex sentences
- More redundant sentences before less redundant sentences
- Sentences with familiar information before those with unfamiliar information
- Use the sentence verification or recognition format:
 - Present various pictures that include similar elements (e.g., pictures of a dog chasing a man,

a man chasing a dog, a dog chasing a cat, and a cat chasing a dog)

- Say a sentence and ask the individual to show the picture that represents the sentence; for example, “Show me the man chasing a dog” or “Show me the dog chasing the man”

Target Comprehension in Conversation and Narration. Targeting comprehension during conversation and discourse may be the best strategy, as it combines production as well as comprehension; target such skills as:

- Understanding conversation
 - Hold typical conversations and frequently check for comprehension of what you say
 - Reinforce for correction statements that imply good comprehension
 - Reiterate the statements that are misunderstood or not comprehended
- Understanding narratives
 - Tell a brief story and ask the individual to retell it
 - Read aloud a brief story and ask the individual to retell it
 - Ask questions about the details and sequence of the story to assess and positively reinforce comprehension
 - Prompt details and sequences to reinforce narrative skills

Treatment of Aphasia: Verbal Expression

Treatment of Naming: General Considerations.

Select both the target words and intervention strategies that are client specific and functional:

- Select words that are most commonly used; make the list individual specific (e.g., nouns related to the individual’s hobbies, interests, and occupation; names of family members, friends, and pets)
- Select the names of manipulable objects; select objects that are relevant to the individual (e.g., a

- hammer in case of a carpenter, a painting brush in case of a painter, a book or a pen in case of a writer)
- Select names of objects rather than pictures; because not all important stimuli are picturable, start with objects before you move on to pictures
- Use realistic drawings or colorful photographs rather than line or abstract drawings
- Ask the individuals to generate their own cues, based on their personal experience, that help name the objects or persons
- Let the individual regulate the rate of stimulus presentation (e.g., slower rate the individual may prefer)
- Give extra time to respond
- Display the stimuli for longer duration if necessary
- Present simultaneously visual and auditory stimuli

Treatment of Naming: Targets and Techniques

Confrontation Naming: Treatment Procedure.

Confrontation naming is naming an object when asked, “What is this?”

- Start with more familiar objects and move on to less familiar objects; generate the list based on the individual’s own experiences
- Place a picture or an object in front of the individual
- Ask, “What is this?”
- Prompt the correct response; give cues as described in a following section
- Reinforce the correct response; give corrective feedback on incorrect responses

Naming in General: Treatment Procedure

- Use cueing hierarchies (response-evoking stimuli arranged in hierarchies as described next)
- Find a stimulus (cue) that evokes the response; use phonemic, personalized, and other kinds of cues as described
- Use a stronger cue only when weaker cues do not evoke the response

- Start with a single cue and add additional cues only when necessary
- Use different types of cues
- Fade the cue so that natural stimuli come to evoke the response
- Regardless of the type of cue, first ask the relevant question (e.g., “What is this?”) and then give the selected cue

Types of Cues

- Modeling: Model the entire word
 - Ask a question (“What is this?”)
 - Immediately model the response (“Say, *a book*”)
 - Let the individual imitate
 - Reinforce the individual for correct imitation
- Sentence completion tasks as cues: Give parts of sentences as cues.
 - Clinician (CN): “You write with a _____?”
 - Individual (CT): “Pen.”
 - CN: “You write with a ball-point _____?”
 - CT: “Pen.”
- Initial sound of words as cues: Give initial sounds as cues.
 - CN: “You write with a (pause); the word starts with a p____.”
 - CT: “Pen.”
- Syllables as cues: Give syllables of words as cues when the sound cue is not effective.
 - CN: “This is a spoo____.”
 - CT: “Spoon.”
- Silent phonetic gestures as cues: Give articulatory postures without vocalizations as cues.
 - CN: “This is a ____.” (silent articulatory posture for *p*).
 - CT: “Pen.”
- The individual-selected personalized experiences as cues: Ask the individual about his or her personal experience with the object and formulate a question based on that experience.

- CN: [showing the picture of a hammer] “Once you hit your finger with it. What is this?”
- CT: “Hammer.”
- CN: [showing the picture of a pen] “This was your retirement gift. What is this?”
- CT: “Pen.”
- Functional descriptions as cues: Give a description of the use of an object as its cues.
 - CN: “This is a round object that you roll or kick. What do you call it?”
 - CT: “Ball.”
- Description and demonstration of an action as a cue: Request the target name, describe its use, and demonstrate an action as a cue.
 - CN: “What is this? You use this to write” (demonstrate writing).
 - CT: “Pen.”
- Individual’s description as cues: Ask the individual to first say what an object is used for and then name it.
 - CN: [showing a pen] “Tell me what you use this for and then tell me its name.”
 - CT: “I use it to write. It is a pen.”
- Individual’s demonstration of functions as cues: Ask the individual to first demonstrate the function of an object and then name it.
 - CN: “Show me how you use this and then tell me the name.”
 - CT: Demonstrates the action of drinking and then says “cup.”
- Objects or pictures with their printed names as cues: Present an object or a picture with its printed name and ask the individual to name it.
 - CN: Presents a book (or a picture of a book), the printed word *book*, and then asks the individual, “What is this?”
 - CT: “Book.”
- An associated sound as a cue: Present a sound associated with an object and then ask the individual to name it.

- CN: “This goes *urf, urf*; what is this?”
- CT: “Dog.”
- An associated smell as a cue: Present an object and let the individual smell the fragrance typically associated with it and then ask the individual to name it.
 - CN: [Presents a fruit or a flower] “Smell this first and then tell me what it is.”
 - CT: Names the object after smelling it.
- A synonym as a cue: Say a commonly used word that has a synonym and ask the individual to say a word that means the same.
 - CN: “Please say the word that means the same as the word *dwelling*”
 - CT: “House.”
- An antonym as a cue: Say a commonly used word that has an antonym and ask the individual to say a word that means the opposite.
 - CN: “Please say the word that means the opposite of the word *woman*”
 - CT: “Man.”
- A typically associated word as a cue. Say a common word with which another word is typically associated.
 - CN: “What word usually goes with the word *spoon*?”
 - CT: “Fork.”
- A superordinate as a cue: Prompt the class (category) name to evoke a specific name within the class.
 - CN: [Showing the picture of a cake] “It is a food item. What do you call it?”
 - CT: “Cake.”
- A rhyming word as a cue: Say a word that rhymes with a target word to evoke the target word.
 - CN : “It rhymes with *hog*. What is it?”
 - CT: “Dog.”
- Fade the special cues: Gradually reduce the amount and extent of cues to have the individual

respond to more natural stimuli; reinstate previously successful cues when necessary; again, try to fade them out; fade cues in various ways (e.g., fading a verbal cue into a mere gestural cue; fading multiple cues into a single cue).

- CN: [presents a picture of a ball; models the word in a soft voice, barely audible] “What is this? Say, *ball*.”
- CT: “Ball.”
- CN: [presents the same picture and mouths the word ball with no voice] “What is this? Say, *ball*.” [the word *ball* is only mouthed]
- CT: “Ball.”
- CN: [presents the same picture; omits all cues] “What is this?”
- CT: “Ball”
- Teach self-cueing: Select naming strategy that has worked for the individual (e.g., production of an antonym, an associated word, or description and demonstration of the use of an object to be named); teach the individual to self-generate effective cues.
 - CN: “What do you think works well for you in recalling the names?”
 - CT: “My own experience associated with a word helps me the most.”
 - CN: “I agree, you had the most success when you could think of your own cues, based on your experience. Every time you have difficulty with a name, think of something special about it. Think how you can relate to it. Then try to say the word.”
 - Show a few new stimuli not used in treatment, ask the individual to think of a personal cue, say the cue, and then say the word; use this strategy to teach self-cueing, using the cues that have worked for the individual.

Treatment of Aphasia: Expansion of Verbal Expressions. Expand words into phrases, phrases

into sentences, and sentences into narratives and conversational speech. For expansion, select verbal expressions that:

- Are most useful to the individual and his or her caregivers
- Are most effective in expressing personal experiences, basic needs, emotions, and thoughts
- Are most meaningful in social contexts to sustain conversation and promote social participation
- Generate target expressions in consultation with the individual, the family members, and health care staff, if involved

Teach Verbal Expressions

- While asking the individual to describe scenes in a picture, model simple sentences for the individual to imitate
- Fade the model and ask a question to evoke the sentence the individual imitated
- Ask questions about the individual's daily activities
- Ask the individual to describe actions in a picture
- Supply functional words and ask the individual to make sentences with those words
- Tell a story and ask the individual to retell it
- Tell a story and ask questions about the details
- Show sequenced pictures and ask the individual to construct a story
- Give such cues as "say more" or "elaborate on that" to have the individual expand limited expressions
- Ask the individual to describe such familiar tasks as making an omelet or a favorite sandwich, planting flowers, mowing the lawn, changing flat tires or light bulbs
- Engage the individual in more naturalistic conversation (e.g., ask about what he or she did that day, what activities are planned for the weekend)
- Have family members engage the individual in conversation in and out of the treatment setting
- Reinforce the individual for correct or functionally appropriate productions

- Give corrective feedback (“That is not correct,” “That was a wrong word,” “How about this _____?”)
- Repeat successful trials several times to strengthen the responses

Treatment of Aphasia: Functional Communication and Maintenance.

Communication skills that are effective in social situations are *functional*; final targets of aphasia treatment; use the following generic treatment procedure and select one of several special programs described later in this section under Treatment of Aphasia: Specific Techniques or Programs; integrate compatible procedures.

- Target effective communication, as opposed to linguistic accuracy
 - Do not correct grammatical errors if the verbal expressions communicate effectively
 - Accept any mode of communication (words, phrases, gestures)
- Select socially effective words, phrases, and sentences:
 - Generate functional target in consultation with the individual and the family members and, possibly, with health care staff
 - Pay special attention to the individual’s need to express personal experiences, bodily needs, emotions, and thoughts
- Design individual-specific treatment programs in which you shape progressively longer utterances
- Start with what the individual can say, perhaps a few words or even syllables
- Add other syllables to create words, or words to create phrases
- Add additional words to create sentences
- Evoke a variety of sentence structures
 - Noun and verb combinations
 - Active declarative sentences
 - Requests, commands, demands
 - Wh-questions
 - Sentences with adjectives and comparatives
 - Yes/no questions

- Sentences with prepositions, pronouns, present progressives, and so forth
- Use special stimuli only when necessary (pictures, modeling, prompting, and so forth)
- Fade-out the special stimuli and fade-in the naturalistic stimuli:
 - Stop modeling the full word (such as, “Say, *pen*”), but provide only the first letter of the word; then show only the articulatory posture for the initial sound of the word; eventually just ask a question, such as “What is this?” or “Can you tell me something about it?”
 - Withdraw pictures the individual has been describing; ask the individual to describe something without the aid of external stimuli
- Reinforce all correct productions
- Move to conversational speech
 - Engage the individual in meaningful, functional conversation
 - Ask the individual to describe personal experiences, hobbies, professional experiences, family-related events, favorite foods, entertainment, books read, vacations taken, and so forth
 - Narrate a story and ask the individual to retell it
 - Role-play Turn-Taking
 - Reinforce the individual for staying on a topic; extend the duration of Topic Maintenance
- Implement a maintenance program
 - Train the individual to generate his or her own cues for better speech
 - Teach the individual to self-monitor
 - Implement treatment in naturalistic settings
 - Use natural response consequences
 - Conduct group sessions in which the individuals learn to monitor and reinforce each other’s verbal or nonverbal expressions and to strengthen social communication; use the following two ways of structuring the group sessions
 - More tightly structure the groups to directly teach the skills or to reinforce the previously taught skills

- More loosely structure the groups to incidentally and indirectly reinforce skills that the clients exhibit in interactions with each other
- Teach the group members to reinforce each other in a positive and supportive manner
- Train health care professionals to support and socially reinforce the communicative behaviors
- Train family members to
 - Evoke and reinforce speech; teach them to prompt, occasionally model, and provide cues that have worked
 - Reduce demands when it is appropriate (e.g., teach the to ask one question at a time)
 - Pay attention to the individual's strengths (e.g., if the individual is very good with gestures, promote their use)
 - Express emotional support for the individual
 - Include the individual in communicative and other social activities

Treatment of Aphasia: Reading Problems.

Treatment of reading problems may or may not be needed for all individuals. When it is, use the following guidelines:

- Assess whether reading skills are important for the individual
- Consider the level of premorbid reading skills and the current need to read
- Teach first the functional reading skills to persons who have mild or moderate aphasia
- Target comprehension of silently read material rather than oral reading
- Target newspaper- and book-reading skills only when functional and basic reading skills are intact or have been regained
- Teach comprehension of printed words in the beginning
 - Have the individual read aloud selected printed words
 - Model and prompt the responses
 - Repeat successful trials for each word

- Have the individual read the words silently and state their meaning
- Provide positive reinforcement and corrective feedback
- Construct phrases and sentences with words already comprehended
- Have the individual read them aloud with the help of modeling and prompting
- Have the individual read them silently and state their meanings
- Present progressively complex reading material and assess comprehension at each level of complexity

Treatment of Aphasia: Writing Problems. When intervention for writing problems is needed, use the following guidelines:

- Consider the level of premorbid writing skills and the current need to write
- Teach functional writing skills to persons who have mild or moderate aphasia
- Consider the preferred hand and whether it is free from neuromuscular disorders
- Consult with the individual, family members, and other caregivers to select words, phrases, and sentences that are important to the individual and are useful in his or her daily living (e.g., names of family members, address and phone numbers, grocery lists, short letters, filling-out forms, writing down appointments)
- Target correct spelling of words and grammatical accuracy of sentences
- Have the individual trace printed letters and words
- Have the individual copy letters and words
- Have the individual write letters and words to dictation
- Have the individual copy sentences
- Have the individual spontaneously write sentences
- Have the individual write paragraphs, short letters, lists, and so forth
- Give writing homework the individual completes
- Train family members to help sustain the writing skills at home

Treatment of Aphasia: Apraxic Speech in Persons with Aphasia.

Persons with aphasia, especially those who have Broca's aphasia, are likely to exhibit apraxia of speech; use the treatment procedures described under Apraxia of Speech.

- Brookshire, R. H. (2015). *An introduction to neurogenic communication disorders* (8th ed.). St. Louis, MO: Mosby Year Book.
- Chapey, R. (Ed.). (2008). *Language intervention strategies in adult aphasia and related neurogenic disorders* (5th ed.). Baltimore, MD: Wolters Kluwer/Lippincott Williams & Wilkins.
- Davis, G. A. (2014). *Aphasia and related cognitive-communicative disorders*. Boston, MA: Pearson.
- Elman, R. J. (Ed.) (2007). *Group treatment for neurogenic communication disorders: The expert clinician's approach* (2nd ed.). Boston, MA: Butterworth-Heinemann.
- Freed, D., Celery, K., & Marshall, R. C. (2004). Effectiveness of personalized and phonological cueing on long-term naming performance by aphasic subjects: A clinical investigation. *Aphasiology*, 18(8), 743–757.
- Hallowell, B. (2017). *Aphasia and other acquired neurogenic language disorders*. San Diego, CA: Plural.
- Hegde, M. N. (2018). *A coursebook on aphasia and other neurogenic language disorders* (4th ed.). San Diego, CA: Plural.
- Helm-Estabrooks, N., Albert, M. L., & Nicholas, M. (2014). *Manual of aphasia therapy* (3rd ed.). Austin, TX: Pro-Ed.
- Milman, L. (2016). An integrated approach for teaching discourse in aphasia: Bridging the gap between language impairment and functional communication. *Topics in Language Disorders*, 36(1), 80–96.
- Payne, J. C. (2014). *Adult neurogenic language disorders: Assessment and treatment* (2nd ed.). San Diego, CA: Plural.

Treatment of Aphasia: Specific Types. Experimentally validated treatment procedures that are specific to certain types of aphasia are limited. In individuals with any type of aphasia, clinicians use behavioral procedures to teach the particular set of skills that are impaired. There are no special procedures for each type of aphasia; for all specific types, use the procedures described under the main entry, Aphasia. The following highlights a few unique concerns for each type.

Broca's Aphasia. In treating clients with this type of nonfluent, effortful, agrammatic speech, use the procedures described under Aphasia and consider the following:

- Increase length of utterances in gradual steps; model progressively longer utterances and ask the individual to imitate
- Increase complexity of responses in gradual steps
- Decrease grammatical errors by reinforcing the correct production of grammatical morphemes in sentences and conversational speech
- Treat naming difficulties
- Decrease stereotypic utterances by giving corrective feedback
- Provide immediate, positive feedback
- Encourage pointing, gestures, drawing, writing, and reading to improve verbal expression
- Teach a sign language system (e.g., AMER-IND) if necessary
- Combine gestures with verbal expressions, as this combination is known to facilitate naming and other verbal expressions
- Find out the compensatory strategies an individual uses (e.g., singing, gesturing, writing key words to effectively communicate) and incorporate them into training; reinforce their use in and out of the clinic
- Reinforce even telegraphic productions and then model more complete productions using the individual's telegraphic productions
- Reinforce the individual's imitation of more complete productions
- Select one of the special programs described under Aphasia; Treatment of Aphasia: Special Programs (e.g., the Helm Elicited Language Program for Syntax Stimulation, Promoting Aphasics' Communicative Effectiveness, Response Elaboration Training)

Crossed Aphasia. To treat this type of atypical aphasia in right-handed individuals with left-language dominance caused by right hemisphere injury, use the

procedures described under [Aphasia](#) and consider the following:

- Treat individuals with crossed aphasia as you would those with [Broca's Aphasia](#) if nonfluent, agrammatic, effortful speech dominates the clinical picture
- Treat individuals with crossed aphasia as you would those with [Wernicke's Aphasia](#) if fluent but jargon-filled speech and poor auditory comprehension are dominant problems
- Note that individual differences are significant, and general statements are difficult to make; tailor the treatment to the deficits found

Global Aphasia. To treat this type of severe aphasia with impaired comprehension and expression, use the procedures described under [Aphasia](#) and consider the following:

- If the traditional aphasia therapy proves ineffective, train the health care staff and family members to:
 - Be alert to the individual's communicative efforts of any kind, including eye contact, head nodding, facial expressions, postures, simple gestures, and so forth
 - Eliminate distractions while talking to the individual
 - Face the individual while talking to him or her
 - Draw the individual's attention before each attempt at communication
 - Speak slowly
 - Pause at syntactic junctures and between stimulus presentations
 - Use appropriate stress and intonation
 - Use short, simple sentences; simplify all messages
 - Pause between sentences
 - Use nonverbal cues to improve communication
 - Allow extra time for the individual to respond
 - Be unhurried
 - State verbally the guessed response from the individual so he or she can confirm it

- Let the individual know if there is difficulty in understanding him or her (e.g., “I am sorry, I do not understand”)
- To implement a more formal treatment program:
 - Establish realistic goals for the individual
 - Select basic, simple, functional words and phrases for initial treatment
 - Teach *yes/no* responses to basic questions
 - Teach a few unequivocal gestures to express basic needs
 - Teach simple line drawing to express basic needs
 - Select words and phrases that express basic needs
 - Accept any mode of response: verbal, gestural, or signed
 - Provide both auditory and visual stimulation; combine verbal responses with gestures
 - Provide multiple stimuli (modeling, pictures, written stimuli, objects, gestures)
 - Model target responses and require immediate imitation; fade modeling
 - Shape the response to achieve more complex forms if found appropriate
 - Provide manual guidance in shaping gestures, nodding the head, and pointing to objects
 - Give prompt, natural, and social reinforcement
 - Teach responses to simple questions
 - Teach simple requests
 - Teach simple descriptions
 - Move to basic conversational skills training if judged appropriate
 - Teach an organized gestural system and consider techniques described under Augmentative Communication (including AMER-IND, Communication Boards, and Blissymbolics)
 - Consider one of the special programs (Aphasia; Treatment of Aphasia: Special Programs, including Visual Action Therapy)
 - Counsel the family about the effects of stroke, the communication problems and prospects of

treatment, home strategies to enhance communication, and so forth

Isolation Aphasia. To treat this rare type of nonfluent aphasia which resembles global aphasia, use the procedures described under [Aphasia](#) and consider the following:

- Design a comprehensive treatment program as individuals with isolation aphasia have skills impaired in many domains except for better preserved repetition skills
- Use the treatment procedures described under [Global Aphasia](#)

Transcortical Motor Aphasia. To treat this type of nonfluent, agrammatic, paraphasic, and telegraphic speech with intact repetition, use the procedures described under [Aphasia](#) and consider the following:

- Use imitation and naming to improve speaking
- Select pictures as stimuli
- Ask the individual to say or write nouns and verbs that the pictures suggest
- If the individual fails, point out dominant aspects of the stimulus or prompt nouns and verbs
- Obtain from the individual or supply three or more words for each picture
- Ask the individual to form sentences with one of the words produced or supplied
- Ask the individual to expand the sentence with other words
- Reinforce all attempts in the right direction

Transcortical Sensory Aphasia. Treat this type of fluent aphasia as you would [Wernicke's Aphasia](#), and address the following special features:

- Target naming skills; see [Aphasia](#), main entry
- Reduce echolalia; use such procedures as nonexclusion time-out and extinction; do not pay attention or reinforce them otherwise
- Reduce neologism and paraphasia with extinction and by teaching specific words in conversational speech

- Improve listening skills and reduce the rush of speech by giving prompts to stop or slow down

Wernicke's Aphasia. To treat this type of fluent aphasia with good or even excessively fluent but meaningless speech, use the procedures described under Aphasia and consider the following:

- Reduce the impulsive and incessant talking
- Structure the treatment sessions and reduce distracting stimuli
- Ask the individual to listen
- Use gestures and manual guidance to stop the individual from talking (touch your lips with your index finger to suggest "be quiet," touch the individual's hand to make him or her stop talking)
- Ask yes/no questions and accept only such answers, not elaborate utterances
- Expand utterances gradually
- Train the individual to listen carefully, instead of rushing to speak
- Reduce the rate of speech directed toward the individual, as this helps improve comprehension
- Train the individual to reduce his or her rate of speech and self-monitor the rate
- Treat auditory comprehension deficits as described under Treatment of Aphasia; Auditory Comprehension
- Use one of the special programs described under Aphasia; Treatment of Aphasia: Special Techniques or Programs, including Treatment for Wernicke's Aphasia (TWA)

Treatment of Aphasia: Specific Techniques or Programs

Communication Drawing Program. A program that teaches drawing to people with aphasia with extremely limited oral language skills so they can express their basic needs; may be considered an augmentative and alternative form of communication, but the interesting part of the program is that a number of individuals studied so far who drew pictures of objects also began to verbally name them; may be appropriate for

individuals who cannot verbally express their basic needs but can use a felt-tip pen with their preferred or nonpreferred hand; the program consists of 10 steps (not all the details given here) of drawing, and to use the procedure, consult the cited source; essentially, the method involves:

- Step 1. Classifying objects into superordinate categories (e.g., placing the picture of a *hammer* into the *tool* category)
- Step 2. Color selected line drawings with their appropriate color (e.g., red for watermelon)
- Step 3. Outlining pictures of objects on paper, without touching the lines of the picture
- Step 4. Copying geometric shapes without touching the exemplars
- Step 5. Completing drawings with missing elements
- Step 6. Drawing objects from memory
- Step 7. Drawing objects to command when a printed word is shown (e.g., present the printed word *eyeglasses* and ask the individual to draw one)
- Step 8. Drawing objects in superordinate category when presented a printed word (e.g., present a printed word *vegetables* and ask the individual to draw a vegetable)
- Step 9. Drawing as many types of animals and modes of transportation as possible on blank paper
- Step 10. Drawing one-, two-, and three-paneled cartooned stories first by showing such cartoons for study, and then asking the individual to draw each from memory

Helm-Estabrooks, N., Albert, M. L., & Nicholas, M. (2014). *Manual of aphasia therapy* (3rd ed.). Austin, TX: Pro-Ed.

Constraint-Induced Language Therapy. This approach limits (constrains) to verbal responses only; designed to remove reinforcement for nonverbal responses and strengthen verbal responses; gestures, drawings, and any other forms of communication are strongly discouraged and verbal responses are reinforced; an intensive treatment program, with up to

fours hours of daily treatment for up to two weeks is typical; a more comprehensive enhanced protocol (CAIT II) targets speech repetition, activities of daily living phrase repetition, language card game, picture description, role play with caregivers to promote everyday verbal interactions, and a generalization; another modified constraint approach, called the *Intensive Language Action Therapy*, accepts and reinforces nonverbal behaviors which help evoke verbal behaviors; to implement the basic program:

- Have two to three clients play a language card game
- Distribute to the participants duplicate (paired) picture cards showing action, colors, number, phonemically similar words, and nouns of both high and low frequency
- Raise cardboard barriers between the participants so that the cards are invisible to each other
- Have a client request a card (e.g., “Mr. Jones, do you have a card with a blue mug?”) and ask the client who has a duplicate of that to give it to the requester (saying, “Yes, Mrs. Smith, I do have it”)
- Have the client who doesn’t have the card respond verbally (“No, Mrs. Smith, I don’t have it”)
- Do not allow any type of nonverbal responses or clues
- Prompt and shape more complex, more polite, and expanded utterances

Johnson, M. L., Taub, E., Harper, L. H., Wade, J. T., Bowman, M. H., Bishop-McKay, S., . . . Uswatte, J. (2014). An enhanced protocol for constraint-induced aphasia therapy II: A case series. *American Journal of Speech-Language Pathology*, 23, 60–72.

Kurland, J., Stanek III, E. J., Stokes, P., Li, M., & Andrianopoulos, M. (2016). Intensive language action therapy in chronic aphasia: A randomized clinical trial examining guidance by constraint. *American Journal of Speech-Language Pathology*, 25, S798–S812.

Melodic Intonation Therapy (MIT). An intervention program for alert and motivated individuals with severe nonfluent aphasia with good auditory

comprehension; developed by M. Albert, R. Sparks, and N. Helm; uses musical intonation, continuous voicing, and rhythmic tapping to teach verbal expression; hierarchically structured; appropriate for individuals with Broca's aphasia who produce some words accurately in a familiar song they sing; contraindicated for individuals with Wernicke's, transcortical motor or sensory, and global aphasia; the program is detailed with a specific scoring procedure not summarized here; the clinician should consult the cited source to use it.

General Procedures

- Select a wide variety of high probability words (minimally two syllables), phrases, and sentences
- Use pictures or environmental cues for each target utterance
- Intone each word, phrase, or sentence slowly and with constant voicing; intoning involves singing the words, phrases, and sentences with melodic patterns
- Maintain pitch and stress variations of normal speech
- Hold the individual's left hand with your right hand, and raise it to tap on the table top once for each intoned syllable
- Signal with your left hand when to listen and when to intone
- Generally, move to the earlier step when the individual fails at a step
- Level I
 - Humming: Show a picture, hum the target item, and tap; no response required
 - Unison singing: Intone in unison with the individual and tap
 - Unison with fading: Intone, tap, and fade half-way through the phrase
 - Immediate repetition: Ask the individual to listen to you as you intone the phrase and tap; let the individual imitate

- Response to a probe question: Following a correct imitation, intone a probe question (e.g., “What did you say?”)
- Level II
 - Introduction of item: Intone the phrase twice and tap; no response required
 - Unison with fading: Intone, tap, and fade half-way through the phrase
 - Delayed repetition: Intone and tap, and after 6 seconds of delay, let the individual tap with assistance; ask the individual to intone without help
 - Response to a probe question: Six seconds following the individual’s response, intone the probe question; do not hand tap; let the individual intone the phrase
- Level III
 - Delayed repetition: Tap and intone and let the individual intone the phrase after 6 seconds and give tapping assistance
 - Introducing *sprechgesang* (speech song) with fading: Present the target phrase twice slowly, without singing, much like in choral speaking, but with exaggerated rhythm and stress; tap the hand but require no responses
 - *Sprechgesang* with fading: Begin the target phrase intoning along with the individual, but fade your voice, so the individual continues alone and completes the phrase
 - Delayed spoken repetition: Present the phrase in normal prosody, without hand tapping and let the individual imitate after 6 seconds in normal prosody
 - Response to a probe question: Ask a probe question with normal prosody after a 6-second delay; let the individual respond with normal prosody

Helm-Estabrooks, N., Albert, M. L., & Nicholas, M. (2014). *Manual of aphasia therapy* (3rd ed.). Austin, TX: Pro-Ed.

Promoting Aphasics' Communicative Effectiveness (PACE).

A program designed to promote face-to-face conversation; developed by G. A. Davis and J. Wilcox; emphasis on exchange of new information, functional communication (as against linguistic precision) with turn-taking, free choice for the individual to communicate in any modality; and natural feedback.

- Use a large number of stimulus cards that contain pictured objects, actions, and stories; stack the cards face down on the table
- Take turns drawing cards from the stack; communicate information about the stimulus
- Encourage any mode of expression (words, gestures, drawings, writing, pointing, or a combination of these)
- Add new stimulus cards to promote the exchange of new information
- Provide natural consequences (e.g., What did you say? Do you mean ____? I am not sure . . .)
- Acknowledge the individual's message while suggesting the correct word or words (e.g., "I understand. You mean *book*, right?")
- Make variations and adaptations
- Exchange the roles of speaker and listener with the individual

Davis, G. A. (2005). PACE revisited. *Aphasiology*, 19(1), 21–38.

Response Elaboration Training. A program designed to expand utterances of individuals with aphasia; emphasis is on shaping and chaining utterances the client, rather than clinician, initiates; allows a wide variety of responses as against a predetermined correct response; may be used in group therapy; may be useful with clients who have both aphasia and apraxia; developed and researched by K. Kearns and his associates.

- Select line drawings to stimulate speech
- Show a stimulus card and evoke an initial response, any response (e.g., the individual may say "Man

. . . sweeping” to a line drawing of a person with a broom)

- Reinforce the individual; also, shape and model the individual’s response (e.g., say, “Great. The man is sweeping”)
- Ask a Wh-question to evoke an elaboration of the initial utterance (e.g., ask “Why is he sweeping?”)
- Reinforce the individual’s elaboration and shape and model the initial response combined with the subsequent elaboration (e.g., the individual may answer by saying “wife . . . mad” and you say, “Way to go! The man is sweeping the floor because his wife is mad”)
- Model the longer response a second time and ask the individual, “Try and say the whole thing after me. Say . . .”
- Ask the individual to imitate after a delay if the individual is successful at the previous step
- Continue until the individual fails to elaborate any more
- Introduce another picture for a similar sequence or initiate a different initial response for the same picture

Kearns, K. P., & Scher, G. P. (1989). The generalization of response elaboration training effects. In T. E. Prescott (Ed.), *Clinical aphasiology* (Vol. 18, pp. 223–245). Austin, TX: Pro-Ed.

Wambaugh, J. L., Wright, S., & Nessler, C. (2012). Modified response elaboration training: A systematic extension with replications. *Aphasiology*, 26(12), 1407–1439.

Script Training. A method to promote conversational skills in clients with aphasia by using scripts the client and the clinician collaboratively write; scripts may be monologues or dialogues; in using this approach:

- Ask the client to suggest personally meaningful topics for monologues or dialogues (e.g., personal stories or experiences); suggest topics that might be of interest to the person
- Help the client generate a script on the selected topic

- Read the script aloud with the person; then have the client read it aloud; have the person read the script at home
- Have the client practice the script in social situations and with conversational partners

Cherney, L. R. (2012). Aphasia treatment intensity, dose parameters, and script training. *International Journal of Speech-Language Pathology*, 14(5), 424–431.

Sentence Production Program for Aphasia. Based on the earlier *Language Program for Syntax Stimulation*, this revised program is designed to increase the production of varied sentence types by individuals with agrammatic aphasia; especially appropriate for clients with agrammatic speech (limited verbal expressions); the method helps teach the following 8 sentence types, each with 15 target sentences:

1. Imperative Intransitive (“Lie down”)
 2. Imperative Transitive (“Wash the dishes”)
 3. Wh-interrogative (“What” and “Who” questions)
 4. Wh-interrogative (“Where” and “When” questions)
 5. Declarative Transitive (“She cleans teeth”)
 6. Declarative Intransitive (“She skates”)
 7. Comparative (“They’re funnier”)
 8. Yes/No Questions (“Did you buy the paper?”)
- Level A involves teaching the target sentences in an imitative form within a story context
 - In Level B, the client is told a brief story and answers questions about the story
 - The criterion is for the client to produce sentences at Level B (more spontaneous than in Level A)
 - Consult the program manual for procedural details
- Helm-Estabrooks, N. Albert, M. L., & Nicholas, M. (2014). *Sentence production program for aphasia* (3rd ed.). Austin, TX: Pro-Ed.

Schuell’s Auditory Stimulation Approach for Aphasia. The method concentrates on intensive auditory stimulation or auditory bombardment; developed by H. Schuell; the method hardly has any evidence in

its favor; it may be more efficient to address auditory comprehension in the context of production training.

- Find varied and abundant stimulus materials (objects and pictures of objects)
- Design a sequence of auditory stimulation
- Work systematically and intensively
- Begin with easy and familiar tasks and increase their complexity; ask the individual to:
 - Point to objects named, described, spelled, and so forth
 - Follow directions (simple to more complex)
 - Answer yes/no questions
 - Respond to alternate items (switch responses) (e.g., “Show me the horse/Tell me your name”)
 - Repeat words, phrases, and sentences
 - Complete your sentences
 - Answer different kinds of questions
 - Form simple sentences
 - Retell stories
 - Describe pictures and events
 - Engage in conversation
 - Copy and write words
- Combine auditory stimulation with visual stimulation
- Evoke responses to each stimulation, but do not force them
- Evoke many and varied responses
- Do not correct responses; instead repeat stimulation
- Give such feedback as visual charting of progress made in treatment sessions
- Introduce new materials that contain or extend old materials

Duffy, J. R. (1994). Schuell’s stimulation approach to rehabilitation. In R. Chapey (Ed.), *Language intervention strategies in adult aphasia* (3rd ed., pp. 146–174). Baltimore, MD: Williams & Wilkins.

Social Approaches to Aphasia Rehabilitation. An overall philosophy of treatment that emphasizes social communication, social integration of the individual

with aphasia, and training family and friends to become conversational partners of the individual; questions the usefulness of teaching grammar or extremely limited, nonfunctional speech; approaches do not discard the traditional language treatment and functional communication skill training, but go beyond them to promote social integration of treated individuals; group therapy is one of the key elements of social approaches; although the approaches have clinically valid arguments, the treatment methods the approaches use need more methodological clarity.

- Target conversational skills in social or real-life situations as the ultimate goal of aphasia treatment
- Design treatment that includes the family members who take active roles in it; let the individual and the family set treatment goals and suggest treatment strategies based on the individual's strengths and needs
- Have the individual, the family members, and caregivers evaluate the individual's improvement in social and natural communication skills and overall satisfaction with the treatment and rehabilitation
- Address social isolation of people with aphasia; in addition to language treatment, target social communication and integration; get the individual involved in community activities, to volunteer in hospitals, participate in social clubs and organizations, speak at support groups, talk to students in academic classes on aphasia, and so forth
- Help rejuvenate such premorbid social activities as playing cards with friends, participating in cook-outs and picnics, watching sports or games, and so forth
- Help create new social skills such as painting with friends or playing musical instruments, going to movies with spouses or friends, tending a garden, and so forth
- Try to move treatment sessions out of the clinic room and into the more natural settings where the skills are prompted or reinforced in subtle ways

(e.g., conduct a conversational skill training session at the individual's home or in a restaurant)

- Train both the person with aphasia and those who interact with him or her in effective social communication skills
- Train health care workers and volunteers in nursing homes where the individual may reside in getting the individual involved in social activities and to support any effective form of communication
- Train the individual as well as the family members and other caregivers in adaptive strategies, as the deficits may not be completely eliminated even with effective treatment (e.g., talking slowly to the individual, accepting gestures from the individual, minimizing demands while not ignoring the individual)

Elman, R. J. (2007). *Group treatment of neurogenic communication disorders: The expert clinician's approach*. Boston, MA: Butterworth-Heinemann.

Simmons-Mackie, N., King, J. M., & Beukelman, D. R. (Eds.) (2013). *Supporting communication for adults with acute and chronic aphasia*. Baltimore, MD: Paul H. Brookes.

Treatment for Wernicke's Aphasia (TWA). A method of aphasia treatment developed by N. Helm-Estabrooks and P. Fitzpatrick to treat auditory comprehension and verbal expression problems; appropriate for individuals with severe Wernicke's aphasia who can read and understand single picturable words; reinforced responses to written stimuli help strengthen verbal comprehension:

- Select a corpus of words printed in lowercase that the individual can read aloud and point to pictured stimuli
- Present a printed word and ask the individual to select a picture that goes with the word (one correct and five foils)
- Ask the individual to read the word aloud
- Turn over the card so the printed word is invisible, say the target word, and ask the individual to repeat after you

- Say the target word and ask the individual to select the picture that goes with the spoken word from among an array of six pictures
- Introduce new words as the individual shows progress
- If new words cannot be introduced by about the fifth session, reevaluate the procedure; select another procedure
- Chart correct and incorrect responses on a recording sheet

Helm-Estabrooks, N., Albert, M. L., & Nicholas, M. (2014). *Manual of aphasia therapy* (3rd ed.). Austin, TX: Pro-Ed.

Visual Action Therapy (VAT). A nonvocal, visual/gestural communication approach for individuals with extremely limited oral or written language skills due to global or mixed nonfluent aphasia, accompanied by some degree of limb and oral apraxia; should be able to spontaneously produce a few everyday gestures (e.g., pointing, waving good-bye); developed by N. Helm-Estabrooks and her associates; neither the clinician nor the individual talk during treatment; an individual who cannot match an object with the tracing of that object is not a good candidate for VAT; more treatment efficacy data are needed; there are three levels, each level has 12 steps; to implement the procedure, consult the cited source.

- Select 15 each of real objects, shaded line drawings of these objects, and action pictures involving the objects
- Select some contextual props (e.g., a screw in a block of wood to use a screwdriver)
- Level I
 - This initial level uses real objects, pictures of objects, and action pictures (e.g., someone sawing a piece of wood, a hand holding a cell phone); steps include placing objects on pictures, placing pictures on objects, pointing to objects, pointing to pictures, object use training, action picture demonstration, following action picture com-

mands, pantomimed gesture demonstration and recognition, representation of hidden object demonstration, and production of gestures for hidden objects

- Level II
 - The second level involves the steps of action picture demonstration, following action picture commands, pantomimed gesture demonstration and recognition, pantomimed gesture production, representation of hidden action pictures demonstration, and production of gestures for action pictures
- Level III
 - The third level involves object picture demonstration, following picture commands, pantomimed gesture demonstration and recognition, pantomimed gesture production, pantomimed hidden object pictures demonstration, and production of gestures for hidden object pictures

Helm-Estabrooks, N., & Albert, M. L. (2014). *Manual of aphasia therapy* (2nd ed.). Austin, TX: Pro-Ed.

Aphonia. See [Voice Disorders](#) to treat loss of voice.

Apraxia. Speech-language pathologists treat disordered volitional movement in the absence of muscle weakness, paralysis, or fatigue mostly in the context of apraxia of speech; see [Apraxia of Speech \(AOS\) in Adults](#).

Apraxia of Speech (AOS) in Adults. Treatment of this neurogenic speech disorder with documented neuropathology in the left cerebral hemisphere with no muscle weakness, paralysis, or muscle fatigue targets speech movement patterns that are impaired, presumably because of central speech planning and programming deficits; treated primarily as a speech articulation and prosodic disorder; see the companion volume, *Hegde's PocketGuide to Communication Disorders* for epidemiology, etiology, and symptomatology of AOS; see also [Childhood Apraxia of Speech](#) in this volume.

Treatment of Apraxia of Speech:

General Considerations

- Make a thorough assessment of apraxia of speech; see the sources cited at the end of this main entry and *Hegde's PocketGuide to Assessment in Speech Language Pathology* for details
- Assess the frequently associated aphasia, dysarthria, or both
- Note that treatment of AOS is essentially behavioral, highly structured, focused on speech production, repetitive, and intensive
- Note that treatment efficacy research on AOS lags behind descriptive research; many treatment procedures have not been experimentally evaluated for their effectiveness; nonetheless, there is clinical evidence that treatment produces improvement, although effectiveness cannot be claimed from such evidence
- Although articulatory accuracy is a technical goal of treatment, consider effective and functional communication as the more practical and important goal of treatment; within this framework, include all communication styles and targets, including gestures and facial expressions, as appropriate communication targets; a combination of such treatment targets may be expected to reduce the impairments to the greatest possible extent
- Ascertain the suitability and acceptance of prosthetic or technological devices to enhance communication, as some individuals may reject them; if individuals are agreeable, carefully selected prosthetic devices (e.g., palatal left prosthesis, nasopharyngeal obturator to reduce hypernasality) may be of benefit, even for a short duration, to achieve functional communication
- Consider a suitable augmentative or alternative communication technique in cases of severely apraxic individuals with extremely limited oral communication potential
- Do not spend time on oral sensation and form recognition exercise in treatment, as they are unlikely to result in improved speech production

- Consider individual preference of certain techniques (e.g., some may not like melodic speech or clinician's manipulation of articulators)
- Drop techniques that do not produce results or modify them to increase their effectiveness
- Defer treatment for AOS until treatment for a severe aphasia produces some language production; defer treatment for individuals who are scheduled for medical or surgical treatment (e.g., removal of a brain tumor) until after such treatments are completed and for those who are depressed or seriously cognitively impaired; counsel the family and reassess treatment candidacy at a later date
- Counsel the individual and the family on the nature of apraxia of speech and expected treatment efforts and possible outcomes; counsel them about the degenerative neuromuscular diseases that may not permit full restoration of communication skills
- Start management as early as possible, but not when the individual is just coming out of an acute phase of the disease (e.g., soon after a stroke); such a individual may be weak and unable to concentrate on treatment tasks
- Select treatment targets and strategies that are functional for the individual, medical caregivers, and the family; make all treatment decisions in consultation with the individual, family members, and other caregivers; consider the typical speaking situation and people who regularly communicate with the individual
- Hold frequent treatment sessions, but determine the frequency based on the individual's physical condition and concentration during treatment
- Organize sessions to move from easy to difficult tasks
- End sessions with success; if errors persist on a certain treatment target, go back to the skills that have been learned better, and end the session with those relatively successful targets
- Emphasize communicative efficiency and naturalness as you would with most individuals in communicative disorders

- Target articulatory accuracy, but only in the context of effective and functional communication
- Carefully sequence the speech tasks; train:
 - Automatic speech before spontaneous speech
 - Frequently occurring sounds before less frequently occurring sounds
 - Stimulable sounds before nonstimulable sounds
 - Sounds in word-initial positions before those in other positions
 - Visible before nonvisible sounds
 - Oral-nasal distinctions before voicing distinctions
 - Voicing distinctions before manner distinctions
 - Manner distinctions before place distinctions
 - Bilabial and lingua-alveolar sounds before others
 - Singletons before clusters
 - High-frequency words before low-frequency words
 - Meaningful words
 - Single-syllable words before multisyllable words
 - Single words before phrases or sentences
- Teach self-monitoring skills

Treatment of Apraxia of Speech:

General Procedures

- Provide counseling and support for the individual and family and discuss the treatment details with the individual and the family
- Always establish systematic, reliable baselines of the individual's skills that will be treated:
 - Note that a description of disorders and impairments does not constitute a baseline
 - Baselines are quantitative, reliable, and detailed measures of skills and the frequency at which they are exhibited; it is the positive change (increase) in the frequency of behaviors during treatment over the baseline that will support a claim of improvement
 - Baselines may include a rating of speech intelligibility, although rating scales are less objective than frequency measures
- Try to improve physiologic support for speech production before starting direct treatment for speech

(e.g., consider modifying poor posture or improving respiratory support for speech); but do not waste too much time trying to improve respiratory support that is unnecessary for speech

- Use effective behavioral treatment procedures:
 - Give instructions on how to produce the desired target and demonstrate the actions
 - Frequently model the target skills
 - Ask the individual to immediately imitate the modeled productions
 - Positively, verbally, and naturally reinforce correct responses; specify why the response is correct
 - Give immediate corrective feedback for errors; experiment with instrumental or biofeedback to judge their effectiveness
 - Offer consistent and repeated trials, especially when the individual begins to produce the target behavior more accurately
- Provide systematic practice in producing the target speech sounds (drill); use discrete trials to repeatedly evoke the same target responses
- Reduce the speech rate initially to improve articulatory accuracy and to convince the individual of the desirable effect of slower rate on intelligibility; increase the speech rate as articulatory accuracy improves and stabilizes
- Use Phonetic Placement Method and Phonetic Derivation
- Use a variety of sounds and sound combinations to promote generalization and maintenance
- Have the individual practice sound productions with meaningful material; avoid nonsense syllables or words that are not functional to the individual
- Target speech tasks during treatment, instead of unrelated nonspeech tasks (e.g., questionable oral-motor tasks unrelated to speech)
- Use contrastive stress tasks in which the linguistic stress is varied within an utterance; see Contrastive Stress Drills for details
- Use the Key Word technique; select words in which a typically misarticulated sound is correctly produced;

use such words to facilitate the correct production of the sound in other words

- Use cueing techniques as a means of fading modeling; for example, show a tongue tip placement or bilabial lip posture for a phoneme production instead of orally modeling the sound or the sound in a word
- Use automatic speech tasks (e.g., counting, reciting the days of the week) to initially evoke speech and quickly move to more spontaneous word, phrase, and sentence productions
- Push on abdomen to achieve vocal fold closure and phonation for the speechless individual
- Use the Shaping method to promote natural prosody; a gradual increase in speech rate while still maintaining intelligibility, varied stress patterns, variations in intonation will help achieve a more natural-sounding speech
- Emphasize total communication (combined use of verbal expressions, gestures, writing, augmentative devices)
- Teach Self-Control (Self-Monitoring) skills
- Use the techniques of teaching Speech Sound Disorders as found appropriate

Treatment of Mild Apraxia of Speech

- Note that persons with mild AOS are:
 - Good candidates for massed-trial treatment of articulatory accuracy
 - Likely to learn to communicate well in social and occupational situations
 - Not likely to exhibit severe forms of aphasia
- Counsel the individual and the family; tell them about the good prospects of recovered or vastly improved communication if warranted
- Keep the focus on articulatory accuracy, good prosody, and appropriate rate of speech
- Select meaningful and personally relevant words for treating sound productions
- Model sound productions in words initially

- Fade modeling in gradual steps
- Move to modeled and evoked productions of phrases and sentences in graduated steps
- Use visible and simple utterances in the beginning
- Give visual feedback of movement of articulators (e.g., “see how I produce the sound and do the same”)
- Encourage the individual to listen to his or her own sound productions to judge their adequacy
- Require immediate imitation of modeled productions because it is easier than delayed imitation
- Reduce the individual’s rate of speech
- Extend treatment to utterances that are more complex and sound productions that are less visible in carefully graded steps
- Use the Phonetic Placement Method
- Use Contrastive Stress Drills to promote articulatory proficiency and prosodic features of speech; in constructing contrastive drill materials:
 - Use a single sound target initially in any phrase or sentence
 - Use simpler and more familiar sounds initially
 - Use shorter phrases or sentences initially
 - Use longer words and sentences subsequently
 - Add more sound targets to each utterance
 - Use infrequently occurring words later
 - Increase rate of speech gradually
- Use the Eight-Step Continuum Treatment, described under Treatment of Apraxia of Speech, Specific Techniques or Programs following this section
- Use the Darley, Aronson, and Brown Procedure for AOS, described under Treatment of Apraxia of Speech, Specific Techniques or Programs following the end of these sections
- Encourage the individual to create original sentences
- Ask open-ended questions
- Encourage the individual to ask questions to practice normal rhythm
- Encourage the individual to read aloud and self-correct mistakes

- Improve ability to talk under stress or interference
- Encourage self-correction
- Increase speed of response (reduced reaction time)

Treatment of Moderate Apraxia of Speech

- Note that persons with moderate AOS are likely to be:
 - Hemiparetic or hemiplegic
 - Mildly or moderately aphasic
 - Apraxic in other respects (e.g., limb apraxia, buccofacial apraxia)
- Counsel the individual and the family about:
 - Variability in symptoms
 - Faster recovery of speech during the earlier weeks and slower recovery later
 - Prospects for improved communication
 - Potential need for long-term speech treatment
 - Need to work hard in treatment
 - Coping strategies
- Encourage the individual to make decisions about the future (returning to work, changing assignments at work, driving, and so forth)
- Use modeling to promote imitation of carefully selected speech sound contrasts
 - Provide individuals with auditory and visual cues
 - Ask the individual to imitate a model
 - Place a single target in varied linguistic contexts (e.g., for the target /t/, a typical list of stimuli might be *tea*, *tie*, *toe*, and *two*)
 - Ask the individual to contrast the target with other sounds
 - Replace single-syllable words with polysyllabic ones
 - Construct phrases and sentences out of practiced words for more practice
 - Make contrasts harder
 - Use slow rate initially with difficult targets
 - Use varying rhythm and stress (e.g., begin with equal and even stress and progress toward normal)
 - Use multiple contrasts
 - Encourage greater independence
- Use Contrastive Stress Drills

- Use Melodic Intonation Therapy (MIT)
- Use imitation initially
- Use a question-and-answer dialogue

Use Reading in Treatment

- Ask the individual to read aloud
- Fade the printed stimuli by having the individual:
 - Look at the text and talk about it
 - Look at the text and wait before talking about it
- Teach the individual to Self-Monitor rate, rhythm, stress, and errors

Treatment of Severe Apraxia of Speech

- Note that persons with severe AOS:
 - May not have much spontaneous speech at all
 - May have other kinds of apraxia (often limb and buccofacial)
 - May be moderately aphasic
 - May be hemiparetic or hemiplegic
- Counsel the family members and the individual
 - Give the family a reasonable statement of prognosis
 - Discuss the severity of accompanying aphasia and how it might complicate apraxia treatment
 - Ask the family members and health care workers to speak slowly, use shorter sentences, reduce background noise, talk only when the individual is focused, and use Total Communication
 - Teach family members and health care staff to use various prompts (cues), including the use of the Cloze Procedure, suggesting the first letter of the word, the first syllable of a word, paraphrasing what the individual may have said for the individual to indicate *yes* or *no*, and so forth.
 - Ask the family and the individual to allow for some failures
 - Ask the family to observe treatment so they may learn to reinforce speech or other types of communication attempts at home
 - Tell the individual what the family members are asked to do and what he or she can expect from treatment and with what efforts

- Educate the other members of the team about the individual's communication problems, strengths, and the treatment program
- Begin direct treatment with modeling and ask the individual to imitate; be aware that it may not work very well with severely apraxic individuals who tend to persevere
- Use the Phonetic Placement Method to help improve articulatory accuracy; encourage the individual to
 - Use manner distinctions (especially plosive and fricative)
 - Use simultaneous manner and place distinctions
 - Make voicing distinctions (evoke any kind of sound, including humming or grunting and then shape it)
 - Make oral-nasal distinctions
- Use Phonetic Derivation (Shaping or progressive approximation) if other techniques fail
- Combine modeling, phonetic placement, and shaping (phonetic derivation) techniques
- Use rhythm to evoke speech sounds, syllables, and words; use aspects of Melodic Intonation Therapy described under Aphasia; Treatment of Aphasia: Specific Techniques or Programs
- Use the Key Word technique to have the individual practice correct articulation
- For the most severely apraxic, consider teaching Augmentative or Alternative Communication skills

Duffy, J. R. (2013). *Motor speech disorders: Substrates, differential diagnosis, and management* (2nd ed.). St. Louis, MO: Elsevier Mosby.

Freed, D. (2012). *Motor speech disorders* (2nd ed.) Clifton Park, NY: Thomson Delmar Learning.

Wambaugh, J. L., Nessler, C., Cameron, R., & Mauszycki, C. (2013). Treatment for acquired apraxia of speech: Examination of treatment intensity and practice schedule. *American Journal of Speech-Language Pathology*, 22, 84–102.

Treatment of Apraxia of Speech: Specific Techniques or Programs

Darley, Aronson, and Brown's Procedure for AOS.
A procedure for treating AOS by systematically shap-

ing speech sound productions and capitalizing on automatic productions. It includes intensive trials for phoneme productions (phonemic drills).

- Shape vowels, consonants, and consonant-vowel (CV) syllables:
 - Identify a vocal response the individual can make (perhaps an “ah”; if not, a voluntary cough or a hum that can be shaped into a vocal response)
 - Have the individual repeat the “ah” or any other phonated sound several times, varying its duration and intensity
 - Shape different vowel sounds from the phonated and repeated sound productions (e.g., “ee,” “oh,” “oo”)
 - Move to consonants; model /m/ for the individual to imitate; give repeated trials
 - Shape syllables out of /m/ (e.g., *me*, *moe*, *moo*)
 - Have the individual produce other combinations of consonants and vowels
- Move to automatic responses; have the individual:
 - Count to 10
 - Recite days of the week and months of the year
 - Produce such routine expressions as “Hello,” “How are you?” “Fine,” “Good morning,” “Thank you,” and so forth
 - Sing songs
- Move to intensive practice of phoneme productions (“phonemic drills”)
 - Produce or hum /m/ and ask the individual to imitate it
 - Model CV combinations with initial /m/ and ask the individual to imitate them (e.g., *my*, *moe*, *maw*, *moo*, *may*, *me*)
 - Model doubled CV combinations and ask the individual to imitate each at least 20 times (e.g. *my-my*)
 - Model consonant-vowel-consonant (CVC) combinations and ask the individual to imitate them (e.g., *mom*, *moom*, *meem*); when successful, evoke CVC combinations without modeling

- Model simple words with initial /m/ and have the individual imitate each 10 to 20 times (e.g., *man*, *mine*, *moon*, *more*); when successful, evoke the words
- Model two-word phrases, both the words with initial /m/, and have the individual imitate them multiple times (e.g., *my mom*, *miss me*, *much more*); when successful, evoke the phrases
- Model two-word phrases, both the words ending in /m/, and have the individual imitate them multiple times (e.g., *come home*, *name him*); when successful, evoke the phrases
- Model two-word phrases with /m/ in the initial position of the first word and in the final position of the second and have the individual imitate them (e.g., *my home*, *meet them*); when successful, evoke the phrases
- Model longer phrases containing multisyllabic words (e.g., *moment by moment*, *Monday morning*)
- Introduce other sounds and sound combinations; use essentially the same procedure to create syllables, words, and phrases
- Introduce sentences containing practiced words and phrases

Darley, F. L., Aronson, A. E., & Brown, J. R. (1975). *Motor speech disorders*. Philadelphia, PA: W. B. Saunders.

Eight-Step Continuum Treatment for AOS. A procedure for treating AOS developed by J. C. Rosenbek and associates. The main feature of this technique is the systematic shaping of words with the use of cues that are faded out; developed by J. C. Rosenbek and associates.

- Select the sounds and words for training
 - Select words for initial training that contain easier sounds in them (vowels, nasals, and stops as against fricatives and affricates)
 - Select words that contain sounds produced with closer articulatory contacts for initial training (e.g., words with only bilabial sounds) and increase the distance between the articulatory

- contact gradually (e.g., words with bilabial and lingua-alveolars moving on to velar sounds)
- Select short, simple, and frequently used words for initial training and increase the length of words gradually
 - Apply a sequenced treatment
 - Say “Watch me” and “Listen to me” and then model a target word
 - Ask the individual to say the word along with you; this is unison production
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word aloud as you silently mouth the word
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word without any cues
 - Say “Watch me” and “Listen to me” and then say the word; ask the individual to produce the word several times without any cues
 - Present the target word printed on a card and ask the individual to produce it
 - Show the printed target word briefly, remove it, and then ask the individual to produce it
 - Ask a question to evoke the target word and let the individual respond with it
 - Arrange role-playing conversational situations in which the individual has opportunities to produce target words; get family members and others involved in this role-playing

Rosenbek, J. C., and associates (1973). A treatment for apraxia of speech in adults. *Journal of Speech and Hearing Disorders*, 38, 462–472.

Prompts for Restructuring Oral Muscular Targets (PROMPTS). A procedure for treating AOS in children and adults that emphasizes the use of tactile-kinesthetic cues in teaching correct production of phonemes; uses manual guidance to position articulators or suggest their appropriate movement; developed by P. Square-Store and D. Hayden.

- Note that kinesthetic and tactile cues may be effective only in the context of speech production; by themselves, such cues may be ineffective
- Learn the cues that are described in the cited source; note that cues are given for articulatory positions, voicing, extent of jaw opening, syllable timing, manner of articulation, coarticulation, and so forth
- Learn the many contact positions on the face, neck, jaw, and so forth
- Provide cues by touching the individual's face, neck, jaw, and other structures
- Manually guide the articulators to correct positions
- Model target words for the individual to imitate
- Give cues if the imitative production was incorrect
- Manually guide the articulators to correct positions
- Ask the individual to produce the sound as the articulators are being moved to correct positions
- Give other cues to suggest articulatory movements that cannot be manually guided

Square-Store, P., & Hayden, D. (1989). PROMPT treatment. In P. Square-Store (Ed.), *Acquired apraxia of speech in aphasic adults* (pp. 165–189). London, UK: Taylor and Francis.

Sound Production Treatment. A treatment procedure developed by Wambaugh and associates; has been researched and supported by evidence, although generalization and maintenance need to be better researched; to use this procedure:

- Select the sounds that are consistently difficult for the individual with AOS
- Develop monosyllabic and multisyllabic words, phrases, and sentences that contain those sounds
- Model the target words, phrases, or sentences
- Use the Minimal Contrast method
- Provide cues for correct articulatory placement (see Phonetic Placement Method)
- Provide repeated practice
- Provide reinforcement for correct responses and feedback for incorrect ones

- Consider using Biofeedback via Electropalatography to improve articulatory accuracy

Wambaugh, J. L., Nessler, C., Wright, S., Mauszycki, S. C., DeLong, C., Berggren, K., & Bailey, D. J. (2017). Effects of blocked and random practice schedule on outcomes of Speech Sound Treatment for acquired apraxia of speech: Results of a group investigation. *Journal of Speech-Language Hearing Research*, 60, 1739–1751.

Artificial Larynx. Mechanical larynges used in the communicative rehabilitation of individuals with laryngectomy; generates a mechanical sound that is articulated into speech with people whose larynges have been removed because of such life-threatening diseases as cancer; the same as Electronic Device for Alaryngeal Speech or Pneumatic Device for Alaryngeal Speech.

Aspiration. To treat aspiration, a condition in which food, fluid, and secretion penetrate below the true vocal folds, causing asphyxiation and aspiration pneumonia, see Dysphagia.

Aspiration Pneumonia. To prevent lung infection due to aspiration associated with swallowing disorders, see Dysphagia.

Assessment. Procedures that include (a) description of an individual's existing and nonexistent communicative behaviors, background variables, and associated factors to evaluate or diagnose a communicative problem; (b) clinical measurement of a person's communicative behaviors.

- Obtain case history
- Interview individual, the caregivers, family members, and others concerned
- Conduct an orofacial examination
- Make individual-specific judgments on use of standardized or nonstandardized measures
- Use measures appropriate to the individual and his or her ethnic, cultural, and linguistic background
- Screen hearing

- Obtain a speech-language sample
- Administer the selected tests
- Analyze results
- Draw conclusions; make a diagnosis; recommend treatment; counsel the individual and his or her family
- Consult the cited sources and the *PocketGuide to Assessment in Speech-Language Pathology*

Hegde, M. N., & Freed, D. (2017). *Assessment of communication disorders in adults: Resources and protocols* (2nd ed.). San Diego, CA: Plural.

Hegde, M. N., & Pomaville, F. (2017). *Assessment of communication disorders in children: Resources and protocols* (3rd ed.). San Diego, CA: Plural.

Assistive Listening Devices. See [Hearing Assistive Technology Systems](#).

Ataxic Dysarthria. To treat this type of motor speech disorder resulting from damage to the cerebellum. See [Dysarthria: Specific Types](#) under [Dysarthria](#).

Attention Disorders. It is essential to improve attention in treating any disorder of communication; if the individual does not attend to treatment stimuli and tasks, the clinician will have to integrate activities to improving attention with communication training; attention deficits affect a person's reaction to stimuli in the environment; they may reduce the duration for which the individual can attend to stimuli; because attention is a prerequisite for all kinds of learning, disturbed attention makes learning difficult or impossible and compounds existing memory deficits; intervention for attention is often necessary while treating individuals with aphasia, dementia, traumatic brain injury, intellectual disabilities, autism spectrum, and similar disorders.

- During assessment and early treatment sessions, take note of such attentional deficits as slow reaction, lack of sustained attention to assessment materials and tasks, frequent shift in attention from one stimuli to the other, distractibility, losing track of topics of conversation, and so forth

- Integrate attentional training activities with other functional activities such as communication training
- Work directly on functional attentional tasks (e.g., paying attention to speech or instructions; to environmental cues; to treatment stimuli) instead of meaningless and nonfunctional attentional tasks (e.g., crossing out circles intermixed with squares on paper; counting backward; listening to auditory clicks and signaling when there is a shift in the pattern of clicks)
- Consult with the family and health care staff in identifying attentional deficits of concern and functional treatment activities and goals
- Design a treatment program that is individual specific and functional and helps the person pay attention to environmental events; consider strategies to improve the individual's attention to listening to speech and speaking, paying attention to conversational topics, and so forth
- Work with the family and health care staff to ensure that they, too, use the same strategies as you do
- Select treatment stimuli that are relevant to the individual's interests and premorbid hobbies; prepare colorful and attention-getting stimuli for treatment
- Begin treatment in a quiet, simple, clutter-free place with minimal stimuli; gradually extend the treatment to more natural settings in which you reinforce attentional behaviors
- Conduct treatment sessions sitting in front of the individual; make sure there is enough lighting on your face and on the treatment stimuli and activities
- Give all treatment instructions in simple, brief, and direct language; repeat, if necessary; break down the instructions to smaller steps; talk slowly; and make sure the individual understands your instructions by asking questions about them
- Draw the individual's attention before presenting all treatment stimuli by giving alerting signals and frequently using his or her name (e.g., "Watch for this now, Mr. Triumph," "Here it comes, Mrs. Robinson," "I am going to show you a picture now, Ms. Lopez," "Mr.

Rodriguez, look at me now,” “Mr. Woo, please pay close attention”); make sure the individual has established eye contact with you or with the stimulus before you present the stimulus

- Verbally reinforce the individual for maintaining eye contact, looking at the treatment stimuli, concentrating on treatment tasks, performing accurately on tasks at hand; paying attention for longer durations
- Give corrective feedback; stop the individual as soon as you find his or her attention wandering; reinstruct, reorient, draw attention, and continue with treatment trials
- Experiment with the loudness of your voice; some individuals may need a louder and more intoned voice; others may react better to soft voice and speech
- Limit choices offered in treatment tasks (e.g., put only a few stimulus item in front of an individual with brain injury)
- Use manual guidance in the case of severely impaired individuals to orient their face toward the stimuli; use touch to draw attention; guide the individual's hand toward required manual tasks (e.g., pointing to a correct picture in a stimulus array)
- Give necessary intertrial interval time; do not rush stimulus presentations; but do not wait too long between stimulus presentations, as this would give room for competing responses
- Judge the individual's physical endurance; hold brief sessions if the individual fatigues easily; take short breaks during treatment; hold sessions at times when the individual is well rested
- Shape attending behaviors; increase the duration of required attention gradually (e.g., initially set a realistic goal of attending to a task for only a few seconds and gradually increase the duration); reinforce for all on-task behaviors
- Teach the individual to make appropriate requests that help him or her concentrate (e.g., requesting others to turn down the music, close a door or a window to reduce outside noise, speak louder or softer)

- Use high probability behaviors to reinforce attending behaviors (e.g., good attending behavior in an earlier portion of treatment may mean that the individual can have a rest or watch TV)
- Teach the individual to assess and verify his or her work to promote self-correction (e.g., ask the individual how she or he did in concentrating on a task on hand; give feedback; encourage realistic self-evaluations)
- Measure duration of attending behaviors and the frequency of inattentive behaviors to document changes and improvements; give this informational feedback to the individual (e.g., “Yesterday you could concentrate for only 2 minutes; today you concentrated for 3 minutes”)
- Teach the individual to cue him or herself (e.g., by repeating instructions and reading a list of steps to be taken in completing a task before beginning the task)
- Teach family members to prompt and reinforce attending behaviors; to draw attention; to repeat instructions; and so forth

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Audience Generalization. A treatment target; production of unreinforced responses in the presence of persons not involved in training; to promote this important type of generalization:

- Invite persons not involved in training to treatment sessions conducted in later stages of therapy
- Evoke and reinforce the target behavior (e.g., fluency, correct production of speech sounds, naming, appropriate vocal qualities) in their presence
- Have the visitor engage the individual in conversational speech
- Have the visitor reinforce the target communication skills

- Take the individual to nonclinical situations and evoke and reinforce target communication skills in the presence of other persons

Auditory Discrimination Training. Treatment designed to teach individuals to distinguish between correct and incorrect articulation of speech produced by the clinician and other persons; used on the assumption that misarticulations are due to a failure to hear differences between different speech sounds and that auditory discrimination training is a precursor to speech sound production training; assumption contradicted by some experimental evidence showing that production training will induce discrimination as well; it is likely that in most cases, production training will be sufficient to generate auditory discrimination as well; the same as Perceptual Training; a clinically practical strategy is to train production first and then probe for discrimination to see if that, too, needs to be trained with the following procedures:

- Describe the target sound, how it is made, and how it sounds
- Produce words and phrases that contain the target sound as well as nontarget sounds
- Ask the individual to respond in some way to the production of the target sound (by raising the right hand or by pointing to the drawing of a smiling face)
- Produce a word with the correct sound and a word with a sound the individual substitutes for a correct sound (e.g., *radio* and *wadio*)
- Ask the individual to respond to the correct production (/r/ in *radio*) in one manner (pointing to the drawing of a smiling face) and to the incorrect production (/w/ in *wadio*) in a different manner (pointing to the drawing of a frowning face)
- Produce many words and phrases that are loaded with the target sounds to provide intense auditory stimulation
- Move on to production training, as there is little or no evidence that auditory discrimination training will result in correct production of speech sounds

Augmentative and Alternative Communication (AAC).

Providing augmentative and alternative modes of communication to individuals with limited oral communication potential is an important area of rehabilitation; augmentative communication enhances and expands extremely limited oral means of communication by non-vocal means; alternative communication replaces oral communication by non-oral modes of expression; the two methods may be integrated in rehabilitation; both include various means, some of which are more technologically oriented than others; appropriate for persons who have limited oral communication skills because of severe clinical conditions, including aphasia, autism, cerebral palsy, and other neurological disorders, severe hearing loss, dementia, dysarthria, glossectomy, intubation, laryngectomy, intellectual and developmental disabilities, tracheostomy, and traumatic head injury; aided and unaided are the two kinds of systems available to teach AAC.

Basic Principles of Selecting an Augmentative Communication Mode or System

- Use the *participation model* in assessment that evaluates the opportunities and access barriers an individual with limited oral communication skills faces
- Assess the individual's existing speech as well as non-speech communication skills
- Consider the individual's strengths and limitations
- Sensory and motor capabilities and limitations
- Spoken language comprehension
- Select a mode or system that gives the maximum advantage to the individual
- Consider cost of the device
- Consider the individual's acceptance of the mode or system
- Consider the communicative demands the individual faces
- Consider the amount of training required
- Consider how the individual and the family will use the mode or system

Augmentative Communication—Aided. Methods of communication in which an individual (a) selects or scans messages displayed on a nonmechanical device (e.g., a communication board), (b) displays messages on a mechanical device (e.g., a computer monitor, a smartphone, a table screen), or (c) uses a dedicated speech synthesizer that “speaks” on his or her behalf; specialized AAC software are available for electronic devices.

Use Pictures and Symbols to Teach Functional Communication

- Teach the individual to communicate with photographs and drawings that may be displayed on a communication board
 - Teach the individual with limited cognitive functions to communicate basic needs with regular or miniaturized objects (e.g., the individual points to a fork to indicate he or she wants to eat)
 - Teach the individual to express a particular message through a picture (e.g., teach the individual to point to or look at a picture of a person sleeping to communicate that he or she is tired or sleepy)
 - Teach the individual to express bodily states (e.g., pain in a certain part) by pointing to or looking at specific body parts on a line drawing
- Teach the individual to communicate with various symbols that may be displayed on a communication board; select among many symbol systems that are available on the market; for instance:
 - Picture Exchange Communication System (PECS), in which a person picks up a picture and hands it to a communication partner to express his or her needs or wants
 - Picsyms, a set of graphic symbols that represent nouns, verbs, and prepositions
 - Pic Symbols (Pictogram Ideogram Communication), which are white drawings on a black background
 - Premack-type Symbols, or the Carrier Symbols which are abstract plastic shapes; associate words and phrases with each shape; teach the individual to

arrange and rearrange the plastic shapes like printed words

- Traditional orthography (e.g., the English alphabet); teach the individual to spell out the word (by pointing to or scanning) along with the alphabet, display digits 1 through 10 and a set of common phrases or sentences so that not every word has to be spelled out or scanned (Scanning in Augmentative Communication)
- Teach the individual to communicate with Rebuses
 - Use rebuses (pictures that represent objects or events along with words, grammatic morphemes, or both)
 - Teach the individual to add grammatic morphemes to a picture or a word (e.g., adding *s* to the picture of a book to suggest *books*)
 - Combine rebuses to form more complex utterances

Use Nonelectronic Communication Boards to Teach Functional Communication

- Teach the individual to communicate with messages on a nonelectronic communication board
 - Design a board of paper, cardboard, fabric, wood, or cork; if practical, prepare a book of symbols and written messages; select a board that all conversational partners can see simultaneously; portable, if necessary; attractive to look at; big enough to contain critical elements of the system; not overwhelmingly big or complex
 - Write symbols (alphabets, orthographic messages, pictures, various kinds of symbols) on separate cards that can be mounted on the board
 - Teach the individual who cannot point (because of extremely limited motoric performance) to scan the message; offer selections and have the individual indicate *Yes* to the right selection (e.g., You point to the word “food,” or a symbol for it, or a picture of a food item; the individual indicates *Yes* or *No*)
 - Teach the individual to encode a message by pointing to a number printed on a separate, smaller,

portable selection chart; have the messages on a larger communication board numbered; let the individual point to a number on the selection chart; decode the number into the message on the board (e.g., if the individual points to #5, it may mean “I am hungry” as per the communication board)

- Teach the individual to directly select the message on the board, instead of a number which stands for a message; teach individuals to select by means of pointing and other hand gestures, finger movements, eye gestures, gaze, headpointers, or headsticks.
- Teach the individual to communicate by drawing symbols or pictures; have the individual draw simple line drawings to communicate; let the individual use paper, magic slate, or any other convenient surface
- Teach the individual to communicate by writing (Traditional Orthography); teach conventional writing to nonverbal children who can master it; teach them initially to write simple, functional messages; increase response complexity

Use Electronic Communication Systems to Teach Functional Communication

- Select an appropriate system for the individual; consider the cost, ease of use, and efficiency of the system
- Select an appropriate and practical switching mechanism that the individual can use with little effort and learn to generate signals for the electronic device (such as those that are specially constructed or a modified or regular microcomputer); consider push switches, push plates (plate-like structures that when touched will generate a signal), large and specially designed keyboards, joy sticks, squeeze bulbs, and several other available types of selection devices
- Select an appropriate display system to show messages when the individual activates the switching mechanism; these may be computer screens, liquid-crystal displays (found on calculators), printed outputs (as with a computer printer), and many other kinds of special displays

- Select an appropriate control electronic unit (a dedicated augmentative communication unit or a computer)
- Teach the individual to use the device; start with simpler messages; give plenty of practice in using the switching mechanism; increase the complexity of messages in gradual steps; train the communicative partners in the environment

Augmentative Communication—Unaided. Methods of communication that use patterned muscle movements (gestures) to enhance oral communication but do not use instruments or external aids; gestures play a crucial role in conveying the speaker's message; appropriate for all persons with severely impaired oral, expressive communication.

- Teach gestures for *Yes* and *No* to all speakers with extremely limited expressive oral communication because of severely impaired motor performance but relatively intact receptive language
 - Teach the individual to gesture *Yes* or *No* in response to a series of common questions with the carrier phrase "Do you want _____?"
 - Shape a clear gesture that all communication partners can understand
 - Model the gesture if necessary
 - Reinforce consistently discriminated responding (individual always gives the gesture that is meant)
 - Consider the following gestures: head movements (side to side for *No* and up and down for *Yes*); eye movements (looking up for *Yes* and down for *No*; blinking once for *Yes* and twice for *No*; blinking the right eye for *Yes* and the left eye for *No*); hand movements (thumbs up for *Yes* and thumbs down for *No*); feet movements (right foot movement for *Yes* and left foot movement for *No*)
- Teach a pattern of eye-blinks that convey certain basic messages; for instance, beyond the *Yes* and *No* teach the individual to:
 - Blink three times to say *I am hungry*
 - Blink four times to say *I am thirsty*

- Blink five times to say *I need to go to the bathroom*
- Teach pointing to objects needed
 - Teach finger pointing
 - Teach pointing by directing gaze
- Teach the Left-Hand Manual Alphabet
 - Consider teaching the left-hand manual alphabet for individuals whose right hand is paralyzed
 - Consider teaching the Manual Shorthand, which combines gestures with letters from the left-hand manual alphabet (*talking hand system*)
- Teach Pantomime
 - Teach the individual to use pantomime along with speech
 - Teach initially a few mimed concepts that help communicative basic needs
 - Expand the mimed repertoire as the individual becomes more competent in its use
 - Fade mimes if and when the individual regains or improves oral speech
- Teach American Indian Hand Talk (AMER-IND)
 - Teach first the signs that express mands (basic needs, requests)
 - Teach the one-hand version for those with one paralyzed hand
 - Teach signs that express concrete ideas first and those that express abstract ideas later
- Teach American Sign Language (ASL or AMESLAN)
 - Select initially the signs that express Mands (basic needs, requests)
 - Teach signs that express concrete ideas first and those that express abstract ideas later

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Augmentative and Alternative Communication, High Technology. Methods of communication that enhance or expand (and rarely substitute for) vocal communication by external means that use sophisticated electronic technology, including computers; generate speech or printed messages; usually software-run; more versatile than low-technology augmentative communication, but more demanding of the user.

Augmentative and Alternative Communication, Low Technology. Methods of communication that enhance or expand (and rarely substitute for) vocal communication by external means that use no or limited electronic technology; there is no message storage, printed output, or speech output; a communication board with letters and words on it is an example.

Aural Rehabilitation. An educational and clinical program implemented, for the most part, by audiologists; includes the assessment of hearing loss in adults and children; counseling; selection and fitting of hearing aids and auditory training; use of group amplification systems in educational and communication training sessions; often implemented by a team of specialists, including an audiologist, otologist, special education specialists, psychologists, and speech-language pathologists; for speech-language pathologists' treatment of communication disorders in persons with hearing problems, see Hearing Loss; the main components of aural rehabilitation include the following:

- Counseling individuals with hearing loss
 - Giving information to persons with hearing loss on available services
 - Helping individuals make appropriate decisions regarding services
 - Educating them about different types of hearing aids and assistive listening devices (hearing assistive technology systems)
 - Answering questions the individuals may have about their problems and needs

- Educating them about available sources of financial help to meet the cost of hearing rehabilitation
- Informing individuals about the nature of services offered by other professionals and making referrals to appropriate selected professionals (e.g., otologists, speech-language pathologists, educators of the deaf)
- Counseling parents of children with hearing loss
 - Giving information on hearing loss, its causes, and its effects on the child's communication, education, and eventual occupation
 - Informing them about the various services needed and available to them and to their child, including special services at the school
 - Helping them with their search for private and public sources of financial help
 - Helping the family develop an aural rehabilitation program for the child through such counseling
- Hearing evaluation
 - Testing hearing and establishing thresholds
 - Testing speech reception and discrimination skills
 - Evaluating such complex functions as central auditory processing
 - Making an accurate assessment of the type and degree of hearing loss, impairment, or disability; and if possible, the causes of hearing loss
- Communication needs assessment to help design the rehabilitation program
 - Assessing the differential communication needs of individuals with hearing loss
 - Evaluating the personal, social, and occupational needs of the individual
 - Using interviews and questionnaires to assess the communication needs
 - Evaluating the emotional reactions to communication problems
 - Understanding the individual's coping strategies
- Hearing aid selection and fitting
 - Conducting a hearing aid evaluation which includes an assessment of acceptability of an aid

- Trying different hearing aids on the individual to evaluate their suitability
- Selecting an aid that is most suitable to the individual's pattern of hearing loss
- Hearing aid orientation
 - Educating the individual about the use of the selected (and purchased) hearing aid (e.g., wearing it, adjusting the volume, changing the battery)
 - Educating the individual about the care of the aid (e.g., protecting the aid from the elements, keeping the ear mold clean, storing it at night time)
- Auditory Training
 - Letting the individual experience various amplified sounds so he or she can get used to it
 - Helping the individual discriminate amplified environmental sounds as well as speech sounds
 - May involve face-to-face training, electronic auditory training with the help of computers, or both types of training
 - Auditory training may be phoneme based, word based, or sentence based; a combination may be effective
- Speechreading
 - Training the individual to understand speech as much as possible by all available visual as well as auditory clues by watching the lips, face, and associated gestures of the speaker
 - Taking into account the individual differences in learning to speechread and adjusting the duration of training accordingly
- Teaching the individual effective communication strategies
 - Teaching the client to exhibit appropriate assertiveness to seek help from others, request modification of speaking styles (e.g., to request louder speech or getting closer before talking)
 - Teaching individuals to position themselves correctly so they can see the face of the speaker
 - Teaching the person conversational repair strategies (e.g., letting the speakers know that he or she was not understood)

- Teaching the client to request clarification, repetition, restatement of what the speaker said to make sure the understanding was correct
- Teaching effective strategies to communication partners
 - Teaching the client's partners to modify their speech to promote effective communication
 - Teaching them to not shout from a distance, face individuals while talking, turn the lights on before talking, not cover their mouths while talking, and so forth
 - Teaching them to provide background information to the topic of conversation, not to change the topic abruptly, and so forth
 - Teaching them to avoid excessively loud, overly articulated speech
- Working with other professionals on interdisciplinary teams
 - Helping speech-language pathologists design and implement appropriate oral communication training programs and serving as a resource and consultant
 - Helping educators of the deaf in designing and implementing appropriate educational goals and serving as a resource and consultant
 - Working with otologists and helping make decisions about surgical or medical interventions
 - Working with pediatricians, psychologists, and other professionals as found necessary in individual cases
- Selecting Hearing Assistive Technology Systems
 - Counseling individuals about the availability of various assistive listening devices
 - Helping the individual select one or more devices that best improve the individual's communication in social situations, helps the client watch television and speak on the telephone; see Hearing Assistive Technology Systems
 - Helping the individual understand use and care of devices
- Offering clinical speech-language services
 - A responsibility of the speech-language pathologist
 - See Hearing Loss
- Follow-up and continued support

- Keeping in touch with the individuals and their families
- Periodically evaluating the individual's hearing status and communication needs
- Updating hearing aids or assistive listening devices
- Helping the individual and the family meet the changing needs of aural rehabilitation

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Autism Spectrum Disorder (ASD). Treatment of ASD requires a team of specialists, including speech-language pathologists, applied behavior analysts, and special educators; this entry is concerned with speech-language services but it should always be coordinated with the services of other specialists; severe general behavioral impairments may make it difficult to teach communication skills until those impairments are addressed; see *Hegde Pocket-Guide to Communication Disorders* for epidemiology, etiology, symptomatology, and theories of autism; see also *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for evaluating communication disorders; many of the treatment procedures for Language Disorders in Children are applicable; as per the DSM-5, entries for Asperger's disorder, childhood disintegrative disorder, and the Rett disorder have been eliminated, as they are not separate disorders within the spectrum; note that the treatment procedure described here is comprehensive and integrative of most of the specific evidence-based approaches (e.g., Milieu Therapy or Discrete Trials); when their effective components are analyzed, many treatment programs described as unique or special contain behavioral procedures described under this entry.

Treatment of Autism Spectrum Disorder: General Guidelines

- Note that behavioral methods of teaching appropriate behaviors, including communicative behaviors, have received the most extensive experimental support

- Note also that such popular procedures as sensory integration therapy, auditory integration training, and facilitated communication have not proven effective and should be avoided; be wary of many “new and revolutionary” methods periodically advocated and sometimes hailed in public media with no experimental evidence
- Integrate communication training with other skill training by working with behavior analysts, special educators, and other professionals
- Use a consistent set of goals that all members of an interdisciplinary team will support
- Let other professionals know the target skills you will be teaching and ask them to incorporate your goals into their treatment (e.g., if you are teaching a set of basic words, give the list to other professionals so that they can prompt and reinforce the production of those words)
- Seek information on what other professionals are doing and support their efforts by integrating their goals into your work (e.g., if the educator is teaching the child to answer questions appropriately, teach correct responses to different types of questions in language therapy)
- Make language therapy relevant to the child’s educational program; select language therapy goals in consultation with the regular and special education teachers
- Teach observable, measurable skills so the progress or lack of it can be evident
- Consider the three types of teaching, all known to be effective in teaching skills to children with ASD: direct teaching, activity-based teaching, and incidental teaching
- Use direct teaching to establish initial communication skills (e.g., words and phrases; grammatic morphemes; articulation of speech sounds); use the discrete trial methods in which target skills are taught with massed trials; use modeling, prompting, shaping, fading, and similar behavioral techniques; positively reinforce correct responses

- Use activity-based instruction to teach more advanced language skills or to have the child expand established language skills; in a play-oriented setup, teach advanced language skills as you engage the child in conversation (e.g., the production of plural or other grammatic morphemes in sentences)
- Use the Incidental Teaching Method to promote even more naturalistic language production; reinforce a child's attempt to communicate in naturalistic contexts (e.g., when a child points to banana on top of the refrigerator, say the word *banana* several times and give it to the child); see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques
- Expect resistance, interfering behaviors, aggressive behaviors, self-stimulation, inattention, stereotypic behaviors—all interrupting and disrupting your planned communication treatment; acquire the basic skills in behavioral techniques that help control such problems (e.g., extinction, differential reinforcement of desirable behaviors that replace problem behaviors, prevention of escape, simplifying tasks, teaching requests that will reduce uncooperative behaviors, and so forth; see Direct Methods of Response Reduction, Indirect Methods of Response Reduction, and Differential Reinforcement)
- Keep the child motivated and on-task by structuring short sessions; positively reinforcing even small improvements in behaviors, keeping quiet, sitting even for as short a duration as 1 minute; allowing frequent breaks in which the child is free to play; demanding less and demanding what the child is capable of doing
- Increase gradually the task complexity, the demand level, and session durations; decrease gradually the frequency and duration of breaks; begin to control what the child does during breaks
- Sit in front of the child, with the child's legs between yours; gradually, move the chairs closer to the table and begin working off the table top
- Select appropriate and strong reinforcers because motivating children with autism spectrum disorders is

extremely important; much of the success may depend on whether you have a functional reinforcer for the child; use tokens and have a variety of backup reinforcers available; select activities the child chooses on his or her own; make access to those activities the backup reinforcer for tokens

Teaching Language and Communication Skills

- Use a variety of stimuli to promote generalized production of target skills in natural settings:
 - Use objects whenever possible
 - Use colorful and clear pictures
 - Ask parents to bring the child's toys, books, hobby materials, and so forth to be used in treatment sessions
- Teach in a variety of linguistic contexts:
 - Teach grammatic morphemes in several phrases and sentences with varied linguistic contexts
 - Teach the same sentence structure (e.g., simple active declarative sentence form) with different words in them
- Teach multiple exemplars of the same target skill; use the Discrete Trials procedure:
 - Select at least 20 exemplars for each language target (e.g., 20 phrases or sentences that contain the present progressive *ing* or one of the regular plural morphemes)
 - Teach multiple exemplars, probe, and teach additional exemplars if the probes do not show generalized productions of untrained exemplars
- Teach in a variety of environments; use the Milieu Teaching approach (see under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs)
 - Train parents to conduct home treatment sessions
 - Teach initially in a quiet room, but shift training to less formal situations and everyday environments of the child
 - Take the child to the cafeteria or library or for a walk on the campus to informally reinforce the target communication skills

- Train teachers to praise the child for appropriate communication skills in the classroom
- Reduce Echolalia:
 - Find out if echolalia is functional; hand the object to the child or ask if he or she wants what the echolalic response included or suggested; if the child agrees or accepts what is offered, perhaps echolalia is functional
 - Ignore echolalia completely; it may have been maintained by attention
 - Give corrective feedback whenever the child echoes
 - Teach alternative, communicative skills
- Give direct, intensive training:
 - Use the method of Discrete Trials, known for its effectiveness
 - Repeat trials on the same target skill until the behavior is reliably produced
 - Provide frequent training; the more frequent the sessions per week, the better the effects
 - Target useful skills and teach them directly
- Reinforce any attempt at appropriate communication:
 - Accept any form of appropriate communication, especially in the initial stages of treatment
 - Teach words, phrases, gestures, functional echolalia
 - Gradually shift the reinforcement contingency to more precise, socially appropriate, complex, verbal responses
- Reduce autistic leading (tendency to grasp an adult's hand and leading to a desired object):
 - First, teach the child to point to objects desired; give the object pointed to
 - Next, teach the child to request the object verbally by modeling it
 - Withdraw reinforcement for pointing, but maintain reinforcement for verbal requests
- Target eye contact during conversation:
 - Periodically, say "look at me!" during treatment sessions
 - Stop talking when the child averts eye contact
 - Start talking when the child resumes eye contact

- Praise the child for making and maintaining eye contact
- Reduce pronoun reversal by teaching the correct use of *I* and *you*:
 - To teach the pronoun *I*, ask the child to perform an action such as clapping hands
 - Ask, “What are you doing?”; prompt the correct response: “*I* am clapping”
 - To teach the pronoun *you*, perform a similar action and ask, “What am I doing?”; prompt the response, “You are clapping,”
 - Fade the modeling
- Teach conversational turn-taking:
 - Use a toy microphone to play the role of a speaker; teach the child to exchange the microphone with the statement, “It is your turn”
 - Teach the child to speak only when holding the microphone
 - Request the microphone and talk; and hand the microphone to the child by saying, “It is your turn”
 - Stop the child if he or she interrupts you
 - Fade the use of microphone (e.g., by keeping it in front, but not holding it; eventually removing it from view)
- Teach verbal requests through the mand-model method; the procedure has experimental support:
 - Teach parents to observe the child as he or she approaches or looks at a desired object (placed out-of-reach)
 - Teach parents to immediately prompt a verbal request by saying, “Tell me what you want”
 - Teach parents to expand the child’s response if necessary (e.g., the parents might expand the child’s single word response, “ball,” into a sentence by saying, “Tell me, *I want the ball*”)
 - Use the same procedure in treatment sessions
- Teach appropriate social behaviors with the help of a *video modeling technique*; the method has experimental support:

- Show videotaped social interactions among children to the child with an autism spectrum disorder
- Show multiple and varied patterns of appropriate interactions
- Instruct, model, and reinforce attentive watching of the videotaped interactions
- Measure the effect of such watching on the child's subsequent social interactions
- Use the *time delay* procedure to teach spontaneous requests or other kinds of verbal behaviors; the method has experimental support:
 - Present a picture to the child that is expected to evoke a response
 - Do not immediately prompt or model the response; instead, delay your prompt a few seconds to see if the child would give a response
 - Immediately reinforce the response; however, prompt if the child did not respond within a few seconds
 - Once again, delay prompts; progressively, increase the delay duration as the child begins to respond spontaneously to stabilize such spontaneous requests, comments, or kinds of verbal behaviors
- Use the *Visual Schedule* method, which shows a pictorial representation of scheduled activities at certain times; such schedules help regulate the flow of behavior:
 - Teach parents to prepare visual display of daily activities (breakfast time, dinner time, homework time, bedtime); have them pictorially represent the different actions involved in that activity (e.g., setting the table, sitting on the chair, eating during breakfast time); train parents to draw the child's attention to the schedule and use it to regulate the child's behavior
 - Have the teacher prepare a visual display of daily classroom activities (to be used in school); and train the teacher in teaching the child to use the schedule
 - Prepare a treatment session activity schedule into which communication treatment is integrated (e.g.,

- painting, working with puzzles, listening to stories); teach the child to consult the schedule and move from one activity to the other in an orderly fashion
- Teach topic maintenance by using multiple topics:
 - Ask the child to talk about an event or a weekend activity
 - Prompt the child to say “more” about the topic
 - Require the child to say three or four new things about the topic
 - Extend the required number of new pieces of information
 - Generously reinforce the child for talking on the same topic
 - Stop the child immediately when he or she abruptly switches a topic or interjects extraneous comments
 - Consider using the Picture Exchange Communication System (PECS), in which the child is taught to exchange a picture for a desired object; the method has experimental support:
 - Display several pictures in front of the child; have the objects displayed available to you
 - When the child picks up a picture that shows a desired item, extend your hand to receive the picture
 - As soon as the child hands you the picture, give the object depicted on the picture to immediately reinforce the child
 - Model the word the picture represents, but initially give the picture even if the child did not say the word; subsequently, reinforce the child for verbal expressions
 - Teach other forms of nonverbal communication (e.g., American Sign Language) if necessary:
 - Present an object, and sign the word for it; say the word as you sign it
 - Manually guide the sign if necessary (forming the finger shapes with)
 - Initially, reinforce the sign response even if the child did not produce the word
 - Later, emphasize the vocal expression and reinforce it

- Reduce socially inappropriate verbal expressions
 - Analyze and target a sample of socially inappropriate and tactless verbal expressions of the child
 - Target reductions in such inappropriate expressions by replacing them with appropriate expressions
 - Give prompt corrective feedback for all inappropriate expressions; make sure that you do not inadvertently reinforce them because they are funny or “cute”
 - Try Extinction, in which you completely ignore inappropriate behaviors (this means not giving corrective feedback)
 - Use Differential Reinforcement techniques to teach alternative, desirable expressions
- Reduce undesirable behaviors by positively reinforcing alternative, incompatible, desirable behaviors that will replace the undesirable behaviors; note that especially verbal requests, when taught, will reduce many undesirable behaviors:
 - Make a functional analysis of undesirable behaviors to find out the cause that maintains such behaviors; eliminate those causes to reduce the undesirable behaviors; see Differential Reinforcement for several specific methods of indirectly reducing undesirable behaviors; see also Direct Methods of Response Reduction
 - Ask the child to put together a difficult puzzle; when the child shows signs of getting upset, ask the child to say, “Help me!”; when the child makes the request, immediately offer help by showing how to put a few pieces together; repeat the procedure several times before the puzzle is completed
 - Let the child play with another child in a small group treatment session; teach each child to request (instead of grabbing) a toy, a book, or some play or project material
 - Keep a desired object on a shelf the child cannot reach; teach the child to request the object by modeling a form of a request; reinforce the child immediately by handing the object to the child

- See Differential Reinforcement to teach many other forms of alternative behaviors that replace undesirable behaviors
- Pay special attention to generalization and maintenance strategies:
 - Note that working with the families is the most critical element in generalization and maintenance of clinically established skills
 - Whenever possible, conduct a few treatment sessions at home
 - Train parents to prompt and reinforce desirable behaviors at home; if possible, train them to conduct brief treatment sessions at home
 - Periodically assess the child and offer booster treatment
 - Work closely with other specialists and family members:
 - Review parents' home session, give feedback
 - Review the progress of other professionals (e.g., teachers, behavior analysts); consult or give feedback as necessary
 - Supply periodic progress data to other professionals and consult with them about the future course of intervention

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Washington, DC: Author.

Wetherby, A. M., & Prizant, B. M. (2000). *Autism spectrum disorders*. Baltimore, MD: Paul H. Brookes.

Automatic Reinforcers. Sensory consequences of responses that reinforce those responses (e.g., the sensation a child with autism derives from banging his or her head); because the consequences are automatic, reinforcement is swift; to reduce behaviors that are automatically reinforced:

- Prevent automatic sensory consequences of undesirable behaviors during treatment
- Make the child wear a helmet that eliminates sensory stimulation from head-banging

- Cover the table with a soft and padded material so that a table banging would not produce the reinforcing auditory stimulation
- See if providing stimulating and reinforcing activities would reduce undesirable behaviors maintained by automatic reinforcers because often such behaviors may be due to sensory deprivation

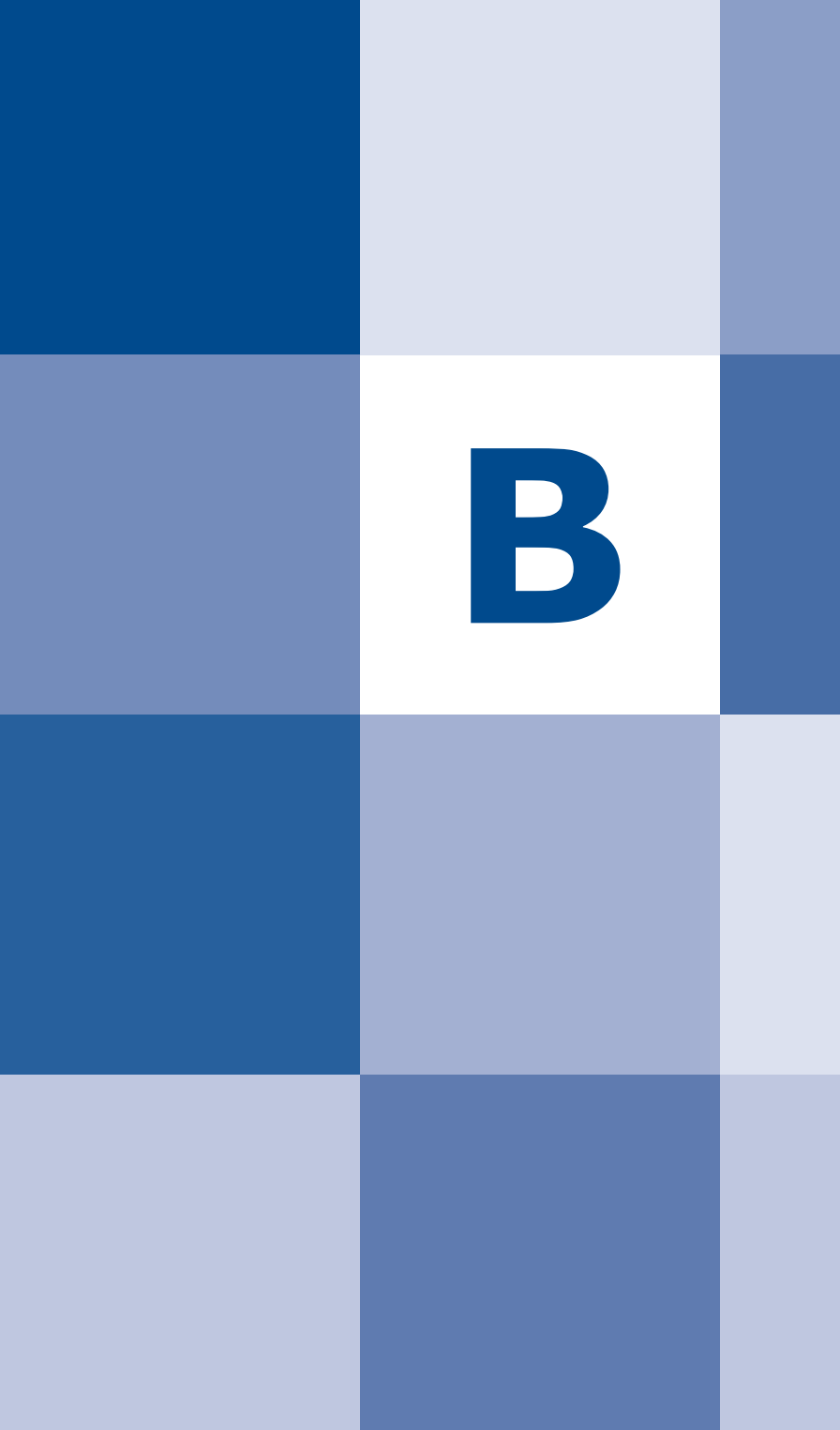
Aversive Stimuli. Events that people work hard to avoid or move away from; reduction in aversive stimulation is the essence of negative reinforcement; a behavior that reduces negative experiences tends to increase in frequency; in treatment, positive reinforcement is preferable to negative reinforcement; however, negative reinforcement derived from avoiding aversive treatment may maintain many kinds of undesirable behaviors children exhibit during treatment; to control them:

- Make a functional analysis of the child's undesirable behaviors (find out what maintains them)
- For example, if the child is leaving the seat during treatment—a negatively reinforced behavior—it may be due to aversive treatment trials from which the child is seeking escape
- Reduce treatment aversiveness; present more powerful reinforcers; simplify treatment tasks
- Alternatively, physically prevent escape

Avoidance. A behavior that prevents the occurrence of an aversive event and hence is reinforced; negatively reinforced behavior; in treatment, target is to reduce avoidance if judged undesirable, a typical target in persons who stutter; typically, the individual is made to face previously avoided situations with appropriate clinical support; for instance, in reducing avoidance of certain speaking situations by persons who stutter:

- Build a hierarchy of most to least frequently avoided speaking situations or tasks
- As the individual becomes more fluent during treatment, introduce the individual to least frequently avoided situations first and move up the hierarchy

- Offer training in situations the individual avoids; for instance:
 - Take the individual to a restaurant and have him or her order food (an avoided responses); monitor fluency and provide subtle reinforcement and corrective feedback
 - Have the individual make phone calls as you monitor fluency (e.g., prompt the person to slow down)
 - Arrange group situations in which the individual will speak or make brief presentations
 - Introduce strangers to the treatment setting and let the individual practice fluency skills in front of them
- Generally, reinforce the individual for facing previously avoided situations and tasks



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Backup Reinforcers. Events, objects, and opportunities for selected actions that become available to individuals who exchange their earned tokens in treatment sessions.

- Have a collection of backup reinforcers
- Have the child select a backup reinforcer for that particular session (e.g., a child might select a toy car, a chance to paint)
- Give tokens to reinforce target responses (e.g., for correct articulations, language responses, fluent productions)
- Exchange tokens for selected events, objects, or opportunities for actions at the end of the session (e.g., the child gets the toy or a chance to paint for 3 minutes at the end of the session)

Baseline Evoked Trials. Establish baselines of target skills on discrete trials on which the expected responses are not modeled but evoked with natural kinds of interactions (e.g., such questions as “What is this?”); measure each attempt to produce a target behavior separately; do not offer consequences for the correct or incorrect responses.

- Place stimulus item in front of individual (e.g., a picture of a *ball*) or demonstrate an action (e.g., moving a toy car)
- Ask the relevant predetermined question (e.g., “What is this?” “What am I doing?” or “What is happening?”)
- Wait a few seconds for the individual to respond
- Record the individual’s response on the recording sheet
- Remove the stimulus item (move it toward you, away from the individual)
- Wait 2–3 seconds to signify the end of a trial
- Begin the next trial with a different item

Baseline Modeled Trials. Establish baselines of target skills on discrete trials in which you model the correct response for the individual to imitate; measure each attempt at target behavior imitation separately; offer no consequences for the correct or incorrect responses.

- Place a stimulus item in front of the individual or demonstrate an action
- Ask the predetermined question (e.g., “What is this?”)

- Immediately model the correct response (e.g., “Johnny, say *ball*”)
- Wait a few seconds for the individual to respond
- Record the individual’s response on the recording sheet
- Remove the stimulus item (move it toward you, away from the individual)
- Wait 2–3 seconds to signify the end of a trial
- Begin the next trial with a different stimulus item

Baselines. Establishing baselines of target behaviors is the initial step in treating all individuals; baselines are recorded rates of responses in the absence of planned intervention (absence of reinforcement for correct responses and corrective feedback for incorrect responses); reliable (repeated) measures help establish the need for treatment and demonstrate improvement during treatment; in experimental treatment research, baselines help rule out extraneous variables; see Baselines, Conversational Speech and Discrete Trials, Evoked and Discrete Trials, Modeled for specific baseline procedures.

Baselines, Conversational Speech. Establish a measure of target behaviors produced in conversational speech in the absence of treatment before starting treatment; do not reinforce or offer corrective feedback.

- Record a conversational speech sample in as naturalistic a manner as possible
 - With children, have toys, pictures, books, and other materials to evoke speech; engage the child in conversational speech with the help of the materials; if necessary, focus on the target features to be measured (e.g., drawing the child’s attention to actions you perform to evoke the *ing*)
 - With adults, hold conversation on their favorite topics
 - In most cases, the individual interview might also be used to establish baselines of target behaviors (e.g., language characteristics, fluency or stuttering, vocal quality)
- Measure the correct and incorrect productions of the target behaviors in the sample
- Calculate the percent correct baseline response rate

Baselines, Discrete Trials. Establish baselines of target behaviors in discrete trials in which an individual's multiple attempts to produce a target response are counted separately; trials are separated in time, hence the name; establish discrete trial baselines in both the evoked trials and modeled trials; do not offer reinforcers or corrective feedback for the responses.

- Select target behaviors (phoneme productions, grammatic morphemes, sentence structures, pragmatic skills, fluent productions, naming skills, etc.)
- Specify target behaviors in measurable terms; for instance
 - Production of /s/ in word initial positions
 - Production of present progressive *ing*
 - Naming pictures
 - Reduced rate of speech
 - Elimination of hard glottal attacks
- Prepare stimulus items to evoke target responses; in the case of speech and language targets, prepare 20 stimulus items for each target response; for instance
 - Twenty pictures that help evoke 20 words with /s/ in the initial position
 - Twenty sentences with the present progressive feature in them (e.g., The boy is walking)
- Prepare questions to be asked to evoke the response, and the exact way of modeling the response
- Prepare a recording sheet
- Administer the two types of trials: Baseline Evoked Trials and Baseline Modeled Trials
- Analyze data to calculate percentage of correct responses (e.g., 50% correct production of the /s/ in word initial positions; 75% correct production of *ing* in sentences)
- Repeat measures; compare the discrete trial and conversational speech measures
- When measures are stable, begin treatment

Behavioral Contingency. In behavioral analysis and treatment, a dependent relationship between Antecedents, responses, and Consequences; the clinician manages this contingency by:

- Providing antecedents (stimuli, modeling, instruction, demonstration, etc.)
- Requiring a specified response
- Providing immediate consequences in the form of positive reinforcers or corrective feedback

Behavioral Momentum. A behavioral treatment procedure in which the clinician rapidly and repeatedly evokes a high-probability response and then immediately commands a low-probability response; often used to reduce non-compliance; in increasing the frequency of a low-probability response:

- Find a response the individual readily performs (e.g., hand clapping)
- Model and have the child imitate that high-probability response repeatedly and in rapid succession
- While the child is still performing the high-probability action, quickly interject a request to perform a low-probability target response (e.g., ask the child to open his or her mouth, a low-probability response)
- Reinforce the occurrence of the low-probability response

Binswanger Disease. To treat communication disorders associated with this type of vascular dementia, see [Dementia](#) and [Vascular Dementia](#).

Biofeedback. A method used to reduce incorrect responses or shape and increase desirable responses in treatment; includes mechanical feedback given to the individual on vocal pitch and intensity, respiration, electropalatography, and muscle action potential level.

Bite Block. A custom-made small block of acrylic or putty for an individual who holds it between the lateral upper and lower teeth; observed to improve speech intelligibility in individuals who have abnormal jaw movements; recommended for some individuals with dysarthria.

Blissymbolics. A nonverbal communication system for individuals with severe communication deficits; a set of symbols used to communicate non-orally; originally meant to be an international language; more widely applied and researched than other symbol systems in teaching communication to severely handicapped individuals; symbols may be combined to form complex expressions; developed by C. Bliss.

Booster Treatment. Give booster treatment any time after the individual was dismissed from the original treatment to help maintain clinically established skills; part of response maintenance strategy.

- Conduct periodic follow-ups
- If the follow-up measures show decline in response rate, give booster treatment
- Use the original or newer, more effective, procedures

Botulinum Toxin Injection. A medical treatment procedure for neurogenic or idiopathic adductor spasmodic dysphonia and adductor spasmodic dysphonia that does not respond to behavioral treatment; botulinum toxin is injected into the thyroarytenoid muscle unilaterally or bilaterally; effects last about three months.

Bound Morphemes. Significant treatment targets for children with language disorders; teaching grammatic morphemes that are inflected with words (hence the name, *bound*) is essential to remediate childhood language disorders; intervention targets include such bound morphemes as the present progressive *ing*, the allomorphic variations of the regular plural and regular past, various prefixes (e.g., *pre-*, *post-*, *anti-*) and various suffixes (which include the regular plural and past inflections); to teach bound morphemes:

- Select the morphemes to be taught
- Develop stimulus materials (words, phrases, or sentences) and pictures or objects
- Present a stimulus item and ask a relevant question (e.g., present the picture of two books and ask the question,

- “What are these?”) and immediately model the correct response (e.g., “Say, *these are books*”)
- Reinforce with verbal praise if the imitated response is correct, and give corrective feedback if the response is absent or incorrect
 - Fade modeling, train multiple exemplars, and probe to see if the child would produce the trained morpheme in untrained contexts (words, phrases, or sentences)
 - See [Language Disorders in Children](#) for additional information

Breathiness. Target this voice quality for reduction during treatment; it is a voice quality that results when there is excessive air leakage during phonation because of inadequate approximation of the vocal folds; caused by various factors; treatment varies by cause; see [Voice Disorders](#).

Broca’s Aphasia. To treat this type of nonfluent and agrammatic aphasia with effortful speech production, see [Aphasia: Specific Types](#).

Buccofacial Apraxia. Treat the difficulties in buccofacial movements when requested in the context of [Apraxia of Speech](#).

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Carrier Symbols. Use these plastic symbols (adapted from the Premack symbols) to teach nonverbal communication to children and adults with limited oral communication skills; they are used as a part of the Non-SLIP (Non–Speech Language Initiation Program); once learned, the program helps initiate oral speech training; developed and researched by J. Carrier.

Carryover. Generalized production of clinically established behaviors in natural and untreated settings, typically as responses to novel stimuli; an important goal of clinical intervention; the same as Generalization.

Cerebral Palsy. Treatment of communication disorders associated with cerebral palsy requires a thorough understanding of the consequences of the neuromotor problems of the disorder and how disorders affect speech production; cerebral palsy is a congenital, nonprogressive neurological disorder that affects motor control, caused by injury to the cerebral levels during the prenatal or perinatal period; not being a degenerative disorder, the symptoms tend to improve with growth; causes speech disorders classified as Dysarthria; symptoms related to speech include respiratory control problems, laryngeal dysfunction resulting in voice problems, possible velopharyngeal inadequacy, potential language disorders, and significant speech sound disorders (dysarthria); may involve cognitive functions; may be associated with feeding problems; see the cited sources at the end of this main entry and *Hegde's PocketGuide to Communication Disorders* for the epidemiology and ethnocultural factors, symptomatology, and classification of cerebral palsy and its etiology.

Treatment of Cerebral Palsy:
General Principles

- Make a thorough assessment of communication disorders and the associated neuromuscular problems; see the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*
- Work closely with the team of specialists serving children with cerebral palsy; the work of physical thera-

pists may be especially useful in improving muscle strength and controlling unwanted reflexes

- Counsel parents about the effects of cerebral palsy on communication and their role in stimulating language at home
- Work closely with parents throughout the treatment duration
- Make a thorough assessment of communication problems and design treatment to suit the child's problems, needs, and strengths
- Consider educational demands made or to be made on the child in planning treatment; work closely with educators
- Borrow techniques from other communicative disorders in children (e.g., language disorders, Speech Sound Disorders, Dysarthria, and Voice Disorders) as cerebral palsy is not the name of a unique speech disorder; modify the standard techniques to suit the individual child and his or her specific symptom complex

Treatment Procedures

- The issue of non-speech oral-motor exercises
 - Note that some experts advocate the use of feeding and oral-motor exercises to improve non-speech neuromuscular control, while others reject this because non-speech oral movements are mediated at a lower level in the central nervous system than the speech movements which are mediated at the cortical level
 - There is no controlled evidence to show that oral-motor exercises are effective in promoting better speech production
 - It is generally more efficient to improve speech-related oral movements through speech training, instead of non-speech movement training
- Treatment of language disorders
 - Train parents to stimulate language at home; see Language Disorders in Infants and Toddlers; Language Stimulation by Parents
 - Assess the child's language development periodically to determine the need for formal clinical treatment

- Implement formal language treatment if necessary
- Use the treatment procedures described under [Language Disorders in Children](#) and [Language Disorders in Infants and Toddlers](#); modify the procedures to suit the individual child with cerebral palsy
- Treatment of voice disorders
 - Diagnose the specific voice disorder; when appropriate, use one or more treatment techniques described under [Voice Disorders](#) with suitable modifications; be aware that voice disorders may be due to respiratory problems associated with cerebral palsy
 - Treat associated respiratory problems; prescribe exercises to improve breath support for speech; use techniques described under [Dysarthria, Treatment](#)
 - Treat velopharyngeal incompetence only if there is enough tissue mass, and behavioral training thus is likely to be effective; see [Treatment of Voice Disorders](#) and [Treatment of Disorders of Resonance](#)
- Treatment of speech sound disorders
 - Assess the child's specific sound errors and error patterns
 - Assess the compensatory articulatory postures the child uses
 - Modify or eliminate inappropriate and ineffective compensatory postures
 - Teach the specific phonemes or classes of phonemes based on distinctive features or phonological patterns
 - Use the treatment techniques described under [Speech Sound Disorders](#) with appropriate modifications; see also [Dysarthria](#) for additional treatment suggestions
- Managing associated attentional problems
 - Behaviorally treat the attention deficits the child may exhibit during treatment sessions
 - Positively reinforce progressively longer periods of attention to treatment stimuli; also consider using a token system to reinforce attending behaviors
 - Draw attention to treatment stimuli as soon as the child shows signs of distraction; praise the child for paying attention
- Managing associated sensory problems

- Refer the child to an audiologist for any needed auditory diagnosis
- Check the child's hearing aid to ensure that it is working properly
- Use FM and other auditory trainers if needed
- Ask the child to wear any prescription glasses during treatment
- Make needed accommodations for any auditory or visual limitations (e.g., conducting treatment in well-lit rooms, making sure that the light is on your face when you talk)

Yorkston, K. M., Beukelman, D. R., Strand, E. A., & Hakel, M. (2010). *Management of motor speech disorders in children and adults* (3rd ed.). Austin, TX: Pro-Ed.

Chaining. A behavioral technique of linking elements of a complex skill; similar to shaping; useful in teaching a variety of nonverbal and verbal skills, although most frequently employed in teaching self-help skills; includes forward chaining and backward chaining.

- Initially, make a task analysis and identify the steps involved in achieving a complex task; e.g., in teaching a child to tie his or her shoelaces, identify such different individual skills as (1) inserting the foot into a shoe, (2) pulling the lace ends together, (3) making a bow, and (4) making the final knot
- Use backward chaining to teach the skills in the reverse order and then practice the skill in an integrated manner:
 - Prompt and manually guide the child through skill 3; stop direct assistance and prompt the child to take the final step of making a knot (skill 4)
 - Starting all over, prompt and manually guide the child through skill 2; prompt the child to perform skill 3, making a bow out of the two lace ends
 - Starting over, prompt and manually guide the child to perform skill 1; stop and prompt the child to perform skill 2
 - Starting over, prompt the child to perform all four skills, ending in a knotted shoelace

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- Use forward chaining to teach skills in their sequence of first skill component to last skill and practice the total skills in an integrated manner
 - Use essentially the same procedure as in backward chaining except begin with skill 1 and move through the sequence

Changing Criterion, Research Design. A single-subject research design to evaluate treatment effects; documented changes in target behaviors that approximate a changing criterion of performance show the treatment effectiveness; in successive stages of treatment, the behavior is held to a lower or higher criterion; undesirable behaviors are held to progressively lower criteria; desirable behaviors are held to progressively higher criteria.

Changing Criterion, Treatment Procedure. A method of shaping desirable behaviors by using performance criteria that change every time the individual meets a certain criterion; the criterion may change in either direction (lower or higher) depending on the target behavior; in reducing the speech rate of individuals with certain communicative disorders, the criteria are progressively lower; in shaping longer utterances, the criteria are progressively higher; for instance:

- Use changing criteria to reduce undesirable behaviors
 - To reduce misarticulation of a phoneme, set a target for the child (a criterion such as “Try *not* to make more than five errors in this session; in the last session, you made eight errors”) and state it at the beginning of each session
 - Decrease the number of errors allowed by setting progressively smaller criteria of performance (e.g., “Try *not* to make more than three errors today; last time you made five errors,” “Try *not* to make more than one error today; last time you made three errors”) as the previous criterion was met and sustained
 - Use similar criteria to reduce or eliminate the frequency of asking irrelevant questions, leaving the chair, dysfluencies, naming errors, and so forth

- Use progressively smaller temporal criteria to reduce such problem behaviors as high-pitched voice (e.g., “Try *not* to produce a high-pitched voice for more than five minutes; last time your pitch was too high for seven minutes,” “Try *not* to produce a high-pitched voice for more than three minutes; last time your pitch was too high for five minutes”); use similar criteria to reduce other behaviors measured on the basis of duration (including hoarse or harsh voice, resonance disorders, and such undesirable behaviors as looking away from the stimulus material or fidgeting)
- Use changing criteria to increase desirable target skills
 - To increase the frequency of correct production of phonemes, grammatic morphemes, naming skills, and so forth, require progressively increasing criteria of performance; set a target (a criterion of performance) at the beginning of each session and state it (e.g., “Let us see if you can increase correct responses to 80% today; you were 76% correct last time”).
 - Progressively increase the criterion of performance as the previous one is met and sustained
 - Use similarly progressive temporal criteria to increase behaviors measured for their duration (e.g., “Let us see if you can sit quietly and look at the pictures for two minutes today; you did it for one minute last time,” “Let us see if you can talk fluently for five minutes today; you did it for three minutes last time”)

Chant-Talk. To use this voice therapy technique to reduce problems associated with hyperfunctional voice, see [Voice Disorders: Specific Normal Voice Facilitation Techniques.](#)

Chewing Method. To use this voice therapy technique, see [Voice Disorder: Specific Normal Voice Facilitation Techniques.](#)

Child-Centered Approach. A general intervention philosophy that assigns a more active role to the child; treatment sessions may be play oriented and the treatment may be indirect; the clinician takes the child’s lead

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in targeting language structures for intervention; see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Childhood Apraxia of Speech (CAS). Treatment of apraxia of speech in children has been controversial for many years, as this diagnostic entity was both vigorously advocated and seriously questioned by experts; in recent years, it has been generally agreed that the problem does exist, although its theoretical explanation is still unclear; also called *developmental apraxia of speech*, but the term *childhood apraxia of speech* is gaining greater acceptance; it is a speech disorder in children that shares some common characteristics with Apraxia of Speech (AOS) in Adults, but without documented neuropathology (hence the controversy); treatment of CAS is primarily concerned with the articulatory (speech sound) disorders characterized by problems in positioning and sequentially moving muscles for the volitional production of speech; treatment should address associated prosodic problems and the possibility of coexisting language disorders, reading problems, and writing deficiencies; controlled treatment efficacy data remain extremely limited; most treatment programs are only suggestive and are often modeled after treatment of apraxia of speech in adults.

Treatment of CAS: General Guidelines

- There is much controversy about the best approaches to treating childhood apraxia of speech
- Most clinicians tend to use the well-established methods of treating Speech Sound Disorders, but some experts believe that such methods are inappropriate and ineffective and may give false hopes and disappoint children and their families; others believe that certain aspects of traditional articulation treatment are useful but certain other aspects are unnecessary
- Possibly, a strictly phonological approach that emphasizes patterns and rules and auditory discrimination drills may be unnecessary, whereas imitation of articulatory postures, correct placement and movement of articulators, and repeated practice of motor produc-

tion are useful; emphasis on sequencing the target sounds may be more productive than drills on individual sounds

- There is a clinical opinion that AOS is much more difficult to treat than the regular disorders of speech production; if AOS is very severe, clinicians tend to judge the prognosis for significant improvement as poor, although much treatment research is needed to establish prospects of improvement or lack thereof
- Several treatment programs or suggestions described in the literature are not detailed or specific enough to be useful to most clinicians; in addition, most are untested for their effectiveness
- The need for oral motor exercises and oral sensory training that are unrelated to speech production is unclear, and in spite of their popularity their effectiveness remains undocumented; therefore, it is more efficient to concentrate on stabilizing movements that are involved in speech production
- Effective behavioral treatment to address the dominant symptoms that seem to affect intelligibility the most seems to be a pragmatic approach to treating CAS

Treatment Approaches for CAS

- Teach correct articulatory placement and movement to improve speech production
- Avoid auditory discrimination training, as the main difficulty is sequencing the sounds
- Avoid nonspeech oral-motor exercises
 - Plan on providing intensive treatment to children with CAS; two or three weekly sessions conducted over several months may be needed to show improvement in speech production and intelligibility
 - Move hierarchically from easy to difficult tasks; use a behavioral shaping procedure in which progressively more complex patterns of speech movements are targeted; consider using the Kaufman Speech to Language Protocol
 - Determine at what level the child will respond (phonemes, syllables, words); start with the level

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- the child is proficient at; move through more complex skill levels (shaping)
- Use meaningful speech stimuli in treatment instead of nonsense syllables or nonspeech movements
- Concentrate on vowels and consonants that children normally produce early; alternatively, concentrate on the more advanced or more difficult sounds that if learned to produce correctly, may make easier and earlier-acquired sounds more readily taught
- Teach consonants that are visible before you teach those that are not visible
- Teach phonemes that occur often or the phonemes the child misarticulates most consistently
- Teach voiceless consonants before voiced consonants
- Model the productions
- Cue the production of the sound or sound sequences before the child's attempted production
- Teach progressively more complex utterances
- Fade modeling and cueing in gradual steps
- Reduce the speech rate if necessary to improve speech intelligibility
- Reinforce the correct responses with effective reinforcers
- Use multiple repetitions of speech productions
- Use extensive drill; stress sequence of movements involved in speech production
- Use selected aspects of treating Speech Sound Disorders
- Note that such techniques as melodic intonation and rhythmic foot or finger tapping have been advocated; it is not clear whether they produce long-standing effects when such external stimuli are faded out or forgotten
- Provide visual, auditory, kinesthetic, or tactile cues or a combination that works best with a child
 - Give touch, pressure, kinesthetic, and proprioceptive cues as in the PROMPT program, developed by Hayden and associates

- Give visual cues that may be hand shapes and gestures or may involve biofeedback through computerized programs and electropalatography
- Address associated problems:
 - Treat such resonance problems as hypernasality or nasal emission; see [Cleft Palate](#) and [Voice Disorders](#) for details
 - Treat prosodic problems by teaching variable intonation; teach normal stress patterns with selected sentences in which the stress may be varied; consider using [Contrastive Stress Drills](#)
 - Treat language problems; see [Language Disorders in Children](#) for specific procedures for teaching semantic, grammatic, and pragmatic aspects of language production
 - Treat fluency disorders if they are clinically significant; see [Fluency Disorders](#) for treatment options
- Teach [Self-Control \(Self-Monitoring\) Skills](#)
- Consider the needs of a bilingual child with CAS
 - Determine the dominant language of the child
 - Do not expect generalized production across the two languages of treated sounds that are unique to one or the other language
 - If practical, treat the child in both the languages, which may promote greater improvement in speech sound production
 - Target sounds that are common to the two languages of the child
 - Use stimuli that are familiar to the child, especially in the bilingual environment
 - Errors unique to a given language may need to be targeted, but do not expect generalization to the other language; for instance, teaching word-final consonants taught in English may not generalize to Spanish if those consonants are not in that position in Spanish
 - See [Language Disorders in Ethnoculturally Diverse Groups](#) for additional suggestions
- Consider the need for augmentative and alternative communication (AAC)

- Children with extremely severe forms of CAS may be candidates for AAC either as a temporary form of communication at the beginning of verbal speech training or as a potentially permanent form
- See Augmentative and Alternative Communication (AAC)

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Yorkston, K. M., Beukelman, D. R., Strand, E. A., & Hakel, M. (2110). *Management of motor speech disorders in children and adults* (3rd ed.). Austin, TX: Pro-Ed.

Cleft. An opening in an anatomic structure, such as the upper lip or the palate, that is normally closed.

Cleft Lip. To treat communication disorders associated with it, see Cleft Palate.

Cleft Palate. Treatment of children with cleft palate with or without cleft lip is a team effort, with speech-language pathologists attending to the needs of communication; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology, etiology, symptomatology, and communication impairments associated with clefts of the palate with or without the clefts of the lips.

Surgical Management of the Clefts

- *Lip Surgery.* Surgical methods to close unilateral or bilateral clefts; usually done when the baby is about 3 months old or weighs about 10 pounds.
- *Palatal Surgery.* Surgical procedures performed to close the cleft or clefts of the palate; done when the baby is between 9 and 24 months, many between 9 and 12 months; the earlier the closure of the cleft, the better the speech development.
- *Variations in surgical methods.* Lip and palatal surgical methods vary across surgeons; consult the sources listed at the end for an overview of surgical methods.
- *Delayed Hard Palate Closure.* A surgical sequence to close the cleft in which the soft palatal cleft is closed first and the hard palatal cleft is closed later.
- *Palatal lengthening.* Surgical procedures to increase the length of the soft palate to improve velopharyngeal function.
- *Primary Surgery for the Clefts.* The initial surgery in which the clefts are closed.
- *Posterior Pharyngeal Flaps.* A secondary palatal surgical procedure designed to improve the velopharyngeal functioning for speech; a muscular flap is cut from the posterior pharyngeal wall, raised, and attached to the velum; the flap is open on either side to allow for nasal breathing, nasal drainage, and production of nasal speech sounds; helps close the velopharyngeal port and thus reduce hypernasality.
- *Pharyngoplasty.* A surgical procedure designed to improve velopharyngeal incompetence; such substances as Teflon, silicone, Dacron wool/silicone gel bag, and cartilage may be implanted or injected into the posterior pharyngeal wall to make it bulge and thus help close the velopharyngeal port.
- *Secondary Surgeries for Clefts.* Surgical procedures done after the primary surgery to improve functioning and appearance.

Cleft Palate: Treatment for Communication Disorders

General Principles

- Establish a good working relationship with all the members of the cleft palate team; in public school setting, stay in touch with the child's physician, the plastic surgeon, the otolaryngologist, the audiologist, and educational specialists that serve the child
- Make a thorough assessment of articulation (speech sound production) skills and velopharyngeal function before starting treatment; see the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for details
- Treat resonance disorders if the child has at least a marginal velopharyngeal competence
- Refer children with significant velopharyngeal incompetence to a laryngologist for possible surgery, prosthetic assistance, or both
- Offer treatment as early as possible
- Emphasize speech production (not auditory discrimination)
- Conduct trial therapy to determine prognosis
- Use behavioral principles and procedures in teaching correct speech sound production
- Integrate speech sound and language interventions, as this combination is known to be effective
- Use the techniques described under Speech Sound Disorders in children without clefts to promote better sound production in children with repaired clefts; modify the procedures or the targets to suit children with repaired clefts
- Work with teachers or special education specialists serving the child to meet the communication as well as educational needs of the child; select targets, especially language targets that support the child's academic success in the classroom
- Integrate literacy goals (e.g., reading and writing skills the child may be deficient in) with communication treatment

Treatment Procedures: Speech Sound Disorders.

Use the procedures of treating Speech Sound Disorders; consider the following suggestions, some of which are unique to children with repaired clefts.

- Educate the parents about the speech mechanism, the effects of the clefts, and speech problems that accompany the clefts
- Educate the parents about communication treatment, the speech and resonance targets that will be the main concerns in treatment
- Ask parents to withhold reinforcement for undesirable compensatory behaviors, the need for which has been eliminated by medical or speech treatment
- Teach the more visible sounds before the less visible except for the linguadentals
- Teach stops and fricatives before other classes of sounds
- Avoid or postpone training on /k/ and /g/ if the velopharyngeal functioning is inadequate
- Teach fricatives, affricates, or both if they are stimutable or after stops are mastered
- Teach linguapalatal sounds, lingua-alveolars, and linguadentals in that order
- Progress from syllables to words, phrases, and sentences
- Give auditory and visual cues; model the correct productions frequently for the child to imitate
- Provide systematic practice and reinforce correct productions; provide corrective feedback for incorrect productions
- Introduce compensatory articulatory positioning where appropriate
- Teach the individual to direct the breath stream orally; let the child feel the airstream on hand or see the movement of a piece of tissue
- Teach the child to avoid posterior articulatory placements
- Teach the child to articulate with less effort and facial grimacing

- Give tactile cues and instruction to improve tongue positioning
- Work on generalization and maintenance; train parents to reinforce correct articulation at home
- Use the Enhanced Milieu Teaching with an emphasis on phonological (speech sound production) skills

Cleft Lip and Palate: Treatment of Language Disorders

General Principles

- It is essential to determine if language stimulation by parents is all that is available; if so, the parents need to be trained on providing language treatment at home
- It is necessary to conduct a few language treatment sessions in the clinic to demonstrate the procedure to the parents
- A formal language treatment may be necessary if assessment data warrant it
- There may be a need to work with the parents from early infancy to establish a long-term rapport
- Counseling parents about language development is essential
- The basic language treatment procedures are not much different from those used with Language Disorders in Children without clefts
- The Enhanced Milieu Teaching method with an emphasis on speech sound production, helps target both speech and language skills

Treatment Procedures: Language Disorders. Use the procedures of treating Language Disorders in Children without clefts; consider the following suggestions, some of which are unique to children with clefts.

- Teach individuals to stimulate language at home
- Ask parents to encourage free verbal expression in their child
- Ask parents to integrate stimulation for articulation and language
- Integrate information about all aspects of rehabilitation in your discussion with the parents

- Ask parents to socially reinforce the child's spontaneous verbal productions
- Teach parents to reduce negative feedback, and make more positive statements about the child's communicative attempts
- Meet with parents regularly to review progress and modify their home language stimulation program
- Assess the child's language skills at regular intervals
- Start formal language treatment when one of the periodic assessments warrant it
- Consider the imminent educational demands and plan language intervention to help meet them

Cleft Lip and Palate: Treatment of Phonatory Disorders

General Principles

- Phonatory problems may be due to compensatory behaviors or may be independent of velopharyngeal insufficiency
- Techniques described under Voice Disorders may be appropriate if the problems are independent of velopharyngeal insufficiency (and due to vocal abuse)
- Phonatory treatment is diagnostic; to be discontinued if there is no improvement in phonatory problems or other speech symptoms worsen
- Children with a clear diagnosis of velopharyngeal incompetence are not candidates for phonatory treatment
- Follow-up of treated children is essential because some improve, some deteriorate, and some stay the same

Treatment Procedures for Hyperfunctional Voice

- Describe how voice is produced to the child and the parents
- Reduce Vocally Abusive Behaviors in the child and in other members of the family
- Counsel the family about good vocal behaviors (e.g., talking less in noisy environments, practicing soft speech, good conversational turn-taking, clapping instead of shouting or yelling)

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- Contrast the child's voice with those of children without vocal nodules or other vocal pathology
- Use such biofeedback instruments as the VisiPitch in training
- Train healthy voice production by teaching the child to
 - Reduce vocal loudness
 - Eliminate hard glottal attacks
 - Initiate words that start with vowels
 - Use easy, gentle onset of phonation
 - Self-monitor voice

Cleft Lip and Palate: Treatment for Resonance Disorders

General Principles

- Hypernasality that is a result of velopharyngeal incompetence is not a treatment target
- Only the child who is capable of achieving velopharyngeal closure should be offered intervention for hypernasality
- Certain surgical procedures may reduce or eliminate hypernasality; improvement may continue for up to a year following surgery

Treatment Procedures: Resonance Disorders

- Do not treat hypernasality and nasal emission if they are due to insufficient muscle mass (physiologically insufficient velopharyngeal mechanism), as speech treatment cannot increase the muscle mass
- Treat hypernasality and nasal resonance only if they are functional, following surgical or prosthetic intervention, or only as trial therapy to assess the suitability of continued treatment
- See Velopharyngeal Dysfunction for additional suggestions and guidelines on treating hypernasality and audible nasal emission
- Use techniques described under Voice Disorders to reduce hypernasality, increase vocal loudness, promote discrimination between oral and nasal resonance, lower the pitch, and increase oral opening
- Use respiratory training to improve loudness

- Attempt articulation with the nares occluded
- Decrease intra-oral breath pressure on stop consonants and fricatives while simultaneously using loose articulatory contacts
- Use such biofeedback instruments as Tonar II to reduce hypernasality

Bzoch, K. R. (2004). *Communicative disorders related to cleft palate* (5th ed.). Austin, TX: Pro-Ed

Kaiser, A. P., Scherer, N. J., Frey, J. R., & Roberts, M. (2017). The effects of enhanced milieu teaching with phonological emphasis on the speech and language skills of young children with cleft palate: A pilot study. *American Journal of Speech-Language Pathology*, 26, 806–818.

Peterson-Falzone, S. J., Hardin-Jones, M. A., & Karnell, M. P. (2001). *Cleft palate speech* (3rd ed.). St. Louis, MO: Mosby.

Zajac, D. J., & Vallino, L. D. (2017). *Evaluation and management of cleft lip and palate*. San Diego, CA: Plural.

Client-Specific Strategy. Use this strategy to select target communication skills for individuals with all kinds of disorders; it is a method of selecting target behaviors that are relevant, useful, and functional for the individual in that it promotes effective communication in social contexts.

- Observe the individual's environment for clues to functional targets; interview family members, colleagues, teachers, peers or friends, professional caregivers, and others who interact with the individual regularly to select treatment targets
- Study the educational, occupational, and social demands made on the individual
- Select targets that are useful and relevant to the particular individual; select targets that help meet the educational, occupational, and social demands made on the individual
- Select targets that will immediately enhance the individual's communication in natural settings
- Select targets that have potential for generalized productions
- Select targets that serve as building blocks for more complex communicative behaviors

Hegde, M. N. (1998). *Treatment procedures in communication disorders* (3rd ed.). Austin, TX: Pro-Ed.

Closed-Head Injury. To treat communication disorders and related deficits associated with closed-head injury, also known as nonpenetrating head injury, see [Traumatic Brain Injury](#).

Cloze Procedure. Use this treatment procedure in which you model parts of an utterance and pause for the child to produce words and phrases to complete the utterance; it is essentially a sentence completion method; the same as [Partial Modeling](#) and [Completion](#).

- Model only the initial portion of a target response (e.g., say “The boy is . . .” and wait for the response)
- Let the child complete the partial model (e.g., the child says, “Walking”)
- Reinforce the child’s response

Collagen Injection. A medical treatment procedure for individuals with paralyzed vocal cords; injected into the middle third of the cord, collagen increases the bulk and the chances of adduction to produce phonation.

Communication. Social behavior that affects other individuals; need not always be produced under social conditions, as in writing meant for others to read or messages sent via electronic media; traditionally described as exchange of information through various verbal or non-verbal actions; more or less organized behavior; target of treatment in individuals with communicative disorders; its various forms include:

- *Aided communication:* Communication achieved through the assistance of such external devices as paper and pencil, communication boards, and computers.
- *Alternative communication:* Communication achieved by non-oral means; all modes other than verbal.
- *Augmentative communication:* Oral or verbal communication that is in some way limited but enhanced or

expanded by aided or unaided alternative communication means, including electronic devices, communication boards, and paper and pencil.

- *Manual communication:* Communication achieved by signs, gestures, and symbols and without oral speech.
- *Nonverbal communication:* Communication achieved without oral speech; may be in the form of signs, gestures, facial expressions, and symbols.
- *Simultaneous communication:* Communication achieved through multiple means, including oral speech, signs, symbols, and gestures.
- *Total communication:* Communication achieved through the simultaneous use of verbal expressions as well as a sign language (e.g., American Sign Language).
- *Unaided communication:* Communication achieved without the help of external means; normal oral communication that is accompanied by typical gestures and expressions.
- *Verbal communication:* Communication achieved through spoken words and language; may be accompanied by culturally appropriate normal gestures and facial expressions.

Communication Boards. Use this low-technology augmentative or alternative communication system to teach individuals with limited oral skills; it consists of boards on which letters, words, phrases, sentences, symbols, or pictures are pasted for the individual to point to, touch, or select in any manner possible to communicate; see [Augmentative and Alternative Communication](#) for procedural details.

Compensatory Strategies. Teach these means of achieving communication in unusual or atypical means to compensate for residual deficits; individuals in whom certain undesirable effects of disabilities linger and cannot be eliminated are candidates for compensatory strategies; the strategies help achieve communication in spite of organic, intellectual, or other kinds of limitations; in articulation, methods of producing speech sounds in atypical means

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because of neuroanatomic deficiencies; in adult communication rehabilitation (such as in individuals with dementia), communication through gestures may compensate for oral communication deficits; making a list of things to do may compensate for memory problems.

Completion. Use this sentence completion method in language treatment involving both adults and children; it is the same as Cloze Procedure and Partial Modeling; for instance, presenting two pictures, one showing a small house and the other a big house, you might say, “This is a small house. But this one is a . . .” and look expectantly; the individual is expected to complete the sentence by saying “big house.”

Conditioned Generalized Reinforcers. Use these powerful reinforcers to teach various skills, especially to children; they include Tokens, money, opportunities to indulge in favorite activities, which are effective reinforcers in a wide range of conditions; token reinforcement is known to be effective in teaching correct articulation of speech sounds and in increasing fluency in young children; they are Secondary Reinforcers that have a generalized effect; use them to:

- Promote generalized productions of target behaviors
- Enhance the effectiveness of the reinforcers used in treatment (e.g., combine tokens with verbal praise)
- Always exchange tokens for a prize the child will have selected at the beginning of the treatment session; see Tokens; see also Direct Stuttering Reduction Strategy under Fluency Disorders for an example of its application; see Response Cost to understand the use of token reinforcement in teaching various target skills.

Conditioned Reinforcers. Use these kinds of reinforcers when primary reinforcers (e.g., food, drink) are not appropriate or not needed; conditioned reinforcers are events that reinforce behaviors because of past learning experiences (e.g., verbal praise or tokens); the same as Secondary Reinforcers; see Unconditioned Reinforcers; they

are effective except for certain kinds of mands (e.g., request for food or drink cannot be reinforced by such conditioned reinforcers as verbal praise; they can be reinforced only by primary reinforcers—food and drink).

Conditioning and Learning. Use this behavioral method of treatment to teach virtually any type of communication skill; the method involves selecting and strengthening behaviors in individuals; technique of teaching new behaviors; techniques to decrease or increase the probability of existing behaviors by arranging different consequences for behaviors:

- *Avoidance conditioning:* Learning or teaching behaviors that help avoid aversive conditions, stimuli, and persons; once learned, hard to extinguish; often found in persons who stutter; modifying such avoidance behaviors as not talking on the phone is a treatment target.
- *Classical conditioning:* Also known as Pavlovian conditioning, classical conditioning involves systematic pairing of two stimuli—one, an unconditioned stimulus (UCS) and the other a conditioned stimulus (CS) so that the CS begins to elicit the response typically given to the UCS; most fears and anxieties associated with speech or speaking situations are learned through classical conditioning; pairing relaxation with feared speech situations in persons who stutter is an example of the use of the classical conditioning principle in treatment.
- *Operant conditioning:* Creating, shaping, selecting, strengthening, or weakening behaviors of an individual by arranging different consequences for those behaviors; the most researched and effective techniques known to teach new skills, including communicative skills; includes such well-known principles of treatment as positive reinforcement, negative reinforcement, differential reinforcement, shaping, prompting, modeling, corrective feedback, discrimination training, generalization, and so forth.

Conduction Aphasia. To treat this type of fluent aphasia, see [Aphasia](#) and [Aphasia: Specific Types](#).

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Conductive Hearing Loss. To treat communication deficits associated with inefficient conductance of sound to the middle or inner ear due to the abnormalities of the external auditory canal, the ear drum, or the ossicular chain of the middle ear, see Hearing Loss.

Confrontation Naming. Treat confrontation naming deficits in individuals with Aphasia, Dementia, and Traumatic Brain Injury who typically have this skill impaired; it is naming a stimulus when asked to do so; to teach this skill:

- Select functional and individual-specific names to teach; consult with the individual and the family members for useful words
- Show objects, pictures, or photographs
- Ask the common question, “What is this?”
- Model the correct response and have the individual imitate it
- Fade modeling by giving a cue (e.g., such a phonetic cue as “The word starts with a *p*” to evoke the word *pen*; such a personalized cue as “It was your last birthday gift . . . ” to evoke “book”)
- Reinforce the correct response
- Give multiple trials on the same word
- See Aphasia for details

Congenital Palatopharyngeal Incompetence. To treat communication disorders associated with an inadequate velopharyngeal mechanism that cannot close the velopharyngeal port for the production of non-nasal speech sounds, see Velopharyngeal Dysfunction; see also Cleft Palate and Voice Disorders.

Consequences. Treatment procedures the clinician uses in almost all treatment sessions; events that immediately follow responses and thus increase or decrease the future probability of those responses; in treatment, clinician’s differential response to individual’s correct, incorrect, and lack of response; technically known as reinforcers (both

positive and negative) that typically increase behaviors and punishers (corrective feedback) that decrease them.

Constituent Definitions. Dictionary definitions of terms with no reference to how the defined skills are measured (e.g., *the goal of treatment is better communicative competence*) contrasted with Operational Definitions (e.g., *the goal of treatment is to increase the production of selected words; to reduce specified dysfluencies to below 3% of the words spoken*).

Contact Ulcers (Contact Granuloma). Benign lesions on the posterior third of the glottal margin; possibly due to trauma, reflux, or vocally abusive behaviors; voice symptoms include low pitch, effortful phonation, and vocal fatigue.

- Do not recommend complete vocal rest or surgical treatment
- Do not recommend forced whispering
- Ask the individual to talk less
- Reduce Vocally Abusive Behaviors
- Teach the individual to speak with less effort and force
- Teach relaxed phonation and speaking
- Teach the individual to speak more softly
- Eliminate glottal attacks

Contingency. Essence of behavioral treatment of communication disorders; to use the procedures effectively, manage the treatment contingency efficiently; it is an interdependent relation between events or factors; in behavioral analysis and treatment, it is the dependent relation between antecedents (stimuli), the individual's responses, and the clinician-provided response-contingent consequences (feedback) to the individual; see Environmental Contingency and Genetic/Neurophysiological Contingency.

Contingent Consequences. During treatment, consequences are what the clinician says or does as soon as the individual gives a response; consequences are *contingent* when the clinician's verbal or nonverbal responses occur

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immediately after the individual's response; consequences that closely follow behaviors help change their frequency; in treatment, the feedback clinicians give their clients immediately after the latter produce correct or incorrect responses helps increase desirable behaviors and decrease undesirable behaviors.

Contingent Queries. Questions the clinician asks immediately following an unclear or truncated statement from the individual in language therapy; lead to more specific or elaborate responses from the individual.

- Ask a question immediately following an unclear or telegraphic response from the child (e.g., the child says “kick ball”; you ask, “Who is kicking the ball?”)
- Vary the type of question you ask (“Do you want to kick the ball?” or “What is happening to the ball?”) to encourage different kinds of elaborations or differently clarified productions from the child

Continuous Airflow. Use continuous airflow as a stuttering treatment target while using either the regulated breathing or a comprehensive fluency shaping in which it is an element; require the individual to maintain uninterrupted and even airflow throughout an utterance; for procedures, see [Stuttering, Treatment of Stuttering: Specific Techniques or Programs](#).

Continuous Reinforcement. To initially establish a target behavior in the beginning stage of treatment, use this schedule of reinforcement in which you reinforce every occurrence of a response; effective in establishing new skills, but inefficient in maintaining already established skills; therefore:

- Use this schedule only in initial stage of treatment
- Gradually shift from continuous to [Intermittent Reinforcement](#)

Contrastive Stress Drills. A treatment method used to promote both articulatory proficiency and natural prosody, especially the stress and rhythm aspects of spoken lan-

guage; used in treating Apraxia of Speech (AOS) in Adults; using different phrases and sentences, the clinician teaches the individual to stress different words; stressed words may be used to promote articulatory proficiency or simply to vary prosodic features of speech.

In Teaching Articulatory Proficiency

- Construct phrases and sentences preferably with a single target sound in them (e.g., “My name is Peter” for /p/; “Sam did it” for /s/)
- Ask a series of questions such that the individual will respond with the target phrase placing extra stress on the target word (e.g., “Is your name Tom?”; individual will respond “No, my name is *Peter*”; the individual is likely to stress the word *Peter*, especially the initial sound, and thus improve the articulatory precision of /p/; similarly, ask, “Tom did it?”; the individual will respond “*Sam* did it”)
- Reinforce the individual for articulatory proficiency

In Teaching Prosodic Features

- Create a series of phrases and sentences (e.g., “Tom does not read mystery novels”)
- Ask questions that will force stress on different words in target phrases and sentences (e.g., “Does Tom read *romance* novels?” may evoke “No, Tom reads *mystery* novels.” “Does Tom *never* read mystery novels?” may evoke “Tom *reads* them all the time.”)
- Reinforce the individual for varying stress on different words

Control Group. In group treatment research designs, it is the group that does not receive treatment and hence shows no change in the target disorder or disease; part of the Group Design Strategy that includes an Experimental Group to evaluate treatment effects.

- Select subjects randomly (Random Selection)
- Assign subjects into control and experimental groups randomly
- Alternatively, match subjects in the experimental and control groups (see Matching)

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- Assess the control group
- Withhold treatment to the control group while the experimental group receives treatment.
- Demonstrate that the control group did not change (improve) while the experimental group did

Controlled Evidence. In selecting treatment procedures, look for controlled evidence, which constitutes data that show that a particular treatment, not some other factor, was responsible for the positive changes in the individual's behavior; evidence gathered through controlled experimentation with either group or single-subject design strategy; data that show that treatment is significantly better than no treatment; evidence that supports the use of a treatment technique; one of several Treatment Selection Criteria.

Controlled Sentences. Specific sentences that contain target speech or language features the clinician asks the child or an adult to produce; may be modeled; use of pictures and other clinical stimuli may be used to evoke them; less spontaneous than speech produced in naturalistic settings; essential to establish target skills; eventually, modeling, pictures, and other stimuli are faded to make the individual's productions more natural and less clinician controlled.

Conversational Probes. Use conversational probes, which are methods to assess generalized production of clinically established behaviors in conversational speech and language; conversational probes help assess whether the individual has met the final goal of treatment: production of the target skills in naturalistic conversational speech.

- Take a naturalistic conversational speech sample
- Do not direct the conversation or do so only minimally to adequately sample the production of speech or language behaviors under probe
- Count the number of opportunities for producing the skill under probe
- Calculate the percent correct production of probed behaviors

- Give additional training at the conversational level if the adopted probe criterion is not met (e.g., 90% accuracy)
- End intervention only when the individual meets the criterion

Conversational Turn-Taking. Teach this important social communication skill to children and adults with language disorders; turn-taking is a pragmatic language skill and treatment target; involves appropriate exchange of speaker and listener roles during conversation; to teach appropriate turn-taking:

- Teach the individual to alternate between the roles of a speaker and a listener
- Use such verbal cues as “It is your turn to talk” and “It is my turn to talk”
- In case of a young child, use a toy microphone; and exchange it between the two of you; the one who holds it speaks, the one who does not have the microphone listens
- Teach the individual to speak for a few minutes and then yield the floor by saying, “It is your turn”
- Stop the individual if he or she interrupts your speech
- Reinforce the individual for not interrupting, taking an appropriate turn, and yielding the floor
- See also Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Corrective Feedback. Use response-contingent feedback to reduce the frequency of undesirable responses of individuals; frequently used in treatment.

- Give corrective feedback as soon as you detect an incorrect response
- Give Verbal Corrective Feedback (“No” or “That is not correct”) for all incorrect responses
- Give Nonverbal Corrective Feedback when appropriate (gestures that show disapproval of a response)
- Give Mechanical Corrective Feedback or Biofeedback whenever possible
- Measure the frequency of incorrect responses to see if the feedback is effective

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- Replace ineffective forms of corrective feedback with other, potentially more effective forms
- Minimize the use of corrective feedback by giving more positive feedback for correct responses and by Shaping complex skills

Craniocerebral Trauma. To treat communication disorders associated with brain injury due to external forces, see Traumatic Brain Injury.

Criteria for Making Clinical Decisions. Rules to make various clinical judgments; includes such treatment-related rules as when to model, when to stop modeling, and when a behavior is considered trained.

- Model most target behaviors for most individuals, especially in the initial stages
- Discontinue modeling when the individual gives five consecutively correct, imitated responses
- Reinstate modeling if errors persist
- Consider an exemplar of a target behavior (e.g., a target speech sound in a word, a grammatic morpheme in a sentence) trained when the individual gives 10 consecutively correct responses
- Consider a behavior (e.g., correct production of the regular plural or phoneme /s/) tentatively trained when the individual gives 90% correct responses on untrained exemplars on an Intermixed Probe
- Consider a behavior clinically trained when the individual gives 90% correct responses on untrained exemplars on a Pure Probe
- Consider a behavior functionally (socially) trained when the individual gives 90% or better correct responses in a Conversational Probe produced in clinical as well as extraclinical situations

Cued Speech. Teach this system of nonverbal communication as a supplement to speech reading; consists of eight hand shapes that represent categories of consonants and four positions about the face that represent categories of vowels; these hand shapes and positions suggest speech

sounds in running speech; known to promote better reading skills in children who are deaf.

Cultural Diversity and Treatment Procedures. In designing treatment programs, consider factors related to ethnic background, culture, and linguistic status that may influence treatment outcome; selection of individual- and family-specific treatment targets that are consistent with the family's ethnocultural background and use of non-biased treatment materials are important elements of this approach; see Ethnocultural Variables in Treatment.

Cysts. Treatment for acquired or congenital, fluid-filled lesions of the larynx caused by trauma is surgery; unilateral cysts can occur contralaterally to a unilateral Vocal Nodule.

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Dedicated Systems of Augmentative Communication. Computers and other electronic devices designed and built exclusively to augment limited oral communication; see Augmentative and Alternative Communication.

Deglutition. Swallowing; see Dysphagia for treatment of abnormal swallow.

Delayed Auditory Feedback (DAF). Use this treatment procedure to slow down speech in an individual when a slower rate of speech is a treatment target; procedure in which a speaker's speech is fed back to his or her ears through headphones after a delay; most speakers slow their speech down under DAF; technique is used in reducing the speech rate in persons who stutter or clutter and those who have dysarthria; see Fluency Disorders for treating Cluttering and Stuttering with DAF; see also Dysarthria.

- Select one of the several DAF machines available on the market
- Experiment with different durations of delay that induce speech that is free from stuttering or cluttering or speech rate that improves intelligibility in dysarthric speakers
- Train and stabilize the target speech skills with the selected delay
- Fade DAF and maintain the desired rate without the use of DAF
- Note that a significant negative side effect of DAF is monotonous speech
- Shape normal prosodic features to make the speech sound more natural

Dementia. Treatment of communication disorders associated with neurological diseases and certain medical conditions resulting in dementia has been advancing in recent years; as medical management of dementia improves, so does the prospect for sustaining or improving behavior and communication for relatively longer periods of time; recent research supports vigorous efforts to treat the behavioral and communicative deficits in people with dementia to

improve their quality of life and to offer sustained support to the family members; a combination of behavioral intervention, physical exercise, latest and effective medical treatment to minimize memory and behavioral deficits holds the greatest promise; sustained clinical efforts to slow down the rate of regression as long as possible are now considered worthwhile; in all cases, both the individual and his or her family need counseling, support, and direct intervention; see the cited sources at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology, etiology, and symptomatology of Dementia, Progressive and Dementia, Reversible; before starting a treatment program, make a thorough assessment of dementia and the family communication patterns; see the cited sources and *Hegde's Pocket-Guide to Assessment in Speech-Language Pathology*.

Counsel and Educate the Family Members.

Dementia affects the family members and other caretakers as much as it affects the persons who have it; therefore, counseling the family and offering them emotional, financial, and professional support are important.

- Educate the family members about:
 - The causes, symptoms, and course of dementia
 - The specific type of dementia and its neurological basis
 - General medical and behavioral management procedures
- Give family members a realistic picture of what is ahead; discuss with them the need to cope with a difficult, prolonged, and expensive problem, the nature or even existence of which may not be understood by the individual himself or herself, especially in later stages
- Tell them that as time passes, they are most likely to be concerned with the individual's:
 - Physical violence
 - Memory deficits
 - Catastrophic reactions
 - Incontinence
 - Delusions and hallucinations

- Accusative and suspicious behaviors
- Uncooperativeness at bathing and at meal time
- Communication problems
- Demanding, critical behaviors
- Unsafe driving, cooking
- Hiding things
- Daytime wandering and night walking
- Let them understand that individuals with dementia, especially in the beginning states, are frustrated, worried, anxious, and angry
- Let them understand that in due course, the individual is likely to be depressed and may withdraw from the family
- Prepare the family members for:
 - Potential emotional outbursts and angry exchanges over trivial matters
 - Abusive, aggressive, and violent behaviors in later stages
 - Constant supervision or institutionalization of the individual in the later stages of dementia
 - The eventual need to feed, clothe, and take care of the person in all respects, including oral hygiene
 - The eventual incontinence that will increase the burden of care tremendously
 - Extended period of home care (from 1 to 2 years to 10 to 15 years)
 - Needing emotional, social, financial, and psychological help for themselves, not just the individual with dementia
- Impress upon the family members that rehabilitative efforts are still very important to slow down the process of deterioration in the physical, social, and intellectual status of the person with dementia
- Ask the family members and caregivers not to automatically assume that a person with dementia is incapable of making decisions in the early and middle stages of the disease
- Let them understand that some individuals with Alzheimer's disease may have a slow progression with

several years of relatively stable behavior patterns, suggesting both a need and the usefulness of prolonged rehabilitation efforts

- Help the family members cope financially with the long-term care of the individual with dementia:
 - Let them understand the long-term cost and efforts involved in managing a person with dementia
 - Discuss the family's available resources to care for and pay for the services
 - Help them find and contact local, state, and federal agencies along with private sources that offer financial support to families who have an individual with dementia
- Offer them help in finding appropriate nursing home placement when the time comes
- Give them consumer-oriented printed information on dementia and its management
- Give them names and addresses of local professional and social associations and organizations concerned with aging and dementia
- Introduce them to local support groups
- Give them information on any accessible specialty clinics, research centers, and universities where unique intervention programs or newer medical treatment evaluations are being conducted
- Give them information on websites that offer suggestions, newsletters, and information on new developments in medically or behaviorally managing persons with dementia
- If not already served by a team of specialists, including those who can counsel the family members (e.g., psychiatrists, psychologists, social workers), make referrals
- Arrange for the team to help the family members make rational and acceptable decisions about making or not making heroic efforts to sustain life in the terminal stage of dementia
- Arrange for continued counseling after the demise of the individual to help the family members regain their

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energies, rebuild deteriorated relationships among surviving members, deal with grief and loss, and manage financial burden of extended care

Clinical Management of Individuals with Dementia. Design a program to help the person with dementia sustain skills and behaviors to the extent possible; cope with progressive deterioration in skills and behaviors; teach compensatory strategies.

- Establish a simple routine for the individual and the family
- Design and implement a program to minimize the effects of memory problems and to help manage daily activities:
 - To help persons with failing memory, design various kinds of memory aids, including calendars, cue cards, diaries, written to-do lists, weekly planners, written daily reminders, watch alarms, timers, vibrating signals, prominently displayed display materials, orientation boards and signs, and so forth; see also *Memory Impairments*
 - Design *memory wallets*; these are sets of index cards on which various kinds of personal and useful information may be written; teach the client to frequently consult the memory wallets
 - Use the *spaced retrieval* procedure in which you teach the person to recall specific information with progressively longer intervals between recalls; for example, ask the client, “What do you do when you want to know what time is your doctor’s appointment?” The client says, “I look into my list of daily activities.” Reinforce the response, and then talk on some other topic (for a specific duration, such as 1 minute, 2 minutes, and so forth); then ask the question again; increase the duration between the repeated questioning
 - Direct individuals who repeatedly ask questions seeking information of various sorts to memory books, index cards, and bulletin boards where such information is made available

- Teach the client to write down instructions and needed information; teach the individual to ask others to write down information for his or her use
- Note that external cues are more effective than internal cues (self-monitored) in improving memory skills, although teaching self-monitoring skills is recommended
- Note that any kind of prompt that reminds the individual of an activity or encourages an individual to do something is better than no prompts
- Give verbal prompts that remind individuals of scheduled activities (e.g., a social gathering, a game, a party), which may be more effective than other kinds of reminders (e.g., a bright poster printed in large letters)
- Note that just making materials and opportunities for activities may not be sufficient to prompt actions and activities; individuals need verbal prompts and reinforcement for following through
- Reinforce institutionalized individuals for paying attention to reminders, signs, posters, announcements, and scheduled activities, as these may not be effective without such reinforcement
- Train individuals by reinforcing them to use, and prompt them to consult, a diary that they keep about appointments and schedules of events because it is not sufficient to have them simply write them down on a piece of paper
- Have caregivers in an institution wear name tags printed in larger letters and train the caregivers to draw attention to their name tags, and have individuals read them or caregivers read them aloud; train the caregivers to reinforce the individuals for saying the name or reading it aloud
- Have individuals keep a list of activities that are more immediate (i.e., *today's* activities as against this *month's* activities), and train caregivers to prompt the individual to consult the list at appropriate times

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- Teach the individual to use portable alarms that remind him or her of appointments and scheduled activities; monitor the use of the devices and reinforce the individual for maintaining this skill
- Give written instructions on daily living chores (closing the windows, locking the doors, turning the stove and the lights off); teach the individual to follow the instructions and reinforce him or her for doing so
- Train staff members in health care facilities to give frequent and systematic reminders to the individuals and to reinforce the individual in appropriate ways for exhibiting the required skills or following directions
- Teach the individual to rehearse information (e.g., just before leaving, rehearsing the names of people to be encountered at a party or class reunion; visualizing the faces of people to be encountered)
- Teach the individual to self-monitor; reinforce the individual for doing this and evaluate its maintenance
- Create a naming wallet containing pictures of family members, their names, and sentences and phrases about them, and train the individual to carry it and use it
- Teach individuals to make a written list of *what to do* every day; train the individual to use the list by frequent prompts and positive reinforcements
- Teach the individual to keep personal belongings (keys, clothing items, eyeglasses, pens) in a specific, invariable place; frequently monitor and reinforce this skill
- Teach the individual to keep related objects together (e.g., paper and pencil; socks and shoes; coffee and sugar); frequently monitor and reinforce this skill
- Train the individual to carry a card that contains the name, address, and telephone numbers of a family member and a health care professional; frequently check the individual to see if this is maintained
- Teach the individual to wear a bracelet that contains personal identification; check its continued use

- Instruct the individual to exploit his or her strengths to compensate for weaknesses (e.g., writing down everything when memory tends to fail)
- Teach individuals to ignore relatively minor problems (e.g., forgetting certain words)
- Teach the individual to get enough rest so fatigue will not further complicate memory problems
- Control disorientation and confusion
 - Place a large calendar in the individual's bedroom and cross off the current date every night; draw the individual's attention to this activity and tell the individual what you are doing (remember that you are not doing it for your benefit)
 - Make the individual wear a digital calendar watch with large display of AM and PM, date and day; frequently draw attention to it; have the individual use it in realistic situations (e.g., instructing the individual to look at the watch and saying that it is now 12 noon and time to have lunch)
 - Provide maps of frequently visited places (e.g., homes of relatives or friends, doctor's office, shops, favorite restaurants); before visiting the places, prompt the individual to consult the maps
- Offer direct intervention for communication and related problems
 - Note that direct intervention for clients with dementia can be useful until the late stage; instead of judging usefulness of treatment on a priori grounds, offer effective treatment and measure progress
 - Offer systematic language stimulation, physical strength exercise and fitness training, and social interactions involving service at social agencies, and supervised recreational activities; all have been found to be effective
- Provide cues to evoke words and then teach self-cueing techniques, especially in the early stages of dementia intervention; see [Aphasia](#) for details and additional examples:
 - Give phonemic cues for words (e.g., the clinician says, "The word starts with an s" to evoke *spoon*)

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- Give such semantic cues as a generic class (e.g., the clinician says “woman” to evoke the word “wife”); a synonym (e.g., the clinician says “dwelling” to evoke “house”); an antonym (e.g., the clinician says “good” to evoke “bad”); a category name (e.g., the clinician says “it is a fruit” to evoke “apple”); or an associated word (e.g., the clinician says “cup” to evoke “plate”)
- Train the individual to use description as cues in which he or she describes an object before naming it
- Teach self-cueing by having the individual produce the successful cues to generate the intended word
- Provide contingent consequences (positive reinforcement and corrective feedback) because they are known to be effective in modifying communicative behaviors in individuals with dementia
- Ask orientation questions (questions about time and place), model the correct responses, and verbally reinforce the individual for imitating the correct responses; gradually fade the modeling and have the individual respond to questions
- Train the individuals to initiate conversation and reinforce them for doing so
- Ignore inappropriate, irrelevant, vulgar, delusional, and any other type of unacceptable or bizarre verbal behaviors and reinforce any appropriate verbal responses (differential reinforcement)
- Reinforce the individuals continuously (reinforcement for all desirable responses); if needed, reinforce the individuals with tangible reinforcers
- Teach gestures as a means of communication even before the oral skills begin to fade so that nonverbal means of communication is established
- Arrange group therapy sessions involving family members or other individuals with similar problems to sustain social and communicative skills as long as possible; direct the sessions and manage the behavioral contingencies to promote the selected goals of the group session; provide refreshments during these ses-

sions, as some data suggest their potentially positive effects on interaction; arrange the sessions to have the individuals:

- Tell stories in group sessions
- Maintain topics of conversation
- Take appropriate turns in conversation
- Express their own feelings and thoughts about their disease
- Arrange for generous verbal reinforcers for all targeted expressions and any other appropriate expressions
- Ignore and teach the group members to ignore inappropriate responses
- Remind the individual of the topic of discussion anytime he or she switches topics inappropriately
- Teach compensatory strategies for lost functions; for instance, teach individuals who can still read to have memory wallets—index cards on which various kinds of information is written; individuals are known to consult the cards and maintain meaningful conversation
- Sustain skills as long as possible, even if they have to be progressively simplified:
 - Encourage the individual to continue to cook but with support
 - Arrange for cooking simpler foods
 - Sustain reading skills with routine materials (e.g., TV listings, newspapers, labels on packaged foods, restaurant menus)
 - Encourage participation in social and family group activities as long as possible
 - Get the individual to continue the previous social activities as long as possible (e.g., volunteering in social clubs, seeing movies with friends)
 - Continue to reinforce previously established physical exercise programs (resulting in improved mood and a sense of physical well-being)
 - Reinforcer the individual's effort to sustain skills and social behaviors

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Management Strategies for the Individual's Caregivers, Including Family Members.

Family members and other caregivers need help in managing the persons with dementia; ask all those who care for and regularly interact with the individual to:

- Be observant about changes in the individual's behavior that might signal a significant change in health status
- Take steps to sustain the individual's physical health by making sure that the individual:
 - Takes the prescribed medications
 - Eats properly
 - Exercises regularly
- Reduce distractions
- Design and maintain a consistent routine for the individual
- Reassign household chores the individual cannot perform anymore, but do not expropriate the individual's responsibilities too soon
- Design and maintain a constant and simplified environment
- Make environmental modifications that help prevent social isolation and more rapid deterioration in behavior and communication:
 - Create and maintain communication opportunities for the individual
 - Incorporate as much relevant speech as possible with daily routines
 - Do not restrict communication with the fear that the individual may be irrelevant
 - Remove unduly restrictive rules against talking and social interactions in institutional settings
 - Find regular conversational partners for the individual at home and in institutional settings; recruit friends, grandchildren, family members who do not see the individual regularly, neighbors, former colleagues, and volunteers (especially in an institution)
 - Help the individual maintain contact with grandchildren and other younger acquaintances; work

with both the younger persons and the individual to foster this relationship

- Encourage individuals to observe social activities even if they do not participate in them
- Encourage individuals in the early stage of dementia to offer help in some designated activities to their spouses and other family members at home or other individuals in institutions (e.g., helping spouses in cooking, cleaning, or shopping; helping new individuals with routines in an institution)
- Arrange regular small group activities at home; facilitate and reinforce the individual's participation
- Let the individual have a say in arranging or rearranging his or her personal space (room design, color, furniture, decorations)
- Encourage the individual to be responsible for maintaining his or her personal space for as long as possible
- Allow the individual some privacy (do not relentlessly supervise when it is unnecessary)
- Use contrasting colors to enhance orientation to hand rails, hallways, communication boards, table settings, and room decorations; note that just these measures may not promote appropriate behaviors; train the individual to make use of them
- Pay attention to the individual's ethnocultural background in arranging his or her environment
- Pay attention to the individual's ethnocultural background in arranging conversational partners and topics
- Pay attention to a bilingual individual's primary language or the dominant language in arranging communication opportunities and topics; find interpreters or similarly bilingual conversational partners
- Have family members regularly visit the institutionalized individual
- Have family members participate in social events at the institution

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- Have family members take part in social activities arranged at the institution
- Attend to the sensory needs of the individual, especially in the early and middle stages of dementia:
 - Have the individual's vision checked and provide new glasses if need
 - Provide magnifying glasses and printed material with larger print to sustain reading skills
 - Place all visual information at the individual's eye level
 - Keep the individual's living environment visually attractive
 - Have the individual examined by an otologist and take necessary steps to maintain the individual's otological health
 - Have the individual's hearing checked by an audiologist and provide hearing aids as recommended
 - Monitor the use of hearing aids on a daily basis, as assistance may be needed in inserting and removing the hearing aids and in volume adjustments
 - Discuss with an audiologist the need for Assistive Listening Devices described under Aural Rehabilitation and follow the recommendations
 - Reduce or eliminate noise in situations where communication takes place (e.g., turn off the TV or radio while talking to the individual; monitor such mechanical noises as those of air conditioners and fans)
 - Evaluate whether the individual whose visual and auditory acuity is diminished reacts better to improved tactile and olfactory cues
- Approach the individual slowly, with calm and inviting expressions, and within his or her visual field (do not surprise the individual)
- Establish eye contact before speaking
- Let the individual see your face clearly while talking (stoop down if the individual is in a wheelchair)
- Supplement speech with gestures, smiles, and posture
- Speak clearly and directly
- Speak in simple terms

- Use requests instead of commands (e.g., “Should we do this together?,” “Would you give me a hand and help?”)
- Specify referents for speech (e.g., “We need to go to the dining hall” instead of “We need to go over there”); use proper names (e.g., your wife Jane, physical therapist Tom) instead of pronouns (e.g., *he, she, they*)
- Have only one or two people engage the individual in conversation at any one time
- Do not argue with the individual; instead rearrange the environment (e.g., do not try to convince a suspicious person that no one has taken his or her possessions; instead, make sure that the possessions are always kept in one predictable, easily accessible place)
- Record problems that occur with a view to find patterns in them; design strategies to control them (e.g., an episode of aggressive behavior may have been triggered by physical pain that may need medical attention)
- Avoid asking open-ended questions; instead:
 - Ask yes/no questions
 - Ask either/or questions
 - Ask short questions
 - Ask simple questions
- Be specific and direct in your communication with the individual
- Avoid vague references, sarcasm, humor, indefinite referents, analogies, similes, proverbs, and statements with double or abstract meanings
- Say only a little at a time and make sure the individual understands before saying more
- Repeat questions if necessary
- Be redundant, repeat, restate, and paraphrase your descriptions and explanations
- Talk about familiar and concrete topics and directly observable objects
- Use photographs and drawings to improve understanding
- Touch the individual while talking with him or her; gentle and affectionate touch will help draw the individual’s attention and improve communication

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- Praise the individual frequently for appropriate behaviors
- Say good-bye, wave, or give other departing signals
- Use the same phrase to suggest daily routines (e.g., “Let’s eat,” “Let’s go to bed now”)
- Observe what conditions aggravate the individual’s behavioral problems and try to avoid or reduce those conditions
- Look for physical reasons for emotional outbursts (e.g., pain, side effects of medication)
- Look for early warning signs of emotional or aggressive outbursts (e.g., body rigidity, a certain look, crying)
- Eliminate stimuli and situations that trigger emotional and aggressive responses; engage the individual in a distracting activity
- Reduce difficult demands; do not insist on remembering useless facts
- Limit choices about food and clothing so that the individual has fewer choices to make and reduced chances to get confused
- Control feeding problems some individuals exhibit by:
 - Feeding all meals in a constant place with no distractions
 - Placing food in clear view of the individual
 - Placing the eating utensils in the individual’s hand
 - Getting the individual’s attention and modeling eating
 - Manually guiding the act of eating (use touch and hand guidance)
 - Teaching the individual to pace eating (same time between bites)
 - Routinely providing the individual’s preferred beverage
 - Offering finger foods as often as practical and nutritionally appropriate
 - Offering plenty of social reinforcers for eating behaviors
- Install sensors under the rug in the house to monitor individual roaming at inappropriate times
- Install complicated locks on doors to prevent the individual from leaving the house and getting lost

- Control the individual's hostility and emotional outbursts by:
 - Taking note of conditions under which the individual exhibits such reactions (e.g., the individual becomes angry when the spouse takes over bill payment, financial management, grocery shopping, business activities)
 - Modifying those conditions to the extent possible (e.g., ask the spouse to pay the bills when the individual is not in sight or take the individual to the store even if only the spouse manages shopping)
- Control the individual's sleep disturbances by:
 - Controlling the frequency and duration of daytime naps
 - Putting the individual to bed at the same hour every night
 - Requiring and encouraging the individual to exercise every day for about 30 minutes
 - Feeding the individual a light snack an hour before the regular bedtime
 - Having the individual use sleepwear that does not disturb the sleep by twisting or binding around the body
 - Keeping the bedroom quiet by closing the windows and doors
 - Maintaining a night light to avoid confusion or anxiety in the night when the individual wakes up
- Sustain themselves (especially family members) by:
 - Joining support groups for families who have individuals with dementia
 - Taking breaks from caregiving to renew themselves
 - Maintaining certain hobbies and recreational activities
 - Recognizing their own need for professional counseling
 - Seeking financial support to meet the cost of caring for the individual at home and to be able to take a break from their demanding caregiver duties

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Demonstration. A stimulus procedure used in treatment; usually preceded by instructions on how to produce a target response.

- Describe the target behavior the individual is expected to produce
- Model the response for the individual
- Show how the response is produced (e.g., the tongue position for the production of /k/)
- Give maximum feedback (use a mirror if necessary)
- Reinforce the correct response or an approximation of it

Denasality (Hyponasality). Treatment for hyponasality is often medical or surgical, because the typical cause is some kind of an obstruction (e.g., nasal polyps or

overgrown adenoids) in the nasal or nasopharyngeal passage; in some cases, it may be a temporary condition due to edema following severe nasopharyngeal infection that requires medical treatment; in a few cases, hyponasality may need a speech-language pathologist's intervention, but none without a medical examination; in some individuals, hyponasality may persist even after a physiological cause (e.g., chronic allergies) has disappeared; in such cases, behavioral treatment is appropriate; a few treatment suggestions follow; see also [Voice Disorders](#) and [Cleft Palate](#) for additional information.

- Refer the individual to an otorhinolaryngologist for an examination before starting treatment for denasality
- Select words, phrases, and sentences that contain the nasal sounds and have a few words, phrases, and sentences without the nasal sounds to provide contrast to the individual
- Model the production of utterances with and without nasal sounds and point out the contrast
- Ask the individual to imitate, alternatively, utterances with and without nasal sounds
- Experiment with nasal sounds placed in different phonetic contexts, as some contexts may prove to be better than others in given individuals
- Provide differential feedback to the individual; reinforce the individual for even a slight increase in nasal resonance on nasal sounds; provide corrective feedback on denasalized production of nasal sounds

Deteriorating Baselines. It is essential to establish stable and reliable [Baselines](#) before treatment; if the behaviors are variable across sessions, treatment should not be introduced; an exception to this rule is deteriorating baselines that show the problem is getting worse; desirable behaviors (e.g., fluency) may be lower each time they are measured.

- Measure baselines repeatedly
- If the desirable behavior decreases (or the undesirable behavior increases) across baseline sessions, initiate treatment immediately

Developmental Apraxia of Speech (DAS). To treat apraxic speech in children, see Childhood Apraxia of Speech.

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Dialect. Treatment offered to change the dialect of a person should be considered carefully, as a dialect is a normal spoken form of a language with its own phonologic, semantic, grammatic, and pragmatic properties and rules; a dialect is any natural variation of a language; a particular variation (e.g., the general American English dialect) may be considered standard, although all variations are acceptable forms of communication; dialects of the secondary language in bilingual speakers may be influenced by their primary language; in African Americans, the English dialect may be the product of their cultural and linguistic heritage; such dialectal variations are not a basis to diagnose a communication disorder; see Ethnocultural Variables in Treatment for more detailed guidelines on treating children and adults with varied dialects.

Differential Reinforcement. Treatment procedures that help establish discriminated responding by reinforcing a response in the presence of one stimulus and not reinforcing the same response in the presence of another stimulus; it is also a collection of indirect methods of reducing undesirable behaviors by increasing another, desirable behavior; all are especially effective in reducing undesirable behaviors in nonaversive ways because the methods target desirable behaviors that replace undesirable behaviors; specific techniques include Differential Reinforcement of Alternative Behaviors (DRA), Differential Reinforcement of Incompatible Behaviors (DRI), Differential Reinforcement of Low Rates of Behaviors (DRL), and Differential Reinforcement of Other Behaviors (DRO); use these experimentally well-supported procedures to increase desirable behaviors while decreasing undesirable behaviors simultaneously:

Differential Reinforcement of Alternative Behaviors (DRA). One of the Indirect Methods of Response Reduction in which an undesirable behavior is reduced by reinforcing a specified desirable behavior

that serves the same function as the one to be reduced; also known as Functional Equivalence Training.

- Find out what function (purpose) the undesirable behavior to be reduced seems to serve (e.g., fussing in treatment sessions may serve the purpose of requesting help, when the child finds the task too difficult and cannot verbally request help)
- Select a behavior that is a desirable alternative to the behavior to be reduced (e.g., the response “help me,” if the child could make it, may serve the same function as fussing)
- Reinforce the production of the alternative, desirable response (e.g., teach the child to say “help me” instead of fussing)

Differential Reinforcement of Incompatible Behaviors (DRI). One of the Indirect Methods of Response Reduction in which an undesirable behavior is reduced by reinforcing a behavior that is incompatible with the one targeted for reduction.

- Specify the behavior to be reduced (e.g., leaving the chair and walking in the therapy room)
- Specify a behavior that is incompatible (e.g., sitting quietly and looking at the stimulus items presented)
- Systematically reinforce the child for exhibiting the desirable behavior (for sitting quietly and looking at the stimulus items) that is incompatible with the undesirable
- Suspend training on the target communicative skill for a while if necessary and until the prerequisite desirable behaviors (e.g., quiet sitting) are stabilized

Differential Reinforcement of Low Rates of Responding (DRL). One of the Indirect Methods of Response Reduction in which an undesirable behavior is reduced by reinforcing its progressively lower frequency of occurrence; the method shapes down an undesirable behavior.

- Specify the undesirable behavior to be reduced (e.g., interrupting treatment by irrelevant questions)
- Specify an acceptable level of the undesirable behavior (e.g., two questions in a 10-minute period)

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- Reinforce the individual for not exceeding the set level (“Good! You asked only two questions during the last 10 minutes!”)
- Specify a new, more stringent criterion in successive stages until the behavior is eliminated or kept to a minimum

Differential Reinforcement of Other Behaviors (DRO).

One of the Indirect Methods of Response Reduction in which an undesirable behavior is reduced by reinforcing any one of many unspecified behaviors; the behavior that will not receive reinforcement is clearly stated.

- Specify the undesirable behavior to be reduced (e.g., leaving the chair and walking around)
- Tell the individual that he or she will not receive reinforcers for that behavior; also say that he or she will receive a reinforcer as long as the undesirable behavior is not exhibited
- Periodically reinforce the child’s desirable behavior, and for not exhibiting the undesirable behavior (perhaps for sitting quietly, reading, coloring, working on other assignments, but none specified as the response to be reinforced)

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Digital Manipulation. Physical manipulation of the larynx during voice therapy to produce changes in vocal quality; for the procedure, see Voice Disorders: Specific Normal Voice Facilitating Techniques.

Direct Language Treatment Approaches. Clinician-planned and implemented language treatment with specified target behaviors; structured treatment sessions; requirement that the child first imitate and then spontaneously produce the selected target behaviors; described under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Direct Methods of Response Reduction. Procedures to reduce undesirable behaviors by directly placing a contingency on them; contrasted with Indirect Methods of Response Reduction.

- Specify the undesirable behavior to be reduced
- Place one of the following contingencies on it:
 - Corrective Feedback (e.g., say “No”)
 - Time-Out (say “Stop,” turn your face away for 5 seconds and then re-establish eye contact and resume conversation)
 - Response-Cost (give a token contingent on every correct response, and take back a token contingent on every incorrect response)
 - Extinction (ignore the response)
 - Imposition of Work (ask a child who disrupts your stimulus materials to reorganize them for you)

Direct Stuttering Reduction Strategies. Use these treatment techniques to reduce stuttering by placing behavioral contingencies directly on stuttering itself; effective procedures include Time-Out and Response Cost; see also Stuttering under Fluency Disorders.

Direct Treatment for Swallowing Disorders. Treatment of swallowing disorders by feeding the individual small amounts of food or liquid; see Dysphagia for treatment of swallowing disorders; see also Indirect Treatment for Swallowing Disorders.

Discrete Trials. Use these structured treatment or probe trials that are temporally separated to offer the individual separately measured opportunities for producing responses; they are useful in establishing target skills but not efficient in promoting generalized and maintained production; include Baseline Evoked Trials which do not include modeling or feedback (reinforcement or corrective feedback), Baseline Modeled Trials, which include modeling but no feedback, Treatment Evoked Trials, which include feedback but no modeling, and Treatment Modeled

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Trials, which include both modeling and feedback; the general structure of a discrete treatment trial includes the following steps:

- Present a physical stimulus to evoke a response (e.g., a picture, an object)
- Ask a question that evokes a response (e.g., “What is this?”)
- Model the correct response initially; fade the modeling gradually (e.g., “Johnny, say I see a *rabbit*” on the initial trials and just the question on later trials)
- Reinforce the correct response and give corrective feedback for incorrect responses
- Move the stimulus away from the individual
- Record the response as correct, incorrect, absent, and so forth
- Wait for a few seconds to mark the end of a trial
- Represent the stimulus and start a new trial
- Adapt this basic structure to suit the different target behaviors that are taught with discrete trials

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Discrimination. Teach discrimination between speech or language elements that are confused but need to be distinguished; discrimination is a behavioral process of establishing different (and appropriate) responses to different stimuli; opposite of generalization in which similar responses are given to different (though similar) stimuli; needed to teach such discriminated responding as plural words to plural stimuli and singular words to singular stimuli; nasal resonance on nasal sounds, but oral resonance on oral sounds; and so forth.

Distinctive Features. Analysis and treatment of speech sound disorders based on distinctive features was once a recommended practice; when the phonological pattern analysis became the standard in the assessment and treatment of multiple speech sound errors, the distinctive feature approach has faded; in any case, whether a clini-

cian uses a distinctive feature analysis or phonological pattern analysis, each individual error sound may have to be treated, unless there is generalization of learning one sound production to other sounds within a defined class; see [Speech Sound Disorders](#).

Dysgraphia. To treat writing problems associated with recent neurological injury and unrelated to poor instruction or lack of schooling, also called [Agraphia](#), see [Aphasia: Treatment of Writing Problems](#).

Dysarthrias. Treatment of a group of motor speech disorders resulting from disturbed muscular control of the speech mechanism due to damage of the peripheral or central nervous system is a multifaceted endeavor; treatment targets oral communication problems due to weakness, incoordination, or paralysis of speech musculature; to varying extents, all aspects of speech production, including respiratory control, articulation, resonance, and prosodic features need to be addressed in treatment; unique treatment considerations that are relevant to the different subtypes follow this main entry that gives the general guidelines, goals, and procedures that apply to treatment of dysarthrias in general; see the sources cited at the end and *Hegde's PocketGuide to Communication Disorders* for epidemiology, etiology, and symptomatology of dysarthria.

Treatment of Dysarthria: General Guidelines

- Conduct a thorough assessment of dysarthria and its type; consult the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.
- Note that dysarthria treatment efficacy research involving controlled experimentation is limited; nonetheless, there is enough evidence to support treatment of dysarthria
- Set the treatment goal as increased efficiency, effectiveness, and naturalness of communication; improving intelligibility of speech is an overriding concern; select goals that are appropriate for the individual

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- Be fully knowledgeable about medical, surgical, pharmacological, and prosthetic management, their limitations, and how they affect communication training
- Consider the complicating medical condition, associated conditions, and their prognosis in planning treatment
- Finalize the treatment plan only after a thorough discussion with the client and family members
- Consider the individual's environment and typical communication partners in planning treatment goals and procedures
- Exploit the individual's strengths (e.g., residual physiological support)
- Start management early, but consider the individual's physical status; postpone treatment initiation until the acute phase is over and the individual is more receptive of treatment efforts
- Provide treatment frequently
- Organize sessions to move from easy to difficult tasks
- End sessions with success
- Spend time on activities that focus on improvement of communication
- Increase physiologic support for speech initially
- Give intensive, systematic, and repetitive treatment trials
- Avoid activities or exercises that do not include speech; movements targeted for treatment should include speech production
- Use modeling (followed by imitation), shaping, prompting, fading, differential reinforcement, and other proven behavioral management procedures
- Use phonetic placement and its variations
- Provide instruction and demonstration
- Teach self correction, self-evaluation, and self-monitoring skills
- Provide immediate, specific, and social and natural feedback
- Use instrumental feedback or biofeedback when necessary

- Give repeated trials on the same target skill in the initial stages; subsequently, give trials on different speech tasks
- Emphasize accuracy initially, and as accuracy is achieved, emphasize increased speech rate while still maintaining intelligibility
- Restore lost function to the extent possible; teach compensatory behaviors for lost or reduced functions; reduce dependence on lost or reduced function
- Avoid nonspeech exercises designed to increase oral muscle strength because of lack of evidence; target muscle strength in the context of speech production
- Implement alternative or augmentative communication systems, if necessary

Provide Counseling and Support

- Teach the individual to inform listeners at the outset of an interaction how to effectively communicate with him or her (e.g., demonstrating use of an Alphabet Board)
- Train the individual to set the context and topic before beginning a conversation
- Train the individual to modify content and length of utterances
- Teach the individual to monitor listener comprehension
- Teach significant others to modify physical environment, be active listeners, and maximize their own hearing and visual acuity
- Teach the individual and significant others to maintain eye contact, establish effective communication strategies, and determine methods of feedback

General Treatment Goals for Individuals with Dysarthria

- Modification of respiration
- Modification of phonation
- Modification of resonance
- Modification of articulation
- Modification of prosody

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General Treatment Procedures. Because of the variability of dysarthria, its subclassifications, and varied neuropathology, some techniques produce temporary effects; others are contraindicated for certain individuals; many are suggested based on clinical experience without controlled experimental evidence; continue to use a technique only when it produces a clear and positive effect on the individual's behavior; abandon ineffective procedures and modify those that seem to hold promise.

Modification of Respiration

- Train consistent production of subglottal air pressure; use manometer or air pressure transducer
- Train maximum vowel prolongation; reinforce sufficient loudness as the person learns to sustain the vowel production for longer durations
- Shape production of longer phrases and sentences produced on a single breath
- Shape a greater number of words and phrases said on a single breath (referred to as *optimal breath groups*)
- Cue the individual to inhale before speaking and reinforce the individual for inhaling more deeply and exhaling with more force to produce speech; push manually the abdomen to induce a more forceful expiration
- Teach controlled exhalation to stabilize even speech production
- Teach sustained exhalation throughout utterances to maintain loudness and other characteristics of speech
- Teach prompt initiation of phonation after inhalation to make the air supply for speech last longer
- Teach pausing and breathing at appropriate junctures in speech
- Teach the individual to push, pull, or bear down during speech production; such activities tend to increase the breath support for speech
- Find a normal or an unusual posture that promotes respiratory support and teach it (e.g., some individuals' speech improves in a supine position)

- Let the individual use neck and trunk braces if found to be helpful
- Use adjustable beds and wheelchairs to make postural adjustments
- Use girdles and wraps around the abdominal area to increase muscle strength for respiration; do not use this for extended periods of time, as such devices limit inhalation and pose the risk of pneumonia
- Use an Expiratory Board to stabilize the abdominal muscles for respiration
- Teach the individual to exhale slowly and in a controlled fashion after deep inhalations
- Train the individual to terminate speech earlier during exhalation to prevent voice fading

Modification of Phonation

- Discuss with medical staff the need, effects, and effectiveness of medical treatments, including Laryngoplasty, Teflon or Collagen Injection, Recurrent Laryngeal Nerve Resection, Botulinum Toxin Injection, and pharmacological measures; consider them in the total management of the individual and in treating communication disorders
- Increase vocal intensity (loudness of speech) and production of relatively longer utterances per each breath
- Use biofeedback devices to give the individual immediate feedback on vocal intensity to effect changes in excessive or too little loudness; always also provide verbal feedback along with biofeedback; see Voice Disorders
- Shape vocal intensity to desirable levels with modeling, reinforcement and corrective feedback; this avoids the use of biofeedback units that need to be faded out
- Train the individual with excessively soft voice in using a portable amplification system if behavioral methods to increase vocal intensity have not been successful or as a temporary measure until the behavioral methods produce desirable results

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- Train aphonic individuals in the use of Artificial Larynx
- Ask individuals with aberrant neck movements or neck muscle weakness to wear Neck Braces
- Teach Effortful Closure Techniques for individuals with vocal cord paralysis (e.g., pulling or pushing while phonating)
- Teach individual to initiate phonation at beginning of exhalation
- Teach the individual to turn head toward weak side during speech; try digital manipulation of the thyroid cartilage to increase loudness; be aware of temporary effects of these
- Try relaxation exercises and laryngeal massage to increase loudness
- Teach the individual to tilt head back, initiate speech after a deep inhalation, and increase pitch to reduce strained voice quality
- Teach the individual with vocal fold hyperadduction to initiate phonation with breathy onset or a sigh

Modification of Resonance

- Discuss with medical staff the need, effects, and effectiveness of medical treatments, including pharyngeal flap surgery, Teflon injection into the posterior pharyngeal wall, and palatal lift prosthesis to treat velopharyngeal incompetence
- Provide feedback on nasal airflow and hypernasality by using a mirror, nasal flow transducer, or a Nasendoscope
- Train the individual to open the mouth wider to increase oral resonance and vocal intensity
- Teach the individual to articulate more precisely to increase oral resonance as well as to improve intelligibility
- Use nasal obturator or nose clip; have the individual speak in the supine position; be aware of temporary improvement
- Avoid such nonspeech exercises as sucking, blowing, whistling and icing, stroking, or brushing the

structures to improve velopharyngeal closure and to reduce hypernasality, as such exercises are ineffective

Modification of Articulation

- Discuss with medical staff the need, effects, and effectiveness of medical treatments including Neural Anastomosis, botulinum toxin (Botox) injection to orofacial or mandibular muscles to decrease abnormal movements, and pharmacological treatment in relation to communication training
- Analyze the error patterns and their potential reasons before developing a treatment program
- Encourage the individual to assume the best posture for good articulation
- Use the Bite Block to improve jaw control in individuals whose jaw control is severely impaired; it is a custom-fitted putty or acrylic material the individual holds between the upper and lower teeth
- Use behavioral methods to treat articulation disorders with individuals for whom articulatory modification is a main target
 - Provide instructions and demonstrations
 - Simplify the task, use shaping
 - Model frequently for the individual to imitate
 - Use phonetic placement techniques
 - Reduce speech rate to improve intelligibility
 - Ask the individual to exaggerate the production of medial and final consonants
 - Give immediate reinforcing and corrective feedback
 - Use minimal contrast pairs (e.g., *peet-beet*; *stop-top*)
 - Teach self-monitoring skills
 - Modify techniques in light of data
 - Move from simpler level of training to more complex levels
- Use electromyographic biofeedback to reduce hyper-tonicity and spasm of speech muscles
- Teach compensatory articulatory movements (e.g., use of tongue blade to make sounds normally made with tongue tip)

- Use intelligibility drills
 - Ask the individual to read texts or describe pictures you are not familiar with
 - Retell what you hear
 - Let the individual work on improving his or her articulation to promote better understanding on your part

Speech Rate Control

- Note that a reduced rate of speech is an important treatment target aimed at improving speech intelligibility of dysarthric speakers; the strategy is especially needed for individuals who have a more rapid or variable rate than normal (e.g., individuals with hypokinetic dysarthria)
- Do not teach a slower rate if the individual's speech intelligibility is acceptable
- Teach the individual to slightly increase pause durations in speech; when the individual learns to pause between words, phrases, and sentences, intelligibility is likely to improve
- Use such prosthetic devices as Delayed Auditory Feedback (DAF), a Pacing Board, an Alphabet Board, or a metronome to induce Metronome-Paced Speech; because these procedures negatively affect prosodic features and induce an unnatural-sounding speech, fade their use
- Prefer to teach rate reduction without external mechanical help; modeling a slower rate, letting the individual determine a rate that is slower but sounds less unnatural, reinforcing the individual for a slower but still less unnatural speech may be a better strategy than using mechanical devices
- Use hand or finger tapping to reduce rate, but fade this, as tapping also induces an unnatural speech rhythm
- Provide visual feedback from computer or storage oscilloscope on the desired rate of speech
- Use rhythmic or metered cueing; point to words in a passage in rhythmic or metered fashion

- Fade all external cues, mechanical devices, finger tapping, and cueing in gradual steps while the individual still maintains a slower rate and improved intelligibility
- Teach individuals to self-monitor rate because slow speech in the absence of external cues is difficult to sustain; have the individual monitor his or her own rate; reinforce the individual for slowing down when the rate goes up without your help; ask the individual to frequently evaluate his or her rate of speech

Modification of Prosody

- Reduce the rate of speech and increase the frequency and duration of pauses at linguistic junctures to achieve improved prosodic features
- Modify pitch with the help of instruction, modeling, differential feedback, or such instruments as Visi-Pitch; be aware that direct work on pitch modification may not be needed in many cases because of successful modification of rate, intonation, and stress
- Shape louder speech through modeling, shaping, and differential reinforcement of greater inhalation, increased laryngeal adduction, and wider mouth opening
- Teach the individual to chunk utterances into natural syntactic units to promote more natural sounding speech
- Increase breath control to extend breath groups
- Use Contrastive Stress Tasks (sentences with the same words that change meaning when different words are stressed)
- Teach the individual to signal stress by using other means (e.g., prolongation of syllables or pausing before a stressed word)
- Teach the individual to vary the number of words per breath group
- Begin treatment with structured tasks and make transition to conversational speech
- Teach self-monitoring skills

Modification of Interactions Between the Individual and Communication Partners

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- Go beyond direct speech modifications and work on how people with dysarthria and their communication partners interact with each other in natural settings
- Promote two kinds of strategies: those that the individuals themselves use to help their listeners better understand their speech and those that the listeners use to better comprehend the speech of dysarthric speakers
- Teach individuals with dysarthria who have intact intellectual and good language skills to:
 - Say, “I am going to say something to you” or “Excuse me, I have something to tell you” before actually saying something
 - Establish firm eye contact with the listener to signal readiness to speak
 - Make a hand gesture that commands attention from listeners in a group or from a single conversational partner
 - Tell the listeners that the client has a neurological problem that the conversational partners may need to consider
 - Request the conversational partners to speak slowly, write the key words, repeat what they say, and so forth
 - Introduce the topic by name or tell when topics are about to be shifted from one to the other (e.g., “I am going to tell you about . . . ,” “My topic of discussion is . . . ,” “I am now going to change the topic to . . . ”)
 - Speak in the most effective and efficient manner (e.g., if a single word response or a shorter response will do, just say that, instead of unnecessary and possibly difficult-to-understand elaborations)
 - Use gestures and facial expressions to supplement oral communication

- Learn to understand the cues listeners give when they don't understand the message; get sensitized to facial gestures, general postures, signs of boredom, and so forth to gauge the listener's difficulty
- Modify statements when listeners fail to understand; say it differently, slowly, loudly, and so forth
- Request that distractions (e.g., TV, radio) and extraneous noise be minimized or eliminated before speaking; and make sure that the room is well lit
- Request or carry a personal amplification system when speaking to a group of listeners
- Teach strategies to the individual's typical conversational partners to enhance communication effectiveness; ask them to:
 - Pay close attention to the speech patterns of the dysarthric speaker to get familiarized with the speech pattern
 - Maintain eye contact and watch the face of the speaker to pick up facial expressions that may help understand the speech
 - Ask questions and request clarifications when communication breaks down
 - Minimize or eliminate extraneous noise and distractions; get closer to the speaker and hold conversations in well-lit places
 - Use amplification when talking to a group of people

Treatment of Dysarthria: Specific Types. To treat a particular type of dysarthria, determine the type and the associated neurological and physical conditions by a thorough assessment; consult the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*; tailor the treatment to the unique symptom complex of the individual; consider first the general guidelines and procedures described in the previous section along with those that follow for specific types; if a definitive diagnosis of a particular type is not made,

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carefully evaluate the communicative deficits and design a treatment program to remediate or modify those deficits (respiratory, articulatory, phonatory, and so forth).

Ataxic Dysarthria. In treating this type of dysarthria dominated by articulatory and prosodic problems, including imprecise consonants, excess and equal stress, and irregular articulatory breakdowns, consider the following that apply especially to it:

- Modify respiratory problems associated with speech production:
 - Teach the individual to inhale more deeply and exhale in a slow and controlled manner to sustain speech
 - Measure the duration of exhalation and reinforce progressively longer (more controlled) exhalation
 - Teach the individual to start speaking as soon as the exhalation begins to avoid wasted airflow; reinforce prompt phonation upon initiation of exhalation
 - Teach the individual to end an utterance well before running out of air; stop the individual when signs of airflow dissipation are evident, and ask the individual to inhale again
 - Shape longer utterances per breath
 - Teach the individual to stop and inhale at natural junctures in a sentence (e.g., at the beginning of a grammatical clause)
- Modify prosodic problems that result mostly from irregular, slow, or even rapid rate of speech:
 - Slow the rate of speech with the help of metronome beats; fade the metronome use
 - Use finger or hand tapping to generate an even and appropriate rate of speech; fade the tapping
 - Provide such cues as pointing to a printed word to generate an appropriate and even oral reading rate; teach the client to point to each word before reading; fade the cues
 - Teach appropriate stress on words in sentences; use Contrastive Stress Drills

- Teach variations in pitch by using both printed sentences and conversational speech
- Control excess loudness variation by reinforcing normal variations
- Modify articulation problems that may persist even at a slower rate of speech:
 - Ask the individual to produce words on a list while you judge their intelligibility without looking at the list or the individual's face
 - Give corrective feedback to encourage appropriate production of sounds in words you do not understand
 - Teach correct production of sounds by using the Phonetic Placement Method
 - Reinforce overarticulation or exaggerated articulation of medial and final consonants
 - Use the Minimal Contrast Method, to improve intelligibility of words that differ by only one phoneme
- Reinforce more natural sounding conversational speech
- Implement a Maintenance Strategy to train family members and other caregivers who will help sustain treatment gains

Flaccid Dysarthria. In treating this type of dysarthria dominated by breathy and harsh voice quality, hypernasality, nasal emission, imprecise production of consonants, audible inspiration, monopitch and loudness, and short phrases, consider the following that apply especially to flaccid dysarthria:

- Avoid exercises to strengthen muscles of the velum; such nonspeech muscle exercise as pushing and pulling techniques and whistle or bubble blowing activities are ineffective in improving the velar function; efforts to increase muscle strength while speaking is preferred to nonspeech strengthening exercises; asking the individual to increase effort while speaking may be sufficient in some cases
- Modify respiratory problems

- Make postural adjustments by asking the individual to sit straight in the chair to increase breath support for speech
- Place a padded lap tray on the wheelchair, positioned next to the abdomen, and ask the individual to lean against it to increase breath support
- Teach the individual to phonate at the beginning of exhalation to conserve air supply for speech
- Teach the individual to inhale deeply and exhale in a controlled manner
- Increase breath group durations
- Increase the number of words per breath group
- Modify phonatory problems
 - Increase loudness by modeling louder speech and reinforcing increase in individual's vocal loudness; use computer programs to give feedback on loudness
 - Ask the individual to push on the arm of the chair while phonating to promote *effortful vocal fold closure*; note that this is different from the ineffective nonspeech pushing and pulling
 - Ask the individual to inhale deeply and hold the breath to promote firmer approximation of folds
 - Teach phonation initiation with hard glottal attacks
 - Ask the individual with a unilateral vocal fold weakness or paralysis to turn head toward the *affected* side or manually push the larynx toward the *unaffected* side to help achieve better closure
- Modify resonance problems
 - Note that hypernasality is the main resonance problem caused by damage to the pharyngeal branch of the vagus nerve; the soft palate may be weak or paralyzed
 - Discuss the suitability of the Pharyngeal Flap Operation with a surgeon to improve velopharyngeal closure; note that the results have been inconsistent
 - Discuss the suitability of Pharyngoplasty with a surgeon to improve velopharyngeal closure; note that the results have been inconsistent

- Recommend Palatal Lift Prosthesis, as this has been more effective in reducing hypernasality than pharyngeal flap or pharyngoplasty; recommend this device to individuals who (a) are severely hypernasal and have not improved with behavioral treatment; (b) do not have a spastic velopharyngeal mechanism which may dislodge the device; (c) have teeth for anchoring the device; (d) have good articulation and phonation without which the device will not improve speech intelligibility; (e) do not have hyperactive gag reflex or swallowing problems; and (f) are motivated to wear it and care for it
- Use modeling, shaping, positive reinforcement, and corrective feedback to progressively decrease hypernasal voice
- Modify articulatory problems
 - Reduce the rate of speech to improve speech intelligibility (and to some extent, hypernasality); use finger tapping to cue a slower rate by having the individual produce a syllable or a word per tap; give frequent instructions to slow down and reinforce the desired rate
 - Reinforce improved articulation of speech sounds; systematically model and reinforce correct articulation in words, phrases, sentences, and conversational speech
 - Consider using the Intelligibility Drills in which the individual reads aloud a list of words when the clinician has turned his or her back; when the clinician does not understand a word, he or she will turn around to face the individual, analyze the errors, give suggestions on correct productions, and reinforce correct productions
 - Use the Phonetic Placement Method to teach correct placement of articulators in producing target sounds
 - Teach exaggerated production of consonants to improve speech intelligibility; ask the individual to fully articulate consonants, especially in medial and final positions of words

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- Use the Minimal Contrast Method to further improve articulation
- Modify prosodic problems
 - Target improved pitch, stress, and rhythm to reduce abnormal prosodic features in individuals with flaccid dysarthria
 - Ask the individual to discriminate pitch changes as you model different levels of pitch
 - Ask the individual to prolong an /a/ with lower and higher pitch
 - Have the individual read printed sentences that indicate higher and lower pitch (arrows above and below words)
 - Model different and appropriate patterns of intonation in phrases and sentences and have the individual imitate them
 - Monitor pitch in conversational speech and reinforce or give corrective feedback
 - Teach the individual to chunk utterances into syntactic units; reinforce pauses (and inhale if necessary) at appropriate junctures in sentences
 - Use Contrastive Stress Drills

Hyperkinetic Dysarthria. In treating this type of dysarthria dominated by prosodic disturbances, prolonged intervals, variable rate, monopitch, loudness variations, inappropriate silences, imprecise consonants, and distorted vowels, consider the following that apply especially to hyperkinetic dysarthria:

- Medications that control involuntary movements
 - Haloperidol controls chorea and tics
 - Clonazepam and valproic acid control myoclonic jerks
 - Botox injections control dystonia (more effective than other drugs listed in treating individuals with hyperkinetic dysarthria)
 - Note that medical treatment does not always eliminate the need for behavioral management of dysarthria

- Use a Bite Block (a small plastic cube the individual bites down on) to inhibit or reduce interfering jaw movements during speech in individuals with mandibular Dystonia
- Teach easy onset of speech, as this can help reduce involuntary movements that disrupt laryngeal movements, especially in individuals with mild hyperkinetic dysarthria
- Teach slower rate and increased vocal pitch when appropriate
- Try relaxation therapy to control involuntary movements

Hypokinetic Dysarthria. In treating this type of dysarthria dominated by voice, articulation, and prosodic problems that include monopitch, monoloudness, reduced stress, imprecise consonants, variable rate of speech, increased speech rate in some cases and a slower rate in a few, short rushes of speech, inappropriate silences, and harsh and breathy voice, consider the following that apply especially to hypokinetic dysarthria:

- Medical treatment for individuals with hypokinetic dysarthria
 - L-dopa, a dopamine precursor, helps replace dopamine in the substantia nigra and thus reduces tremor, bradykinesia, akinesia, and muscle rigidity, although its positive effects on speech are negligible
 - Anticholinergic drugs decrease the acetylcholine activity in the brain and thus reduce hypokinetic movement problems
 - All drugs have bad side effects, eventually become ineffective, and no drug replaces behavioral interventions, especially for communication problems
 - Deep brain stimulation, in which electrodes are implanted into the globus pallidus or subthalamic nucleus, spreads low-level current in the surrounding tissue; improvements in muscle function have been notable

- Modify respiratory behaviors by teaching the individual to:
 - Inhale deeply before speaking
 - Start speaking when inhalation begins
 - Exhale slowly and in a controlled manner
 - Stop talking well before exhausting the air supply
 - Gradually increase the number of words spoken per breath
- Modify phonatory problems
 - Use voice therapy techniques to increase vocal loudness and to decrease breathiness; use various biofeedback instruments (e.g., the VisiPitch)
 - Use pushing and pulling while phonating to increase the movement range of laryngeal muscles (e.g., having the individual push down on the arm of the chair while phonating)
 - Use portable voice amplifiers to increase loudness
 - Consider teaching initiation of phonation with hard glottal attacks
 - Consider using the Lee Silverman Voice Treatment Program, whose emphasis is on teaching louder speech by intensive training that includes modeling louder speech for the individual to imitate, pushing and pulling while phonating, wider mouth opening, and massed practice trials
- Modify articulatory problems
 - Reduce the speech rate in individuals who speak rapidly; use hand or finger tapping to cue-in production of syllables or words; use Delayed Auditory Feedback; induce Metronome-Paced Speech; use a Pacing Board or an Alphabet Board if necessary; none may be effective in all individuals, so experiment; fade the use of all such devices
 - Use Intelligibility Drills in which the individual reads aloud printed words; judge the accuracy solely on the basis of phonatory cues and give corrective feedback or positive reinforcement
 - Teach correct articulation by Phonetic Placement Method, which shows correct placement of articulators for producing target sounds

- Improve speech intelligibility by asking the individual to produce word-medial and final consonants with a certain degree of exaggeration
- Use the Minimal Contrast Method in which the individual is taught to produce clearly pairs of words that differ by only one phoneme (e.g., *pat–bat*)
- Modify prosodic problems
 - Note that a slower rate can improve the individual's prosody
 - teach proper intonation through printed sentences that show rising and falling pitch by arrows
 - Use Contrastive Stress Drills
 - Teach appropriate chunking of words according to syntactic units (e.g., pausing at the end of a grammatic clause and a sentence)

Mixed Dysarthria. In treating this type of dysarthria dominated by varied symptoms that depend on the types of pure dysarthrias that are mixed, consider the following that apply especially to mixed dysarthrias:

- Identify the dominant type, if any, and target the major speech problems for intervention
- Select speech targets that when treated will immediately improve communication
- Treat those targets like you would in the case of pure dysarthrias
- Treat the most severe problem first if multiple problems exist in a single category (e.g., prosody); find out the individual's preference to determine which problems should be addressed first in treatment
- Recommend Augmentative Communication devices for individuals who need them; note that individuals whose mixed dysarthria is due to Amyotrophic Lateral Sclerosis (ALS) are likely candidates for augmentative communication.

Spastic Dysarthria. In treating this type of dysarthria dominated by imprecisely produced consonants, monopitch, monoloudness, reduced stress, hypernasality, slow rate, strained-strangled-harsh voice, pitch

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breaks, and breathy voice, consider the following that apply especially to spastic dysarthria:

- Consult with the individual's physician about medically controlling pathological crying, which might interfere with treatment
- Consider behaviorally modifying crying in treatment sessions by reinforcing noncrying (alternative) behaviors
- Do not teach pushing or pulling exercises that only aggravate hyperadduction
- Use relaxation and stretching exercises with caution because their effects on speech have not been documented
- Note that modification of respiratory behaviors is typically not a major concern in individuals with spastic dysarthria; any apparent respiratory problems may be due largely to such phonatory problems as hyperadduction of vocal folds
- Modify phonatory problems
 - Note that efforts to reduce hyperadduction of vocal folds have not been especially successful; nonetheless, these efforts may be made with caution
 - Teach the individual head and neck relaxation by instruction, modeling, and manual guidance; stand behind the individual, take the individual's head between the two hands, and gently tilt it back, move it forward, and to the sides; ask the individual to move the head in the same manner without manual guidance
 - Teach easy onset of phonation to an individual who has learned to relax the neck and head muscles; model soft glottal closure and ask the individual to imitate it; begin with an exhaled sigh and add a prolonged /a/; shape this relaxed production of /a/ into words, phrases, sentences, and spontaneous speech
 - Teach the yawn-sigh motion before starting gentle phonation; ask the individual to inhale through open mouth, exhale, and begin phonation; shape

the phonated speech into words, phrases, sentences, and spontaneous speech

- Modify resonance problems
 - Increase vocal loudness to control the extent of hypernasality, as louder speech tends to be perceived as less nasal
 - Discuss the usefulness of Pharyngeal Flap Operation with the individual's physician or a Palatal Lift Prosthesis with a prosthodontist
- Modify articulatory problems
 - Avoid tongue and lip stretching exercises that have not been shown to be effective in controlled studies
 - Use the traditional articulation treatment which may be more effective than stretching exercises; use Intelligibility Drills in which the individual reads a list of words and the clinician judges the accuracy of production solely on the basis of phonetic cues and gives appropriate feedback on correct and incorrect productions
 - Use Phonetic Placement Method to teach correct placement of articulators; show articulatory placements in a mirror if necessary; model and reinforce imitated and eventually evoked productions of target words, phrases, and sentences
 - Teach the individual to produce the medial and final consonants in words in an exaggerated manner
 - Use Minimal Contrast Drills in which pairs of words that differ by only one phoneme are used to teach correct productions of target sounds
- Modify prosodic problems
 - Ask the individual to vary pitch on a prolonged vowel production (e.g., /a/); if necessary, model pitch variations as you prolong the target vowel; reinforce correct imitations; fade modeling and ask the individual to vary his or her pitch
 - Teach the individual to vary intonation in sentences; use printed sentences that indicate rising or falling intonation by arrows; model if necessary and fade modeling as the individual becomes

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more successful in imitating suggested patterns of intonation; move on to conversational speech in which varied intonation is reinforced

- Use Contrastive Stress Drills to place stress on different words in questions and answers; model stress on specific words in questions and sentences and ask the individual to imitate; fade modeling; teach appropriate stress patterns in conversational speech
- Teach the individual to chunk utterances into syntactic units by modeling and reinforcing pauses at appropriate junctures in speech (e.g., at the end of grammatic clauses and sentences); ask the individual to inhale at such junctures

Unilateral Upper Motor Neuron Dysarthria. In treating this type of dysarthria dominated by imprecise production of consonants, harsh voice quality, slow, imprecise, or irregular Alternating Motion Rates, slow rate of speech with increased rate in segments, mild hypernasality, and excess and equal stress, consider the following that apply especially to unilateral upper motor neuron dysarthria:

- Treat the associated language deficits (aphasia) and apraxia on a priority basis if warranted
- Modify articulatory problems
 - Use traditional methods to treat articulation disorders
 - Use Intelligibility Drills in which the individual reads a list of words and the clinician judges the accuracy of production solely on the basis of phonetic cues and gives appropriate feedback on correct and incorrect productions
 - Use Phonetic Placement Method to teach correct placement of articulators; show articulatory placements in a mirror if necessary; model and reinforce imitated and eventually evoked productions of target words, phrases, and sentences
 - Teach the individual to produce the medial and final consonants in words in an exaggerated manner

- Use Minimal Contrast Drills in which pairs of words that differ by only one phoneme are used to teach correct productions of target sounds

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Dysfluencies. Behaviors that interrupt fluency; measured in diagnosing and treating Fluency Disorders, especially stuttering; treatment target is to reduce such specific forms as repetitions of sounds, syllables, words, and phrases; prolongations of sounds and articulatory postures; inter- and intralexical pauses; interjections of syllables, words, and phrases; revisions; and incomplete phrases; see the two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, for additional information, including descriptions and assessment procedures; see Fluency Disorders for treatment procedures.

Dysphagia. Treatment of swallowing disorders (dysphagia or deglutition disorders) is an important duty of speech-language pathologists working in medical settings; associated with several medical conditions, including neuromuscular disorders, cancer of the larynx, the surgical removal of structures in the throat, strokes, traumatic brain injury, among others; may occur at any age, although more common in the elderly; see the cited sources and

Hegde's PocketGuide to Communication Disorders for epidemiology and ethnocultural factors and normal swallow and swallowing disorders; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for evaluation of dysphagia.

Treatment of Dysphagia. Speech-language pathologists manage swallowing problems of mastication and those of the preparatory, oral, and pharyngeal phases; disorders of the esophageal stage are treated with either surgical or medical methods, although speech-language pathologists can offer suggestions to symptomatically manage those disorders.

General Guidelines

- Consider the following factors in deciding whether to treat and with what procedures:
 - Diagnosis of the swallowing disorder and related medical condition of the individual; if the disorder is likely to be cleared within a week or two, do not initiate an exercise program; teach a few compensatory strategies; if the individual has significant motor neuron disease, do not initiate range-of-motion exercises that may only tire the individual; if the individual cannot follow directions because of dementia, do not select procedures that require the individual to comprehend instructions
 - Prognosis for swallowing therapy is good and hence recommended for individuals who have had strokes, traumatic brain injury, gunshot wounds, radiation therapy for neck and head cancer; prognosis is not favorable for individuals whose degenerative neurological disease has advanced to a stage where most forms of therapy are ineffective or contraindicated
 - Success in food intake with compensatory strategies suggests that an exercise program may not be needed
 - Severe dysphagia suggests that the individual needs indirect therapy; exercises to increase mus-

cle movements and their range in the absence of food in the mouth may be necessary.

- Poor respiratory function indicates a need to postpone certain therapy procedures until this function is improved
- Caregiver support to help complete prescribed exercises at home is essential; caregivers should be available and willing to remind the individual to perform the exercises and supervise them when necessary
- The individual must have motivation and interest, without which no treatment program will work
- Discuss with the individual and the family the swallowing process and the treatment procedure to be implemented
- Give written instructions to the individual and describe the steps to be followed
- Ask the individual to first practice swallow (without solid or liquid food)
- Note that individuals are likely to reject or eat only a small amount of a variety of food if its bolus takes more than 10 seconds for pharyngeal and oral transit; such individuals need to use at least a supplemental non-oral feeding program to sustain themselves
- Note that individuals who aspirate 10% of each bolus and are aware of it will reject food that causes it
- Introduce only a small amount of food during direct treatment
- Show the individual the amount to be swallowed
- Instruct the individual to cough to clear the airway, and reinforce the individual's coughs
- Initiate indirect treatment if the individual aspirates 10% of each bolus and the direct methods do not progressively reduce aspirations (intake of food into lungs); be aware that only radiographic data show aspiration
- Concentrate on increasing muscle control during indirect treatment
- Reduce distraction during treatment

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Compensatory Treatment. Procedures that promote swallowing without modifying the physiological status of the individual; mostly clinician- or caregiver-managed; requires little or no individual effort and thus does not tire the individual; includes a variety of procedures.

- *Modification of individual's posture:* Use postural modification until swallowing improves or direct treatment may be initiated; instruct the individual to assume a posture that promotes better swallow and reduces or eliminates aspiration
 - The *chin-down posture* that widens the valleculae; teach the individual to tuck the chin to the chest during swallow while maintaining a straight cervical spine; recommended for individuals with delayed triggering of the pharyngeal swallow; may be effective with individuals who have reduced posterior motion of the tongue base; individuals with inadequate laryngeal elevation, poor vocal fold closure, and absent cough reflex may not benefit from chin tuck
 - The *chin-up posture* that helps drain food from the oral cavity because of gravity; teach the individual to tilt the head back by extending the neck by leaning the head back; most effective with individuals in whom the oral transit of the bolus to the pharynx is difficult; not recommended for individuals with neurogenic dysphagia
 - *Head rotation toward the weaker side* that helps direct food to a more efficient side of the pharynx; teach the individual to rotate the head during swallow toward the damaged, weaker, or hemiparetic side of the pharynx, which then gets narrowed; recommended for individuals with unilateral laryngeal dysfunction resulting in aspiration
 - *Combined chin-down and head rotation* may reduce aspiration in some individuals with inadequate laryngeal closure; teach this if a trial application supports this combination
 - *Head tilt to the stronger side*, which directs food to that side; teach this strategy to individuals with uni-

lateral oral and pharyngeal weakness; the individual may need consistent cues to do this

- *Lying down on one side*, which helps control diffuse residue in the pharynx; teach this to individuals in whom pharyngeal wall constrictions are bilaterally reduced, causing aspiration after the swallow; after eating lying down and before sitting up, teach the individual to cough to clear the traces of food in the pharynx
- *Increased oral sensory awareness*: Procedures that enhance oral sensory awareness prior to swallow may be beneficial and include a variety of procedures; recommended for individuals with swallow apraxia, delayed onset of the oral swallow, reduced oral sensation, delayed triggering of the pharyngeal swallow, and tactile agnosia for food; in all the techniques, a sensory stimulus is applied before an attempted swallow; presumably, the stimulus alerts the swallowing centers of the brain; use one or more of the following sensory enhancement techniques
 - Apply a downward pressure on the tongue while presenting food with a spoon
 - Present a sour bolus (50% lemon juice, 50% barium) before presenting the bolus
 - Present a cold bolus
 - Present a bolus that the individual needs to chew
 - Present a bolus of larger volume
 - Present a thermal-tactile stimulation before presenting a bolus; using a size 00 laryngeal mirror that has been dipped in crushed ice for several seconds, firmly rub the anterior faucial arch four or five times before presenting a bolus; this is expected to sensitize the system so that initiation of oral swallow will trigger pharyngeal swallow; note, however, there is controversy about its effectiveness
- *Modification of volume and speed of food presentation*: Procedures that change the amount of food presented and the rate of presentation that facilitates swallowing; use the following techniques

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- Try a larger bolus to trigger pharyngeal swallow in individuals
- Try a smaller bolus to trigger swallowing in some individuals
- Present smaller boluses at a slower rate to trigger swallowing in some individuals
- *Modification of food consistency:* Procedures in which the food consistency or viscosity is changed to promote better swallow; try the following modifications
 - Use thin liquids with individuals who have oral tongue dysfunction, reduced tongue base retraction, reduced pharyngeal wall contraction, reduced laryngeal elevation, and reduced cricopharyngeal opening
 - Use thickened liquids with individuals who have oral tongue dysfunction and delayed pharyngeal swallow
 - Use purees and thick foods (including thickened liquids) with individuals who have delayed pharyngeal swallow, reduced laryngeal closure at the entrance, and reduced laryngeal closure throughout
 - Eliminate a certain food consistency only as a last resort
- *Use of intra-oral prostheses:* Artificially fashioned devices that help compensate physiologic deficiencies; have them constructed with the help of a maxillofacial prosthodontist
 - Use a palatal lift prosthesis, which helps lift a paralyzed soft palate to close the velopharyngeal port
 - Use a palatal obturator, which closes a surgically resected soft palate in individuals who have had oral cancer
 - Use a palatal augmentation or reshaping prosthesis, which gives a new and more normal shape to the hard palate in individuals who have had oral surgery

Swallow Maneuvers. Techniques designed to help individuals gain voluntary control over certain involuntary aspects of swallow; to be effective, the individual should follow the directions and exert some muscular

force, which may be impractical in some cases; use one of the four most commonly used maneuvers.

- The *supraglottic swallow*, which helps close the airway at the level of vocal folds before and during the swallow; it involves holding the breath during swallow to close the vocal folds; to implement this maneuver, place a bolus in the mouth of the individual and ask the individual to:

- Hold the food in the mouth
- Take a deep breath and hold the breath
- Swallow while holding the breath
- Cough soon after the swallow

Note that some individuals may hold the breath by not closing the vocal folds but by stopping the chest wall movement. Ask these individuals to inhale deeply, exhale slightly, and then hold the breath and swallow while holding the breath; holding the breath on exhalation closes the folds more readily.

- The *super-supraglottic swallow*, which helps close the airway entrance before and during the swallow; it helps close the false vocal folds by tilting the arytenoid cartilage anteriorly to the base of the epiglottis before and during the swallow; arytenoids are tilted when the breath is held and the individual bears down; to implement this maneuver, ask the individual to:
 - Inhale and hold the breath tightly by bearing down
 - Swallow while holding the breath and bearing down
- The *effortful swallow*, which helps increase the posterior motion of the tongue base during the pharyngeal swallow; to implement this maneuver, ask the individual to:
 - Squeeze as hard as possible while swallowing
- The *Mendelsohn maneuver*, which helps elevate the larynx more and for longer duration, resulting in an increased width and duration of the cricopharyngeal opening; in implementing this maneuver:
 - Educate the individual about the elevation of the larynx (tell him or her about the Adam's apple or voice box going up)

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- Have the individual palpate the elevation of the larynx when he or she swallows saliva several times
- Instruct the individual to hold the larynx up for a longer duration (several seconds) as he or she swallows; give such instructions as “swallow long and strong” or “stretch out the swallow”

Direct Treatment of Dysphagia. Treating swallowing disorders by placing food or liquid in the individual’s mouth and then shaping and reinforcing swallowing behaviors.

Disorders of Mastication

- Instruct the individual with limited lateral tongue movement to mash food by pressing the tongue against the hard palate or by keeping the food on the more mobile side of the tongue
- Teach the individual with reduced buccal tension to:
 - Apply a gentle pressure with one hand on the damaged cheek to increase cheek tension
 - Put food on the normal or stronger side
 - Keep the head tilted to the stronger side to maintain food on that side
- Teach the individual with limited lateral movement of the mandible to mash food by pressing the tongue against the palate
- Design a Palate Reshaping Prosthesis for the individual with limited vertical tongue movement when indirect treatment (exercises) fail
- Gradually reshape the prosthesis by reducing its size as the individual’s vertical tongue movements improve

Disorders of the Preparatory

Phase of the Swallow

- Teach the individual with problems in forming and holding the bolus due to reduced tongue movement and coordination to
 - Tilt the head forward to keep the food in front of the mouth until ready to swallow
 - Tilt the head back to promote the swallow

- Consciously hold the bolus in the anterior or middle portion of the mouth
- Teach the individual with reduced oral sensitivity to place food on the side of the oral cavity with better sensitivity
 - Better appreciate the placement of food by placing cold or spicy food in the mouth

Disorders of the Oral Phase of the Swallow

- Teach the individual who has developed a tongue thrust to:
 - Place the tongue on the alveolar ridge and initiate a swallow with an upward and backward motion
 - Compensate by placing food at the back of the tongue and then to initiate a swallow
- Teach the individual with reduced tongue elevation to:
 - Compensate by placing food posteriorly in the oral cavity
 - Place the straw almost at the level of the faucial arches to help swallow liquid
 - Tilt the head back and let gravity push the food from the oral cavity into the pharynx
 - Use the Supraglottic Swallow Maneuver to voluntarily protect the airway, if aspiration is a concern
- Teach the individual with disorganized anterior to posterior tongue movement to:
 - Hold the Bolus against the palate with the tongue
 - Begin the swallow with a strong, single posterior motion of the tongue
- Teach the individual with a scarred tongue to:
 - Place food behind the scarring
 - Tilt the head posteriorly to allow gravity to help with oral transit

Disorders of the Pharyngeal Stage of the Swallow

- Teach the individual with delayed or absent swallowing reflex to compensate by:
 - Tilting the head forward while swallowing

- Limiting the amount of Bolus that does not overflow into the open airway
- Counseling the family about the delay in initiating the swallow reflex; ask them to allow that much extra time for each swallow
- Teach the individual with reduced peristalsis such compensatory behaviors as:
 - Switching between liquid and semisolid swallows so that the liquid swallows help clear the pharynx
 - Taking only liquids or semisolids
 - Initiating dry swallows after each swallow of food to clear the pharynx
 - The Supraglottic Swallow Maneuver
- Teach the individual with unilateral pharyngeal paralysis such compensatory behaviors as:
 - Turning the head toward the affected side to close the pyriform sinus on the affected side and to direct the food down the normal side
 - Tilting the head toward the stronger side if the individual has a unilateral paralysis in lingual function and the pharynx
 - The Supraglottic Swallow Maneuver
 - Washing away residual thicker food with liquid swallows
- Ask individuals with cervical osteophyte to limit their diet to semisolid or liquid food until surgery corrects the problem and the individual recovers
- Teach the individual with a scarred pharyngeal wall the same compensatory behaviors used for the individual with unilateral pharyngeal paralysis
- Teach the individual with reduced laryngeal elevation to clear the throat after each swallow
- Use the Supraglottic Swallow Maneuver if residual material needs to be removed from the pharynx
- Teach the individual with reduced laryngeal closure to:
 - Use the Supraglottic Swallow Maneuver
 - Tilt the head forward while swallowing
 - Turn the head to the side that is not functioning properly

- Place pressure on the thyroid cartilage on the damaged side to improve closure

Disorders of the Esophageal Phase of the Swallow

- Do not attempt to treat, as these are handled medically; however, counsel the individual to take the following steps to reduce the symptoms due to gastroesophageal reflux:
 - Avoid certain kinds of food, including chocolate, peppermint, fatty foods, coffee, alcoholic beverages, pepper, citrus fruits and juices, spicy food, and acidic or sour foods
 - Eat small portions of food at each meal
 - Eat meals 2 to 3 hours before going to bed
 - Lose weight if overweight
 - Stop smoking
 - Elevate the head of the bed
 - Stay in an upright position for 30 minutes after eating
 - Wear loose-fitting clothes

Indirect Treatment for Dysphagia. Treatment of swallowing problems using exercises designed to improve the muscle functioning; does not involve food.

Oral-Motor Control Exercises

- Treat the individual with reduced range of tongue movements with such exercises as the following; ask the individual to:
 - Open the mouth as wide as possible and raise the tongue in front as high as possible; hold the tongue for 1 second, and then lower it
 - Raise the posterior part of the tongue as far as possible; hold it for 1 second, and then lower it
 - Continue with the stretching exercises for 5–10 times in a session, for 3–4 minutes
 - Repeat the set of exercises 5–10 times per day
- Increase the individual's buccal tension by asking the individual to:
 - Stretch the lips as tightly as possible and say “e”
 - Round the lips tightly and say “o”

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- Rapidly alternate between “e” and “o”
- Instruct the individual with limited lateral movement of the mandible to:
 - Keep the jaw open as widely as possible and hold this position for about 1 second
 - Open and move the jaw sideways and hold the extended position for 1 second
 - Make circular jaw movements
 - Provide Manual Guidance to move the jaw in the desired directions
 - Stop the task if any pain is experienced
- Treat the individual with limited tongue resistance by asking the individual to:
 - Push the tongue against a tongue depressor and hold the pressure for 1 second
 - Push the tongue against the tongue blade, in an upward, forward, and sideways direction; hold the pressure for 1 second
- Shape more firm lip closure by asking the individual with problems in lip closure to:
 - Stretch the lips for 1 second to stimulate the production of /i/; increase the duration gradually
 - Pucker the lips tightly for 1 second initially; increase duration gradually
 - Close the lips tightly for 1 second; increase the duration gradually; provide Manual Guidance if necessary
 - Close the lips around a spoon or an object; reduce the size of the object as the individual’s lip closure improves
 - Hold the lips together for 1 minute once a lip seal is achieved; increase the duration gradually
 - Repeat the exercises 10 times per day
 - Ask the individual to close the lips around a tongue depressor
 - Maintain lip closure when you or the individual tries to open them
- Treat the individual with bolus control problems by asking the person to grossly manipulate materials by:

- Holding a flexible licorice whip in the mouth, with one end on the individual's tongue and the other end in the clinician's hand
- Keeping the licorice stick between the palate and the tongue
- Moving the licorice stick from side to side with the tongue
- Moving the licorice stick forward and backward with the tongue, and then report where the licorice stick is
- Reporting when gross movement of the licorice stick is achieved
- Moving the licorice stick in a circular motion starting from the center of the mouth
- Chewing a piece of gum as manipulation capabilities improve
- Treat the individual with bolus control problems who has learned to grossly manipulate materials by:
 - Experimenting with boluses of different temperature; some may respond better with cold boluses, others may not
 - Placing a small bolus of paste consistency on the tongue
 - Asking the individual to move the bolus around in the mouth
 - Telling the individual not to spread the bolus around in the mouth
 - Asking the individual not to lose the bolus
 - Instructing the individual to cup the tongue around the bolus
 - Expectorating the bolus once the task is complete (inspect the mouth for residue)
 - Varying the consistency of the bolus, once success is achieved
 - Introducing one third of a teaspoon of liquid to the individual's mouth once success is achieved with the paste
- Treat the individual with bolus propulsion problems through posterior bolus propulsion exercises

- Place a long wad of gauze that is dipped in fruit juice in the individual's mouth
- Hold one end of the gauze
- Ask the individual to use the tongue to push the gauze up and back

External and Internal Stimulation

- Improve the timing and trigger the swallowing response by a variety of methods that more or less directly stimulate the structures involved in swallowing; verbal instructions stimulate the behavior of swallowing:
 - Thermal stimulation, which includes touching or stroking of the anterior faucial pillars with a chilled laryngeal mirror; repeat the stimulation 5–10 times; release a small amount of liquid into the individual's mouth with a pipette and ask the individual to swallow by saying "Now"
 - Direct stimulation of oral structures before placing food in the mouth; includes such procedures as massaging the gums; stroking the lips, gums, and the tongue with special brushes; and repeated touching of the posterior faucial pillars
 - Verbal prompts to swallow, which include such instructions as "close your lips," "chew," "swallow now"
 - Reduction of distracting noises and events to help concentrate on swallowing
 - Electrical stimulation of the pharynx may positively affect the cortical structures involved in swallowing and improved swallowing
 - Transcutaneous electrical stimulation may involve stimulation of the muscles of the neck with surface electrodes; more evidence needed for this type of intervention
 - Electrical stimulation of targeted muscle may involve both surface-placed and implanted electrodes; target muscles may include the laryngeal elevator muscles.

- Cortical neural stimulation, especially for stroke patients with swallowing problems, include transcranial magnetic stimulation, and transcranial direct current stimulation; may be used with other behavioral treatment procedures; needs more research

Improving Adduction of Tissues at the Top of the Airway

- Teach lifting and pushing exercises to improve laryngeal adduction to protect the airway during swallowing; ask the individual to:
 - Sit on a chair and hold his or her breath as tightly as possible
 - Use both hands and push down, or pull up on the chair, while holding the breath for 5 seconds
 - Use only one hand while pushing down or pulling up on the chair and to try and produce a clear voice with each trial; repeat this exercise 5 times
 - Use Hard Glottal Attack and repeat “ah” 5 times
 - Repeat the exercises three times in succession, 5–10 times a day for 1 week
 - Lift or push with simultaneous voicing; use both hands, pull on a chair, and use prolonged phonation
 - Use Hard Glottal Attack, commence phonation on “ah,” and sustain phonation with smooth voice quality for 5–10 seconds
 - Practice a Pseudo Supraglottic Swallow; instruct the individual to inhale, hold the breath, and cough

Medical Treatment of Dysphagia. Use of medical, mostly surgical, procedures to treat dysphagia; these procedures are designed to (a) improve a specific anatomic or physiologic deficiency related to swallow, (b) eliminate or reduce aspiration, and (c) provide food and liquid non-orally; medications are limited to treating esophageal swallowing disorders, not for improving oral or pharyngeal swallowing problems.

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Botulinum Toxin Injection. A surgical procedure of injecting botulinum, a toxic substance, into cricopharyngeal muscle to improve swallowing; technically difficult because of the position of the target muscle (hidden behind the cricoid cartilage); misplaced botulinum can paralyze other muscles, resulting in more serious dysphagia.

Cricopharyngeal Myotomy. A surgical procedure of splitting the cricopharyngeal muscle from top to bottom to keep a permanently open sphincter for swallowing; fibers of the inferior constrictor above and the esophageal musculature below also may be slit; eating may be resumed within about a week; recommended for individuals with Parkinson's disease, amyotrophic lateral sclerosis, and oculopharyngeal dystrophy whose main problem is cricopharyngeal dysfunction; not to be performed in early phases of recovery from stroke, head injury, or spinal cord injury, as these individuals are likely to recover normal or near normal swallow.

Epiglottic Pull-Down. A surgical procedure designed to control unremitting aspiration; the epiglottis is sutured to the arytenoids by making incisions around the epiglottis, aryepiglottic folds, arytenoids, and interarytenoid area; the procedure may fail in some cases, as the epiglottis tends to pull away from this attachment.

Esophagostomy. A non-oral, surgical feeding method for individuals who cannot tolerate oral feeding; insertion of a feeding tube into the esophagus and stomach through a hole (stoma) surgically created through cervical esophagus.

Gastrostomy (G-Tube). A non-oral, surgical feeding method for dysphagic individuals who cannot tolerate oral feeding; insertion of a feeding tube into the stomach through an opening in the abdomen; may be inserted under general anesthetic (called operative gastrostomy) or percutaneously with local anesthetic with the help of an endoscope (called percutaneous

endoscopic gastrostomy); blended table food is directly transported to the stomach; recommended when long-term (more than 30 days) non-oral feeding is required; may be more or less permanent; can be removed when no longer needed.

Jejunostomy. A non-oral, surgical feeding method for individuals who cannot tolerate oral feeding; insertion of a feeding tube into the jejunum through the abdominal wall; the jejunum is that portion of the small intestine that extends from duodenum to the ileum; often recommended to reduce reflux; the food needs to be prepared because it enters the body below the stomach.

Laryngeal Bypass or Tracheoesophageal Diversion. A surgical procedure designed to separate the air and food passages to prevent or reduce aspiration; recommended for severely neurologically involved individuals with life-threatening aspiration; the trachea is cut at the third or fourth tracheal ring; the proximal end is sutured into the cervical esophagus, and the distal end is bent forward to bring it up to the skin where an opening is made and sutured to it; prevents phonation; a tracheoesophageal stent voice prosthesis will help develop esophageal speech.

Laryngeal Closures. A surgical procedure to minimize or eliminate unremitting aspiration; in one procedure, the vocal folds are sutured together to prevent aspiration; in the other procedure, false vocal folds are sutured together.

Laryngeal Suspension. A surgical procedure to promote laryngeal elevation in individuals whose larynx does not raise up and move forward to help close the airway during swallow; the procedure raises the larynx and tilts it forward under the base of the tongue; a suture made from the middle of the mandible to the laryngeal cartilage helps accomplish this; head and neck cancer individuals, not neurological individuals, are candidates for this procedure.

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Medialization Laryngoplasty (Thyroplasty). A surgical method now preferred to Teflon injection to the vocal folds; designed to help position a paralyzed vocal fold into a more medial position so the folds can approximate for phonation and swallowing without aspiration; a small silicon prosthesis is placed in a window surgically created in the thyroid cartilage at the level of the paralyzed fold; the prosthesis helps position the paralyzed fold in a more medial position; performed under local anesthesia; has been successful in achieving complete medialization of a paralyzed fold.

Medications. Medications are available for individuals whose swallowing disorders are due to such neurological diseases as Parkinson's disease, myasthenia gravis, multiple sclerosis, and gastroesophageal reflux; note that individuals with oral-pharyngeal swallowing problems typically are not treated with medications.

Nasogastric Feeding (NG Tube). A non-oral feeding method for dysphagic individuals who cannot tolerate oral feeding; a tube inserted through the nose, pharynx, and esophagus into the stomach feeds the individual; recommended when this type of feeding is needed for a short duration (less than 30 days); less acceptable to individuals than some of the other procedures because of constant nasal irritation and social appearance.

Pharyngeal Flap. A surgical procedure to improve the mass and the functioning of the soft palate by raising the flap of the pharyngeal tissue and attaching it to the soft palate; see [Pharyngeal Flap Operation](#).

Pharyngostomy. A non-oral, surgical feeding method for individuals who cannot tolerate oral feeding; insertion of a feeding tube into the esophagus and stomach through a hole (stoma) surgically created through the pharynx; often more acceptable to individuals than the nasogastric feeding.

Surgical Enhancement of the Tongue. A surgical procedure to increase the bulk of the tongue that is

weakened, contributing to swallowing difficulties; tissue flaps added to the tongue may increase its bulk and reduce swallowing problems.

Surgical Reduction of Cervical Osteophytes. A surgical procedure to remove the bony growth on the cervical vertebra that can narrow the pharyngeal area, displace the posterior pharyngeal wall anteriorly, and thus cause swallowing problems; the procedure poses risk of nerve damage and causing a more serious dysphagia.

Teflon Injection into Vocal Folds. A surgical implant method to improve airway closure during swallowing in dysphagic individuals by adding implanted muscle mass that will help close the airway; Teflon may be injected into a normal or reconstructed vocal cord or any remaining tissue on top of the airway; other substances injected include glycerin and Gelfoam; often performed on individuals who undergo partial laryngectomy; also performed on individuals with such neurological disorders as Parkinson's disease; note that aspiration caused by lack of vocal fold closure may occur only in 10% of individuals; its use is on the decline because of such complications as the formation of Teflon granuloma.

Total Laryngectomy. Total removal of the hyoid bone and the larynx to permanently separate the food and air passages; performed on individuals who already have undergone partial laryngectomy and now cannot control aspiration; a last-resort procedure to control aspiration.

Tracheostomy. A surgical procedure to reduce or eliminate aspiration and improve pulmonary toilet; partially protects the lower respiratory tract from material that may pass the laryngeal sphincter; however, the procedure's effectiveness in reducing aspiration is limited.

Leonard, R., & Kendall, K. A. (2018). *Dysphagia assessment and treatment planning* (4th ed.). San Diego, CA: Plural.

Logemann, J. A. (1998). *Evaluation and treatment of swallowing disorders*. (2nd ed.) Austin, TX: Pro-Ed.

Provencio-Arambula, M., Provencio, D., & Hegde, M. N. (2007). *Treatment of dysphagia: Resources and protocols in English and Spanish*. San Diego, CA: Plural.

Dysphonia. A general term that means disordered voice; any voice disorder with the exception of Aphonia; for treatment procedures, see Voice Disorders.

E

Ear Training. Teaching an individual to auditorily discriminate between speech sounds as against teaching production of those sounds; the same as Auditory Discrimination Training or Perceptual Training; see also the Traditional Approach under Treatment of Speech Sound Disorders: Specific Techniques or Programs.

Echolalia. To control parrot-like repetition of what others say, a major characteristic of children with autism, see Autism Spectrum Disorder.

Effectiveness of Treatment. Assurance that treatment, not some other factor, was responsible for the positive changes documented in individuals under treatment; requires controlled evidence gathered through clinical experimentation; data should show that treatment was better than no treatment; may use a group experimental design or a single-subject experimental design to establish this; not the same as Improvement or Functional Outcomes; see Treatment Selection Criteria on judging effectiveness of treatment programs before selecting them.

Effortful Closure Techniques. Use these behavioral treatment techniques to promote laryngeal adduction for individuals with unilateral or bilateral vocal fold paralysis or weakness as found in many individuals with dysarthria and in some individuals with voice disorder without dysarthria; teach the individual to grunt, cough, push, lift, and pull while trying to phonate; the muscular effort helps push the folds together.

Effortful Swallow. Teach this swallowing maneuver to help increase the posterior motion of the tongue base during the pharyngeal swallow; to implement this maneuver, ask the individual to squeeze hard with all of the muscles while swallowing.

Electronic Communication Systems. Use these methods of augmentative communication with persons who have limited or no oral speech; may be of high or low technology; systems use electronic devices with a switch-

ing mechanism to activate a message, control the electronic system, and display the message; used in teaching Augmentative Communication, Aided.

Electronic Device for Alaryngeal Speech (Electrolarynx). Use these hand-held electronic instruments that generate sound to teach alaryngeal speech communication in persons who have undergone laryngectomy; for rehabilitation procedures, see Laryngectomy.

Electropalatography (EPG). A biofeedback tool, EPG is used in treating speech sound disorders found in children as well as those with Apraxia of Speech; requires the fabrication of an electronic pseudopalate, similar to an orthodontic retainer, but embedded with electrodes that record the contact between the tongue and the palate; properly interfaced with a computer that contains special software, the pseudopalate provides visual feedback to the client during speech sound production training.

Elicited Aggression. Aggressive behavior directed against any object or person when an aversive stimulus (as in operant punishment procedures) is delivered; not necessarily directed against the person delivering the aversive stimulus; a potential undesirable side effect of punishment.

- A child may kick the table when “No” is said to a wrong response
- A child may sweep the stimulus materials off the table when a token is taken away in the Response Cost procedure
- To reduce elicited aggression in treatment:
 - Use punishment procedures sparingly
 - Offer more positive reinforcement than corrective feedback
 - Use Differential Reinforcement to indirectly reduce nonaggressive responses; for example:
 - Tell the child that if he or she did not kick the table when “No” is said, you will offer an additional reinforcer (e.g., a separate token for not kicking the table, in addition to what you may offer for correct responses)

- Tell the child that if he or she did not sweep the materials off the table when a token is withdrawn, you will offer a separate token for not doing it
- Role-reverse and ask the child to play the role of a therapist for a few minutes; for example:
 - Exhibit the same kinds of error responses for which the child was told “No” and have the child say “No” to you a few times
 - Simulate stuttering and ask the child to take away a token each time you stuttered
- Reinstate the procedure for the child

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Environmental Contingency. Most easily manipulated behavioral treatment variable; manipulate this interdependent relation between antecedents, responses, and the consequences the responses generate to teach and maintain those target responses; this type of contingency includes environmental events that shape and sustain behaviors, includes the reinforcers and punishers used in treatment in a manner that immediately follows the target behaviors; contrasted with Genetic/Neurophysiological Contingency, with which it interacts.

Escape. Controlling or eliminating escape behaviors is an important component in treating certain kinds of communication disorders; it is a behavior that reduces or terminates an aversive event and hence increases in frequency; escape eventually leads to complete avoidance of situations or persons that are aversive; reducing avoidance of speaking situations a person with a communication disorder may exhibit (e.g., as do people who stutter) is essential to achieve full benefits of treatment; to reduce avoidance reactions:

- Work on eliminating the aversiveness of the event by teaching the needed, alternative skill (e.g., increased fluency in difficult speaking situations)
- In gradual steps, encourage the individual to come in contact with the aversive event (avoided speaking situation)

- Reinforce the approach behavior (e.g., picking up the telephone instead of having someone else respond)

Escape Extinction. Use this procedure to reduce negatively reinforced undesirable behaviors some individuals, especially children, exhibit during treatment; in this procedure, block an escape behavior, which is negatively reinforced because the escape provides relief from aversive treatment; it is useful in treatment sessions in which children exhibit many undesirable behaviors to escape from treatment regimen (e.g., crawling under the table during treatment).

- Prevent the occurrence of an undesirable response (e.g., crawling under the table) exhibited to escape from aversiveness (e.g., training trials)
- Physically restrain the child every time an attempt is made to leave the chair to prevent reinforcement of escape behavior
- Measure the frequency of attempts to crawl; if there is no reduction, use a different technique, such as reducing the aversiveness of treatment (e.g., simplifying the target skills or using more powerful reinforcers)

Ethnocultural Variables in Treatment. In treatment planning, take into consideration an individual's cultural, ethnic, and other personal variables that may affect treatment of communicative disorders; research on the effects of ethnocultural variables on treatment techniques is limited; see under the main entry for each disorder (e.g., Aphasia, Speech Sound Disorders, Language Disorders in Children) for specific guidelines on treating individuals with varied ethnocultural backgrounds; see also the sources cited at the end of this main entry and *Hegde's PocketGuide to Communication Disorders*; a few major and general guidelines that apply to treatment of all disorders of communication include the following:

- Select assessment procedures that are ethnoculturally relevant; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for additional guidelines

E

- Note that many of these suggestions are relevant to any disadvantaged family, not just a minority family; many poor, less educated, rural, and working-class families of any race may face problems similar to those faced by families of ethnocultural minority communities
- Understand the cultural communication patterns of the individual and the family
- Understand the families' views and beliefs about health, wellness, illness, specific diseases, disability, handicap, and impairment
- Find out the family members' beliefs about disorders of communication and the value they place on communicative skills
- Find out about the educational levels and sophistication about health care systems; without imposing a clinical recommendation, educate the individual and the family about clinical resources available to them
- Understand the family environment, living conditions, financial resources to support treatment that needs to be paid for; help them find financial support from local, regional, state, federal, and private sources to help pay for clinical services
- Find out about a family's transportation needs; many do not receive help in speech and hearing clinics because they cannot afford to travel to clinical facilities that often are located in distant areas from where disadvantaged people live
- Assess the time that is available for family members to get involved in clinical treatment, home treatment, and communication skill maintenance over time
- Understand the phonological, semantic, syntactic, and pragmatic rules of the individual's dialect (e.g., African American English or an English dialect influenced by a primary language in a bilingual child)
- Place communication patterns in the larger cultural context; avoid the narrow view of differences in linguistic rules among the languages of different ethnic groups
- Achieve a larger understanding of culture, literature, history, and heritage of the people served; avoid the pit-

falls of stereotypic and narrow linguistic descriptions of differences; note that this is an enriching experience to clinicians themselves

- Do not assume that individuals of different ethnocultural background automatically reject the mainstream communication patterns or that they have not acculturated to mainstream values—many have; the only right thing to do is to find out in an objective and non-evaluative manner
- Do not accept cultural stereotypes about any group; for instance, do not assume that a boy from an Asian background does not maintain eye contact during conversation with his teacher *because of his cultural background*; he may have learned to avoid eye contact *because of his stuttering*
- Work with your colleagues to make sure that needed interdisciplinary services are made available to the individuals and their families
- Receive family input in selecting treatment targets; recommend the targets you think are appropriate for the individual, but discuss them with the individual and his or her family members; make modifications in light of this discussion
- Serve as a resource to other professionals in helping them understand communication patterns of ethnoculturally different individuals and their families so that communication between service providers and receivers is more effective
- Treat communicative disorders that are diagnosed with ethnoculturally appropriate assessment; for instance, treat speech sound disorders in African American children only when they have a disorder in African American English (AAE) or a disorder in their Mainstream American English that cannot be attributed to their AAE rules and patterns; in bilingual children, treat speech sound disorders only if they have a disorder within the phonological system of their primary language or a disorder in English that is not due to the primary language's phonological patterns; treat a dialectal variation to induce

mainstream English patterns only when the individual, family, or both demand it

- Select treatment stimulus materials that are ethnoculturally appropriate; select materials that the individuals are familiar with; in the case of children, find out their preferences in selecting toys, pictures, line drawings, objects, and other stimulus materials; seek family members' input and, if possible, select treatment stimuli from the individual's home environment (e.g., ask the child to bring his or her favorite toys or books to treatment sessions and use them as stimuli for treatment tasks)
- Select treatment procedures that are known to have ethnocultural generality; evaluate treatment studies for their subject selection criteria; apply a given treatment procedure with caution if, in experimentally testing that procedure, an ethnoculturally diverse population had not been adequately sampled
- Select participants from varied ethnocultural backgrounds for your clinical treatment research
- In the absence of treatment research data on a particular disorder in a particular ethnic group (which is typically the case), assume tentatively that generally effective treatment procedures might work with individuals of varied ethnocultural background; note that such treatment principles as modeling, demonstration, instructions, positive reinforcement, corrective feedback, extinction, differential reinforcement, shaping, and so forth have been researched with good generality; nonetheless, expect to modify them to suit individuals
- Gather systematic individual performance data on treatment procedures that are of unknown ethnocultural generality; if possible, publish your data as case studies to help other clinicians select treatment procedures appropriate for their varied individuals
- Modify treatment procedures in light of the performance data and the individual's ethnocultural background; publish information and data on such modifications that were found to be effective with certain individuals

Brice, A. E., & Brice, R. G. (2009). *Language development: Monolingual and bilingual acquisition*. Boston, MA: Allyn & Bacon.

- Hegde, M. N. (2018). *A coursebook on aphasia and other neurogenic language disorders* (4th ed.). San Diego, CA: Plural.
- Kamhi, A. G., Pollock, K. E., & Harris, J. L. (1996). *Communication development and disorders in African American Children*. Baltimore, MD: Paul H. Brookes.
- Payne, J. C. (2014). *Adult neurogenic language disorders* (2nd ed.). San Diego, CA: Plural.
- Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.
- Roseberry-McKibbin, C. (2014). *Multicultural students with special language needs* (4th ed.). Oceanside, CA: Academic Communication.

Event Structures. Use sequentially organized, familiar events taken from daily life and routinized to teach language structures to children; for procedures, see [Language Disorders in Children](#); [Treatment of Language Disorders: Specific Techniques or Programs](#).

Evoked Trials. Often used in the beginning stages of treatment, evoked trials are treatment trials on which the clinician does not model the target response; structured and discrete opportunities to produce under more spontaneous stimulus conditions; most useful in teaching practically all communication target behaviors; contrasted with [Modeled Trials](#); include [Baseline Evoked Trials](#) and [Treatment Evoked Trials](#); in administering an evoked trial:

- Present a stimulus, such as a picture or an object to the individual by placing it on the table
- Ask a relevant question (e.g., “What is this?” “What do you see?” “What are these?”)
- Allow a few seconds for the individual to respond
- Move the picture or object toward you
- Record the response as correct or incorrect; if no response, record this fact as well
- Represent the stimulus and begin another trial
- Note that this is the basic structure of a discrete trial used in establishing evoked baselines; if it is a treatment evoked trial, reinforce the individual’s correct responses and give corrective feedback for incorrect responses

Exclusion Time-Out. Response-contingent exclusion of a person from a reinforcing environment; a variety of Direct Methods of Response Reduction; a form of Time-Out.

- Contingent on an undesirable behavior, remove the child from the stream of activities (e.g., make the child sit outside the classroom or in a corner)
- Bring the child back to the stream of activities after a brief period of time
- Note that this procedure is generally not recommended because it takes time away from treatment
- The method, though used frequently in educational settings, may be ineffective because the period of time-out outside the stream of activities may be reinforcing (enjoyable to the student or may provide escape from aversive classroom)
- Prefer non-exclusion time-out to exclusion time-out; in non-exclusion time-out, impose a brief period of silence (about 5 seconds) with no eye contact on an undesirable behavior; more efficient than removing the child from the stream of activities

Exemplar. An example of a target response being taught in a treatment session; it is a response that illustrates a target behavior; these may be words, phrases, sentences, gestures, and so forth; for example, the word *soup* is an exemplar if used in teaching the /s/ or /p/; thus, all words, phrases, and sentences used in teaching correct production of phonemes are exemplars; the sentence *The boy is running* is an exemplar if used in teaching the noun phrase, the auxiliary, or the main verb plus *ing*; thus, all phrases and sentences used in teaching language structures are exemplars; pictures and objects used in treatment also may be considered exemplars; to develop treatment exemplars:

- Select multiple exemplars for each target behavior to be taught (e.g., 20 words to teach the /p/ in initial positions; 20 sentences that contain the regular plural s)
- Teach multiple exemplars of each target behavior to enhance the potential for generalized production
- Probe after teaching a few (4 to 6) exemplars to see if generalized productions emerge with untrained stimulus items

- Teach more exemplars if the individual does not meet the probe criterion (e.g., 90% accuracy of the target phoneme production in a set of 10 untrained words)
- Note that generalized production often results with a few exemplar training items (typically less than 10 exemplars)

Expansions. A language treatment technique in which an individual's incomplete or telegraphic utterances are expanded into grammatically more complete productions; for procedures, see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

Experiment. A means of establishing cause-and-effect relationships; it includes a controlled condition in which an independent variable (such as treatment) is manipulated to produce changes in a dependent variable (production of speech or language); experiments are needed to establish treatment effects; may use the [Group Design Strategy](#) or the [Single-Subject Design Strategy](#); select treatment procedures that have been experimentally evaluated; see [Treatment Selection Criteria](#).

Experimental Group. In a clinical experiment, the group that receives treatment and hence shows changes in skills taught; part of the [Group Design Strategy](#) for establishing treatment effectiveness; contrasted with a [Control Group](#); see [Treatment Selection Criteria](#) for additional information; to show that a treatment is indeed effective, investigators:

- Select participants randomly ([Random Selection](#), unfortunately, ignored in most experiments)
- Assign participants into control and experimental groups randomly ([Random Assignment](#))
- Alternatively, match subjects in the experimental and control groups ([Matching](#))
- Assess the experimental and control groups to make sure they are equal
- Treat the experimental group while withholding treatment to the control group

- Demonstrate that the experimental group changed (improved), whereas the control group did not
- Conclude that the treatment was indeed effective

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Expiratory Board. Consider using this prosthetic device to improve respiratory muscle strength for speech, although controlled evidence of its effects on speech is limited; individual pulls a board attached to wheelchair toward his or her abdomen and leans against it to stabilize the muscles; often used in treating breathing problems associated with Dysarthria.

Expressive Aphasia. Not a type of Aphasia, but an older and general term for nonfluent aphasia whose main characteristic is difficulty in expressive language and speech with no significant impairment in auditory comprehension, see Aphasia and Aphasia, Specific Types: Broca's Aphasia.

Extension. A language treatment method in which the clinician makes comments on the child's utterances to add additional meaning; for procedures, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Extinction. Use this effective procedure to reduce certain undesirable responses by terminating reinforcement for them; similar to ignoring in everyday life; one of the Direct Methods of Response Reduction; use the following guidelines in implementing an extinction procedure:

- Note that whether and how fast a behavior will be extinguished depends on the:
 - Amount of past reinforcement for that response; a heavily reinforced response is more difficult to extinguish than the one reinforced less heavily
 - Duration of reinforcement; a response reinforced for a longer duration may be more difficult to extinguish than the one reinforced for a shorter duration
 - Previous exposure to reinforcement; a response that has been extinguished in the past will be more read-

ily extinguished now than the one that is being extinguished for the first time

- Effective use of extinction in treatment session:
 - Educate the individual's significant others about extinction before you use this procedure because parents find it difficult to watch their child's behavior (e.g., crying) put to extinction; explain that other methods of temporarily stopping an undesirable behavior (e.g., picking up a child crying in a treatment session) strengthens that behavior and becomes more frequent in future sessions
 - Remove reinforcers for the response because this operation defines the procedure
 - Make an analysis of the reasons for the undesirable behavior you wish to extinguish; if the behavior is maintained by attention, then withdraw attention; see Extinction of Positively Reinforced Behaviors; if the undesirable behavior helps the child avoid working hard in therapy (which is aversive to the child), then physically prevent such escape behaviors as crawling under the table when treatment trials are presented; see Extinction of Negatively Reinforced Behaviors; if the undesirable behavior is automatically reinforced (e.g., head banging), remove the sensory stimulation that results from such behaviors (e.g., make the child wear a padded helmet); see Extinction of Automatically Reinforced Behaviors
 - Do not use extinction to control aggressive, self-destructive, and generally disruptive behaviors because extinction is a slow process and the response is allowed to be made with no reinforcers; allowing such responses to be made may result in injury to others and self; prevent the occurrence of such responses and use Differential Reinforcement to encourage alternative, desirable behaviors
- If the extinguished responses recover sooner or later, a likely event, promptly extinguish again
- Note that undesirable behavior may intensify for a brief duration as soon as you initiate extinction; for instance, a child's crying behavior may intensify when extinction is

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begun; do not give in at this point; stay calm and do not look at the child

- Combine extinction with positive reinforcement for a clearly stated alternative, desirable behavior (e.g., telling the child that as soon as you stop crying, we can go out and see your mother and following through)

Extinction Burst. A sudden, initial, and temporary increase in an undesirable response at the beginning of extinction; for example, when clinician tells a crying child, “We will start working when you stop crying” and turns her back or sits motionless to extinguish that behavior, the child’s crying may show a sudden increase in intensity; it is not a reason to abandon extinction when it is appropriately chosen.

Extinction of Automatically Reinforced Behaviors. Use this procedure to terminate reinforcers that some undesirable responses earn automatically; it is one of the Direct Methods of Response Reduction; especially useful in reducing self-stimulatory behaviors (e.g., head banging or table banging) of individuals with autism spectrum disorder, severe intellectual disabilities, or brain injury because these behaviors are thought to be automatically reinforced by the sensory consequences they generate for those who exhibit them.

- Determine the sensory consequence of the undesirable behavior to be reduced (noise from banging on the table; stimulation from banging the head)
- Reduce or eliminate the sensory stimulation derived from the behavior to be reduced (cover the table with soft material or make the child wear a padded helmet)

Extinction of Negatively Reinforced Behaviors. Use this procedure to terminate negative reinforcers for undesirable responses that need to be reduced; one of the Direct Methods of Response Reduction; also known as Escape Extinction; appropriate to reduce such behaviors as crawling under the table during treatment, leaving the chair and walking around the treatment room, interrupting treatment trials by suddenly interjecting “You know

what?” types of responses, and grabbing the clinician’s pen or the stimulus material; such behaviors are often exhibited because treatment trials are aversive (or boring) to the child and the behavior provides escape from treatment work; the undesirable behavior is negatively reinforced because it offers an escape from aversive work; this escape needs to be prevented to stop reinforcement for it.

- Physically prevent the behavior; sit close to the child and physically restrain when you see an early sign of leaving the chair; sit immediately opposite to the child and hold the child’s chair between your legs; remove stimulus materials from the child’s reach
- Continue to present treatment trials
- Note that by preventing the escape behavior, you stopped its reinforcement as well; this is escape extinction
- Note that negatively reinforced undesirable behaviors cannot be extinguished by simply withdrawing attention or ignoring them because the behavior carries its own negative reinforcement

Extinction of Positively Reinforced Behaviors.

Use this procedure to terminate positive reinforcers for undesirable responses to be reduced during treatment; one of the Direct Methods of Response Reduction; appropriate to reduce such behaviors as crying maintained by positive reinforcement; in the case of children, explain the procedure to the parents or other caregivers and get their approval; this technique should *not* be used to reduce aggressive, self-destructive, and generally disruptive behaviors; and those that are negatively reinforced, automatically reinforced, or are due to physical pain and discomfort.

- At the very outset of extinction, tell the individual that you will pay attention only when the undesirable behavior stops (e.g., say, “As soon as you stop crying, I will take you out to see your Mom”)
- Pay no more attention until the behavior stops; turn your back and sit motionless; do not try to use other means of stopping the behavior; do not peek at the child
- Do not be unnerved when the behavior initially intensifies (Extinction Burst); continue to ignore the behavior

- Pay immediate attention when the behavior subsides or stops (e.g., say, “That is very good! You stopped crying; we can now see your mother”)

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Extraclinical Settings (Training In). Training given in such nonclinical settings as a playground, classroom, home, and other places; essential part of Maintenance Strategy; training is less formal and targets spontaneous, functional communication; in addition to clinician, administered also by such significant others as teachers, family members, and friends; any feedback given to the individual is subtle.

Eye Contact. Looking at the listener’s face during conversation; a pragmatic language intervention target; subject to ethnocultural variations, as in some cultures, eye contact between certain conversational partners (e.g., between a student and a teacher) may not be appropriate unless the child or the parents opt for treatment; for procedures, see Eye Contact under Language Disorders in Children; Treatment of Language Disorders, Specific Techniques or Programs; generally:

- Integrate eye contact training with communication training
- During treatment sessions, give such instructions as “Look at me!”
- Repeat instructions as found necessary
- Reinforce every time the individual looks at you

Eye Gaze. Teach this nonverbal method of communication to individuals with severe neuromuscular impairment; it is a method of Augmentative and Alternative Communication; teach the individual to gaze at a word, a phrase, a symbol, or an object to convey a message.

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Facilitated Communication. A once-popular technique of language treatment for children with autism and others with severe language impairment in which a facilitator maintains physical contact with the hand, wrist, or elbow of the individual to facilitate writing, typing, or pointing on a message board; controlled studies have produced negative evidence; results suggest that the facilitator may be the source of the messages typed; the American Speech-Language-Hearing Association, the American Psychological Association, and the Association for Behavior Analysis all have concluded that the method is ineffective and invalid; not recommended.

Factorial Stimulus Generalization. In treating all individuals with communication disorders, promote this type of a generalized production of unreinforced responses given in relation to new stimuli, settings, and audience; the most complex form of stimulus generalization to be targeted in treatment; to promote this outcome:

- Use a variety of stimuli to evoke target behaviors
- Vary the treatment settings; move treatment out of the treatment room and into more natural settings in later stages of treatment
- Arrange different conversational partners for the individual; invite family members, friends, and colleagues into the treatment sessions
- Teach skills in conversational speech
- Probe for factorial stimulus generalization

Fading. Use this method of reducing the controlling power of such special stimuli as modeling and prompting while still maintaining the target responses the stimuli evoke; essential to teach more spontaneous speech.

- Reduce the frequency of the special stimulus (e.g., modeling) gradually
- Reduce the intensity of the stimulus (e.g., present Prompts in progressively softer voice until it is no longer provided)
- Present only a partial stimulus (as in Partial Modeling)

- Make the stimulus progressively more subtle (e.g., make the hand gesture given to slow down the speech of a person who stutters less and less conspicuous)
- Make a mechanical stimulus nonfunctional (e.g., turn off a microphone that the individual still holds, or turn off a computer screen that remains in front of the individual)
- Increase the distance from the individual and the special stimulus in graded steps (move the microphone or the computer screen away from the individual)

First Words. The first few words a child typically acquires are a basic language treatment target for young children who are nearly nonverbal.

- Select child-specific words; select functional words in consultation with the parents and after an analysis of the child's home environment
- Generally, select the words from the following categories:
 - Names of family members and child's playmates
 - Child's favorite toys (*car, doll, bike, train*)
 - Common food items (*milk, juice, cookie, candy*)
 - Clothing items (*sock, shoe, hat, pant*)
 - Action verbs (*come, go, walk, sleep*)
 - Simple, concrete, and paired adjectives (*big/small, happy/sad; wet/dry*)
 - Pets or other animals (*kitty, doggie, fish, bird*)
 - Household objects (*pen, book, spoon, TV*)
 - Outdoor vocabulary (*grass, tree, flowers, snow*)
- Use the structured, Direct Language Treatment Approaches if the child is nearly nonverbal and has attention deficit:
 - Use the Discrete Trials
 - Use Shaping if necessary
 - Model and reinforce the responses or approximations
- Use indirect language stimulation if the child interacts well and can concentrate on loosely structured treatment activities conducted in play-oriented session:
 - Frequently model the target word productions
 - Use the Mand-Model approach
 - Use the Incidental Teaching Method

- Train parents to stimulate language at home; teach parents to:
 - Have the child label an item before you hand it to him or her
 - Read stories to the child and have the child name pictures
 - Ask questions about the pictures (e.g., “How does the kitty go?”)
- Give training in varied contexts and probe for generalized productions
- Move on to teaching Phrases (Word Combinations)

Fixed Interval Schedule (FI). An intermittent schedule of reinforcement in which an invariable time duration separates opportunities to earn reinforcers; the first response made after the interval is reinforced; responses made during the interval are not reinforced; limited use in treating communicative disorders.

Fixed Ratio Schedule (FR). A schedule of reinforcement in which a certain number of responses are required to earn a reinforcer; an FR1 in which every response is reinforced is a continuous schedule; schedules greater than 1 are intermittent; frequently used in treatment sessions.

- Specify the schedule to the individual (“I will give you a token every time you say it correctly”)
- Reinforce according to the specified schedule

Flaccid Dysarthria. To treat this type of motor speech disorder caused by damage to the motor units of cranial or spinal nerves that supply speech muscles (lower motor neuron involvement), see Treatment of Dysarthria: Specific Types under Dysarthria.

Fluency. A treatment target in individuals who have such fluency disorders as stuttering, cluttering, and neurogenic stuttering; an aspect of speech and language production; flowing and smooth speech without an undue amount of dysfluencies or muscular effort.

Fluency Disorders. Treatment of fluency disorders begins with an assessment and differential diagnosis of three main types: cluttering, neurogenic stuttering, or stuttering of early onset; treatment of stuttering of early onset is the most researched; each fluency disorder has unique characteristics that must be taken into consideration in designing intervention; a common feature to all three is impaired fluency of speech; cluttering is unique in its increasingly rapid but often irregular rate that causes breakdown in speech intelligibility; neurogenic stuttering is of late onset and has obvious neuropathology associated with its onset; treatment procedures are described for cluttering, neurogenic stuttering, and stuttering of early onset; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for details on epidemiology and ethnocultural factors, symptomatology, etiological factors, and theories of fluency disorders; before initiating treatment, make a thorough assessment of the overall symptoms of each disorder; consult the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Cluttering. Treatment of cluttering and stuttering share some common features because stuttering also may be found in individuals who clutter; but in treating cluttering, the clinician needs to consider such additional features as an abnormally fast and irregular rate that reduces articulatory accuracy, potential language problems, disorganized thought, reading and writing problems, impaired prosody (monotonous speech) and lack of concern about one's own speech problem.

- Increase the individual's awareness of his or her problem:
 - Show the individual's video recorded conversational speech sample and discuss the problems that are evident; note that most people who clutter are not convinced of their speech difficulty:
 - Discuss with the individual the erratic rate, excessively fast rate, and lack of intelligibility of speech

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- Discuss such other problems as impulsive talking, incomplete sentences, syllable telescoping, deletion of sounds and syllables
- Discuss any language problems or poor organization of thoughts that may be evident on the taped language sample
- Address the problems in some order of priority:
 - Consider reducing the rate and dysfluencies as a priority
 - Add additional treatment targets (e.g., language, reading, and writing difficulties)
 - Have the individual practice multiple target skills in an integrated manner in later sessions
- Teach a slower, even, and generally better controlled rate of speech to improve speech intelligibility
 - Prefer instructions, modeling, positive verbal reinforcement to delayed auditory feedback devices (DAF) or metronomes to teach a slower rate; use the latter devices if instructions and modeling are ineffective
 - Have the individual prolong syllables and reduce the overall rate of speech
 - Target not only a slower rate of articulation, but distinct, even slightly exaggerated articulation
 - Increase the rate beyond baseline and then slow down to encourage discrimination
- Modify prosodic features that improve speech quality and intelligibility by teaching the individual to:
 - Pause between clauses and sentences
 - Produce syllables with deliberate stress, especially the final and unstressed syllables of words
 - Vary the stress on different words in the same sentence
 - Vary the intonation patterns
- Correct any phoneme-specific misarticulations, if an independent speech sound disorder exists:
 - Target either individual phonemes or a pattern of errors
 - Use the methods described under [Speech Sound Disorders](#)

- Teach self-monitoring skills that may also promote greater awareness of the problem and better maintenance of treatment gains:
 - Give prompt, contingent corrective feedback on cluttered speech to increase awareness
 - Give prompt, contingent positive reinforcement for appropriate and even rate, normal rhythm, prosody, and well-formed verbal expressions
 - Increase the individual's awareness if his or her listeners express their difficulty when they do not understand him or her (e.g., listener's subtle facial expressions that betray a lack of comprehension); show such expressions on your face and teach the individual to pause, slow down, or change the speech in other ways
- Teach the individual to respond appropriately to listeners' request for clarification:
 - Make frequent requests for clarification (e.g., "I am not sure what you just said," "I didn't understand you")
 - If the individual did not modify his way of speaking, model one, and ask the individual to use the strategy
 - Teach a variety of ways in which one might modify one's own production to help listener's understanding
- Treat language and thought impairments in conversational speech:
 - Treat word-finding difficulties by having the individual name rapidly and learn words in semantically varied categories
 - Target specific syntactic errors and correct them with contingent feedback
 - Ask the individual to describe objects, events, or activities
 - Ask the individual to list synonyms and antonyms
 - Ask the individual to describe how two objects are similar or different
 - If necessary and found appropriate, use the techniques described under [Language Disorders in Children](#)
- Teach social communication skills (pragmatic language):

- Teach conversational turn-taking by stopping the individual when he or she interrupts you; ask the individual to speak when he or she is told, “Its your turn”
- Reinforce organized expressions
- Give corrective feedback on coherent talking
- Teach the individual to clearly and slowly greet others, introduce himself or herself, say good bye, and so forth
- Assign a topic for discussion during the following session:
 - Ask the individual to come prepared for the next session with a systematic, well-planned, and well-organized presentation
 - At the very beginning tell the individual not to hurry or talk impulsively
 - Ask the individual to collect his or thoughts and give a systematic speech on the topic
 - Give prompt feedback on coherent, organized expression of ideas on the topic
 - Control the rate if the individual goes too fast or uses an erratic rate
- Teach Self-Control (Self-Monitoring) Skills
 - Ask the individual to rate his or her own intelligibility; let the individual compare the self-rating with your rating
 - When you show a facial expression that indicates lack of comprehension, pause to see if the individual would modify his or her expressions; if not, draw attention to the missed sign
 - Ask the individual to chart his or her own errors along with you; help improve accuracy of self-charting
 - Use procedures described under Self-Control (Self-Monitoring) Skills
- Treat a coexisting stuttering
 - Target stuttering if it is a coexisting disorder, which is likely in many cases
 - Possibly, when the rate control procedure in which a slower rate of speaking is taught and stabilized,

dysfluencies may be reduced to normal levels (less than 5% of the words spoken, when all kinds of dysfluencies are counted; less than 3% of the words spoken when only the part-word repetition and speech sound prolongations are counted)

- If a high dysfluency rate is still evident, use one of the stuttering treatment procedures
 - Consider the Contemporary, Comprehensive Treatment Procedure for Stuttering in Older Children and Adults described under Stuttering
 - Consider the Direct Stuttering Reduction Strategy: Pause-and-Talk (Time-Out) for an older child or an adult
 - Consider combining Pause-and-Talk with a slight prolongation of only the initial syllable of phrases and sentences for an older child or adult
 - Use the Direct Stuttering Reduction Strategy: Response Cost for a younger child (perhaps under age 10, but experiment informally)
- Implement a maintenance program
 - Teach family members on helping the individual slow down the speech and use other skills taught in the clinic
 - Work with teachers to have them prompt and reinforce appropriate speech and language skills
 - Follow up the individual and give periodic booster treatment

Daly, D. A., & Burnett, M. L. (1999). Cluttering: Traditional views and new perspectives. In R. F. Curlee (Ed.), *Stuttering and related disorders of fluency* (2nd ed., pp. 222–254). New York, NY: Thieme.

Myers, F. L., & St. Louis, K. O. (1992). *Cluttering: A clinical perspective*. Kibworth, UK: Far Communications.

Neurogenic Stuttering. Treatment of stuttering with an adult onset and a demonstrated neurological basis has lagged behind descriptive and diagnostic information; there is little or no research on treatment effects; treatment may not be necessary in some individuals in whom the problem is transient; suggested treatment

procedures are based on reported clinical experiences and on the reasonable but untested hypothesis that stuttering treatment procedures may be used with suitable modifications; clinicians should evaluate the results of selected procedures carefully and abandon those that are ineffective.

- Develop a comprehensive treatment program for the neurogenic stuttering as well as any associated disorder such as aphasia, hypokinetic dysarthria, or any other neurogenic communication disorder
- Delay treatment to see if stuttering due to drug toxicity will improve when the toxicity is eliminated or treated effectively
- Develop a symptomatic behavioral treatment program to reduce the symptoms of stuttering or stuttering-like behaviors; use such techniques as the following, but carefully evaluate the course of the disorder and improvement in fluency:
 - Use the rate reduction strategy to minimize the frequency of dysfluencies:
 - Reduce the speech rate; teach a slower and more fluent speech
 - Use such techniques as the Pacing Board to help the individual reduce the speech rate
 - Teach the individual to tap on his or her lap as he or she speaks a syllable to each tap
 - Eventually, fade the use of the pacing board or tapping
 - Consider using Delayed Auditory Feedback (DAF) to see if it is effective in slowing the speech rate; some individuals may benefit, although lack of benefit from such techniques is a reported characteristic of neurogenic stuttering
- Experiment with auditory masking to see if it is helpful, even if a lack of an effect is a suggested diagnostic sign of neurogenic stuttering; some individuals may benefit
- Be aware that individuals who exhibit stuttering along with slow and effortful speech may not benefit from pacing devices, DAF, and masking

- Consider relaxation and biofeedback to reduce speech muscle tension; evaluate the results carefully

Duffy, J. R. (2013). *Motor speech disorders: Substrates, differential diagnosis, and management* (3rd ed.). New York, NY: Elsevier Mosby.

Guitar, B. (2013). *Stuttering: An integrated approach to its nature and treatment* (4th ed.). Philadelphia, PA: Lippincott Williams & Wilkins.

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Stuttering. Treatment of stuttering in young children, including preschoolers, has been shown to be effective in reducing the later and long-standing consequences of stuttering; although there are varied forms of treatment, only a few have been subjected to experimental test; some are questionable; some have uncontrolled clinical support; several are purely rational with no efficacy data to support them; nonetheless, there are several, experimentally supported treatment options for adults and children who stutter; most clinicians combine certain effective components of treatment to create somewhat personal programs; airflow management, gentle phonatory onset, and rate reduction through prolonged syllables are common elements across diverse contemporary treatment programs; these programs are empirically supported with some experimental evidence; such other direct stuttering reduction strategies as pause-and-talk, response cost, and positive reinforcement for fluency also are known to be effective, the latter two especially with young children.

1. A Contemporary, Comprehensive Treatment Procedure for Stuttering in Older Children and Adults

- The goal of treatment is to reduce the rate of dysfluencies in conversational speech to less than 1% in clinic speech samples and no more than 5% in everyday situations by:
 - Teaching the individual to manage his or her speech-related airflow properly
 - Teaching the individual to initiate speech softly and gently

- Teaching the individual to prolong syllable durations to reduce the speech rate
- Shaping normal prosodic features of speech and stabilizing fluency
- Strictly managing the behavioral contingency by giving prompt positive and corrective feedback
- Shifting treatment to more naturalistic settings
- Training significant others to manage the skills in the natural environment to promote maintenance of fluency over time and across situations
- Following up the individual periodically and giving booster therapy when needed
- Before you start treatment, take note of suggestions under *Measure dysfluency rates in all treatment sessions* and *Vary the treatment procedures to suit the individual*, both found at the end of this main entry
- Define stuttering in a measurable way; note that specific dysfluency measures are generally more reliable than such global measures as *stuttering moments* and *stuttering events*.
- Baserate stuttering in the clinic
 - Take extended conversational speech samples
 - Count all types of dysfluencies and the number of words spoken
 - Calculate the percent dysfluency rate
 - Continue to record brief conversational speech samples in the next two sessions during the first 5 minutes
 - Take additional samples if the three-sample dysfluency rates are highly divergent
- Select the three target fluency skills that are incompatible with stuttering, commonly used, and known to be effective in establishing stutter-free speech: Airflow Management, Gentle Phonatory Onset, and Rate Reduction through syllable prolongation; all described later under Treatment of Stuttering: Specific Techniques or Programs.
- Counsel the individual and the family about the treatment program

- Give an overview of the treatment program and its rationale (essentially tell them that the method prevents stuttering and helps practice fluency skills)
- Point out its known effects and research data that support its use
- Point out its drawback, which is initially an unacceptable artificial-sounding fluency
- Point out its advantages: it teaches fluency skills that the individual can fall back on at any time in the future
- Caution about the potential need for Booster Therapy in the future; tell clients that brief periods of repeated booster treatment over a period of several years may be needed
- Impress on the individual and the family that a relapse of stuttering does not mean the treatment has failed; with booster treatment, fluency can be regained and stabilized
- Answer all questions the individual, the family, or both may have about the treatment
- Begin teaching one target skill at a time; complete this skill component training in one session
- Teach airflow management first
 - Model airflow management for the individual
 - Inhale a slightly deeper than usual amount of air through your nose
 - Exhale a slight amount of air through your open mouth as soon as inhaling the air; make sure the air is not impounded in the lungs
 - Ask the individual to imitate your airflow modeling
 - Reinforce correctly imitated responses promptly
 - If the individual has difficulty doing both, model only a deeper than the usual inhalation; reinforce correct imitation; then model exhalation of a small amount of air; reinforce correct imitation; then combine the two sequenced responses
 - Stop the individual at the earliest sign of mismanaged airflow; point out the error (e.g., too deep an inhalation; impounding the air in the lungs;

- exhausting the air supply when asked to exhale a small amount of air; model again for imitation; reinforce the imitated response
- Continue until the individual can, on request and without modeling, inhale through the nose and exhale a slight amount of air through the mouth; give several successful trials
- Introduce the next skill—gentle phonatory onset
 - Model gentle onset and contrast that with hard glottal attacks, especially the kind exhibited by the individual; use short and simple words (e.g., *hi*, *how*, *bye*, *my*)
 - Initiate syllables softly, gently, slowly, and in a relaxed manner
 - Ask the individual to imitate only the just modeled gentle onset; reinforce correctly imitated soft onset productions
 - Continue until the individual can, upon request and without modeling, initiate sounds softly while producing several words
- Combine airflow management and gentle phonatory onset
 - Model both the component skills—airflow management and gentle phonatory onset—for the individual to imitate; use the same words as before
 - Ask the individual to imitate your modeling of airflow management and gentle phonatory onset; reinforce correct productions
 - Go back to single skill training if the individual mismanages a component; concentrate on the mismanaged component; combine them again
 - Continue until the individual can, on request and without modeling, inhale and exhale slightly and initiate sounds softly while producing several words; reinforce such evoked (unimitated) productions
 - Stop the individual at the earliest sign of mismanaged airflow, abrupt or tensed onset of phonation, or stuttering (dysfluency); explain what went wrong, concentrate on the missed target if neces-

- sary, combine the skills again, and give additional practice
- Introduce the next skill—rate reduction through syllable prolongation
 - Prefer the clinician-induced rate reduction; use Delayed Auditory Feedback if necessary (described later under Treatment of Stuttering: Specific Techniques or Programs); note that instructions and modeling and strict management of behavioral contingencies can be very effective in reducing the rate; avoiding mechanical devices is a good idea
 - Model a slow speech through syllable prolongation; use the same words used until this stage; stretch the syllable duration; reinforce correct imitation of slow, prolonged speech
 - Stop the individual as soon as you hear the sign of increased rate of speech or the production of a dysfluency; explain the error, model again, and reinforce correct imitations
 - Continue until the individual can, upon request and without modeling, stretch syllables in all the words practiced
 - Combine all three skills of fluency—airflow management, gentle onset of phonation, and syllable prolongation—into an integrated skill
 - Model inhalation and slight exhalation, gentle onset, and syllable prolongation using the same words practiced until this point
 - Ask the individual to imitate your modeling of all three target skills in words; reinforce correct imitations of airflow management, gentle phonatory onset, and syllable prolongations
 - Monitor all three skills at this stage and stop the individual for mismanaged targets or production of stuttering; pause briefly and continue
 - Provide training in skills with single words initially modeled and later evoked by questions that lead to one-word responses (“What is your first name?” “What is your last name?” and several similar questions to evoke single word names of

family members and friends; “What is the name of the city you live in?” “What is the name of your street?” and several similar questions)

- Continue until the individual can, on request and without modeling, produce all three target behaviors and with stutter-free speech at the word level with 98–100% accuracy
- Note that within the first one or two sessions, the individual should be producing words (some individuals can be moved to phrases even in the first session) with little or no stuttering; if this is not happening, perhaps the skills are not managed properly; make a careful analysis of errors and pay close attention to skill execution; be more prompt in reinforcing stutter-free speech and to stop and give corrective feedback at the earliest sign of a stutter
- Shift training to the phrase level as soon as possible
 - For the initial phrase training, form two-word phrases with the words already trained (e.g., the first and the last name of the individual and his or her family members; name of the town and that of the city)
 - For subsequent training, form phrases with one trained and one untrained word
 - Finally, form phrases with both untrained words
 - Model the target phrases using all three skills in a smoothly integrated manner
 - Make sure that you do not give a phonatory break between the two words; blend the two words; produce the two words as though they are a string of syllables with no break; stretch all the syllables; emphasize this aspect to the individual
 - Ask the individual to imitate the phrase
 - Reinforce correct imitations with good airflow, gentle onset, prolonged syllables, and continuous phonation throughout the utterance
 - Stop the individual at the earliest sign of a trouble (mismanaged airflow, hard glottal attacks, increased speech rate, or a stutter); explain the

error and concentrate on the skills that broke down (e.g., the individual failed to exhale before starting phonation; this would then be pointed out and practiced two or three times)

- When the individual has imitated several phrases with no stuttering and all skills efficiently exhibited, fade modeling; ask questions that can be answered by two-word phrases (e.g., “Tell me your full name”)
- Promptly reinforce the completely stutter-free productions of phrases with inhalation and slight exhalation, gentle phonatory onset, and rate reduction through syllable prolongation
- Stop the individual for mismanagement of any of the targets and production of dysfluencies; make an error analysis, repeat the skill mismanaged, and continue the training
- Shift training to the sentence level
 - For the initial phase of sentence training, expand already trained phrases into sentences (e.g., “I live in Fresno,” “My name is Kopitron Stratofearopolis”)
 - Model and have the individual imitate sentences if necessary; target completely stutter-free productions with efficient use of the skills in a smooth and integrated manner with no phonatory breaks between words; continue to model until the individual imitates several stutter-free sentences with smooth efficiency; use reinforcement and corrective feedback as before
 - Fade modeling and ask questions the individual will answer with complete sentences; monitor all target behaviors and fluency closely
 - Stop the individual promptly for mismanagement of any of the targets and production of dysfluencies; make an error analysis and concentrate on the particular skill breakdown
 - Continue until the individual speaks in sentences with stutter-free speech by using airflow management, gentle phonatory onset, and rare reduction through syllable prolongation

- Note that treatment at the spontaneous conversational speech level may have to be continued the longest; in fact, most individuals should reach this stage soon in therapy and stay on this stage long enough to maintain stutter-free speech with little or no modeling, thinned out contingency management with only an occasional reinforcement and corrective feedback
- During treatment at the conversational speech level, probe periodically; at the beginning of treatment sessions, conduct 5 minutes of probe; do not provide any modeling or positive or corrective feedback; ask the individual to speak in his or her typical manner; the individual will probably use the skills, but neither encourage nor discourage the individual to do so; record these samples and calculate the percent dysfluency rate
- Shape normal prosody
 - Begin shaping normal prosody when three successive probes show a dysfluency rate around 1% of the words spoken
 - Instruct the individual about normal prosodic features; tell the individual that gradually increased rate and typical intonations are the key to normal-sounding speech
 - Model a slightly higher speech rate and ask the individual to imitate it
 - After a few imitated sentences at the new rate, withdraw modeling and engage the individual in conversational speech maintained at the slightly increased speech rate
 - Monitor the skills and dysfluent productions carefully; if dysfluencies appear as the rate is increased, ask the individual to slow down to a rate that eliminates dysfluencies; after some practice at this slower level, ask the individual to increase the rate again
 - Reinforce the individual for increased rate and maintained fluency; note that at this level, the primary criterion to deliver reinforcement is stutter-

free speech that is moving in the direction of more natural speech; airflow and gentle onset are not emphasized, although they should be if fluency begins to break down

- Ask the individual to speak at a progressively higher rate while maintaining stutter-free speech
- Model pitch variations and let the individual imitate and then talk spontaneously with increased intonational patterns
- Model appropriate vocal intensity and let the individual imitate and then talk with sufficient loudness; note that excessively monitored airflow and rate result in too soft speech
- Make continuous judgments about speech naturalness and modify the speech to approximate normal prosodic features
- Reinforce all appropriate productions
- Stop the individual for excessively slow rate and monotonous speech; pause and continue
- Continue until the speech is judged both normally fluent and natural-sounding
- Implement a maintenance program
 - Teach the individual Self-Control (Self-Monitoring) skills by having him or her count dysfluencies, the production of target behaviors, increase in rate, abrupt phonatory onset, and breath holding; let the individual chart these behaviors as you do in treatment sessions
 - Ask the individual to judge the appropriateness of airflow, gentle onset, rate reduction, and prosodic variations to encourage self-evaluation of skills
 - Teach the individual to stop talking and to pause briefly when he or she stutters or mismanages a target skill
 - Hold informal treatment sessions in naturalistic settings; move treatment out of the treatment room, out of the clinic environment
 - Monitor the skills and fluency in naturalistic settings by giving the individual subtle signals: prompt a slightly slower rate by a hand gesture,

- prompt a gentler onset of voice by touching your own throat, and so on
 - Train the family members, teachers, friends, colleagues, and others in prompting and reinforcing the production of target skills and fluency
 - Train teachers and family members to provide opportunities to practice fluency skills
 - Train family members to hold informal treatment sessions at home
- Dismiss the individual only when natural-sounding fluency is established in natural settings and in conversational speech
- Urge the individual and the family to contact you as soon as an increase in stuttering is noticed to schedule a follow-up assessment and booster therapy
- Give a follow-up schedule to bring the individual back to the clinic periodically regardless of the outcome
- Take conversational speech samples during a follow-up assessment and offer booster treatment when dysfluency rates reach or exceed 5% of the words spoken
- Measure dysfluency rates in all treatment sessions
 - Record at least a few minutes of speech sample in every treatment session, perhaps a 2-minute sample in the beginning (after the previously described probe measure) of the treatment session, 2 minutes in the middle of the session, and 2 minutes at the end
 - Calculate the number of words spoken and the number of dysfluencies to derive a percent dysfluency rate for the entire session
 - Check your own reliability by periodically measuring the sample twice; make sure that you measure dysfluency rates with at least 90% reliability
- Vary the treatment procedures to suit the individual
 - Note that the procedures described are highly structured and the steps are specified somewhat rigidly

- Note that a beginning clinician might find it easier or efficient to follow a highly structured format where decision points and decision criteria are specified; however, with some experience, clinicians begin to skip steps, move faster or slower, and even skip a particular target skill if that does not make a difference
- See if only a rate reduction would be sufficient for a given individual; add airflow only if it enhances the rate of improvement (quickly eliminates certain kinds of dysfluencies)
- De-emphasize or stop explicitly monitoring a skill sooner or later if you did employ all the skills to begin with; probe to see if fluency is sustainable without an explicit monitoring of those skills
- Use oral reading initially if that seems to give a better control on the fluency skills; some individuals do better in oral reading where they do not have to think of what to say
- Start treatment at any level the individual can handle the skills; some need not be started at the word level; phrases might work for others, and with a few individuals, even short sentences may be the starting point
- Try moving to the short sentence level briefly if you did start at the word level; if stutter-free speech can be sustained, continue at this level (thus skipping the phrase level)
- Let the individual's performance data dictate the sequence, speed, relative emphasis on skills, and the number of steps involved in treatment

2. A Contemporary, Comprehensive Treatment Procedure for Stuttering in Very Young Children (2- to 5-year-olds)

- The fluency shaping procedure, with airflow management, gentle onset, and slow speech, is freely modified to suit the individual child; the clinician should freely experiment
- Baserate stuttering in the clinic

- Use toys, objects, pictures, storybooks, and a loosely structured play situation to evoke and manage speech from the child
- Experiment informally with all three targets used with older children and adults: airflow management, gentle phonatory onset, and rate reduction through syllable prolongation; possibly, with very young children, only a slower rate may be effective in inducing stutter-free speech; if so, skip the others; most likely to be skipped is airflow management; the next most likely to be skipped is gentle phonatory onset; the younger the child, the more likely it is that you will use only slow speech
- Counsel the family about the treatment program, its known effects, its drawback (initially unacceptable, artificial fluency), and the potential need for booster therapy in the future; impress on them the need to regularly work with the child at home and the critical role they play in fluency maintenance
- Emphasize slow normal speech even if you use gentle phonatory onset and airflow management; if you use all three, teach one target skill at a time; refer to the preceding program for steps involved in implementing the optional airflow management and gentle phonatory onset; model more frequently and use simple language with younger children
- Model rate reduction through syllable prolongation; use a rubber band to show stretching of syllables; use hand gestures to slow speech; use any other means the child will comprehend; model more frequently than you would for older children and adults; reduce your own rate; if selected, add rate reduction to airflow management and gentle phonatory onset; do not use DAF with very young children
 - Begin at the word level
 - Model stretched-out syllable durations with continuous phonation
 - Ask the child to imitate your modeling
 - Reinforce correct imitations

- Stop the child for mismanaged targets or production of stuttering; pause briefly and continue
- Continue until the child can produce slow speech (or with the optional targets) upon request and without modeling, and with stutter-free speech at the word level with 98%–100% accuracy
- Shift training to the phrase level
 - Model often
 - Fade modeling
 - Promptly reinforce stutter-free production of phrases
 - Stop the child for mismanagement of the target or targets and production of dysfluencies; pause briefly and continue
- Shift training to the sentence level
 - Model frequently and have the child imitate sentences; target completely stutter-free productions
 - Ask questions the child will answer with complete sentences; monitor target behavior or behaviors and fluency
 - Stop the child for mismanagement of the target or targets and production of dysfluencies
 - Continue until the child speaks in sentences with stutter-free speech
- Shape normal prosody
 - Model a slightly higher speech rate and ask the child to imitate it
 - Encourage the child to speak at progressively higher rates while maintaining stutter-free speech
 - Model pitch variations and let the child imitate and then talk spontaneously with increased intonational patterns
 - Model appropriate vocal intensity and let the child imitate and then talk with sufficient loudness
 - Make continuous judgments about speech naturalness and modify the speech to approximate normal prosodic features
 - Reinforce all appropriate productions
 - Stop the child for excessively slow rate and monotonous speech; pause and continue

- Continue until the speech is judged both normally fluent and natural-sounding
- Implement a maintenance program
 - Teach the child to stop talking and to pause briefly when he or she stutters or mismanages the target skill or skills
 - Hold informal treatment sessions in naturalistic settings
 - Signal the individual in a subtle manner to use the target skills
 - Train family members, baby-sitters, preschool teachers, and day care workers in prompting and reinforcing slow, normal, and fluent speech in the child
 - Train teachers and family members to provide opportunities to practice fluency skills
 - Ask parents to participate in all treatment sessions you conduct
 - Train family members to evoke a slow, normal rate of speech and to positively reinforce fluency
 - Ask them to withhold random, noncontingent negative feedback to the child
 - Teach them the skills of gently stopping the child when stuttering occurs and to pause and continue
 - Teach parents to hold informal treatment sessions at home; ask them to submit recorded sessions for your analysis and feedback to the parents
 - Teach the parents to monitor fluency in a subtle and nonpunitive manner most of the time
- Dismiss the child only when natural-sounding fluency is established in natural settings and in conversational speech
- Counsel the family about the potential relapse and the need for booster treatment
- Follow up and arrange for booster treatment if the dysfluency rates exceed 5% of the words spoken

3. **A Simplified, Minimal Therapy for Very Young Children (2- to 5-year-olds): An Exclusive Fluency Reinforcement Program.** Treatment of stuttering in very young children can be simplified

greatly; there is evidence that systematic positive reinforcement for fluency in very young children is effective; in many cases fluency shaping that involves such skills as syllable prolongation and airflow management can be avoided; an advantage of an *exclusive* fluency reinforcement program (nothing else is used) is that the parents may be trained more easily to use the technique at home than perhaps any other method; another advantage is that the procedure does not negatively affect the prosodic features of speech; follow the steps outlined here to use a straightforward fluency reinforcement program; see the following entry to combine fluency reinforcement with direct stuttering reduction strategy.

- Explain the method to the parents and recruit them to do treatment at home at least three times a week; each session should last 15 to 20 minutes
- Baserate dysfluencies for at least three consecutive sessions; offer treatment only if stuttering does not show systematic decline over baserate sessions
 - Hold play-oriented conversational speech sessions and tape-record the sample
 - Count all dysfluencies and the number of words spoken
 - Calculate the percent dysfluency rates
 - Do not treat if the dysfluency rates show consistent decline over the three sessions; counsel the parents to return to clinic if the child shows an increase in the frequency of dysfluencies
 - Note that a few children may show such decline and recover without formal treatment
- Begin treatment in conversational speech mode only, although the child's utterances may be words, phrases, or short sentences
- Select stimulus materials the child likes; toys, arranged play situations, big, colorful picture, storybooks with attractive pictures, and so forth; consult the parents before selecting the materials; if practical, ask parents to bring the child's favorite toys and storybooks to the treatment sessions

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- Ask the parents to observe the first few treatment sessions and then have them join you and the child
- Evoke speech from the child in a play-oriented format; use the toys, pictures, and other materials to stimulate speech
 - Show a picture and ask the child such general questions as “What do you see here?” and “What is happening here?”
 - Show a picture and ask such specific questions as “What is the boy doing here?” “What is the girl eating here?” “What is this Mommy doing?” “What is the kitty doing here?” and so forth
 - Tell a short story about a picture and ask the child to retell it
 - Ask the child to tell you a story
 - Let the child engage in controlled play and ask questions about what he or she is doing
 - Comment on what the child is doing to stimulate speech
 - Role-play such situations as cooking or shopping to stimulate speech from the child
 - Use hand puppets and carry on a conversation that is appropriate to the characters
 - Build simple blocks with the child and pretend difficulty on your part and ask questions about how to do it
 - Give slightly more difficult block designs to stimulate questions and requests for help
 - Show an array of toys placed on a high shelf and ask the child to describe them and request specific items
 - Use your imagination to provoke speech in the context of play and storytelling but do not allow the child to get lost in play
- Manage a behavioral contingency as you evoke speech in the play-oriented treatment sessions
 - Positively reinforce the child for all fluent productions, be they single word productions, phrases, or short or long sentences

- Use verbal praise as the main reinforcer (“That was nice speech,” “That was not bumpy! It was smooth,” “I like the way you talk”)
- Ignore all stutterings completely
- If verbal praise is not effective, begin a token reinforcement program; give a token for every fluent production to begin with; use a fixed ratio schedule in which progressively more fluent responses are required to earn a token; exchange the token for a small gift of child’s choice at the end of the session
- Train parents in reinforcing the child for fluent productions
- Ask the parents to hold informal treatment sessions at home
- Ask the parents to submit a tape-recorded sample of home treatment sessions
- Go over the tape and fine-tune the parents’ skill in immediately and positively reinforcing fluent productions and to completely ignore stuttering
- Measure stuttering as described in the previous section
- Vary the treatment as found appropriate in light of treatment data
- Dismiss the child from therapy when the child has sustained fluency at 95% or better in conversational speech produced in naturalistic settings over at least a three-week period
- Tell the parents to contact you if there is an increase in stuttering
- Give them a follow-up schedule and conduct follow-up assessments
- If the dysfluency rate exceeds 5% during any assessment session, schedule booster treatment sessions; offer the same treatment for a few sessions, or use another effective technique if preferred

4. **Fluency Reinforcement Program Combined with Direct Stuttering Reduction Methods.** In treating children who stutter, the effects of positive reinforcement for fluency may be enhanced by adding

an optional direct stuttering reduction program (non-exclusion time-out, called here pause-and-talk or response cost); to do this, take the following steps.

- Use all procedures described under the previous entry, [An Exclusive Fluency Reinforcement Program](#)
- Note that the fluency reinforcement program requires you to react only to fluent productions (with positive consequences) but nothing is done when the child exhibits a stutter; in this combined procedure, a direct stuttering reduction procedure is added to fluency reinforcement

Add Response Cost to Fluency Reinforcement.

Response Cost is a procedure to reduce an undesirable response directly; this procedure includes giving the child a token for every fluent production and removing a token contingent on every stuttering; thus, the procedure is a combination of fluency reinforcement and reinforcement withdrawal (corrective feedback or operant punishment) contingent on stuttering; there is evidence that this is an effective procedure.

- Describe the procedure to the child and the parents; point out its effectiveness
- Baserate stuttering, select stimulus materials, and structure the treatment session as described under the previous entry on [An Exclusive Fluency Reinforcement Program](#)
- Use practically all procedures described in the previous entry; except that to reinforce fluency, adopt a token system; explain the procedure to the child and the parents
- Design a token system; select plastic tokens to be used as reinforcers; assemble a variety of backup reinforcers; note that tokens themselves are not the true reinforcers—the backups are
 - Consult with parents and the child in selecting backup reinforcers
 - Select small toys, stickers, other inexpensive gift items such as pencils, balloons, and crayons

- Use activities as backup reinforcers because they can be more effective than small gifts with many children
- Arrange opportunities to engage in such activities as listening to a story that you read, drawing on the chalkboard, a play activity that the child selects, and listening to music
- Give the tokens to reinforce fluent productions
 - At the beginning of the session, ask the child to select one tangible reinforcer he or she will get at the end of the session in exchange of the tokens; make sure you offer only those choices that you have access to
 - Evoke conversational speech (as described in the previous entry) and give the child a token for every fluent production; continue to use verbal praise as in the previous entry
 - Make sure the child is able to accumulate enough tokens at the end of the session; if necessary, give two tokens for every fluent production; this may be necessary if the child's stuttering rate is very high and opportunities for earning the tokens are somewhat limited
 - At the end of the session, exchange the token for the selected reinforcer (e.g., a sticker or a pencil)
 - Alternatively, give backup reinforcers throughout the session if that is feasible (e.g., after every 5 minutes of work or as soon as the child accumulates a certain number of tokens required to earn the backup reinforcer, take a break for 1 or 2 minutes and allow the child to draw, listen to part of a story, listen to music, etc.); promptly terminate the activity and return the child to conversational speech and fluency monitoring
- Withdraw tokens to directly reduce stuttering while increasing fluency with reinforcement

- Explain this aspect of treatment to the child; inform him or her that you will be taking away a token for every instance of stuttering or bumpy speech
- Model a stutter and immediately remove a token from your own pile to demonstrate the procedure to the child
- Tell the child that his or her goal is to keep as many tokens as possible so as to earn a gift or an activity and not to lose the tokens to stuttering and thus lose the gift or the activity
- Watch for the earliest sign of a dysfluency or stutter (a lip puckering, an increased tension anywhere in the facial region, the beginnings of a sound prolongation or a sound repetition, twitching of the eyebrow)
- As soon as a sign of stuttering or stuttering itself appears, take away a token the child has accumulated by producing fluent speech
- Take the token back in a matter-of-fact manner; do not show unpleasant reactions
- Continue the procedure until fluency is stabilized in conversational speech in natural settings
 - Measure stuttering in all sessions as described in the previous entry
 - Probe fluency and stuttering periodically as described in the previous entry
 - Periodically probe the speech rate to make sure that the child is sustaining fluency without a rate change that affects prosodic features; available evidence suggests that the rate is not negatively affected and may even increase slightly when stutterings decrease
 - Train parents to conduct home treatment sessions
 - Monitor the home treatment sessions with recorded samples and refine the parents' skills in administering the treatment
 - Schedule follow-up and booster treatment sessions

Add Pause-and-Talk to Fluency Reinforcement.

Pause-and-talk is a procedure to directly reduce an undesirable response; more often described as time-out; because of variations in time-out procedures and some common misapplications of them, a more descriptive pause-and-talk is preferred here; there is evidence that pause-and-talk is an effective procedure to reduce stuttering without affecting the natural prosodic features of speech; pause-and-talk can easily be combined with positive reinforcement for fluency.

- Describe the procedure to the child and point out its effectiveness
- Baseline stuttering, select stimulus materials, and structure the treatment session as described under An Exclusive Fluency Reinforcement Program
- Use all procedures described under An Exclusive Fluency Reinforcement Program in the previous entry; as described, reinforce fluent productions with verbal praise
- Add pause-and-talk to fluency reinforcement
 - Continue to engage the child in conversational speech and to reinforce fluent productions with verbal praise
 - At the earliest sign of a stutter, say “Stop,” turn your face away to avoid eye contact with the individual, and freeze for 5 seconds; you may look at your watch to count 5 sec
 - At the end of the pause (time-out) duration, look at the child to re-establish eye contact, smile, and say something that will let the individual continue talking (e.g., “You can talk now,” “You were saying, . . . ” “OK”); note that soon the individual will begin talking again as soon as you re-establish your eye contact
 - Make sure that the child completely ceases talking when you say “Stop”; the method is ineffective if the individual continues to talk even though you have terminated eye contact and have turned away

- Continue the procedure until fluency is stabilized in conversational speech in natural settings
 - Measure stuttering in all sessions as described in the previous entry
 - Probe fluency and stuttering periodically as described in the previous entry
 - Periodically probe the speech rate to make sure that the individual is sustaining fluency without a rate change that affects prosodic features; available evidence suggests that the rate is not negatively affected and may even increase slightly when stutterings decrease
 - Train parents to conduct home treatment sessions
 - Monitor the home treatment sessions with taped samples and refine the parents' skills in administering the treatment
 - Schedule follow-up and booster treatment sessions

5. Direct Stuttering Reduction Procedures Combined with Fluency Shaping Techniques. A novel approach to treating stuttering in which fluency shaping techniques are combined with direct stuttering reduction methods; may be the most suitable for certain individuals with whom pause-and-talk or response cost may not produce the maximal effects partly because of excessive frequency of stuttering; involves only a minimal use of fluency shaping (minimal syllable prolongation, only of the initial syllable of the first word of an utterance); thus it avoids the negative side effects of fluency shaping (excessively slow and monotonous speech); although minimal prolongation may not be effective in itself, it may be effective when combined with pause-and-talk or response cost; experimental evidence is emerging in its favor; note that this technique may be used with adults or older children who stutter; this procedure may be unnecessary for very young children (3- to 5-year-olds) who may benefit the most from An Exclusive Fluency Reinforcement Program, described earlier under #3.

- Select a combination of a *minimal syllable prolongation* (MSP) with *either* pause-and-talk (P-&-T) or response cost (RC); possibly, a child will react better to one combination (e.g., MSP plus RC) than to the other (e.g., MSP plus P-&-T); therefore, be prepared to experiment
- Describe the procedure to the individual and the family; tell them you will be using a slight syllable prolongation with either RC or P-&-T and that you will be initially trying both the combinations to select the one that works best for the child
- Engage the child in conversation within a play-oriented, semistructured situation; see [An Exclusive Fluency Reinforcement Program](#) under #3 for details on structuring play-oriented treatment sessions with young children
- Apply the MSP plus P-&-T combination; begin with MSP (it does not matter what combination you try first; your preference for either RC or P-&-T may influence your initial selection)
 - Model a slight prolongation of the initial syllable of the first word in an utterance; do *not* prolong all the syllables of all the words as it is done in the full-fledged fluency shaping technique
 - Ask the child to imitate your production; make sure the child imitates a slight prolongation of the initial syllable of the first word only
 - Reinforce the child for correct prolongation with verbal praise or with a token backed up with other reinforcers
 - Say “Stop,” turn your face away from the individual, freeze, look at your watch for 5 seconds every time the child stutters or you observe an earliest sign of a stutter,
 - Make sure the individual ceases talking
 - Re-establish the eye contact, smile, and indicate in some way that the individual can now resume talking (e.g., say “You can talk now,” “You were saying . . . ,” “OK”); note that after a few trials

of this kind, most individuals will resume their speech as soon as you re-establish your eye contact and smile

- After a few successful imitations of MSP, drop modeling and prompt the syllable prolongation by a hand signal; continue to apply P-&-T for all stutterings; individual now is talking with MSP and pausing after each stuttering
- Continue this for two or three sessions to evaluate the effects of this combined procedure
- Measure the exact frequency of all dysfluencies and the number of words spoken; calculate the percent dysfluency rate for all sessions so you can later compare the effect of this combination with that of the other
- Do not try the other combination and continue with this combination only if the changes are impressive, consistent, and the individual seems happy with the technique (smiling and cooperative in the sessions and eager to work with you) with no signs of stress or negative reactions to the procedure
- Try the other combination if you do not find the results impressive (reductions in stuttering are minimal, the individual does not seem to enjoy the session, is reluctant to work with you, the individual is not punctual to the sessions, or even misses them)
- Note that tokens, exchanged for backup reinforcers, may be used to reinforce fluent productions; but do not withdraw a token for dysfluent productions, as this would be response cost; in this MSP and P-&-T combination, consequence dysfluent productions only with P-&-T
- Try the MSP plus RC combination (this could very well be the first combination you try); note that RC involves reinforcing fluent productions with a token backed up by other reinforcers and removing a token for every dysfluent production

- Describe the procedure to the individual and the family; tell them you will be using a slight syllable prolongation with RC (or P-&-T if you already have tried the RC combination); tell them that you will select the one that works best with the child
- Engage the child in conversation within a play-oriented, semistructured situation; see [An Exclusive Fluency Reinforcement Program](#) under #3 for details on structuring play-oriented treatment sessions with young children
- Apply the MSP plus RC combination; begin with MSP
- Model a slight prolongation of the initial syllable of the first word in an utterance; do *not* prolong all the syllables of all the words the way it is done in the full-fledged fluency shaping technique
 - Ask the child to imitate your production; make sure the child imitates a slight prolongation of the initial syllable of the first word only
 - Reinforce the child for correct prolongation resulting in stutter-free speech with a token the child will later exchange for a backup reinforcer
 - Withdraw a token from the child every time you hear a stutter or you observe an earliest sign of a stutter (this is the response cost aspect of the combination)
 - Continue this MSP, token presentation for fluent productions, and token withdrawal for stuttering for two or three sessions to evaluate the effects of this combined procedure
 - Administer for another session or two if unsure of the effects
 - Measure the exact frequency of all dysfluencies and the number of words spoken; calculate the percent dysfluency rate for all sessions so you can later compare the effect of this combination with that of the other
 - Compare the rate of stuttering in treatment sessions involving MSP plus P-&-T that you have already tried and MSP plus RC

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- Pick the combination that reduced stuttering faster compared with the other combination
- Consider other factors that may be important in treatment technique selection: individual's intense dislike of the technique, uncooperative behavior during treatment sessions in spite of good treatment effects, poor treatment attendance that may be attributable to a dislike of treatment, strong parental objection in spite of positive effects and a good effort to convince them; in such cases, use an alternative, which may be the other combination as long as it is effective (even if not to the same extent as the just applied procedure); note that the alternative may be a positive reinforcement of fluency, fluency shaping with syllable prolongation and airflow management, or any other techniques described in this section on [Stuttering](#)
- Tell the individual and the family about the combination (or other technique) you have picked; explain why you picked that combination (because of its more impressive effects compared with the other combination, individual satisfaction with the technique as indexed by smiling and cooperativeness in the sessions and an eagerness to work with in the sessions with no signs of stress or negative reactions to the procedure),
- Continue with the MSP plus RC combination (assuming it was selected) until fluency is stabilized in the clinic and in natural settings
- Continue with the other combination selected if MSP plus RC is not selected (also may be a different technique altogether)
- Dismiss the individual when fluency is sustained in the treatment setting as well as in the individual's natural setting
- Arrange for follow-ups and booster treatment sessions

Hegde, M. N. (2007). *Treatment protocols for stuttering*. San Diego, CA: Plural.

Treatment of Stuttering: Specific Techniques or Programs

Airflow Management in Stuttering Treatment.

Regulated airflow used in the treatment of stuttering; also called Regulated Breathing; a component in many fluency shaping procedures; effective in inducing stutter-free speech; supported by clinical research; often combined with other treatment targets, including Gentle Phonatory Onset and Prolonged Speech (described later in this section).

- Combine it with prolonged speech and gentle phonatory onset
- Model Regulated Breathing to produce stutter-free speech
 - Take an inhalation through the nose, slightly deeper than the usual so it is easily observed
 - Exhale a small amount of air through the mouth before initiating phonation
 - Initiate phonation slowly, gently, and softly only *after* the start of exhalation
 - Model the production of single words or phrases
 - Prolong the vowels and reduce the rate
- Ask the individual to imitate your modeled productions
- Reinforce the imitative productions
- Model frequently and stabilize regulated breathing of inhalation and slight prevoice exhalation
- Fade modeling and evoke regulated breathing and speech production
- Move through the sequence of words, phrases, and sentences as you add other components (soft and gentle phonatory onset, prolongation of vowels, and slow rate of speech)
- Fade the explicit management of airflow into a more natural use of airflow to sustain fluency in conversational speech in and out of the clinic
- Reinstate regulated breathing throughout the treatment when found necessary

Continuous Airflow. Maintaining airflow throughout an utterance; not an independent procedure; often a component of the Fluency Shaping Techniques or the comprehensive procedure described at the beginning of this main entry.

- Instruct the individual to take enough air before beginning speech production
- Ask the individual to exhale a slight amount of air before initiating phonation
- Ask the individual to initiate phonation gently and softly
- Ask the individual to maintain an even airflow throughout an utterance
- Model the technique
- Reinforce the individual for correct production of the target behavior

Continuous Phonation. Maintaining phonation throughout an utterance; a stuttering treatment target; not an independent procedure; often a component of the Fluency Shaping Techniques or the comprehensive procedure described at the beginning of this main entry.

- Instruct the individual in maintaining phonation throughout an utterance
- Model continuous phonation in such a way as to blur the word boundaries
- Ask the individual to imitate your modeling
- Reinforce correctly imitated productions
- Begin with shorter phrases and progress to longer sentences
- Combine it with other targets, including syllable prolongation, gentle phonatory onset, and airflow management

Counseling as Treatment for Stuttering. A collection of varied approaches to treating stuttering by giving information, advice, and strategies to deal with the problem; a range of techniques, most of them psychologically oriented; recipients are adults who stutter; often combined with direct methods of

treating stuttering; efficacy of counseling when used exclusively with no direct work with stuttering by either the clinician or the parent is not established; when combined with direct work on stuttering, whether counseling had any effect is unclear.

Counseling Parents of Children Who Stutter to Reduce Stuttering in the Child. Using the psychological methods of counseling to indirectly treat stuttering in their children; an Indirect Stuttering Treatment method (described later in this section); the main approach is talking with the child's parents to change their feelings, attitudes, ideas, and expectations about stuttering and fluency; efficacy of this approach is not established; often combined with Direct Stuttering Treatment (described later in this section); when combined, the usefulness and the effectiveness of counseling is not clear.

- Be a sensitive, uncritical, accepting listener
- Find out what the parents wish to accomplish through counseling
- Let the parents explore their feelings, emotions, perceptions, and expectations relative to their child's stuttering
- Let the parents freely talk about their fears, possible feelings of guilt, and their overt reactions to stuttering in their child
- Let the parents talk about their strategies of dealing with stuttering in their child
- Help the parents find their own solutions by offering professional views and ideas they may not have known or considered
- Express approval of their positive feelings and helpful reactions toward their child
- Help them realize their child's strengths and limitations
- Reduce their negative thoughts and feelings, including potential feelings of guilt by reassuring them that they may not have created the problem

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- Let the parents put their child's stuttering in perspective so that they do not exaggerate its negative effects
- Let the parents realize that no child is fluent all the time
- Explore actions the parents may have taken with positive effects on the child's fluency and encourage them to increase or strengthen them
- Explore actions the parents may have taken that have worsened the child's problems and encourage them to eliminate or reduce them
- Explore the parents' ideas about fluency and stuttering to encourage a more realistic view of them
- Discuss the kinds of communicative demands the parents make and ask them to reduce such demands
- Encourage the parents to create more positive speech experiences for the child by withholding criticism and accepting the child's stuttered attempts at communication

Counseling Persons Who Stutter. Using psychological methods of counseling to indirectly treat persons who stutter; an Indirect Stuttering Treatment (described later in this section); the main approach is talking with the individual to change feelings, attitudes, and expectations; efficacy of this approach is not established; often combined with Direct Stuttering Treatment (described later in this section); successful combination is no evidence for counseling.

- Be a sensitive, uncritical, accepting listener
- Find out what the individual wishes to accomplish through counseling
- Let the individual explore his or her feelings, emotions, perceptions, and expectations relative to stuttering
- Let the individual talk about the difficult speaking situations, listener reactions, and his or her own emotional reactions

- Help the individual find his or her own solutions by offering professional views and ideas the individual may be unaware of or may not have considered
- Discuss the individual's strengths that he or she may not have realized
- Reduce negative thoughts and feelings by having the individual concentrate on positive experiences, including positive speech experiences
- Let the individual put stuttering in perspective so that he or she does not exaggerate its negative effects
- Let the individual realize that no one is fluent all the time
- Explore actions the individual takes that may exacerbate stuttering and encourage the individual to eliminate or reduce them
- Explore actions and situations that enhance fluency and encourage the individual to increase them or strengthen them
- Encourage the individual to talk more positively about him or herself

Delayed Auditory Feedback (DAF). Hearing one's own speech after a delay introduced by a mechanical device; most typical effect is to slow down the rate of speech; used in treating persons who stutter and those who clutter to slow their speech rate; reduces or eliminates stuttering, but induces unnatural sounding speech; once a widely used stuttering treatment technique, but is waning because of the unnatural speech it induces; still useful in establishing Stutter-Free Speech (described later in this section) but needs additional procedures to make the speech sound natural and to make the fluency last over time and across situations; alternative is to induce slow and prolonged speech with instructions and modeling as described under the contemporary, comprehensive procedure at the beginning of the main entry.

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- Assess the individual thoroughly and establish baserates of stuttering and the speech rate
- Select a miniaturized, electronic feedback device that the individual can use in most situations; a brand is not recommended here because most units have not survived in the market
- Ask the individual to wear the portable device (may use a desktop unit with a loss in flexibility)
- Experiment with different delays to set an individual-specific delay that induces stutter-free speech (most individuals are stutter free at 250 milliseconds of delay)
- Begin by asking the individual to respond to questions that evoke two- or three-word phrases or short sentences
- Drop down to word level only if the individual cannot maintain stutter-free speech at the phrase or short-sentence level
- Model slow, prolonged speech if necessary
- Use oral reading to stabilize a slow, prolonged production if necessary (some initially find oral reading under DAF easier than speaking)
- Establish stutter-free speech with the initial delay over a few sessions
- Increase the length of utterances gradually
- Move to more spontaneous conversational speech containing longer and more complex utterances
- Fade the DAF by initially decreasing the delay in gradual steps; reduce it in 50- millisecond intervals or other intervals that still help maintain stutter-free speech
- Reduce the intensity of DAF
- Increase the rate of speech while still maintaining stutter-free speech; reinforce the individual for speaking at progressively faster rates
- Eliminate the delay altogether, but let the individual wear the unit
- Increase the rate further to move it closer to the pre-treatment, judged normal, or natural sounding rate

- Shape the normal prosodic features, including normal rhythm, intonation, intensity variations, emotional connotations, and so forth
- Conduct informal treatment sessions in varied nonclinical settings
- Train family members, teachers, and others to reinforce fluent, natural sounding speech in non-clinical settings
- Teach Self-Control (Self-Monitoring) skills (charting one's own stuttering, stopping soon after a stuttering or at the earliest sign of increased rate)
- Dismiss only after a natural sounding fluent speech is established
- Counsel the individual about the possibility of relapse and the need for Booster Treatment
- Follow up and arrange for booster treatment (relapses are common; follow-up and booster treatment are critical in maintenance)

Direct Stuttering Reduction Strategy: Pause-and-Talk (Time-Out). A brief period of non-reinforcement during which the individual stops talking and the clinician avoids eye contact with the individual; the period is imposed at the earliest sign of an imminent stuttering or associated behaviors or immediately following those behaviors; duration typically does not exceed 10 seconds; effective in reducing stuttering; supported by controlled experimental evidence; has the advantage of not inducing an artificial and unacceptable pattern of fluency; especially effective with older children and adults.

- Assess the individual thoroughly and base rate stuttering, dysfluencies, as defined
- Use pictures, objects, storybooks, and toys to evoke continuous speech from young children
- Use topic cards initially to promote extended monologues from adults
- Introduce natural conversation with both adults and children as soon as practical

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- Instruct the individual about the procedure: “I will be saying ‘Stop’ at the earliest sign of stuttering. I want you to stop talking immediately. I will also look away from you and count to five seconds. I will then look at you again. When I look at you, continue talking.”
- Ask the child to state the rule and repeat the instructions if necessary
- At the earliest sign of stuttering, say “Stop,” look away for 5 seconds, and then re-establish eye contact
- If the individual does not begin talking immediately, prompt verbally or nonverbally (e.g., *You were telling me, Yes, continue*, a hand gesture to continue)
- Stop the individual for every instance of stuttering or dysfluencies; be prompt, forceful, and unambiguous in your feedback
- Ensure that the individual does stop talking when you say so
- Watch for undue emotional responses; they tend to disappear; however, if they persist, switch to another procedure
- Measure the frequency of stutterings or dysfluencies as you have defined them in each session or after the session through recorded samples
- Begin with words and phrases and progress to controlled sentences and natural conversational speech
- Teach Self-Control (Self-Monitoring) skills in which the individual measures and records his or her stutterings and learns to pause at the earliest sign of stuttering (self-imposed time-out)
- Train family members, teachers, siblings, friends, and others to give subtle signals for the individual to stop when they observe stuttering in all situations
- Fade time-out and keep the individual on social, verbal reinforcers for fluency

- Shift treatment to naturalistic settings; give unobtrusive feedback to the individual in such settings
- Train parents or spouses to hold informal training sessions at home; let them use time-out initially for stuttering and only verbal praise for fluency later
- Have the individual or the family members submit recorded home conversational speech samples for your analysis of stuttering frequency
- Dismiss the individual only when the rate of dysfluencies is below the set criterion (e.g., less than 3%)
- Counsel the individual, the family members, or both about possible relapse of stuttering and the need for booster treatment; ask the individual to contact you as soon as stuttering increases
- Follow up the individual and arrange for booster treatment if necessary

Direct Stuttering Reduction Strategy: Response

Cost. Stuttering-contingent withdrawal of a tangible positive reinforcer the child will have received for fluent productions; each stuttering costs the individual a reinforcer he or she has earned for fluent productions; effective in reducing stuttering; supported by controlled experimental evidence; especially effective in eliminating stuttering in young children (below 10 years, roughly); does not induce an artificial pattern of fluency that should be faded out; good maintenance of fluency.

- Assess the individual thoroughly and establish baserates of stuttering or dysfluencies, as defined
- Use pictures, objects, storybooks, and toys to evoke continuous speech from young children
- Use topic cards initially to promote extended monologues from older clients
- Introduce natural conversation with as soon as practical
- Instruct the individual about the procedure: “I will give you a token for every word (later, *phrases*

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and sentences) you speak without stuttering. At the end of the session, you can exchange the tokens for this gift here. You should have at least five tokens (*or any such low number that ensures the gift for the child*) at the end of the session. The main thing is that I will take a token away from you every time you stutter. You should try to keep as many tokens as possible by speaking without stuttering.”

- Ask the child to state the rule and repeat the instructions if necessary
- Reinforce initially for every fluently spoken word; progress to phrases, controlled sentences, and conversational speech
- Take a token away promptly and in a matter-of-fact manner immediately following a stuttering or at the earliest sign of it
- Watch for undue emotional responses at token loss; they tend to disappear; however, if they persist, switch to another procedure
- Measure the frequency of stutterings or dysfluencies as you have defined them in each session or after the session through tape-recorded samples
- Teach Self-Control (Self-Monitoring) skills in which the individual measures and records his or her stutterings and learns to hand you a token at the earliest sign of stuttering (self-imposed response-cost)
- Fade response cost and keep the individual on social, verbal reinforcers for fluency
- Train family members, teachers, siblings, friends, and others to give subtle signals for the individual to stop when observed stuttering in all situations
- Shift treatment to naturalistic settings; give unobtrusive feedback to the individual in such settings
- Train parents or spouses to hold informal training sessions; let them initially use your token system; later let them fade the tokens and use only verbal praise

- Have the individual or the family members submit recorded home conversational speech samples for your analysis of stuttering frequency
- Dismiss the individual only when the rate of dysfluencies is below the set criterion (e.g., less than 3%)
- Counsel the individual, family members, or both about possible relapse of stuttering and the need for booster treatment; ask the individual to contact you as soon as there is an increase in stuttering
- Follow up the individual and arrange for booster treatment, if found necessary

Direct Stuttering Treatment. Methods in which stuttering is reduced by placing a behavioral contingency on it, instead of reducing it indirectly through counseling and other methods; in behavioral treatment, includes Direct Stuttering Reduction Strategy: Pause-and-Talk (Time-Out) and Direct Stuttering Reduction Strategy: Response Cost (described earlier in this section); a response reducing contingency is imposed on stuttering itself, instead of placing a positive contingency on fluency (and thus decreasing stuttering indirectly).

Fluency Reinforcement Techniques. Techniques of stuttering treatment in which durations of fluency or fluent utterances of varied lengths are positively reinforced; may be used exclusively, in which case there is no contingency on rate reduction, airflow management, or other targets; may be more effective with younger children than with older children or adults; when not effective, other targets and procedures may be added.

- Assess the child's stuttering
- Base rate the child's stuttering frequency in the clinic
- Arrange a loosely structured treatment setting in which toys, objects, picture books, and storybooks serve as stimuli

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- Evoke controlled conversational speech in a relaxed, play-oriented manner
- Select duration-based (e.g., fluent speech sustained for 20 seconds) or topographically based (e.g., a word, phrase, or a sentence) fluent responses for reinforcement
- Describe and model fluent utterances for the child; describe and model dysfluent utterances as well
- Describe the contingency in simple terms (e.g., “I will give you a token for easy speech”)
- Evoke controlled, limited utterances; use modeling
- Reinforce promptly and generously for fluent utterances or durations
- Ignore stuttering
- Increase the length of utterances or duration of utterances in gradual steps
- Train at the level of conversational speech
- Shift training to nonclinical settings
- Train parents in similar techniques and ask them to conduct home treatment sessions
- Counsel parents about the possibility of relapse and the need for booster treatment
- Follow up and arrange for booster treatment
- If this exclusive reinforcement procedure is found to be less effective than desired, add token loss (Response Cost) to the procedure; the child will earn a token for a fluent production and lose one for a dysfluent production

Fluency Shaping Techniques. A collection of somewhat varied treatment procedures for stuttering with an emphasis on teaching skills of fluency; contrasted with the Fluent Stuttering (described in this section) approach of Van Riper; the goal is natural-sounding normal fluency in everyday situations sustained over time; children are more likely to achieve this goal than adults; most adults who stutter may achieve only controlled (monitored) fluency; little or no attention paid to feelings and attitudes;

to use a comprehensive fluency shaping procedure that includes airflow management, gentle phonatory onset, and syllable prolongation, see at the beginning of the main entry Stuttering, and follow the procedures described under 1. A Contemporary, Comprehensive Treatment Procedure for Stuttering in Older Children and Adults.

Fluent Stuttering: Van Riper's Approach. An extensive, early, and influential treatment program for stuttering; also described as stuttering modification therapy; goal is to teach less abnormal, socially more acceptable stuttering, not necessarily normal fluency; based on the assumption that normal-sounding fluency is unrealistic for most adults who stutter; contrasted with Fluency Shaping Techniques; includes counseling and psychotherapy to change feelings and attitudes.

- Teach stuttering identification
 - Teach the individual to identify his or her stuttering and all associated problems, including negative feelings, avoidance, word fears, and easy and difficult stutterings, with discussion, demonstration, reading, modeling, and so forth
 - Teach the person who stutters to identify his or her stuttering and associated problems in everyday speaking situations
- Desensitize to toughen the individual to his or her stuttering
 - Encourage the person who stutters to be open and honest with his or her stuttering
 - Ask the person who stutters to freeze stuttering; teach the individual to continue stuttering until told to stop
 - Ask the individual to face different audiences and stutter voluntarily to learn that most people do not react negatively, and if some do, he or she can tolerate it
- Modify stuttering by teaching more fluent, easier, and less abnormal stuttering

- Ask the individual to face all feared and avoided words and begin to use them
- Teach *cancellation* by asking the individual to pause after a stuttered word and say the word again with easy and more relaxed stuttering (soft articulatory contacts and slower rate); do not ask the individual to say the word fluently; ask the individual to use cancellation outside the clinic
- Teach *pull-outs* by asking the individual to change stuttering in its midcourse; let the individual pull himself or herself out of stuttering by slowing down and using soft articulatory contacts; let the individual use them outside the clinic
- Teach preparatory sets by asking the individual to use the techniques of modifying stuttering (easy, relaxed stuttering) as he or she anticipates difficulty on a word
- Stabilize the treatment gains
 - Teach the individual to continue to assign himself or herself speech tasks that help stabilize the use of cancellations, pull-outs, and preparatory sets
 - Ask the individual to constantly practice the stuttering modification skills on difficult words
 - Reduce the frequency of individual contacts
 - Continue to seek out difficult and previously avoided speaking situations

Van Riper, C. (1973). *The treatment of stuttering*. Englewood Cliffs, NJ: Prentice-Hall.

Gentle Phonatory Onset. Soft, easy, slow, and relaxed initiation of sounds as against harsh, abrupt, and tensed initiation; a target behavior in the treatment of stuttering; combined with such other targets and procedures.

- Combine it with Airflow Management and Prolonged Speech, because gentle onset alone is not a sufficient treatment target

- Instruct the individual on the need for gentle phonatory onset; contrast it with its opposite; point out the relationship between abrupt onset and stuttering
- Demonstrate (model) gentle and tensed/abrupt onset and show how speech may be dysfluent with the latter
- Ask the individual to initiate sound softly, gently, with a relaxed posture
- Model soft and easy initiation of some vowels
- Ask the individual to imitate and reinforce correct imitative productions
- Model soft articulatory contacts for consonants and relaxed production in general
- Reinforce imitative productions of soft articulatory contacts and relaxed speech production
- Model a few single-syllable words (e.g., *I*, *bye*, *Hi*) with soft and slow onset and ask the individual to imitate
- Reinforce correct imitative productions of single syllable words
- Ask the individual to produce selected simple words and phrases with gentle onset (evoked, not modeled)
- Add airflow management, prolonged speech, or both to gentle onset
- Continue treatment with the two or three targets; move through the sequence of words, phrases, controlled sentences, and conversational speech

Gradual Increase in Length and Complexity of Utterances (GILCU). One of two highly structured and programmed operant treatment approaches of the Monterey Fluency Program (described later in this section); developed and researched by B. Ryan and B. Van Kirk; involves reinforcing fluent speech starting with single-word productions and ending with conversational speech; the length and complexity of utterances are increased gradually in the intermediate steps; supported by clinical evidence.

Indirect Stuttering Treatment. Methods in which the clinician tries to manage stuttering in the individual without concentrating on reducing stuttering directly; includes Counseling as Treatment for Stuttering (described earlier in this section); there is no direct work on reducing stuttering.

Integration of Stuttering Modification and Fluency Shaping. A dual approach that uses both the Stuttering Modification (essentially Van Riperian) and Fluency Shaping Techniques (both described in this section); a procedure of treatment described by T. Peters and B. Guitar; the dual approach is more forcefully applied to advanced stuttering than to beginning stuttering; uses a variety of handouts (e.g., understanding stuttering, how to be open about stuttering, how to use feared words) during treatment sessions.

- Let the individual understand his or her stuttering
 - Be warm and friendly; describe the treatment program to the individual
 - Ask the individual to read a brief description of stuttering, what it is, and how it develops; use the authors' handout "Understanding Your Stuttering"; answer all questions; share and reinforce the individual's insights
 - Catalog all aspects of the individual's stuttering to give a good understanding of the problem; model stuttering, use videotapes or mirrors to demonstrate stuttering
- Reduce negative feelings and attitudes and eliminate avoidance behaviors
 - Encourage the individual to discuss his or her stuttering openly with family, friends, and acquaintances; use the authors' handout "Discussing Stuttering Openly" in the treatment session
 - Ask the individual to create a hierarchy of feared and avoided words and situations; encourage the individual to use feared words and enter

- previously avoided speaking situations freely and frequently; use the authors' handout "Using Feared Words and Entering Feared Situations"
- Teach the individual the technique of *freezing or holding on to the moment of stuttering*; use the authors' handout; when the individual stutters, ask to continue (to repeat, prolong) until you signal to stop; teach the individual to be calm while doing this
 - Teach the individual Voluntary Stuttering; use the authors' handout "Using Voluntary Stuttering"; explain the rationale for it; model brief, easy repetitions or prolongations for the individual to imitate; take the individual to naturalistic settings where the individual will stutter voluntarily
 - Teach fluency enhancing skills and modify the moments of stuttering
 - Teach Rate Reduction in Treating Stuttering induced by DAF, Gentle Phonatory Onset and Soft Articulatory Contacts; use the authors' handout "Using Fluency Enhancing Behaviors"; fade DAF in gradual steps
 - Stabilize fluency enhancing skills in conversational speech without DAF
 - Initiate activities to generalize fluency to situations outside the clinic and with an audience other than the clinician
 - Teach easy stuttering; teach cancellation, pull-outs, and preparatory sets described under Fluent Stuttering: Van Riper's Approach, teach the person who stutters to integrate fluency enhancing skills with stuttering modification.
 - Initiate activities to generalize stuttering modification skills to situations outside the clinic and with other audiences
 - Help maintain improvement
 - Help the person who stutters become his or her own clinician; use the authors' handout

“Becoming Your Own Clinician”; help the individual learn to design assignments to reduce fear and avoidance (e.g., voluntary stuttering in a difficult situation); encourage the individual to work on stuttering and fluency every day

- Establish long-term fluency goals; use the authors’ handout; help the individual set the goal of spontaneous (unmonitored) fluency whenever possible; controlled (monitored) fluency when it is important to be fluent; and controlled stuttering (mild stuttering with which the person who stutters is comfortable) when it is acceptable

Guitar, B. (2006). *Stuttering: An integrated approach to its nature and treatment* (3rd ed.). Baltimore, MD: Lippincott Williams & Wilkins.

Peters, T. J., & Guitar, B. (1991). *Stuttering: An integrated approach to its nature and treatment*. Baltimore, MD: Williams & Wilkins.

Lidcombe Program. An intervention program for preschool children who stutter; administered mostly by parents; involves the behavioral methods of positive reinforcement for fluency and corrective feedback for stuttering; parents and their stuttering children visit the clinic once a week to get training in 1-hour sessions; includes systematic methods to obtain at-home and in-the-clinic measures of stuttering throughout the treatment phases; problem solving and maintenance procedures are included; developed and researched by Onslow, Packman, and associates; supportive clinical trial data are available; to use this procedure, get proper training; visit the Lidcombe Program website for training options (<http://www.lidcombeprogram.org>).

- Measure stuttering
 - Engage the child in conversation for about 10 minutes and count the syllables stuttered and syllables spoken fluently
 - Obtain percent syllables stuttered (%SS) before the onset of treatment

- Continue to collect %SS throughout the treatment phase
- Train parents to rate their child's stuttering severity
 - Train them to use a 10-point scale (1 = no stuttering; 10 = extremely severe stuttering)
 - Assess agreement between your measures and the parents' ratings by rating the clinic-measured %SS together with the parent
- Train parents to record 5- to 10-minute speech samples at home and count the number of stutterings and the duration of the child's speaking time
 - Obtain such samples from time to time throughout the treatment phase
 - Calculate stutters per minute of speaking time (SMST) from these recordings
- Train parents to deliver treatment in 5- to 10-minute sessions held at home one or two times a day, preferably when the child is better able to cooperate (such as in the mornings and early afternoons); conduct parent training during the first weeks of clinic visits
 - Set the goal of effortless, stutter-free speech for the child
 - Train the parent to engage the child in structured play with storybooks, picture books, selected toys, and so forth
 - Train them to tell the child: "We will be playing a game and let's see if you can say a lot of words smoothly. I will say 'Great talking' when I hear smooth talking."
 - Train parents to evoke single words with the help of flash cards with a child whose stuttering is very severe; train parents to use more play-oriented conversational speech with a child whose stuttering is mild
 - Train parents to decrease the treatment structure as the child becomes more fluent
 - Train parents to praise the child for stutter-free speech with such verbal statements as "Good talking!" "Your speech sounds great!" or "Your

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words are smooth!” and so forth; train them to deliver such praise promptly, consistently, and with enthusiasm and sincerity; train them to supplement praise with such tangible reinforcers as stickers or tokens

- Train parents to react in one of several ways when their child stutters in structured treatment sessions at home:
 - Either ignore stuttering or react to it as specified
 - Say something like “That was a bumpy word” and continue conversation
 - Model a fluent production of the stuttered word and continue conversation
 - Say, “A bumpy word occurred” and ask the child to repeat the word correctly or fluently
 - Ask the child to repeat the stuttered word fluently and, if successful, ask the child to do that one or two more times; reinforce fluent productions
 - Watch for signs of distress in the child and postpone the use of correction if the child reacts emotionally to it; reintroduce correction gradually and after the child experienced success with reinforcement for stutter-free speech
 - Deliver significantly fewer corrections than praise in all sessions
- Train parents to deliver treatment online
 - Train parents to monitor speech in everyday speaking situations (e.g., the parents correct a child’s stuttering while shopping or on a playground)
 - Initiate this step when the parent can measure and correct stuttering reliably and stuttering has begun to decrease as a result of treatment sessions conducted at home
- Train parents to teach their child self-monitoring skills by
 - Periodically asking the child whether a production was smooth or bumpy to encourage self-evaluation

- Praising the child for self-correction of stuttered production
- Initiate a maintenance program when the child's daily SR at home is 2.0 or lower and the clinic %SS is 1 or less for 3 consecutive weeks
 - Decrease the frequency of clinic visits gradually, building such decreases as rewards for maintaining fluency at home
 - Schedule maintenance clinic visits at 2 weeks, 2 weeks, 4 weeks, 4 weeks, 8 weeks, 8 weeks, 16 weeks, and 32 weeks
 - Assess stuttering when they visit the clinic
 - Advance the child to the next visit if the child maintains an SR average of 2.0 and SMST 1.0 or less
 - Discuss reasons for increase in stuttering should this happen and design a strategy for the parent to implement it; set a returning date
 - Continue monitoring as scheduled
 - Train parents to gradually reduce online monitoring while maintaining fluent speech

Metronome-Paced Speech. Speech that is regulated by the beats of a metronome; a form of treatment used for stuttering and cluttering; syllables or word initiations may be regulated; may be used to slow down or accelerate the rate of speech; documented immediate effects of reduced or eliminated stuttering, but timed, rhythmic, and unnatural sounding speech; possibility of individual adaptation to the beats (no more effective); Delayed Auditory Feedback (DAF) (described earlier in this section), with its similar effects, is preferred over metronome speech in the treatment of stuttering.

- Assess and base rate stuttering
- Select a miniaturized, battery-operated, electronic metronome the individual can wear like a hearing aid
- Find the individual-specific beat rate that reduces or eliminates stuttering

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- Have the individual time the production of syllables with the beats in the early stages of treatment
- Have the individual time the production of words with the beats in the later stages
- Have the individual time the production of phrases and sentences as fluency increases and stabilizes
- Increase the rate of beats or vice versa, depending on the starting point
- Ask the individual to initially wear the unit in all situations
- Fade the metronome beats by reducing its intensity in gradual steps
- Ask the individual to wear the unit with the power turned off
- Ask the individual to remove the unit
- Continue conversational therapy without the unit to stabilize fluency
- Conduct informal treatment sessions in varied nonclinical settings
- Counsel the individual about the possibility of relapse and the need for Booster Treatment
- Follow up and arrange for booster treatment

Monterey Fluency Program (MFP). A programmed operant approach to establish, transfer, and maintain fluency in persons who stutter; uses one of two specific methods: Delayed Auditory Feedback (DAF) or Gradual Increase in Length and Complexity of Utterances (GILCU); DAF is often used with older persons who stutter more severely and GILCU is typically used with younger persons whose stuttering is less severe; contains establishment, transfer, and maintenance phases; supported by clinical evidence; developed and researched by B. Ryan and B. Van Kirk.

MFP Delayed Auditory Feedback Method

- Give an overview of the program to the individual, the parents, or both; describe the role

the parents or other family members will play in fluency maintenance at home

- Give a criterion test consisting of 5 minutes of reading, monologue, and conversation to base-rate stuttering; measure stuttering in terms of stuttered words per minute (SW/M)
- Implement the fluency establishment program
 - Teach the individual to identify and measure his or her stuttering with 75% or better accuracy
 - Begin by reading with the child in a slow, prolonged, and fluent manner; reinforce verbally and with tokens and require a 0 SW/M in this and the subsequent steps
 - Instruct the child to read with a 250-millisecond (msec) DAF
 - Reinforce verbally and with tokens for fluent speech and say, “Stop, use your slow, prolonged speech” when the individual stutters
 - Decrease the DAF to 200, 150, 100, 50, and 0 msec in successive steps
 - At each step of the decreasing DAF, require a 0 SW/M (100% fluency) during a 5-minute oral reading
 - Switch to monologue with 250-msec DAF when the individual meets the 5-minute of SW/M criterion in oral reading with no DAF
 - Decrease DAF in steps similar to those for oral reading
 - Switch to conversational speech with 250-msec DAF when the individual meets the performance criterion (0 SW/M in 5 minutes of monologue with no DAF)
 - Repeat the steps to progressively decrease the DAF to zero and have the individual meet the performance criterion
- Implement the fluency transfer program
 - Vary the physical setting; have the individual read for 1 minute and converse for 3 minutes

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with you in each of five physical settings; verbally reinforce for fluency and say “Stop, speak fluently” when stuttering occurs

- Vary the audience; bring in one person (e.g., the child’s classmate), then two persons, and finally three persons; each time, let the child converse with 0 SW/M
- Ask parents to join you in treatment sessions; train them to conduct home reading, monologue, and conversational practice sessions
- Ask parents to conduct practice sessions at home; have the individual read, engage in monologue, or converse at home with increasing audience size as the corresponding steps are completed in the clinic
- Ask the parents to require fluent speech all the time at home and let them reinforce the child
- Transfer training to classroom; initially, let the child read and converse with you in the classroom
- Eventually, have the child give an oral presentation to the class
- Have the child make telephone calls and require a 3-minute fluent conversation on the phone
- Have the child speak to strangers and require 3 minutes of fluent speech
- Instruct the child to speak fluently at all times and in all situations
- Implement the fluency maintenance program
 - Follow up the child for 22 months; schedule follow-up sessions 2 weeks, 1 month, 3 months, 6 months, and 12 months
 - Give the criterion test at each visit (5 minutes of oral reading, monologue, and conversation with 0.5 SW/M or less)
 - If there is regression, recycle through selected steps of the treatment program

- Dismiss the child after 22 months of maintained fluency

MFP Gradual Increase in Length and Complexity of Utterances (GILCU)

- Give an overview of the program to the individual, the parents, or both; describe the role the parents or other family members will play in fluency maintenance at home
- Give a criterion test consisting of 5 minutes of reading, monologue, and conversation to base-rate stuttering; measure stuttering in terms of stuttered words per minute (SW/M)
- Implement the fluency establishment program
 - Teach the individual to identify and measure his or her stuttering with 75% or better accuracy
 - Instruct the individual to “read fluently”; have the individual read one word fluently; reinforce with verbal praise for fluent production; say “Stop, read fluently” when stuttering occurs; obtain 10 consecutive fluently read words
 - Gradually increase the length of orally read responses; steps include 2, 3, 4, 5, and 6 fluent words; 1, 2, 3, and 4 fluent sentences; fluency for 30 seconds and 1, 1.5, 2, 2.5, 3, 4, and 5 minutes
 - Instruct the individual to “speak fluently”; ask the individual to engage in monologue (first step with a nonreader); use pictures and topic ideas and other necessary stimulus procedures with the same gradually escalating steps
 - Engage the child in conversation; use the same gradually escalating steps
 - reinforce fluent productions with verbal praise and tokens
 - Say “Stop, read fluently” or “Stop, speak fluently” when the individual stutters

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- Model the target response when the individual persists with stuttering
- Require 100% fluency (0 SW/M) at each step
- Give a criterion test at the end of the establishment phase (5 minutes of reading, monologue, and conversation with 0 SW/M)
- Implement the fluency transfer program
 - Use the procedure outlined earlier under *MFP Delayed Auditory Feedback Method*; skip or modify steps to suit the individual (e.g., skip telephone training for a young child; select appropriate extraclinical settings for an adult)
- Implement the fluency maintenance program
 - Use the procedures outlined earlier under *MFP Delayed Auditory Feedback Method*

Ryan, B., & Ryan, B. (1995). Programmed stuttering treatment for children: Comparison of two establishment programs through transfer, maintenance, and follow-up. *Journal of Speech and Hearing Research*, 38, 61–75.

Ryan, B., & Van Kirk, B. (1978). *Monterey Fluency Program*. Monterey, CA: Behavioral Sciences Institute.

Prolonged Speech. Speech produced with extended duration of speech sounds; a target behavior in stuttering treatment; not a treatment procedure but the effect of treatment; often combined with such additional targets as Natural Sounding Fluency, Airflow Management, and Gentle Phonatory Onset; supported by clinical evidence, some experimentally controlled; procedurally, either DAF-induced or clinician-induced.

Prolonged Speech, DAF-Induced. Prolongation of speech sounds forced by the Delayed Auditory Feedback (DAF); induces stutter-free speech that sounds fluent but unnatural and socially unacceptable; a target behavior in many stuttering treatment programs; often combined with such other targets as Airflow Management, Gentle Phonatory Onset, Normal Prosody, and Natural-

Sounding Fluency; supported by clinical evidence, some experimentally controlled; for procedural details, see Delayed Auditory Feedback.

Prolonged Speech, Clinician-Induced. Speech that is produced by prolonging speech sounds, especially the vowels, and particularly in the word-initial positions taught by clinicians without mechanical help; the same as Rate Reduction; Instructions, Modeling, and Differential Reinforcement are the most effective techniques to induce it; supported by clinical evidence, some experimentally controlled; induces stutter-free speech that sounds unnatural and socially unacceptable; a target behavior in many stuttering treatment programs; often combined with such other targets as Airflow Management, Gentle Phonatory Onset, Normal Prosody, or Natural-Sounding Fluency; for procedural details, see 1. Contemporary, Comprehensive, Treatment Procedure for Stuttering in Older Children and Adults under Stuttering

Rate Reduction in Treating Stuttering. A speech rate slower than normal or below an individual-specific baserate; a typical target to reduce stuttering; a component of many treatment programs; essentially the same as prolonged speech; supported by clinical evidence; may use Delayed Auditory Feedback to induce rate reduction; preferable to reduce the rate with instructions and modeling; appropriate with young children especially when the DAF is omitted; for procedural details, see 1. Contemporary, Comprehensive, Treatment Procedure for Stuttering in Older Children and Adults under Stuttering

Regulated Breathing. A direct stuttering reduction method in which the individual is asked to modify breathing patterns along with the use of such other strategies as *thought formulation* and *relaxation*; some clinical evidence supports its use but the

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effective component of the eclectic program is not clear; developed and researched by N. Azrin and his associates; only the components, inhalation, and slight exhalation before initiating phonation have been incorporated into several current treatment programs, including fluency shaping and the contemporary comprehensive program described earlier; more effective with older children and adults than with very young children.

- Ask the individual to formulate thoughts before speaking
- Instruct the individual to inhale and exhale a small amount of air before talking; model the target behaviors
- Ask the individual to continue to exhale a little even after the last sound is produced
- Instruct the individual to pause at natural speech junctures and formulate thoughts again
- Ask the individual to stop soon after a stuttering occurs and relax, especially the chest muscles
- Ask the individual to seek out previously avoided speaking situations
- Ask the individual to practice the new method of speaking daily
- Train and ask the individual to measure and record his or her stutterings in natural settings
- Train a family member in the procedure and let the person help the person who stutters at home
- Maintain phone contact with the individual to follow up

Azrin, N. H., & Nunn, R. G. (1974). A rapid method of eliminating stuttering by a regulated breathing approach. *Behavior Research & Therapy*, 12, 279–286.

Shadowing. A stuttering and cluttering treatment technique in which the individual, without seeing the text, repeats (shadows) everything the clinician reads aloud from a book; the individual stays a few words behind the clinician; typical effect is to reduce the frequency of stuttering; popular in the 1960s and

1970s, especially in Europe; some clinical evidence suggests its effect in reducing stuttering; no research on maintenance of fluency.

- Assess the individual and base rate the stuttering frequency
- Select a reading material that is suitable to the individual
- Instruct the individual to say everything that you read
- Give practice by reading a few sentences at a time, stopping, and reinstructing, if necessary
- Do not show the text to the individual
- Read normally; do not change the rate, rhythm, or phrasing
- Record the individual's shadowing to measure the frequency of stuttering during treatment sessions

Stutter-Free Speech. Speech that contains no or few stutterings; not necessarily the same as normally fluent speech because it may not sound natural when achieved by Rate Reduction or Prolonged Speech induced by clinicians, or mechanically induced by Delayed Auditory Feedback or Metronome-Paced Speech; a result of initial stages of such treatment methods; needs additional procedures to make the speech sound naturally fluent and make it last over time and across situations.

Stuttering Modification. A collection of approaches to treating stuttering in which the emphasis is on changing the form of stuttering so that it is less severe and more socially and personally acceptable; the goal is not normal fluency, but less abnormality; approach exemplified by the Fluent Stuttering approach of Van Riper (described earlier in this section); includes attempts to change attitudes and feelings; treatment sessions loosely structured; little emphasis on measurement of behaviors; contrasted with Fluency Shaping Techniques (described earlier in this section).

Stuttering Prevention: A Clinical Method. An early treatment program for children who stutter; developed by W. Starkweather and his associates; based on the Demands and Capacities Model (DCM) of fluency and stuttering; goal is to reduce demands made on the child's fluency and increase fluency capacities.

- Assess the child's capacity for fluency and the demands the child faces
- Counsel the parents
 - Educate the parents about stuttering, the treatment program, and prognosis; give an optimistic outlook on improvement with treatment
 - Change attitudes of parents by discussing their negative feelings and possible guilt
 - Change behaviors of parents; ask them to speak at a slower rate; ask them to use shorter, simpler sentences while speaking to the child; let them know that negative reactions and punishment can worsen stuttering; encourage polite turn-taking in conversation; ask them to arrange a special talking time with the child; ask parents to demand speech less often; teach parents the direct treatment techniques
- Modify directly the child's stuttering and fluency
 - Reduce the tension and struggle behaviors associated with dysfluency
 - Initially, model behaviors (slower rate, less struggled word and phrase repetitions) without necessarily requiring the child to imitate them
 - Later, ask the child to imitate slower rate by syllable prolongation
 - Implement such fluency enhancing strategies as no interruption and no demands for verbal performance (silent periods are fine)
 - Control play activities so that they are appropriate for the child's cognitive level and allow for conversation
- Include direct intervention strategies and fluency shaping procedures as found necessary

- Use gentle phonatory onset and light articulatory contacts
- Time-out contingent on struggle behaviors
- Self-correcting
- Promote a level of language use that is normal for the child's age and gender
 - Model a level of language use that is appropriate for the child
 - Change parents' language as specified earlier
- Dismiss the child only when both the parents' and the child's behaviors have changed

Duffy, J. R. (2013). *Motor speech disorders: Substrates, differential diagnosis, and management* (3rd ed.). New York, NY: Elsevier Mosby.

Guitar, B. (2013). *Stuttering: An integrated approach to its nature and treatment* (4th ed.). Philadelphia, PA: Lippincott Williams & Wilkins.

Hegde, M. N. (2007). *Treatment protocols for stuttering*. San Diego, CA: Plural.

Logan, K. J. (2015). *Fluency disorders*. San Diego, CA: Singular.

Starkweather, W., Gottwald, S. R., & Halfond, M. (1990). *Stuttering prevention: A clinical method*. Englewood Cliffs, NJ: Prentice-Hall.

Yairi, E., & Seery, C. H. (2015). *Stuttering: Foundations and clinical applications*. Boston, MA: Pearson.

Fluency Reinforcement Techniques. Use these techniques to reduce stuttering by increasing fluency through positive reinforcement; fluent intervals or fluent utterances may be reinforced through verbal praise or tokens that are exchanged for small gifts; one of the effective procedures in treating young children who stutter; for procedures, see [3. A Simplified Minimal Therapy for Very Young Children \(2- to 5-year old: An Exclusive Fluency Reinforcement Program\)](#) toward the beginning of the main entry; fluency reinforcement is a part of almost all techniques of stuttering treatment (including Response Cost, Pause-and-Talk, the Lidcombe Program, Fluency Shaping, and so forth).

Fluency Shaping Techniques. A collection of stuttering treatment techniques based on the assumption that

normal-sounding fluency should be the intervention goal; include teaching such skills as airflow management, gentle onset of phonation, and reduced rate of speech through syllable prolongation; contrasted with Fluent Stuttering: Van Riper's Approach; described under 1. A Contemporary Comprehensive Treatment Procedure for Stuttering in Older Children and Adults at the beginning of the main entry.

Fluent Aphasia. To treat this type of aphasia, characterized by fluent but mostly meaningless speech, see the following varieties of fluent aphasia under Aphasia: Specific Types: Wernicke's Aphasia, Transcortical Sensory Aphasia, Conduction Aphasia, and Anomic Aphasia.

Fluent Speech. Treatment to enhance fluent speech is a clinical target for persons who stutter, clutter, or have neurogenic stuttering; it is speech that is smooth, flowing, effortless, and rapid within acceptable limits; negatively defined, it is speech that does not contain excessive amounts of pauses, repetitions, sound and silent prolongations, interjections, and other forms of dysfluencies; speech that is not produced with excessive effort and struggle; see Fluency Disorders for treatment procedures.

Fluent Stuttering: Van Riper's Approach. See Stuttering; Treatment of Stuttering: Specific Techniques or Programs.

Follow-Up. Take this necessary step following treatment to evaluate maintenance of skills established in treatment; it is done according to a schedule (such as 3 months after dismissal or at 6-month intervals).

- Set up a schedule with decreasing frequency (e.g., twice in the first 6 months of dismissal, the next follow-up after 1 year, the next after 2 years)
- Record a speech-language sample
- Measure the frequency of the target behaviors (production of clinically established speech sounds, language structures, fluency or dysfluency, vocal qualities, etc.)

- Calculate the percent correct use of the clinically established target behaviors
- Give Booster Treatment if the target behaviors are below the previously set criterion (such as 90% accuracy)

Frequency of Occurrence of English Consonants.

Use the following frequency of occurrence information in selecting treatment targets for children who misarticulate; note that frequency of occurrence is not an absolute criterion of selection; consider other factors as well (described under Speech Sound Disorder); note that frequency suggests ranking based on the relative frequency of 24 English consonants (e.g., 16th for /p/ means that it ranks 16th in occurrence among the consonants); the lower the ranking, the higher the frequency; two different rankings for the same sound suggest discrepancy among studies:

Sound	Frequency	Sound	Frequency	Sound	Frequency
/t/	1 st or 2 nd	/ð/	8 th or 9 th	/b/	18 th
/n/	1 st or 2 nd	/k/	10 th	/j/	18 th
/r/	3 rd or 4 th	/w/	11 th	/v/	19 th or 21 st
/s/	3 rd or 5 th	/h/	12 th or 13 th	/θ/	20 th or 21 st
/l/	4 th or 9 th	/f/	15 th or 16 th	/ʃ/	20 th or 21 st
/d/	5 th or 6 th	/g/	15 th or 19 th	/dʒ/	22 nd
/m/	6 th or 8 th	/p/	16 th	/tʃ/	23 rd
/z/	7 th	/ŋ/	17 th	/ʒ/	24 th

Delattre, P. (1965). *Comparing the phonetic features of English, German, Spanish, and French*. Heidelberg, Germany: Julius Groos Verlag.

Shriberg, L. D., & Kwiatowski, J. (1983). Computer assisted natural process analysis (NPA): Recent issues and data. *Seminars in Speech and Language*, 4, 397–406.

Functional Equivalence Training. Use this indirect method of reducing an undesirable behavior by reinforcing a desirable behavior that serves the same function as the

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undesirable behavior; for instance, teach a verbal request to a nonverbal child who whines to get adult attention; the verbal request serves the same function as whining and thus is reduced in frequency; the same as the Differential Reinforcement of Alternative Behaviors (DRA); to implement this procedure:

- Find out the functions (causes) of an undesirable behavior that needs to be reduced by analyzing:
 - The conditions under which an undesirable behavior occurs (e.g., fussing when the child needs something, leaving the chair or asking interrupting questions when the treatment task is difficult)
 - The consequences (reinforcers) it seeks (e.g., attention, reduction in needs, reduction in aversive stimulation as in escape from treatment trials, sensory stimulation as in an autistic child's hand-clapping)
- Teach a desirable, alternative behavior that is followed by the same consequence (the child achieves the same thing the undesirable behavior seeks); a few examples:
 - Teach simple requesting instead of fussing; even a nonverbal pointing to things desired
 - Simplify the treatment task to reduce its aversiveness to the child
 - Make treatment activities especially stimulating to the child
 - Give a more effective reinforcer (treatment is often boring to a child who received ineffective consequences the clinician thinks are reinforcers)
- Record the frequency of both the undesirable behavior (which should decrease) and the desirable (which should increase)

Functional Outcome Measures of Treatment.

During and subsequent to treatment, take these measures of treatment outcomes that go beyond counting the correct production of specific target behaviors in treatment sessions; measures that justify treatment outcomes by global, more general, nonclinical outcome such as the following:

- Qualitative measures of changes not only of the clinical target behaviors, but those of related behaviors (e.g., a

child who has been taught language skills may begin to exhibit more cooperative behaviors on the playground)

- Measures of changes in the individual's overall quality of life (e.g., a person with aphasia who has regained communication skills through treatment may increase his or her participation in family conversation, may become socially more active)
- Measures of generalized production of communicative skills in natural environments (e.g., a person who stutters attains fluency in the clinic and subsequently begins to speak fluently in workplace or classroom)
- Measures of functional communication skills as against grammatical accuracy or linguistic complexity (e.g., a person who has been treated may now be able to meet his or her communication needs in social situations with whatever the means)
- Measures that are based on rating of communicative effectiveness of individuals in everyday situations (e.g., in case of more severely affected individuals who use an alternative mode of communication, people may better understand the messages the individual generates)

Functional Outcome of Treatment. Generalized, broader, and socially and personally more meaningful consequences of treatment; measured in more global terms than the effects measured in treatment sessions.

- Document functional outcomes of treatment by measuring:
 - Improvement in certain quality aspects of life of an individual that may be due to treatment (e.g., improved social or academic performance of a child who has received language treatment; an adult's return to work after receiving rehabilitation services following traumatic brain injury; a stuttering male's improved dating skills or his enhanced communicative skills with his boss at work)
 - Generalized and effective production of clinically established communicative skills in social, personal, family, educational, and occupational contexts; note that minimally, individuals should produce at home

and other settings what they have been taught to produce in the clinic

- Functional communication in natural settings (e.g., effective expressions of needs and wants by persons who have received treatment for aphasia; managing daily communication needs by a child who has received augmentative or alternative communication therapy; effective use of a hearing aid in social situations and documented benefits derived by a person who has received aural rehabilitation)
- Expansion of clinically established skills in natural settings (e.g., new and longer sentences produced in natural settings by an individual who has received language treatment)
- Obtain information on functional outcomes from institutional caregivers and family members by:
 - Interviewing them on different domains of skills and communicative behaviors
 - Having them rate an individual's communicative effectiveness in natural contexts
- Note that functional outcomes
 - Are not the same as Effectiveness of Treatment; treatment effectiveness is established in controlled experimental research
 - Are simply measured changes in the individual's communicative behaviors in natural settings; favorable outcomes, if documented, may be due to any factor, including the family involvement, caregiver attention, and so forth; it cannot be claimed that treatment was effective
- Outcomes research makes sense only after the effectiveness of a treatment is established

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Gastroesophageal Reflux. Backward flow of stomach secretions into the esophagus; may lead to a reflux disorder in which the esophagus is irritated; can cause voice problems and swallowing difficulties.

Gender Identification Therapy (Elective). Communication modification programs, designed for individuals who at some point in their lives assume a different gender, known as either transsexual or transgender individuals; all modifications are entirely elective, and the goals are typically set by the individual seeking services; speech-language pathologist may be asked to help achieve communication patterns that are consistent with the newly identified gender; clinicians need to consider masculinization or feminization not just of voice, but a total pattern of communication consistent with the person's new gender identity; generally, more males opt to assume the female gender identity than females opting for male gender identity; therefore, much of the literature emphasizes male-to-female gender identification therapy; to develop a practice in gender identification therapy, follow the standards of care established by the World Professional Association for Transgender Health (<http://www.wpath.org>).

Gender Identification Therapy: General Guidelines

- Ethnocultural considerations in gender identification therapy
 - Make a thorough culturally sensitive assessment of the communication needs of the transgender individual; note that the male and female verbal and nonverbal communication patterns are highly culture bound
 - Avoid stereotypes, because in many societies the traditional male and female social and occupational communication roles have been changing, more rapidly in some societies than in other societies
- The individual's selection of targets for modification
 - Counsel the individual about the typical or traditional differences in the male and female commu-

nication patterns; note that the individual may have researched this information or may have specific communication targets to be achieved:

- If the individual needs it, provide information on resources on male/female and transgender/transsexual communication patterns (e.g., articles, books, and websites that describe gender differences in communication and etiquette)
- Ask the individual to view male and female TV news reporters and anchors to study the gender differences in communication, including nonverbal communication
- Ask the individual to watch lectures given by women and men to study the gender differences in formal communication
- Suggest the range of skills that could possibly be targeted for potential change, but have the individual set the target skills
- Draw attention to both common (even stereotypic) masculine and feminine communication characteristics as well as significant individual differences; some women may have a few stereotypically masculine communication characteristics (e.g., may not ask many questions of their conversational partners), whereas some men may have a few stereotypically feminine communication characteristics (e.g., may ask many questions)
- Assess the extent to which the individual would want to change or retain the previous feminine or masculine communication patterns even though the retained features may be stereotypically uncharacteristic of their new gender identity; for instance:
 - A former woman who has assumed a new male gender identity may wish to speak with more facial expressions, smile as before, and maintain good listening skills even though these are not stereotypical masculine communication characteristics
 - A former man who has assumed a new female gender identity may wish to retain a more objective or factual speaking style and eschew free and

frequent expressions of emotions and feelings, even though the former are not common feminine communication characteristics

- Finalize the features for modification in light of the individual's education and occupation; a transgender woman who is a business executive may set different goals than a transgender woman who is a nurse or a teacher; to the contrary, a transgender man who is a nurse may set different goals than a transgender man who is a business executive; once again, guide the individual to set his or her own targets for modification
- Assess the extent to which transgender individuals wish to generalize their new communication patterns that are consistent with their new identity; many may refrain from using their new style (e.g., their "new voice") with their family members, especially with their parents, spouses, and children; they may code-switch as they find it appropriate
- Assess the factors that determine whether the individual will want to use the new communication style at work, including the employment responsibilities (more or less verbally oriented job), whether the individual has "come out" or not at the work place, the degree of understanding and support the individual has from supervisors and colleagues, and so forth
- Promote good transition at the workplace by offering to work with the human resources department where the individual works and the supervisors of the individual; let the individual make the decision on how to proceed
- Refer the individual to psychological counseling as significant emotional issues involving family members often surround the transgender person

Communication Modification for the Male-to-Female Transgender Person

- Consider the following modification parameters for a woman whose previous gender identity was that of a

man and needs to acquire female communication patterns; for such a transgender woman, teach the feminine communication patterns:

- Assess to what extent the transgender woman has retained the masculine communication patterns; have the woman list aspects of her communication that she would like changed
- Plan an extensive voice therapy for a transgender female because the medical treatment to achieve an overall female pitch in the male vocal mechanism has met with only limited success
- Increase the vocal pitch level by modeling and systematically reinforcing progressively higher pitch levels; if a female pitch range (220 Hz, plus or minus 20) cannot be achieved, target a gender-ambiguous range of about 170 Hz; discourage the individual from raising pitch by raising tension in the vocal folds
- Show videotapes of women talking to each other or giving formal lectures to sensitize the individual to feminine communication patterns, including non-verbal communication and body language
- Find textbooks on women's studies courses at universities and refer them to the newly transgendered woman
- Ask the individual to watch movies in which female characters have a low-pitched voice for imitation
- Teach the individual feminine vocal characteristics that she chooses; generally, offer to teach the individual to:
 - Assess the entire vocal range, identify a pitch that is appropriate, comfortable, and is free of laryngeal tension
 - Use a computerized program that gives visual feedback to have the individual practice the new pitch
 - Teach feminine intonation patterns (e.g., rising intonation at the end of sentences, which characterizes female speech)
 - Discourage falsetto voice

- Teach the person to speak with more mouth openness and more air
- Teach the individual to speak more softly than before, although vocal intensity goals are secondary to other goals
- Avoid such masculine habits as throat clearing and coughing
- Teach the individual feminine speech characteristics that she chooses; generally, offer to teach the individual to:
 - Speak slightly faster and with increased pause durations
 - Articulate speech sounds more precisely and elongate vowels slightly more
 - Initiate vowels easily and smoothly and maintain light contact of articulators while producing consonants
 - Initiate speech with a slightly breathy voice
 - Link or blend words in a smooth manner
- Teach the individual feminine language and general communication characteristics she chooses; generally, offer to teach the individual to:
 - Produce more indirect rather than direct requests (e.g., “Do you mind doing . . .” instead of “Do this” type of command)
 - Emphasize in speech feelings, emotions, and human relationships as against facts
 - Be more open about self-disclosure during conversation
 - Increase the production of tag questions (e.g., “I did a good job, *didn’t I?*”)
 - Increase the expression of tag clauses (e.g., “I did fine on the exam, *I think*”)
 - Increase the production of conditional clauses at the end of sentences (e.g., expressions that qualify statements by such terms as *unless*, *whenever*, *providing*, *depending on*)
 - Decrease productions of contractions and increase the production of full forms
 - Increase the use of pronouns

- Connect expressions with the conjunction *and*
- Generally elaborate expressions
- Generally ask more questions, as this is a feminine characteristic
- Cultivate listening skills, another hallmark of feminine communication characteristics
- Teach the individual more feminine nonverbal communication patterns that she chooses; generally, offer to teach the individual to:
 - Smile more often during conversation
 - Be facially more expressive, including more expressive eyes during conversation
 - Touch the conversational partners
 - Move hands from the elbow, not from shoulders
 - Maintain more even eye contact
 - Maintain a conversational posture that is closer to the conversational partner

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Communication Treatment for the Female-to-Male Transgender Person

- Consider the following parameters for a man whose previous gender identity was that of a woman who now wishes to acquire male communication patterns; for such a transgender man, teach the masculine communication patterns:
 - Assess the transgender male's retained feminine communication patterns and have the man list aspects of his communication that he would like changed
 - Assess whether female-to-male voice change has already occurred because of hormone treatment that will thicken the vocal folds and lower the pitch to suit the new male gender identity
 - Show videotapes of men talking to each other or giving formal lectures to sensitize the individual to masculine communication patterns, including non-verbal communication and body language
 - Ask the individual to watch TV shows or news broadcasts that feature male reporters or anchors to learn an overall masculine style of communication

- Teach the individual more masculine vocal characteristics that he chooses; generally, offer to teach the individual to:
 - Assess the entire vocal range, identify a pitch that is appropriate, comfortable, and is free of laryngeal tension
 - Model and reinforce a lower pitch; see if the male pitch range could be achieved (120 Hz, plus or minus 20 as against 220 Hz, plus or minus 20 for the female voice); if a typical male pitch range cannot be achieved, target a gender-ambiguous pitch (about 170 Hz)
 - Use computerized programs to teach vocal characteristics, including pitch and loudness
 - Teach masculine intonation patterns (e.g., falling intonation at the end of sentences that characterize the male speech)
 - Teach the individual to use limited variation in intonation (a bit more monotonous than the female voice, which is better intoned)
 - Teach stress or increased loudness (masculine characteristics) in emphasizing a word in a sentence
 - Teach natural sounding speech, as the process of learning specific pitch, stress, or intonation may introduce unnatural sounding speech
- Teach the individual more masculine speech characteristics that he chooses; generally, offer to teach the individual to:
 - Speak slightly slower and without much pause durations, as these are characteristics of the male speech
 - Articulate speech sounds less precisely than before and not to elongate vowels as before
 - Use firmer articulatory contacts than before
 - Avoid breathiness in initiating speech
 - Avoid linking or blending words as women do
- Teach the individual more masculine language and general communication characteristics that he chooses; generally, offer to teach the individual to:

- Produce fewer indirect speech expressions and more indirect requests (e.g., “Do this” type of command instead of “Do you mind doing . . .” type of request)
- Use restraint in his expression of feelings, emotions, and human relationships; try to speak with more factual and observational information
- Use restraint in talking about self or being open about personal experiences or family matters
- Decrease the production of tags at the end of questions (e.g., “I did a good job, *didn’t I?*”)
- Decrease the expression of tag clauses at the end of sentences (e.g., “I did fine on the exam, *I think*”)
- Decrease the production of conditional clauses at the end of sentences (e.g., teach the individual not to frequently qualify statements with such terms as *unless, whenever, providing, depending on*, etc.)
- Use contractions more often than before
- Use restraint in the use of pronouns
- Reduce the frequency of connecting expressions that include the conjunction *and*
- Generally elaborate expressions less than before
- Generally ask fewer questions
- Teach the individual more masculine nonverbal communication patterns he chooses; generally, offer to teach the individual to:
 - Smile less frequently, if that is the choice of the individual
 - Be facially less expressive, if that is the choice of the individual
 - Touch the conversational partners less frequently than before
 - Move hands from the shoulders, not from elbow
 - Slightly decrease the frequency of eye contact
 - Maintain a conversational posture that is more distant than before

Adler, R. K., Hirsch, S., & Mordaunt, M. (2012). *Voice and communication therapy for the transgender/transsexual individual: A comprehensive clinical guide* (2nd ed.). San Diego, CA: Plural Publishing.

Coates, J. (2016). *Women, men, and language* (3rd ed., Reissued). New York, NY: Longman Publishing.

The World Professional Association for Transgender Health. (n.d.). *Standard of care for the health of transsexual, transgender, and gender nonconforming people, Version 7*. <http://www.wpath.org>

Wood, J. (2014). *Gender lives: Communication, gender, and culture* (10th ed.). New York, NY: Wadsworth.

G

Gender Reassignment. Sex change through medical and surgical means; someone who has been living as a man may now become a woman or vice versa through sex-change operation; he or she may seek Gender Identification Therapy to achieve the characteristics of the new and intended gender identity.

Generality of Treatment. The applicability of a treatment procedure in a wide range of situations involving other individuals and clinicians; demonstrated through Replication of treatment efficacy research; a Treatment Selection Criterion:

- Select treatment procedures that are known to have generality; consider the following kinds of generality in evaluating and selecting a treatment procedure:
 - Applicability of a treatment procedure by a wide variety of clinicians; has the technique been effectively used by different clinicians?
 - Applicability of a treatment procedure in a variety of individuals; is the treatment procedure known to be effective with individuals of different ages, socioeconomic conditions, educational levels, and ethnocultural backgrounds?
 - Applicability of a treatment across clinical settings; is the treatment known to be effective in such varied clinical settings as a private clinic, hospitals, university speech and hearing clinics, extended care facilities, and rehabilitation facilities?
 - Applicability of treatment across geographic settings; is the treatment known to be effective in clinical facilities across the country?

- Applicability of treatment across response classes; is the treatment known to be effective in treating a variety of disorders? (e.g., a technique such as modeling and reinforcement, known to be effective in speech and language treatment, may be equally effective in treating disorders of swallowing)
- Note that *effectiveness of treatment* implies experimental evaluation; existence of controlled data showing that treatment is better than no treatment; just because a technique is widely used in different clinics by different clinicians, in different geographic locations, and in treating different individuals is no assurance that the technique is effective; no amount of popularity or expert advocacy in the absence of controlled data should convince clinicians to use a technique routinely

G

Generalization. A declining rate of unreinforced responses in the presence of untrained stimuli; a temporary, intermediate goal of treatment; includes Verbal Stimulus Generalization, Physical Setting Generalization, Audience Generalization, Factorial Stimulus Generalization, and Response Generalization; each may be promoted with specific techniques; see Maintenance Strategy to promote lasting treatment effects.

Generalized Production. Production of clinically established behaviors in relation to new stimuli, new audiences, and in new situations (stimulus generalization); production of new kinds of responses related to the clinically established responses (response generalization); both the types measured through Probes.

Genetic/Neurophysiological Contingency. The interdependent relation between genetic and neurophysiological variables that determine or influence behaviors; contingency that interacts with Environmental Contingency.

Gentle Phonatory Onset. A treatment target for people who stutter as well as those who initiate their voice with

hard glottal attacks; teach this target of initiating voice in a gentle, soft, easy, relaxed manner to reduce hard glottal attacks and some dysfluencies that may be associated with them; for procedures, see [Fluency Disorders: Stuttering: Specific Techniques or Programs](#) and [Voice Disorders](#).

Gestural Communication. Use this method of communication that supplements oral communication with smiles and a variety of other facial expressions, body movements including shoulder shrugging, hand movements, pantomime, pointing, and head nodding or shaking to individuals with limited oral communication skills; it is a part of typical oral communication but enhanced in cases of individuals who need it; in gestural communication, expressions are important; gestural communication may be unaided, as in smiling or hand movements; or aided, as in gestures combined with a communication board; procedures described under [Augmentative Communication \(Unaided\)](#) and [Augmentative Communication \(Aided\)](#).

Glossectomy. Partial or total surgical removal of a diseased or severely damaged tongue.

Gradual Increase in Length and Complexity of Utterances (GILCU). A component of the [Monterey Fluency Program](#); for procedures, see [Fluency Disorders: Stuttering: Specific Techniques or Programs](#).

Grammatical Morphemes. It is essential to teach the grammatical morphemes to children who have language disorders: deficient grammatical morphemes is a hallmark of language disorders, resulting in telegraphic utterances; see the companion volume, *Hegde's PocketGuide for Communication Disorders*, for various grammatic morphemes; see [Language Disorders in Children](#) in this volume for treatment details; to teach the various grammatical morphemes:

- Select the grammatical morphemes that are missing in the child's speech:
 - The present progressive *ing*

- Prepositions in, on, under, and behind
- Regular plural morphemes and their allomorphic variations
- Irregular plural words
- Regular past tense morphemes and their allomorphic variations
- Irregular past tense words
- Possessive morphemes and their allomorphic variations
- Conjunctions
- Uncontractible and contractible copula
- Uncontractible and contractible auxiliary
- Regular and irregular third person singular
- Negation
- Reflexive pronouns
- Comparatives/superlatives
- Write about 20 target words, phrases, and sentences that include a specific target morpheme
- Baseline the production of each morpheme in the context of 20 stimuli; use both Evoked Trials and Modeled Trials
- Teach each word, phrase, or sentence on Discrete Trials to begin with; model the target response and reinforce correct imitative responses
- Fade modeling and start teaching the target responses on evoked trials (see Fading)
- Probe for generalized productions after having trained 4 to 6 exemplars; present untrained stimulus items
- Offer additional training when the child does not meet the 90% correct probe criterion,
- Alternate training and probe trials until the child meets the probe criterion for the morpheme being trained at a certain level (e.g., words, phrases, sentences)
- Shift training to the next higher level or select another morpheme for training
- Reinforce all morpheme productions in naturalistic conversation

Hegde, M. N. (2006). *Treatment protocols for language disorders in children, Volume I: Essential morphologic features*. San Diego, CA: Plural.

Group Design Strategy. A research strategy in which the experimental treatment effect is demonstrated by treating individuals in one group (the experimental group) and not treating individuals in another, comparable group (control group); helps demonstrate that treatment was better than no treatment; one of two strategies for treatment evaluation; contrasted with Single-Subject Design Strategy.

- In implementing a basic control-group/experimental-group treatment research in which a single treatment is evaluated, researchers:
 - Identify a population (a large number of subjects with known characteristics) of participants with the disorder for which the treatment to be evaluated has been designed
 - Randomly select a sample of participants from the population
 - Randomly assign the participants to an experimental and a control group
 - Match participants in the two groups on relevant variables if random selection and assignment are not possible
 - Administer pretests of the disorder (or measure specified skills) in the two groups
 - Treat participants in the experimental group; withhold treatment from the control group
 - Administer posttests of the disorder (or measure specified skills) in the two groups
 - Compare the performance of the two groups on the pretest and the posttest
 - Conclude that the treatment was effective if the experimental group improved while the control group did not
- In evaluating the absolute and relative effects of multiple treatments, the researchers:
 - Use multigroup experimental designs in which two or more groups receive treatment (each group receives only one treatment)
 - Use factorial designs to evaluate interaction between treatments and personal characteristics of individuals; such designs help establish the relative effectiveness of

treatments depending on such personal characteristics as age, socioeconomic factors, and the severity of the disorder

Hegde, M. N. (2003). *Clinical research in communicative disorders: Designs and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Н

Hard Glottal Attack. Treat abrupt voice initiation, a form of vocally abusive behavior, in individuals with voice disorders; it is also a treatment target in some persons who stutter.

- Teach gentle, relaxed, easy onset of phonation
- Teach the individual to blend words initially
- Teach gentle onset in persons with stuttering by instructions and modeling
- Use the Chewing technique, Whisper-Phonation, the Chant-Talk, and the Yawn-Sigh, all described under Voice Disorders, Specific Normal Voice Facilitating Techniques to reduce hard glottal attacks
- Contrast the easy- versus the hard-onset of voice

Hard of Hearing. To treat communication disorders in persons who have reduced hearing acuity but nonetheless are able to acquire, produce, and comprehend language primarily with the help of audition, with the help of amplification and visual cues, see Hearing Loss.

Harshness. To treat this voice quality that results from excessive laryngeal tension, effort, and constriction:

- Use relaxation to reduce vocal tension
- Teach soft, easy contact of the vocal folds
- Teach gentle onset of phonation
- Use a combination of Specific Normal Voice Facilitating Techniques described under Voice Disorders

Hearing Aid Orientation. An aural rehabilitation procedure implemented by an audiologist; a set of procedures in which the use and care of a hearing aid is taught to a person with hearing loss who has just purchased one.

Hearing Aids. Electronic devices that amplify sound, prescribed for individuals with hearing loss; may be of analog or digital variety; types include body-worn, behind-the-ear (BTE), in-the-canal (ITC), in-the-ear (ITE), or completely in-the-canal (CIC).

Hearing Assistive Technology Systems. Devices other than the traditional hearing aids that are more or less technologically sophisticated; designed to help persons with hearing loss in various communicative situations; used in situations in which the traditional hearing aids are less effective; most are capable of converting speech and environmental sound signals directly from the source to auditory (amplified), visual, or vibratory signals that the person with hearing loss receives; also known as assistive listening devices (ALDs) or hearing assistive technology systems (HATS); most systems have detached microphones that may be placed near the source of sound or speech signal; the signal is transmitted to an amplifier and the device the individual wears; see also [Aural Rehabilitation](#); systems include the following:

- Frequency modulated (FM) auditory system that offers wireless connection between a speaker and a listener; consists of a transmitter with a microphone and a receiver with earphones; signal is transmitted to the receiver through radio frequency and is unaffected by noise in the room, a problem not solved by traditional hearing aids; may be used in group aural rehabilitation
- Infrared listening systems that include transmitters that send messages on pulses of light and receivers worn by individuals to receive and decode those messages; useful in large listening environments such as concert halls, theaters, houses of worship, and classrooms; smaller transmitters can be used in homes to view television and listen to music
- Alerting systems that include mechanisms to amplify telephone ringing; flashing lights that alert a person with hearing loss to incoming phone calls, smoke or fire alarm, a baby's cry, ringing of a door bell, and so forth; vibratory mechanism that wakes up a person with hearing loss from sleep
- Television or radio amplifiers that include a small amplifier, a microphone that is plugged into the television or radio, and a receiver the individual needing extra amplification of signals wears; a transmitter sends the signal to the receiver;

volume may be adjusted on the amplifier, thus allowing louder signals only for the person using the system

- Telephone amplifiers that help individuals with mild hearing loss who do not use a hearing aid; individuals may fit an amplifying device on the telephone itself
- Telecommunication devices for the deaf (TDDs) that allow two persons communicating over a phone to type their messages; also called text telephones; allow display of typed messages on a small screen on the telephone; may be a part of a relay service that allows a hearing person without the special equipment to speak to a staff person who types in the message, which is transmitted to the TDD used by person with hearing loss

Hull, R. H. (Ed.) (2014). *Introduction to aural rehabilitation* (2nd ed.). San Diego, CA: Plural.

Tye-Murray, N. (2015). *Foundations of aural rehabilitation* (4th ed.). Stamford, CT: Cengage Learning.

Hearing Conservation. An educational and periodic assessment program implemented by an audiologist to prevent or reduce the risk of hearing loss; includes procedures to monitor hearing over a period of time and to educate the individual, family, employers, and employees about protecting their hearing.

Hearing Loss. Treatment of communication disorders associated with reduced hearing acuity requires an audiological examination to determine the level of hearing loss and an aural rehabilitation program in which the individual has been fitted with a hearing aid; hearing loss is a hearing level that is greater than 25 dB HL; includes the Hard of Hearing and the Deaf; classified as shown under Hearing Loss: Classification; oral speech and language disorders are a common concomitant of hearing loss, especially deafness; mostly, the treatment procedures for Language Disorders in Children, Speech Sound Disorders, and Voice Disorders are applicable with the following special considerations:

General Guidelines

- Note that parents of deaf children may or may not opt for oral communication training; most deaf children

learn the American Sign Language or other forms of a manual communication system and become competent communicators

- Parents of children with a diagnosis of mild to moderate hearing loss are likely to opt for oral communication training
- Review the audiological reports, consult with the child's audiologist to fully understand the child's type and level of hearing loss and its educational implications
- Before beginning a home-based language stimulation program or the clinic-based formal teaching program, make sure that the child has been trained well in the use of the hearing aid and that an auditory training program has been completed
- Begin speech and language stimulation training as early as possible
- Have the child under appropriate medical and audiological management
- Get the family involved from the beginning in speech and language stimulation activities
- Have the child fitted with an individual hearing aid; at the beginning of each session, check the child's hearing aid to ascertain that it is turned on and working properly
- Work closely with educators and special educators, especially the educator of the deaf
- Select treatment targets to support the child's academic work and success in the classroom; consult with the child's regular and special education teachers on relevant communication targets, especially the language targets
- Integrate literacy targets with communication training, as children with hearing loss tend to have significant reading and writing problems; work closely with the child's teacher to select reading and writing targets that may be a part of the communication training
- Train family members to conduct home teaching sessions that parallel yours

Teaching Oral Language

- Train parents to start a home language stimulation program; impress on the parents that it is inappropriate

to withhold normal verbal interaction and special verbal stimulation on the mistaken assumption that it is “of no use because the child cannot hear”

- Begin formal oral language training as early as possible
- Use visual cues in all training sessions
- Teach the basic words initially; select functional words; teach parents to introduce new and functional words on a regular basis; pay special attention to verbs which are especially difficult for children with hearing loss
- Teach phrases and sentence structures subsequently
- Correct immediately any non-English word order that you observe in the child’s speech
- Pay special attention to teaching Grammatical Morphemes, as they are especially difficult for children with hearing loss
- Teach longer, complex, and compound sentences; coordinate teaching various syntactic features with the classroom instruction of language skills; teach a variety of syntactic structures
- Pay special attention to pragmatic language skills, as they are especially difficult for children with hearing loss; see Language Disorders in Children for details; teach such skills as:
 - Topic Initiation, as the children with hearing loss are often reluctant to talk and fail to begin conversation; reinforce the child for suggesting topics for conversation
 - Topic Maintenance, as the child tends to speak very briefly; target progressively longer duration of speech on selected topics
 - Turn-Taking, as the child is likely to interrupt or fail to take an offered opportunity to speak; teach the alternating conversational roles of speaker and listener
 - Narrative Speech, as the child’s narratives tend to lack details and have confused temporal order; tell stories and ask the child to retell them; reinforce more elaborate and well-sequenced narratives
 - Teach the child to give sufficient background information before saying something

- Teach correct stress and intonation to improve prosodic features of speech
- Teach Conversational Repair Strategies; reinforce the child for asking questions or requesting clarifications when messages are not understood; ask questions or request clarifications and reinforce the child for appropriately modifying the utterances, descriptions, or explanations
- Teach abstract terms, terms with dual meanings, and the meaning of proverbs, idioms, and similes, and synonyms and antonyms, as they are especially difficult for children with hearing loss
- Refer to specialists who can teach such nonverbal communication systems as American Sign Language if the individuals, families, or both prefer

H

Teaching Speech Production Skills

- Set improved intelligibility as the main speech treatment goal
- Use the treatment procedures described under Speech Sound Disorders with the following modifications:
 - Give ample visual cues in teaching speech sound production
 - Use an FM auditory trainer if working in small groups
 - Use such procedures as the Phonetic Placement Method
- Target the following speech production tasks and teach them with intensive trials:
 - Teach fricatives, stops, and affricates, as these are especially difficult for children with hearing loss
 - Teach voiced and voiceless sound distinctions; eliminate substitutions of voiced consonants for voiceless consonants
 - Teach discrimination between diphthongs and vowels (e.g., /a/ and /ai/ distinctions)
 - Reduce nasalized production of vowels and oral consonants
 - Reduce unnecessarily extended duration of vowels
 - Reduce breathiness before vowel productions

- Teach sounds that are omitted in word initial, and especially word final, positions
- Teach more precise and definitive production of all speech sounds, especially consonants
- Reduce or eliminate additions of unnecessary sounds (e.g., s_{atop} for *stop*)
- Reduce the frequency of inappropriate release of final consonants (e.g., cap^h)

Treating Voice and Resonance Disorders

- Use the standard techniques described under Voice Disorders
- Use mechanical, visual feedback with such instruments as VisiPitch
- Modify such abnormal voice qualities as breathiness, harshness, hoarseness, stridency, and monotone
- Reduce the vocal pitch if it is judged too high
- Teach appropriate loudness of speech if judged too soft
- Modify resonance disorders; modify both hypernasality and hyponasality

Treating Prosodic Problems

- Teach smooth flow of speech
- Reduce pauses that may be too frequent and placed inappropriately
- Teach normal intonation to reduce a monotone
- Teach appropriate breath control to improve phrasing
- Judge the effects of pitch and loudness modification on speech prosody

Hull, R. H. (Ed.) (2014). *Introduction to aural rehabilitation* (2nd ed.). San Diego, CA: Plural.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Scott, D. M. (2002). Multicultural aspects of hearing disorders and audiology. In D. E. Battle, *Communication disorders in multicultural populations* (3rd ed., pp. 335–360). Boston, MA: Butterworth-Heinemann.

Tye-Murray, N. (2015). *Foundations of aural rehabilitation* (4th ed.). Stamford, CT: Cengage Learning

Hearing Loss: Classification. Hearing loss is typically classified as follows:

- Mild hearing loss: 25–40 dB HL
- Moderate hearing loss: 41–70 dB HL
- Severe hearing loss: 71–90 dB HL
- Profound hearing loss: 90 dB and higher

High Probability Behaviors. Use this unusual method of reinforcing target skills by giving access to highly desired and frequently exhibited behaviors only when the child exhibits a certain number of the target responses; a behavior that is exhibited often can reinforce the one exhibited less often if, and only if, the access to the more likely behavior is made contingent on first exhibiting the less likely behavior; an effective treatment method to increase low frequency treatment targets.

- Identify behaviors your client exhibits frequently (e.g., listening to music, looking in a picture book, listening to stories you tell, drawing on the blackboard, watching television, skiing)
- Design a method by which you in the treatment sessions and the family members at home can control opportunities for those behaviors
- Give tokens in treatment sessions for producing the target responses that are initially of low frequency (e.g., correct production of speech sounds, language features, fluency)
- Let the individual accumulate the tokens and exchange them for opportunities to engage in the high-probability behaviors (e.g., brief periods of listening to music or stories in treatment sessions, watching TV at home, going on ski trips)

Hoarseness. Voice quality that results from leakage of air and aperiodic vibration of the vocal folds; pitch may be too low; any condition that changes the mass and size of the vocal folds, including vocal nodules, may cause hoarseness of voice.

- Obtain a medical evaluation and clearance before starting voice therapy

- Modify the vocally abusive behaviors
- Use a combination of Specific Normal Voice Facilitating Techniques, described under Voice Disorders.

Huntington's Disease. To treat communication impairments in individuals with this degenerative neurological disease, see Dementia in this volume for general management procedures and take note of the following unique features of treatment; see the companion volume, *Hegde's PocketGuide to Communication Disorders*, for etiology and symptomatology of Huntington's disease.

- Work with the medical staff who may prescribe drugs to control such psychiatric symptoms as hostility, depression, anxiety, physical and verbal abuse, paranoia, hallucinations, and delusions that will affect communication treatment;
- Target dysarthric speech for treatment; see Dysarthria and Hyperkinetic Dysarthria, which is the likely type of motor speech disorder associated with Huntington's disease
- Target language skills as you would in individuals with Dementia
- Work with the family and other caregivers as outlined under Dementia
- In the late stage, consider simple, no-technology Augmentative and Alternative Communication, including alphabet boards, communication boards, eye-gaze procedures, and *Yes-No* procedures

Hyperadduction. To treat this laryngeal abnormality causing voice problems, reduce the excessive force and tension associated with vocal fold closure.

- Teach laryngeal relaxation
- Teach breathy onset of phonation
- Teach gentle, relaxed, easy phonatory onset
- Massage the larynx
- Use such other specific normal voice facilitation techniques as the Yawn-Sigh Method and the Chewing Technique described under Voice Disorders: Specific Normal Voice Facilitating Techniques.

Hyperkeratosis. Keratotic lesions in the pharynx or larynx; pinkish, rough lesions with horny growth; associated with [Voice Disorders](#); nonmalignant in the initial states; may be a precursor to malignant lesions; due to continued irritation of tissue, including chronic smoking; treatment is to stop behaviors that result in continued irritation (e.g., cessation of smoking).

Hyperkinetic Dysarthria. To treat this motor speech disorder, see [Hyperkinetic Dysarthria](#), under [Treatment of Dysarthria: Specific Types](#).

Hypernasality. To treat hypernasality in children and adults, see [Velopharyngeal Dysfunction](#) as well as [Voice Disorders: Treatment of Disorders of Resonance and Cleft Palate](#).

Hypertonia. Management of excessive muscle tone or tension associated with neurological damage is important in treating many individuals with neurogenic communication disorders (e.g., [Cerebral Palsy](#) and [Dysarthria](#)).

Hypoadduction. To treat inadequate approximation of vocal folds that causes breathiness and weak voice, see [Dysarthria](#), [Spasmodic Dysphonia](#), and [Voice Disorders](#).

Hyponasality. Treatment of reduced or absent nasal resonance in the production of nasal sounds is typically medical, because it is often due to nasal or nasopharyngeal obstruction due to edema, tumors, and other factors; the same as [Denasality](#); see [Velopharyngeal Dysfunction](#) and [Voice Disorders: Treatment of Disorders of Resonance](#) for additional information.

Hypotonia. Clinical management of reduced tone or tension may be a consideration in individuals who have [Dysarthria](#).

Hypokinetic Dysarthria. To treat this type of motor speech disorder, see [Hypokinetic Dysarthria](#) under [Treatment of Dysarthria: Specific Types](#).

I

Iconic Symbols. Use symbols that look like the object they are supposed to represent in teaching Augmentative Communication, Aided; easier to teach and learn than Non-iconic Symbols.

Ideographic Symbols. Use these symbols that are more abstract than pictographic symbols to teach a type of non-verbal communication in which the ideas are expressed through graphic representation; may be line drawings; for details on teaching methods, see Augmentative Communication, Aided.

I

Idiosyncratic Patterns. Target these phonological error patterns, generally not found in children who speak normally, for elimination; see Speech Sound Disorders for details.

Imitation. Use this mode of response in the early stage of treatment for all disorders; it is a response that follows a modeled stimulus and takes the same or similar form of its stimulus; essential to establish all communicative target skills; Modeling is the treatment technique to evoke imitation; to teach imitative responses:

- Model the correct response for the individual; use instructions and demonstrations as found appropriate
- Place extra vocal emphasis on the specific target behavior in a modeled utterance (e.g., “Say *two cups*,” with an emphasis on the grammatic morpheme in teaching the regular plural productions)
- Reinforce the individual for imitating the modeled response
- Reinforce approximations initially, especially in the treatment of correct speech sound productions
- Require greater match to the modeled response in successive trials
- Fade modeling gradually and transition to evoked responses

Imitation of Aversive Control. Imitatively learned use of aversive methods to control others by persons who were subjected to aversive control themselves; a potential,

undesirable side effect of punishment procedures; a reason to limit punishment procedures in treatment by using strong positive reinforcement techniques for desirable target behaviors; to avoid punishment procedures, use Differential Reinforcement in which you positively reinforce desirable behaviors that replace undesirable behaviors.

Imposition of Work. One of the Direct Methods of Response Reduction in which an undesirable behavior is reduced by immediately imposing work designed to reduce or eliminate the negative effects of that behavior; also known as overcorrection; has two components: restitution and positive practice.

Restitution

- Immediately following an undesirable behavior (e.g., throwing toys around), ask the child to neutralize the effects of that behavior (pick up the toys)
- Ask the child to go beyond neutralizing the effects of his or her behavior by improving the situation (ask the child to put the toys on a shelf and then clean up the mess created by another child)

Positive Practice

- Ask the child to practice an incompatible, appropriate behavior repeatedly without reinforcement (ask the child to organize your stimulus materials)

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Improvement. Documented positive changes in the individual's behaviors compared with initial assessment or baseline performance; needed to justify treatment; what a clinician can claim when routine treatment is offered; not the same as Effectiveness of Treatment, which requires controlled experimental evidence that demonstrates that treatment was better than no treatment; to document improvement:

- Define the target skills in measurable terms
- Establish reliable baserates of target skills before initiating treatment

- Measure the frequency of target skills in all treatment sessions
- Compare the base rate and treatment response rates to show systematic positive changes in the skills under treatment.

Incidental Teaching Method. Use this naturalistic language teaching method that targets functional communication skills in the context of everyday verbal interactions; for procedural details, see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

I

Incompatible Behaviors. Behaviors that cannot be produced simultaneously; use this behavioral strategy in which you teach a desirable action that is incompatible with an undesirable action; for example, to indirectly reduce a child's off-seat behavior, systematically reinforce sitting behavior; the strategy helps avoid punishment of the undesirable action; see [Differential Reinforcement of Incompatible Behaviors \(DRI\)](#).

Independent Variables. Hypothesized or demonstrated causes of events that scientists investigate; they are the same as treatment methods clinicians use; anything a clinician does that affects the individual's behavior, including instructions, modeling, demonstration, positive reinforcement, and corrective feedback are potential independent variables of positive changes in the individual; contrasted with [Dependent Variables](#).

Indirect Language Stimulation. A collection of somewhat varied, naturalistic, unstructured, or minimally structured language stimulation procedures based on play activities with no systematic reinforcement for specified target behaviors; for procedures, see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

Indirect Methods of Response Reduction. Use these highly effective and desirable means of indirectly

reducing undesirable behaviors by increasing desirable behaviors; indirect because no contingency is placed on behaviors to be decreased (e.g., there is no punishment for undesirable behaviors); in many cases, more desirable than the direct methods of response reduction because they avoid negative side effects of punishment; see [Differential Reinforcement of Alternative Behavior](#), [Differential Reinforcement of Incompatible Behavior](#), [Differential Reinforcement of Low Rates of Responding](#), and [Differential Reinforcement of Other Behavior](#) for variations and procedural details.

Indirect Treatment for Swallowing Disorders.

Treatment of swallowing disorders in which food is not presented to the individual; mostly involves various kinds of exercises; see [Dysphagia](#) for details and contrast with [Direct Treatment for Swallowing Disorders](#).

I

Individualized Education Programs (IEPs).

Child-specific intervention programs designed for children with disabilities or special needs served in public grade schools.

- Assess the child's communicative strengths and weakness
- Write an IEP for each child you serve; consult with teachers and special educators to include specific goals and objectives they suggest
- State the short- and long-term intervention objectives in measurable terms
- Describe the frequency and duration of your intervention sessions
- Specify the amount of time the child will spend in regular classroom
- Specify the intervention initiation and termination dates
- Justify the need for your services (use the school district's guidelines in determining service eligibility)
- Specify the names of special education or other professionals who also will serve the child
- Talk to family members to get their input and approval of treatment goals
- Hold an IEP meeting to finalize the intervention plan and to get the signatures of all attending, including those of the parents

- Make periodic assessment of the child to see if the goals are being met and to evaluate whether the goals or treatment procedures need to be modified

Individualized Family Service Plans (IFSPs).

Special education programs designed for children with disabilities in the age range of birth through 2 years and their family members.

- Develop a plan similar to IEPs
- Include information on the family's needs and strengths
- Orient the plan toward family involvement

I

Informative Feedback. Use this method of reinforcing target skills by providing information to the individual on his or her performance levels; such information is known to reinforce clinical target skills; information feedback may be verbal or mechanical; contrasted with Mechanical Corrective Feedback, Nonverbal Corrective Feedback, or Verbal Corrective Feedback, in which the information provided is specific to the wrong responses to be decreased.

Verbal

- Periodically, tell the individual how well he or she is doing (e.g., "You have improved to 85% today")
- Show and describe charts and graphs that depict increases in target skills

Mechanical

- Display positive changes and improvement data on computer monitors and other display devices

Infrared Systems. To use this system in the rehabilitation of persons with hearing loss, see Hearing Assistive Technology Systems

Inhalation Method. Teach this method of air intake to produce esophageal speech; a method of teaching speech without larynx; for procedures, see Laryngectomy.

Inhalation Phonation. Use this method of voice therapy to evoke true vocal fold vibration in individuals who are aphonic; for procedures, see [Voice Disorders, Specific Normal Voice Facilitating Techniques](#).

Initial Consonant Deletion. Target this phonological error pattern for elimination by teaching the correct production of initial consonants in word positions; see [Speech Sound Disorders](#) for procedural details.

Initial Response. The first, simplified component of a target response used in [Shaping](#).

Injection Method. Teach this method of air intake to produce esophageal speech; a method of teaching speech without larynx; for procedures, see [Laryngectomy](#).

Instructions. Use these common teaching procedures at the beginning of intervention and whenever needed throughout the treatment phase; they are verbal stimuli that promote the production of target responses; combined with [Demonstration](#), [Modeling](#), and [Manual Guidance](#) (as in [Phonetic Placement Method](#)).

- Design instructions that clarify the target behavior for the individual
- Simplify your instructions and tailor them to the individual client
- Write your instructions and practice their delivery, but deliver them naturally; do not read them
- Repeat instructions until the individual understands them
- Combine them with demonstrations, modeling, and manual guidance

Intellectual and Developmental Disabilities.

Intervention for individuals with intellectual, developmental, social, and adaptive behaviors that are significantly below normal is a multifaceted task; children and adults with intellectual disabilities need a comprehensive educational, social, and clinical rehabilitative program because

of the academic, social, and occupational challenges they face; treatment and rehabilitation are best offered by a team of specialists, including special educators, psychologists, and speech-language pathologists; communicative problems are a significant aspect of these disabilities and are often the early diagnostic markers; intervention should routinely consider both speech and language skills, and in many cases fluency and voice problems; presence of hearing loss and physical disabilities may complicate treatment efforts; see the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology, etiology, and behavioral limitations associated with intellectual disabilities; the treatment procedures for Language Disorders in Children and those for Speech Sound Disorders are applicable with the following special considerations:

- Develop an educational program with the help of other specialists, including special educators and psychologists
- Recommend or initiate communication treatment as early as possible
- Train the family members to implement an early Language Stimulation program at home
 - Depending on the age of the child and the skill level, target social reinforcement of cooing, babbling, other forms of early vocalization, the first few words, and production of phrases and simple sentences at different stages of intervention
 - Conduct a few trial treatment sessions to find out what works best for the child before you recommend a home language stimulation program
 - Ask parents to observe your treatment sessions
 - Have parents participate in the treatment sessions by presenting stimuli, modeling the target language skills, and reinforcing the child
 - Have parents video record sample home language stimulation sessions and review them for accuracy; give constructive feedback to the parents to improve their implementation of the program
 - Schedule formal treatment sessions when warranted
- Design a comprehensive formal treatment program for the child; consult with the teachers, special education

specialists, and psychologists and review the child's academic program and then:

- Select speech and language targets that help meet the child's academic and social demands
- Include voice and fluency targets if warranted
- Select target behaviors that are functional (effective) in educational, occupational, family, and social situations
- Select emergent literacy skills or regular literacy skills that may be integrated with communication treatment
- Have the parents and the teachers review the selected target skills and modify the targets in light of their suggestions or comments
- Select the initial training targets that will be most effective in natural settings and in classrooms in case of children:
 - Select articulation training before language training if this leads to improved communication sooner
 - Select language training if teaching a few functional words will help improve social communication
 - Select a few functional signs before verbal expressions in the case of nonverbal and severely retarded children
- Sequence the target behaviors carefully
 - Start with functional words or phrases if the child already produces a set of words
 - Build on the existing skills, skills just mastered; expand the functional verbal responses from phrases to sentences
 - Keep social and naturalistic communication as the final target
 - Use small step increments; shape responses whenever the individual cannot imitate modeled responses
 - Model the target responses frequently
 - Reinforce approximations of modeled responses in the initial stages of treatment
 - Require better approximations as the skill becomes more reliably produced
 - Require an exact match, if that is practical, as the child's proficiency increases
 - Fade modeling in gradual steps; use prompts to fade modeling

- Structure the sessions initially, but keep the structure flexible
 - Use objects and events more than pictures as treatment stimuli
 - Have parents bring the child's books and toys that may be used as treatment stimuli to promote better generalization to the child's home environment
 - Use the Discrete Trial procedure to initially establish the skills unless the child can engage in more naturalistic interactions
 - Use primary reinforcers initially, but always pair them with verbal praise
 - Fade primary reinforcers as soon as possible to maintain the correct responses on verbal praise
 - Consider using tokens for correct responses that you exchange for a small gift at the end of the session; pair the tokens with verbal praise so the tokens may be faded later
 - Use food and drink to teach mands (e.g., such mands as "I want some juice, please" cannot be effectively reinforced with verbal praise)
 - Stretch the reinforcement schedule gradually; start with continuous reinforcement and shift gradually to such other intermittent schedules as fixed ratio 3, 4, or 5 (reinforcement for the 3rd, 5th, or 7th correct response)
 - Include parents, siblings, or peers in treatment with a view to have them prompt and reinforce target skills in natural settings
- Loosen up the treatment sessions
 - Shift training from the discrete trial procedure to more conversation-oriented procedures
 - Use such procedures as Mand-Model and Incidental Teaching Method described under Language Disorders in Children
 - Train in varied naturalistic settings to promote generalized production
- Teach social communication skills
 - Teach conversational skills; start with such easily learned skills as turn-taking in conversation (see Language Disorders in Children)

- Teach topic initiation, topic maintenance, and narrative skills (see [Language Disorders in Children](#))
- Implement a maintenance program
 - Train parents and teachers in prompting and reinforcing newly acquired communicative behaviors
 - Train skills in conversational speech
 - Train in varied settings
 - Train in the presence of peers and siblings
 - Reinforce only socially and occasionally
 - See [Maintenance Strategy](#) for additional suggestions
 - Follow up and arrange for booster treatment
- Consider nonverbal means of communication
 - Teach a nonverbal means of communication when warranted
 - Consider an appropriate form of [Augmentative and Alternative Communication](#), including [American Sign Language](#)

Hegde, M. N., & Maul, C. M. (2006). *Language disorders in children: An evidence-based approach to assessment and treatment*. Boston, MA: Allyn & Bacon.

Reed, V. (2012). *An introduction to children with language disorders* (5th ed.). Boston, MA: Allyn & Bacon.

Intelligibility. A treatment target for individuals who cannot produce speech sound accurately; to improve impaired speech intelligibility, teach the correct production of phonemes; see [Speech Sound Disorders](#) and [Dysarthrias](#) for procedural details.

Intelligibility Drills. The same as repeated treatment trials. Use this method to help improve intelligibility of speech in speakers who have [Dysarthrias](#).

- Prepare a list of words that contain the target sounds the individual has difficulty with
- Ask the individual to read the list aloud
- Turn back from the individual to judge the accuracy of speech sound productions based only on phonatory cues
- If the production of a sound is unclear, ask the individual to find out why and try again

- If this second attempt also fails, turn around, look at the printed word, and give corrective feedback to the individual
- Ask the individual to try saying the same word again

Interdisciplinary Teams. Teams of different specialists who assess and design treatment programs for individuals; typically headed by one specialist; members finalize assessment and treatment plans after one or more meetings.

Interfering Behaviors. Controlling behaviors that interrupt the treatment process is an essential part of treatment; interfering behaviors include such actions as leaving the chair, asking irrelevant questions during treatment, crying, wiggling in the chair, and inattentiveness; a priority for clinical intervention because speech-language behaviors cannot be taught unless such interfering behaviors are reduced or eliminated.

- Withhold reinforcement for all interfering behaviors
- Use one of the Differential Reinforcement procedures to increase the alternative desirable behaviors, which will then reduce the interfering behaviors

Intermediate Care. A health care facility where persons with disability or chronic illness needing long-term care are admitted; facilities that persons enter after they have been discharged from a hospital and still need professional care.

Intermediate Response. Responses other than the initial and the final responses used in Shaping a target skill.

Intermittent Reinforcement. Use these schedules of reinforcement in which only some responses or responses produced with specified delay are reinforced to produce stronger response rates than Continuous Reinforcement; includes the Fixed Ratio, Fixed Interval, Variable Ratio, and Variable Interval Schedules; useful in promoting response maintenance over time; to be used in the intermediate and

final strategies of treatment, as it is not effective in establishing the target responses.

Intermixed Probes. Use these procedures to assess generalized production of a trained skill by alternating trained and untrained stimulus items; alternating trained and untrained stimuli help prevent extinction of trained responses because responses given to trained stimuli are reinforced (those given to untrained stimuli are not).

- Have at least 10 items not trained (e.g., 10 words or phrases with the plural *s* when this grammatic morpheme was the treatment target; 10 words or phrases with /z/ in the medial position when this phoneme was the treatment target)
- Prepare a Probe Recording Sheet on which you have alternated trained and untrained exemplars; have at least 10 untrained exemplars that may be words, phrases, or sentences used in training phonemes or grammatical morphemes
- Present a trained exemplar on the first trial (e.g., the picture of two *cups* if this word has been trained); evoke the response by asking a question (e.g., “What are these?”); reinforce the correct production
- Present an untrained exemplar on the second trial (e.g., the picture of two *books*, a stimulus item not used in training); ask a question to evoke the response; provide no reinforcement or corrective feedback
- Alternate trained and untrained exemplars on the subsequent trials
- Calculate the percent correct probe responses based only on responses given to the untrained exemplars
- Give additional training when an adopted probe criterion is not met (e.g., 90% accuracy)
- Move on to next level of training or to new target behaviors when the criterion is met

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Intervention. Introduction and manipulation of external variables to affect the course of a disorder, disease, problem

behavior, or a condition with a view to eliminate the problem or effect positive changes in it; the same as *treatment*.

In-the-Ear Hearing Aid. A small hearing aid that fits within the ear canal and concha.

Intonation. Treat normal intonation when prosodic deficits are evident in an individual; a normal aspect of speech and a treatment target in individuals with Dysarthrias or those with Fluency Disorders who have been taught to speak with prolonged syllables; a treatment target for any individual with a monotonous speech.

I

Isolated Therapy Model. A special education service delivery model in which children are taken out of the classroom for special instruction, including speech-language instruction; the same as the Pull-Out Therapy Model.

Isolation Time-Out. Response-contingent removal of a person from a reinforcing environment and placing him or her in a non-reinforcing environment; a variation of Time-Out; involves physical isolation (such as placing the person in an isolation booth); may be considered only in reducing highly abusive, aggressive, or self-destructive behaviors; not especially useful in communication treatment sessions; misuse of this technique is common in educational settings; prefer the Nonexclusion Time-Out.

J

Jargon. Reduction of undesirable and excessive use of terms the individual has invented is a target for individuals with aphasia; see [Aphasia](#) and [Wernicke's Aphasia](#) for details.

Jitter. To treat a cycle-to-cycle variation in frequency of vocal fold vibrations that exceeds 1%, heard as a pitch disorder, see [Voice Disorders](#).

Joint Action Routines or Interactions. Use this child language intervention method in which repetitive, routinized activities are targeted; similar to [Script Therapy](#) or may be a variation of it; for procedures, see [Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs](#).

Joint Attention. Use this language and general communication teaching strategy in which two or more people pay attention to the same event or object simultaneously; to teach joint attention in child language treatment, ask the child to look at what you are looking at, and reinforce the child who immediately complies with your request.

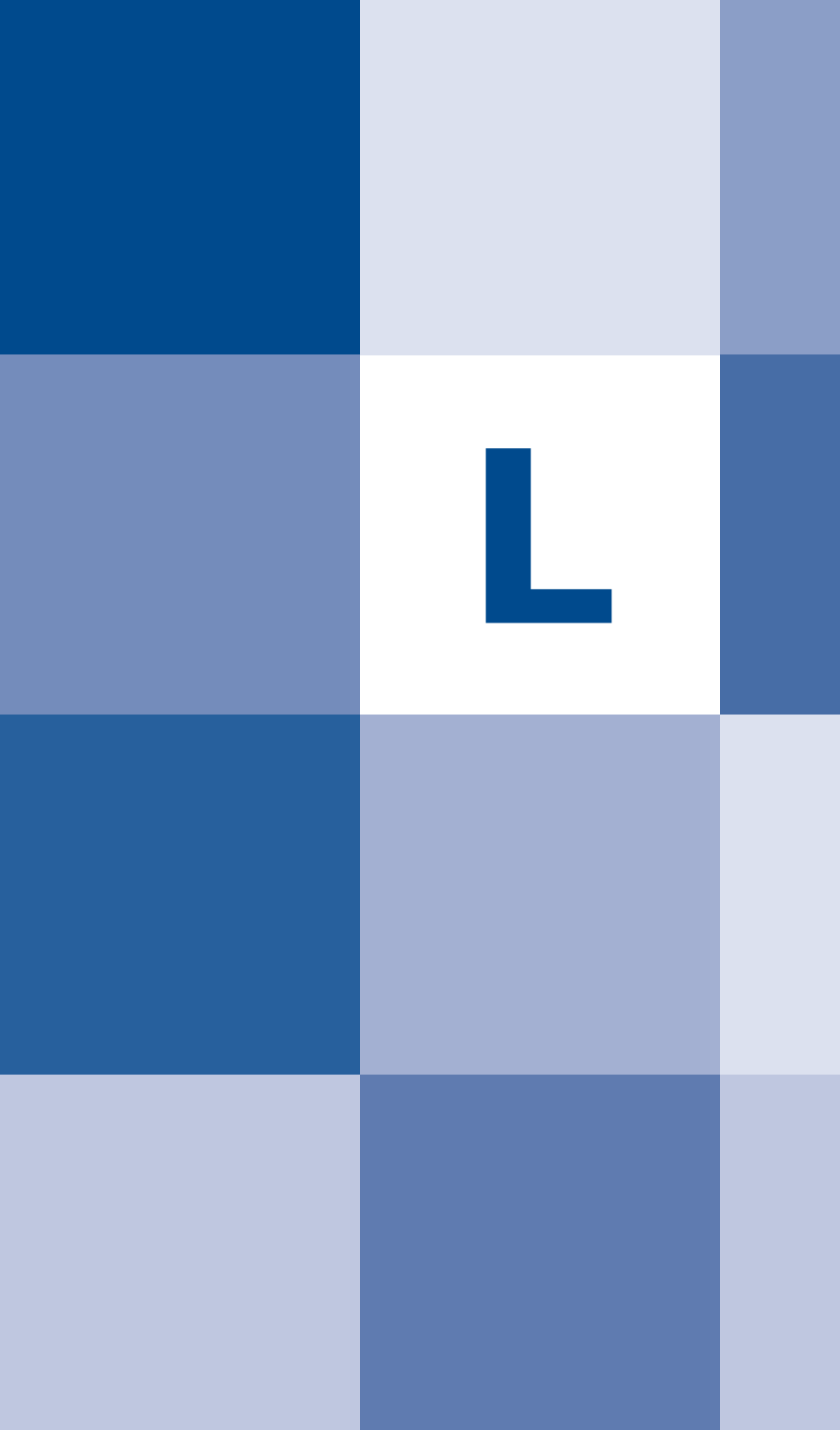
Joint Reference. Use this language treatment strategy, which is essentially the same as [Joint Attention](#), so that you and the individual are looking at the same event or stimulus while teaching specific language skills; in language therapy, ask the individual to look at the stimulus item you just presented before evoking a speech or a language response.

K

Keyboard. Use these electronic devices to teach non-oral communication skills to individuals with limited oral skills; a keyboard is a device that gives input to the computer with different layouts of letters, numbers, and other command inputs; may include key depression, touch membrane, or touch screen surfaces; see Augmentative and Alternative Communication for procedural details.

Keyguard. Use a plastic or Plexiglas material to cover the standard keyboard while teaching non-oral communication to individuals who are minimally nonverbal; a keyguard has holes for each key; the AAC user slides a pointer over the cover; helps prevent miscommunication by accidental hitting of irrelevant keys.

Key Word. Use a word in which a generally misarticulated sound is correctly produced in treating articulation disorders; correct production of a sound in a specific word is contrasted with incorrect production of the same sound in other, non-key words; used in a specific method of articulation treatment called the Paired-Stimuli Approach described under Speech Sound Disorders; Treatment of Speech Sound Disorders: Specific Techniques or Programs.



Language-Based Classroom Model. A model of service delivery in which the speech-language pathologist is in charge of a class organized especially for students with communication disorders, although some normally speaking children also may be involved; the clinician teaches these children all day or part of the day.

Language Delay in Children. Treat language delay as you would Language Disorders in Children; generally the two terms mean the same except for the connotation that children with language delay are slow in learning and that they may catch up with their normally progressing peers; language disorders in children tend to persist, especially without treatment; hence, disorders and delay are not synonyms.

Language Disabilities in Children. For treatment procedures, see Language Disorders in Children, preferred term.

Language Disorders in Adults. Treatment of problems in comprehending, formulating, and producing language in adults, who often have a history of normally acquired and produced language skills that are partially or fully lost because of physical, often neurological, diseases; includes Aphasia, Dementia, and language and other communication disorders associated with Right-Hemisphere Disorders and Traumatic Brain Injury in Adults.

Language Disorders in Children. Treatment of difficulty in learning to comprehend and/or produce language in children is multifaceted because some of them tend to have associated clinical conditions, whereas others show no such conditions; also variously referred to as Language Delay, Language Disabilities, Language Deviance, Language Impairment, Language-Learning Disorders, and Language Problems; until recently, the term Specific Language Impairment (SLI) referred to language disorders with no associated clinical conditions that could explain them; in 2017, a group of experts renamed SLI as *developmental*

language disorder (DLD); neither the term SLI nor DLD is free from disagreement, controversy, and unwarranted theoretical baggage; the new name (DLD) comes with all the questionable assumptions of implied (biological?) *development* of language and an unsupported assumption of a *developmental disorder* that may resolve itself over time; therefore, a theoretically neutral term, *language disorders in children*, is preferred in this guide; language disorders in children pose significant limitations in social communication and academic performance and are unlikely to remediate themselves without professional intervention; to diagnose and treat a language disorder in a child, there is no need to find the discredited discrepancy between the verbal and the nonverbal skills; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology, etiology, and characteristics of language disorders in children; the treatment procedures described under this entry apply mostly to children with language disorders, regardless of whether they are considered *specific* or *developmental*; to treat children's language disorders associated with additional clinical conditions, see Autism Spectrum Disorder, Cerebral Palsy, Intellectual and Developmental Disabilities, Hearing Loss, and Syndromes Associated with Communication Disorders.

Ethnocultural Considerations in Treating Language Disorders in Children

- It is not clear whether children of different ethnocultural groups respond differently to the same treatment procedure
- Well-researched treatment procedures, such as the behavioral methods described in this outline, are likely to be effective because the methods have shown to have generality
- Treatment targets for ethnoculturally diverse children should be selected based on their language characteristics; in the case of bilingual children, selected targets may be those of the dominant language; parental preference based on the academic standing of the

child should be considered; targets must be consistent with the child's dialectal variation (such as the African American English); see [Treatment of Language Disorders in African American Children](#) and [Treatment of Language Disorders in Bilingual Children](#) following the major entry.

- Stimulus materials and reinforcers used in treatment sessions should be effective as well as relevant to the child's cultural and home background
- Family counseling regarding treatment should consider cultural differences and values

Treatment of Language Disorders in Children: General Guidelines.

Issues relevant to the target behavior selection, sequencing, and intervention session structure include the following:

- Selection of Target Behaviors
 - Normative strategy, which requires clinicians to select target behaviors based on language developmental norms; because a child with a language disorder does not meet the normative expectations, the skills that are missing from the standpoint of norms are the treatment targets
 - Client-specific strategy, which emphasizes functional targets; accordingly, target behaviors should be child specific, relevant to the child's educational demands and family communication patterns; they should be the behaviors that, when taught, will make a difference; this view is more consistent with the functional view of language treatment and uniqueness of each individual
- Sequencing the Language Targets for Treatment
 - Normative strategy, which requires that selected target behaviors be taught in the sequence in which children normally acquire them; although some behaviors may be more efficiently taught this way, there is no compelling evidence that this is always the best approach to sequencing target behaviors; rarely put to the test because of the belief that it is the best

- Experimental strategy, which encourages clinicians to experiment with different sequences of teaching the target behaviors and then using the one that produces the best results; the data may show that normative sequence is indeed the best or that other sequences are better or just as good
- Structure of Language Treatment Sessions
 - Tightly structured sessions, which are preferred by some clinicians; tend to involve discrete trial teaching; the clinician controls the stimuli and response consequences; tend to have clearly defined target behaviors for teaching; measurement oriented
 - Loosely structured sessions, which are preferred by some clinicians; tend to involve play-oriented sessions; the clinician does not directly control stimuli and consequences; tend not to have clearly specified target behaviors that are taught systematically; not very measurement oriented
 - The two options are perhaps not a matter of choice; the best approach might be to structure the sessions well in the early stages of treatment and relax the structure in the final stages of treatment; the final stages of treatment should always be less structured, more naturalistic, more conversation oriented
- Treatment efficacy
 - There are many language treatment procedures that have rarely been put to experimental test; techniques that are simply based on expert advocacy should be viewed critically
 - Treatment procedures that have received experimental support should be selected; most behavioral treatment procedures have been experimentally tested; many suggestions that come from structural linguists (and some clinicians as well) are based on speculation and theoretical convictions; see Treatment Selection Criteria

Treatment of Language Disorders in Children: A Comprehensive, Integrated, Procedure

- Steps in developing a language treatment plan for a child:

- Select target behaviors for training
- Plan a sequence of treatment
- Select stimulus materials
- Establish baselines
- Write a treatment and maintenance plan
- Implement the treatment plan
- Implement the maintenance plan
- Select language intervention targets that:
 - Are child specific and ethnoculturally appropriate
 - Are useful in natural settings
 - Can make an immediate and socially significant difference in the child's communicative skills
 - Help meet the academic and social demands the child faces
 - Help expand communication skills into conversational speech in natural settings
 - Are within the child's reach as judged by current performance (words, phrases, sentences, conversational speech)
- Design a flexible Sequence of Treatment that allows for experimentation; generally, move from:
 - Words to phrases
 - Phrases to controlled (less spontaneous) sentences
 - Controlled sentences to spontaneous conversational speech
 - Treatment in clinical settings to treatment in more naturalistic settings
 - More structured sessions to progressively less structured sessions
 - Continuous reinforcement to intermittent reinforcement
 - Primary reinforcers to social reinforcers
 - Social reinforcers to natural consequences inherent in communication
- Prepare stimulus materials for treatment; select at least 20 exemplars to teach each selected target behavior (e.g., 20 phrases that contain the plural s)
 - Select ethnoculturally appropriate, client- and target-specific stimuli that are colorful, attractive, and realistic; prefer objects to pictures

- Obtain stimuli from the child's home (the child's favorite books, toys, and objects)
- Prepare a Response Recording Sheet on which:
 - You can write target behaviors
 - Record the occurrence of each behavior
- Establish Baselines of target behaviors through:
 - Repeated conversational language samples that help reliably document the occurrence of language targets
 - A set of modeled discrete trials and a set of evoked discrete trials that (a) help capture the production of specific language targets that may not be adequately sampled in conversational speech; (b) are necessary in case of children with no or minimum conversational skills
- Administer Modeled Baseline Trials (withhold reinforcers or corrective feedback):
 - Place a stimulus picture or object in front of the child or demonstrate an action or enact an event (e.g., a picture of two books)
 - Ask a question to evoke the target response ("Johnny, what do you see?")
 - Model the response ("Johnny, say, 'I see two books'")
 - Record the response on a recording sheet (note that there is no reinforcement or corrective feedback)
 - Present the next picture (e.g., that of two cups); repeat the procedure until all the 20 exemplars or 30 or 50 basic words are baserated (basic words to teach a set of core vocabulary for a nonverbal child)
- Administer Evoked Baseline Trials (withhold reinforcers or corrective feedback):
 - Place a stimulus picture or object in front of the child or demonstrate an action or enact an event
 - Ask a question to evoke the target response; do not model the response
 - Record the response on the recording sheet
 - Repeat the procedure for all the 20 or the total number of selected exemplars
- Calculate the percent correct baseline response rate in conversational samples and on discrete modeled and evoked trials
- Write a treatment and maintenance plan

Word and Phrase Level of Training

- Begin treatment by teaching a set of functional words to a nonverbal or minimally verbal child; select at least 20 functional words, most of them may be among the First Words children acquire (e.g., such words as *mommy* and *daddy*; food items like *milk*, *candy*, *juice*, *apple*, *banana*; clothing and personal items like *socks*, *shirt*, *pants*, *shoe*; simple adjectives like *big*, *little*; animals like *kitty*, *doggie*; household objects like *spoon*, *chair*, *book*, *pen*); select child-specific words (e.g., names of siblings and those of pets) after consulting with the parents; establish the target words with the discrete modeled trials:
 - Place a stimulus picture or object in front of the child
 - Ask a question to evoke the target response (e.g., “Johnny, what is this?”)
 - Model the response (“Johnny, say *sock*”)
 - Positively reinforce the correctly imitated response; accept an approximation of the correct response
 - Give corrective feedback for wrong response by saying “No, that is not correct; it is a *sock*, not a ____.”
 - Move on to the next trial if the child did not say anything
 - Move the stimulus picture or item toward you to signal the end of a trial
 - Record the child’s response on the recording sheet (correct, incorrect, or no response)
 - Wait for a few seconds, draw the child’s attention, and present the next trial
 - Stop modeling and present evoked trials when the child correctly imitates the target word on 5 consecutive trials
 - Follow the same procedure as before: present the stimulus, ask the question (“What is this?”), but do not model the response
 - Reinforce the correct response and give corrective feedback for the wrong response

- Reinstate modeling when the child gave three or four wrong responses on the first introduction of evoked trials;; again withdraw modeling or fade modeling with Partial Modeling when the child correctly imitates the word on five consecutive trials
- Continue training in this manner until the child gives at least 9 correct responses out of 10 evoked (no modeling) trials (a 90% accuracy in producing the word without modeling)
- Reinforce continuously in the initial stages; in gradual steps, reduce the amount of reinforcers by switching to intermittent schedules
- Give social reinforcers for all responses, even when using tangible reinforcers
 - Fade tangible reinforcers if used
- Train four to six exemplars (words at this level) to a training criterion of 90% correct on a set of 10 trials before you probe
- Present the same trained words on the probe trials, ask the same question, and record the response; do not model, reinforce, or give corrective feedback; just ask the questions and record the response
- Consider the words tentatively trained when the child produces them without modeling or reinforcement on at least four of the five consecutive presentations
- Train four to six more (new) words; train the new words as you did the original four to six words; administer probe trials (no modeling, no reinforcement, no corrective feedback)
- Continue to train the selected functional words (some 30 to 50); make sure that the words trained include nouns (*car, cup, sock, ball, kitty, candy, man, woman, baby*), adjectives (*big, small, blue, red, thin, thick, smooth, rough, happy, sad, hot, cold*), and main verbs (*walking, sleeping, eating, running, jumping, smiling*)
- Form two-word phrases out of already trained words that have met the probe criterion, (e.g., *big car, red*

sock, small kitty; man sleeping, woman walking, doggie eating)

- Begin training the phrases with the same modeled discrete trials; fade modeling and move on to evoked discrete trials
- Probe the phrase production without modeling, reinforcement, or corrective feedback when you have trained 4 to 6 phrases; if probe criterion (90% correct) is met, begin training on new phrases; if not, continue training on the original set of phrases
- Shift training to the level of grammatical morpheme and sentence training when about 20 trained phrases have met the probe criterion,

Teaching Grammatical Morphemes and Sentence Varieties

- Note that one grammatical morpheme, for instance the present progressive *ing*, may have been taught in the context of main verbs (e.g., *walking, eating*); similarly, irregular plural words (*men, women, children, teeth, feet*) also may have been trained as words; note also that some grammatical morphemes can be trained only in sentences (e.g., the auxiliary *is*, as in *The boy is running*) others could be trained in phrases (e.g., the regular plural *s*, as in *Two books*); therefore, phrases and sentences are both involved at this level of training
- Select grammatical morphemes that can be trained in phrases to begin with, as these are syntactically simpler and presumably easier for the child; select the regular plural inflections *s* (e.g., *Two books, blue blocks, three cats, green plants*) and plural *z* (e.g., *four bags, two dogs, red balls, long trains*) for the initial training; other morphemes in phrases include the irregular plurals (*two women, big men, white teeth, big feet*); prepositions *on* (*on the table, on the bed*) and *in* (e.g., *in the hat, in the box*); possessive inflection (e.g., *mommy's hat, doggie's tail*) and so forth
- Use the discrete trial procedure to establish the morphemes; show a picture, ask a question, and model

- the phrases with an emphasis on the grammatical morpheme (e.g., “Johnny, What is this? Say two books”); fade modeling and train with evoked trials
- Probe for generalized production of the morpheme when four to six exemplars are trained to the training criterion of 90% correct on a set of 10 evoked trials;; note that at this stage, probes are different than they were at the word level
 - Use initially the Intermixed Probe Procedure in which you alternate trained and untrained exemplars to assess the generalized productions of the morpheme (such as the plural *s*) in *untrained* phrase; use at least 10 untrained stimulus items; administer intermixed probes as follows:
 1. First, present a stimulus, object, or event used in training; ask a question to evoke the response; reinforce or give corrective feedback; record the response on a Probe Recording Sheet; note that the trained exemplar in an intermixed probe sequence is presented as it was during training
 2. Next, present a stimulus, object, or event not used in training (the *probe item*); ask a question; do not reinforce or give corrective feedback; record the response
 3. Present another stimulus used in training; use the same procedure as in #1
 4. Present another stimulus not used in training and use the same procedure as in #2
 - Calculate the percent correct probe response rate; score responses given only to the untrained stimulus items (exemplars) for this calculation (e.g., if the child gave five correct responses out of 10 untrained stimulus presentations, the probe response rate is 50%)
 - Train additional phrases with the same morpheme when the child does not meet the intermixed probe criterion of 90% correct production of the grammatical morpheme in phrases,
 - Shift training to the sentence level when the child meets the intermixed probe criterion; expand items

already trained into short sentences (e.g., *I see three cats; I see big men*).

- Continue to train the grammatical morphemes in sentences; once again, use the same sequence as the modeled trials leading to evoked trials, intermixed probes, and additional training if the probe criterion is missed
- Move training to conversational speech when the child meets the probe criterion for morphemes in sentences:
 - Evoke speech in naturalistic contexts with pictures, toys, books, and so forth
 - Engage the child in conversation
 - Reinforce the production of grammatical morphemes
 - Give corrective feedback for errors
- Select new grammatical features or sentence types for training; train each in an appropriate entry level (most of them in phrases or sentences; some in words); then move them through other levels (sentences, conversational speech) as described; for example, teach the production of
 - Auxiliaries *is, are, was, were*, and so forth in such sentences as *girl is reading, boy is running; women are writing, dogs are chasing; he was sleeping, she was smiling; they were working, men were digging*
 - Copulas *is, was, are, and were* (e.g., *girl is nice; boy was happy; turtles are slow; plants were green*)
 - Regular past tense inflection (e.g., *he mailed; he mailed yesterday; he painted; he painted yesterday; she walked; she walked yesterday*)
 - Irregular past words (e.g., *he came, she went; he came home; she went to the store*)
 - Articles *a* and *the* (e.g., *the girl, the boy; the girl is reading, the boy is running*); note that phrases that were trained earlier to induce the production of another phoneme may be used to add additional grammatic elements

- Pronouns *he, she, and it* (e.g., *he is playing, she is riding, it is hopping*)
- Conjunction *and* (e.g., *milk and cookies; I like milk and cookies; lions and tigers; I like lions and tigers*)
- Comparative *-er* and superlative *-est* (e.g., *dark; this is darker; this is the darkest; warm; this is warmer; this is the warmest*)
- Adjectives *big* and *small* (e.g., *big house, small house; this is a big house, that is a small house*)
- Tacts or simple active declarative sentences (e.g., *This is a nice house; that is a pretty flower*)
- Mandates or requests (*Juice, please; Crayon, please; Please give me that ball; I want that book*)
- Questions (e.g., *What is this? Where is it? Who is it? When do you eat breakfast? Where is it?*)
- Negation (e.g., *she is not sleeping, he is not eating, they are not walking*)
- Administer the pure probe when a target behavior meets the intermixed probe criterion of 90% correct on at least 10 exemplars taught in sentences; on a pure probe, do not present trained exemplars; present only untrained exemplars to evaluate whether a trained grammatic morpheme or a feature is produced in untrained contexts:
 1. Present a stimulus, object, or event not used in training (e.g., the picture of several elephants, not used in training)
 2. Ask a question to evoke the response (e.g., “What do you see?”)
 3. Do not reinforce or give corrective feedback for the response; the child’s response may be correct (e.g., “I see many elephants”) or incorrect (e.g., “I see elephant”)
 4. Record the response as correct or incorrect
 5. Present another stimulus picture not used in training and follow the same procedure
 6. Present at least 10 untrained stimulus events
 7. Calculate the percent correct pure probe response rate

- If the pure probe response rate is less than 90% correct, give additional training using untrained stimulus items
- When a target behavior meets a pure probe criterion of 90% correct on at least 10 untrained exemplars produced in sentences:
 - Begin training on a more complex response level (e.g., longer sentences or conversational speech)
 - Select another behavior for training if the treatment time permits

Teaching Conversational Skills and Pragmatic Features

- Discard discrete trials when you reach the conversational speech level:
 - Loosen the training structure
 - Use more spontaneous speech
 - Talk about events (e.g., while the child builds some blocks, talk about it); stimulate speech about events and complex stimuli (such as pictured sequences in a story book)
 - Tell or read aloud stories the child retells
 - Use more natural consequences (instead of verbal praise for correct productions), including agreement (affirmation), negation, smile, handing requested objects, meaningful responses to questions, and so forth
- Integrate conversational skills (Pragmatic Language Skills) into training at the spontaneous, natural, conversational speech training level:
 - Teach maintenance of eye contact during conversation by prompting and reinforcing the child for doing so
 - Teach Topic Maintenance (described later in this section under Treatment of Language Disorders: Specific Techniques or Programs) by progressively increasing the duration for which the child talks about a topic
 - Teach Turn-Taking (described later in this section under Treatment of Language Disorders: Specific

Techniques or Programs) by reinforcing the child to alternately play the roles of a listener and a speaker

- Teach Conversational Repair (described later in this section under Treatment of Language Disorders: Specific Techniques or Programs) strategies, including asking questions when statements are not understood
- Teach narrative skills: Tell at least three stories and ask the child to retell it; ask the child to tell a story; prompt longer stories, more details, correct sequences, feelings and thoughts of characters, and so forth; see also Narrative Skills Training (described later in this section under Treatment of Language Disorders: Specific Techniques or Programs)
- Teach discourse skills: Ask the child to talk about a topic of his or her choice; suggest such topics as organizing a birthday party, going on a camping trip, buying school supplies, or any other topic that might be of interest to the child; prompt more details, correct temporal sequences, more complete and longer sentences; more formal or academic words, and so forth

Decreasing Undesirable Behaviors

- Decrease undesirable behaviors through Response Reduction Strategies; note that these include the wrong verbal responses or such interfering behaviors as leaving the chair, interrupting treatment trials with irrelevant questions, not looking at the stimulus materials, and so forth
 - Give corrective verbal feedback (“No” or “Not correct”) for incorrect language responses (e.g., omitting the plural s in words and phrases)
 - Use Extinction for such behaviors as crying or asking irrelevant and interrupting questions
 - Prevent responses that seem to be related to difficult demands you make and thus negatively reinforced (e.g., prevent the child from leaving the

chair when difficult trials are presented; continue to present the trials)

- Reinforce an alternative, desirable behavior that replaces an undesirable behavior (e.g., give extra tokens for sitting to reduce the off-seat behaviors)
- See Differential Reinforcement for more details and other procedures

Implementing a Maintenance Program

- Implement a maintenance procedure
 - Have family members, teachers, caretakers, peers, and others observe the treatment sessions
 - Train the significant others, especially the family members and teachers, in evoking, prompting, and consequenceing target behaviors
 - Have family members conduct informal therapy sessions at home and bring recorded evidence to that effect
 - Ask the child's teacher to provide opportunities for communication in the classroom and to praise the child for producing targeted and other language skills
 - Hold informal training sessions outside the clinic room, outside the building, in other parts of the school or campus, and at other settings to the extent practical
 - Use intermittent reinforcement schedule and natural, social reinforcers
 - Delay reinforcement in later stages of treatment; increase the delay in gradual steps
 - Take training to the natural conversational level
 - Teach Reinforcement Priming by training the child to draw attention to his or her newly acquired communicative skills
 - Teach Self-Control (Self-Monitoring) by training the child to recognize and measure his or her right and wrong responses
 - Continue treatment until the language skills stabilize in the natural environment

- Provide Follow-Up and Booster Treatment
- See Maintenance Strategy

Bishop, D. V. M., Snowling, M. J., Thompson, P. A., Greenhaig, T., & the CATALISE-2 Consortium (2017). Phase 2 of CATALISE: A multinational and multidisciplinary Delphi consensus study of problems with language development: Terminology. *Journal of Child Psychology and Psychiatry*, 58, 1068–1080.

Hegde, M. N. (2006a). *Treatment protocols for language disorders in children: Volume I. Essential morphologic skills*. San Diego, CA: Plural.

Hegde, M. N. (2006b). *Treatment protocols for language disorders in children: Volume II. Social communication*. San Diego, CA: Plural.

Paul, R., Norbury, C., & Gosse, C. (2017). *Language disorders from infancy through adolescence* (5th ed.). St. Louis, MO: Mosby.

Reed, V. (2018). *An introduction to children with language disorders* (5th ed.). Boston, MA: Pearson.

Treatment of Language Disorders: Specific Techniques or Programs.

Several specific treatment programs are available; because most programs are not comprehensive, they do not replace the treatment outline provided in this guide; some have better evidence than others; some have little or no evidence; clinicians need to check the evidence before selecting a technique.

Activity-Based Language Intervention. Treating language disorders with the help of various activities designed to promote language production; a classroom-based approach in which each child has an individualized educational plan; activities are part of classroom activities and use natural antecedents and consequences.

- Plan activities that promote the production of specific language structures in children
- Plan activities around a theme if appropriate (e.g., activities related to going on a camping trip)
- Read stories, narrate events, and sing songs about the theme; include the various language targets for the children in the class

- Ask questions about the presented information
- Forget to give needed objects during activities, leading the child to request them
- Give needed items one at a time so the child requests each one
- Put needed things out of reach so the child asks for them
- Put needed things in a clear jar that the child can see but cannot open to gain access and hence has to request help
- Hide the child's belongings to encourage requests
- Introduce novel items (e.g., wear a funny hat) and let those who talk about it wear it
- Pause during verbal or nonverbal actions so the child will request that you continue

Johnson, J., Rahn, N. L., & Bricker, D. (2015). *An activity-based approach to early intervention* (4th ed.). Baltimore, MD: Paul H. Brookes.

Conversational Repair. Skills in handling breakdown in communication, a treatment target during conversational skill training; refers to such skills as asking questions when messages are not clear and responding to requests for clarification.

Teach the Child to Request

Clarifications from a Speaker:

- Make ambiguous statements (e.g., say “Give me the car” when you have displayed several toy cars)
- Wait for the child to request clarification
- Say “no” if the child does not request clarification and responds anyway (such as picking one of the cars)
- Wait for the child to request clarification
- Model a response for the child who fails to request clarification, (e.g., “When you are not sure, I want you to ask me ‘What do you mean?’ OK?”)
- Make another ambiguous statement
- Model immediately the request for clarification for the child
- Reinforce the child for imitating the request for clarification (e.g., “What do you mean?”)

- Make another ambiguous statement
- Prompt (don't model) a request for clarification (e.g., "What do you ask me?")
- Reinforce the child for asking for clarification (e.g., "What do you mean?")
- Introduce varied ambiguous statements
- Fade modeling and prompting
- Train parents in teaching the child to request clarification
- Probe the generalized repair skill by presenting untrained messages
- Continue training until a set probe criterion is met (such as 90% accuracy in responding with the target skill)

Teach the Child to Vary Unclear Expressions

- Play the role of a listener who does not fully understand the expressions of the child:
- Ask the child to repeat
- Ask the child, "What do you mean?"
- Tell the child, "I do not understand"
- Negate a child's utterance so the child will clarify by assertion ("You did not go on the roller coaster 20 times did you?"; the child might say, "No, I went on it two times")
- Model the clarified statement by modifying what the child said ("You mean you went on the roller coaster 2 times, right?")
- Rephrase the child's utterance into a question and say it with a rising intonation ("You went on the roller coaster 20 times?")
- Model different ways of saying the same thing
- Ask the child to say it differently; reinforce varied phrases or sentences
- Periodically stop responding (e.g., to the child's request) to prompt the child to rephrase
- Train parents to prompt the child to vary expressions and to reinforce the child for compliance
- Probe the generalized repair skill by presenting untrained messages

- Continue training until a set probe criterion (such as 90% accuracy in responding with the target skill) is met

Conversational Skill. A language skill in maintaining a dialogue with one or more partners; an intervention goal for all clients with language disorders; collection of skills that include Topic Initiation, Topic Maintenance, and Turn-Taking (all described later in this section).

- Use Peer Modeling (described later in this section); recruit peers who have good conversational skills to model those skills to the client
- Train the peers to model and have the client participate in conversation
- Monitor closely the behaviors of the peer models and the client
- Train peer models to be the hosts of a mock talk show
- Ask open-ended questions (you or the peer model)
- Ask follow-up questions (you or the peer model)
- Prompt the peer and the client for appropriate behaviors
- Train the client to ask questions
- Reinforce the client for new topic initiations, appropriate turn-taking, and topic maintenance; if necessary, train these skills separately
- Show videorecorded model interactions between adults, between children, and between children and adults
- Let the children analyze the recorded interactions
- Let the children recreate what they saw on the recording
- Train parents to conduct informal conversational skills training sessions at home

Delayed Stimulus Presentation. A child language intervention procedure in which the clinician delays providing such special stimuli as modeling for about 15 seconds to see if the child responds without such

stimuli; provides the stimuli only when the child does not respond within the time limit.

- Establish joint attention regarding a stimulus (e.g., hold an object; establish eye contact with the child; look questioningly or expectantly)
- Do not speak for 15 seconds when the child approaches you or looks at you
- Model a mand or a name after the 15-second delay
- Give the object when the child imitates your modeling
- Give the object anyway when you have modeled three times and the child has not yet imitated

Direct Language Treatment Approaches. Intervention approaches in which the clinician selects specific language targets, designs a treatment environment and implements the treatment; uses specific stimuli, including modeling, prompting, and manual guidance; uses explicit reinforcement contingencies; expects the child to imitate or produce specific targets upon stimulation; moves through a planned sequence of treatment stages.

Enhanced Milieu Teaching (EMT). A collection of child language intervention procedures that emphasize natural, functional, and conversational communicative contexts for teaching language; based on the earlier milieu teaching research; includes three components: (1) environmental arrangement, (2) responsive interaction (RI), and (3) milieu teaching (MT); appropriate for children in the early stages of language learning with an MLU of 10-3-5; also applicable for children with autism spectrum disorders and those from low-income families with a high risk for language disorders; mostly parent administered with 24–36 training sessions; a behavioral method that uses modeling, prompting, shaping, and positively reinforcing specific language targets evoked in natural settings; professionals also may administer MT; EMT may also include the Mand-Model and Time-Delay procedures; supported by controlled evidence.

- Environmental rearrangement; teach the parents to:
 - Select toys and other stimulus materials that are of interest to the child; prefer toys that could be manipulated and are interactive
 - Arrange the materials to evoke language responses from the child; limit the number of stimulus materials; keep some out of reach or in containers
 - Manage the materials to maintain the child's interest and interaction; control the access to the materials; give only a few at a time
- Responsive interaction; teach the parents to:
 - Listen to the child and respond quickly, positively, and warmly; stay at the child's eye level
 - Let the child lead the interaction; initiate language
 - Engage the child in conversation; talk about the child's play
 - Ask for clarification when the child is unclear
 - Expand the child's productions
- Milieu teaching; teach the parents to:
 - Model responses at the child's language level (e.g., if the child looks at a toy car, the mother might model, "Say 'want car'"; if the child said "Car," the mother might model, "Say, 'want car'")
 - Expand the child's production (e.g., parent might say, "You want the red car," and hand it to the child); note the natural consequence
 - If the child did not imitate, model the response and give the reinforcer (e.g., if the child did not respond, the mother might model, "Want car," and give the child the car)
 - Prompt no more than one a minute; stop the prompts when the child is no longer interested

Hancock, T. B., & Kaiser, A. P. (2006). Enhanced milieu teaching. In R. J. McCauley & M. E. Fey (Eds.), *Treatment of language disorders in children* (pp. 203–236). Baltimore, MD: Paul H. Brookes.

Hart, B. M., & Risley, T. R. (1975). Incidental teaching of language in the preschool. *Journal of Applied Behavior Analysis*, 8, 411–420.

Environmental Language Intervention Strategy (ELIS). A language intervention method for pre-school children; developed and researched by J. D. McDonald and associates; a structured, direct treatment strategy; takes a semantic approach to teaching grammar; emphasizes generalized production at home; especially useful in training parents to conduct language stimulation sessions at home.

- Establish baselines of selected target language structures
- Structure treatment in three phases: imitation, conversation, and play
- Introduce the three procedures in the first three individual training sessions
- Train parents to record responses and administer the treatment program at home; train them in the Management of Behavioral Contingencies
- Integrate imitation, conversation, and play into a single session after the first three training sessions
- Spend the first 15 minutes on imitative productions
 - Present a nonlinguistic stimuli (throw a ball)
 - Present a linguistic stimuli (e.g., “Say, throw ball”)
- Spend the next 15 minutes on conversational speech
 - Present the same nonlinguistic stimuli
 - Ask a question (e.g., “What am I doing?”)
 - Model if necessary (e.g., “Say throw ball. What am I doing?”)
- Spend the final 15 minutes on play activity during which the production of the target behavior is reinforced
 - Let the child play with the material used in imitation and conversation
 - Evoke responses from the child that are relevant to the child’s actions (e.g., if the child throws the ball, ask, “What are you doing?”)
- Give such positive reinforcers as tokens and verbal praise for correct responses
- Give such corrective feedback as Time-Out for incorrect productions

- Ask parents to conduct at home three weekly sessions similar to yours
- Review the records of home training in each session and suggest modifications

MacDonald, J. D., Blott, J. P., Gordon, K., Spiegel, B., & Hartman, M. (1974). An experimental parent-assisted treatment program for preschool language-delayed children. *Journal of Speech and Hearing Disorders*, 39, 395–415.

Event Structures in Language Treatment. Use of repetitive, sequentially organized, familiar events from daily life to teach language structures to children; an event structure may be the same as a Script used in Script Therapy (described later in this section); the two may be used in conjunction; also similar to Joint Action Routines or Interactions (described later in this section); needs more evidence to support its practice.

- Select a common event the child has repeatedly experienced (e.g., shopping for a toy, eating in a restaurant, taking part in a birthday party)
- Describe the event verbally
- Assign roles to yourself and the child (e.g., customer and the store clerk)
- Use props to act out the event
- Reverse roles and act out the events; repeat until the various language structures of interest are rehearsed
- Evoke words and phrases as you act out the event by using pauses at junctures (using the Cloze Procedure)
- Evoke increasingly complex or longer description of events
- Vary the sequence and event elements (e.g., after having worked with the event shopping for a toy, have the child work with the event shopping for clothing)
- Violate expected events or sequences and let the child question you or correct you (e.g., go to the sales clerk without trying on the clothes)

Expansions. Expanding a child's incomplete or telegraphic statements into grammatically more complete productions; part of Indirect Language Stimulation.

- Arrange a play situation that provides opportunities for language production
- Engage in parallel play with the child or take part in the child's activities
- Expand the child's structurally incomplete productions into more complete sentences (e.g., a boy says "baby cry" as he looks at a picture; you expand it into "The baby is crying")
- Do not ask the child to imitate your expansions

Expatiations. The same as Extensions.

Extensions. Comments on the child's utterances to add additional meaning; part of Indirect Language Stimulation; also known as Expatiations; the method does not require the child to imitate the clinician's extended utterances, a potential limitation.

- Arrange a play situation that provides opportunities for language production
- Engage in parallel play with the child or take part in the child's activities
- Extend the child's semantically limited productions into semantically richer, structurally complete sentences through comments (e.g., a girl says "play ball" as she plays with a ball; you extend it to include additional meaning: "Yes, you are playing with a big ball"; "You are playing with a blue ball")
- Do not ask the child to imitate your extensions

Eye Contact. A potential pragmatic communication target behavior for certain children who do not look at the listener while speaking or at the speaker while talking; potentially culturally determined; needs culturally sensitive assessment and treatment.

- Target eye contact from the beginning and especially during conversational speech training
- Use simple instruction and verbal praise
- Prompt the child by saying, "Look at me" before you present stimuli, when you begin to talk, and when the child begins to talk
- Hold the stimulus parallel to your face so the child looks at the face and the stimulus simultaneously

- Praise the child for maintaining eye contact
- Track progress by measuring the duration for which eye contact was maintained at the beginning (baseline) and throughout the treatment phase
- Fade the prompts or other cues used

Focused Stimulation. A technique of language intervention in which the clinician repeatedly models a target structure to stimulate the child to use that structure; usually a part of play activity; appropriate for toddlers and preschoolers and early school-age children; repeated modeling without requiring correct imitations and withholding systematic feedback to the child are potential weaknesses.

- Design a play activity to focus on a particular language structure (e.g., the plural morpheme *s*)
- Collect various stimulus materials (books, cups, hats)
- Talk about the materials and repeatedly model the plural constructions (e.g., “I see two *books* here. The *books* have pictures. Here are two red *cups*. You can drink out of these *cups*. There are some *hats*. The *hats* are big”)
- Do not correct the child’s wrong productions
- Respond to the child’s nontarget responses without insisting on the correct response (e.g., the child says, “The *book* is nice”; the clinician says, “Yes, the books are nice”)
- Continue until the child begins to produce the target structure

Weismer, S. E., & Robertson, S. (2006). Focused stimulation approach to language intervention. In R. J. McCauley & M. E. Fey (Eds.), *Treatment of language disorders in children* (pp. 175–203). Baltimore, MD: Paul H. Brookes.

Incidental Teaching Method. A naturalistic child language teaching method that uses typical, everyday verbal interactions to teach functional communication skills; the child often initiates an interactional episode; the clinician turns such episodes into opportunities to teach language; emphasis is on communication;

effective when the child and the teacher interact for extended periods of time and in natural settings (e.g., in special education classrooms, institutions for the intellectually disabled and for those with autism spectrum); an effective method for parents to learn and use at home; a part of Milieu Teaching and Enhanced Milieu Teaching.

- Select certain functional communicative skills for teaching (e.g., requests)
- Arrange therapy situations such that the child is likely to initiate a conversational exchange (e.g., place attractive toys on a shelf the child can see but cannot reach; arrange a child's clothing items or some desirable food items)
- Stay close to the arranged materials and give non-verbal cues to speak (focused attention on the child and a questioning look)
- Give a verbal cue (e.g., ask, "What do you want?") only if the nonverbal prompts fail to evoke a response from the child
- Give cues that evoke more complex responses (e.g., "Ask me in a sentence" if the child gives only single-word responses)
- Reinforce the child with natural consequences (e.g., "Good, here is the car," "Very good, here is your sock," "Fine, have some juice")
- Arrange as many such teaching episodes as possible in a day

Hart, B. B., & Risley, T. R. (1982). *How to use incidental teaching for elaborating language*. Lawrence, KS: H & H Enterprises.

Indirect Language Stimulation. A collection of language stimulation procedures that are part of a play-oriented approach to teaching language disorders; also called Child-Centered Approach; less structured and more naturalistic; thought to be especially suitable for children who are passive, reluctant, or unmotivated to communicate; based on the assumption that variables observed in normally developing children are effective

clinical treatment strategies; needs more controlled evidence to support this approach.

- Arrange a play situation that provides opportunities for language production
- Choose the play materials that are relevant for the targeted response
- Let the child lead the interaction
- Engage in parallel play with the child or take part in the child's activities
- Talk about what the child is doing, looking, playing, or talking
- Describe what you do (e.g., "See, I am drawing a face; I am drawing"; Self-Talk); (described later in this section)
- Describe or comment on what the child is doing (Parallel-Talk, described later in this section; e.g., you say "You are drawing; you are drawing a face; you are making a nose")
- Imitate the child's production (do *not* ask the child to imitate)
- Use Expansions (described earlier in this section; expand the child's telegraphic speech into grammatically more complete sentences; e.g., the child says "Mommy hat"; you expand this to "That is Mommy's hat")
- Use Extensions (described earlier in this section; comment on the child's utterances to add additional meaning; e.g., the child says "Mommy hat," and you say "Yes, it is a big blue hat")
- Recast (described later in this section) the child's utterances (expand the child's utterance type into a different kind of sentence; e.g., the child says "Mommy hat"; you ask a question, "Is this Mommy's hat?" or make a statement "This is not Mommy's hat")
- Do not ask the child to imitate; do not target specific language structures; do not explicitly reinforce correct productions

It Takes Two to Talk: The Hanen Program for Parents. A language intervention program in which

parents are trained to facilitate language development in their toddlers and preschoolers under naturalistic communication contexts; the target population is late-talking toddlers in the age group of 18 to 30 months and their parents; also applicable to preschool children and their parents; clinicians who wish to implement this program should get trained from the program developers.

- Structure the program to:
 - Conduct an initial orientation meeting
 - Assess the child
 - Conduct group parent training sessions
 - Offer three individual video feedback sessions for a total time commitment of 30 hours
- Train the parents, often the mother of the child to:
 - Arrange natural and interactive occasions with the child (e.g., any daily situation, including hand washing or brushing at a sink, eating, playing with toys, bedtime reading, sibling interactions)
 - Pause and wait for the child to initiate an utterance
 - Imitate (repeat) the child's utterance, comment on it, or interpret it (e.g., during hand washing, the child might say such words as *water*, *cup*, and *drink* and the mother might repeat each word several times to emphasize it)
 - Expand the child's utterances, extend the topic of the child's utterances, and ask questions to teach turn-taking in conversational exchanges

Girolametto, L., & Weitzman, E. (2006). It takes two to talk—the Hanen Program for parents. In R. J. McCauley & M. E. Fey (Eds.), *Treatment of language disorders in children* (pp. 77–101). Baltimore, MD: Paul H. Brookes.

Joint Book Reading. Systematic use of storybook reading to teach or stimulate language in children; allows for repetitive use and practice of the same phrases and concepts; helpful in establishing joint attention as well.

- Select storybooks that are linguistically and culturally appropriate for children under treatment
- Select books with colorful pictures

- Read the same story several times during a few sessions so that children memorize it
- Use prosodic features frequently to draw attention to specific language structures
- When the children know the story well, pause at points containing target language structures and prompt the children to supply the words, phrases, or sentences
- During different readings, pause at different junctures so the children produce different language structures
- Manipulate and vary pause locations that prompt progressively longer utterances from the children
- Ask a child to “read” (recite from memory, but looking at the text and the pictures) and pause
- Let the other children supply the words, phrases, and sentences

Kirchner, D. (1991). Reciprocal book reading. A discourse-based intervention strategy for the child with atypical language development. In T. Gallagher (Ed.), *Pragmatics of language: Clinical practice issues* (pp. 307–332). San Diego, CA: Singular.

Whitehurst, G., Falco, F., Lonigan, C., Fischel, J., DeBrayshe, B., Valdez-Menchaca, M., & Caulfield, M. (1988). Accelerating language development through picture-book reading. *Developmental Psychology*, 24, 552–558.

Mand-Model. A part of the Incidental Teaching Method; uses typical adult–child interactions in a play-oriented setting to teach language; supported by controlled evidence.

- Select a variety of attractive toys, pictures, and other stimulus materials
- Design a naturalistic interactive situation
- Establish a joint clinician–child attention to a particular material (such as a toy); if necessary, direct the child’s attention to a stimulus
- Mand a response from the child (e.g., say “Tell me what you want,” “Tell me what this is”)
- Model the correct, complete response if the child fails to respond or gives a limited (e.g., single word) response

- Prompt if the child does not imitate the whole sentence you modeled (e.g., “Tell me the whole sentence”)
- Praise the child for imitating or for responding correctly without modeling
- Give the material the child wants as you praise

Rogers-Warren, A., & Warren, S. (1980). Mands for verbalization. *Behavior Modification*, 4, 230–245.

Milieu Teaching. A collection of child language intervention procedures that emphasize natural, functional, conversational communicative contexts for teaching language; a naturalistic child language teaching method; uses natural consequences as reinforcers; includes the Mand-Model and Incidental Teaching Method and Time Delay; supported by controlled evidence; for its updated version, see Enhanced Milieu Teaching.

Hancock, T. B., & Kaiser, A. P. (2006). Enhanced milieu teaching. In R. J. McCauley & M. E. Fey (Eds.), *Treatment of language disorders in children* (pp. 203–236). Baltimore, MD: Paul H. Brookes.

Narrative Skills Training. A speaker’s description of events (stories, episodes) and experiences in a logically consistent, cohesive, temporally sequenced manner; analyzed in terms of a Story Grammar (described later in this section); an advanced language skill targeted during the final stages of intervention.

- Use the Event Structure (described earlier in this section) approach to give children experience in establishing Scripts (schemes of events)
 - Play such scripts as grocery shopping, eating in a restaurant, birthday parties, camping trips, vacations, playing certain games, and so forth
 - Play daily routine scripts (get children involved in daily activities)
 - Read repeatedly or tell the same stories so the children memorize the words, temporal sequences, characters, and events
 - Let the children act out the stories

- Let the children switch the roles on repeated scripts
- Ask children to narrate experiences as they play out scripts and assume different roles
- As you retell stories, pause before important phrases or critical descriptions so the children supply them
- Prompt the phrases and descriptions as the children hesitate; fade the prompts
- Ask the children to tell the stories or narrate events without enacting the scripts but with the help of pictures or slides
- Ask the children to tell stories or narrate events without scripts, pictures, or slides
- Ask the children to narrate new events or experiences (not rehearsed or scripted)

Nippold, M. A. (2014). *Language sampling with adolescents: implications for intervention* (2nd ed.). San Diego, CA: Plural.

Paul, R., Norbury, C., & Gosse, C. (2017). *Language disorders from infancy through adolescence* (5th ed.). St. Louis, MO: C. V. Mosby.

Naturalistic Child Language Teaching Method. An approach that emphasizes natural, functional, and conversational communicative contexts for teaching language to children; uses loose training structure; uses natural consequences as reinforcers; includes Incidental Teaching Method, Joint Action Routines or Interactions, Mand-Model, and Script Therapy (all described in this section).

Peer Modeling. A child language intervention method in which the peers are trained to model the target skills for the child.

- Select a peer who agrees to help and is acceptable to the client
- Have the peer observe your treatment sessions
- Describe the target behaviors, modeling and imitation sequence, and reinforcement procedures
- Let the peer model and reinforce the child's productions in your presence

- Refine the peer's skills in modeling the target behaviors
- Ask the peer to submit recorded language samples that document appropriate modeling outside the clinic
- Periodically assess the results of peer modeling and provide additional training to the peer

Recast. Repeating some or all of the words in a child's incorrect or inadequate utterance into a more accurate or more complete utterance while retaining the child's presumed meaning; recast also may just add new information, without using the child's words; a method of play-oriented, unstructured, Indirect Language Stimulation; some supportive evidence exists.

- Arrange play activity designed to enhance opportunities for language production
- Play with the child
- Expand the child's utterance into a sentence type that may be different from the child's presumably intended sentence type (e.g., the child says "Big ball"; you recast it into a negative sentence, "No, it is not a big ball"; or expand it into a question form, "Is this a big ball?")
- Recast the child's grammatically wrong utterance into a correct utterance (e.g., the child says, "Him want cookie," and you recast it as "He wants cookie")
- Recast to add new information (e.g., the child says, "He eat a lot," and you recast it "Yes, he is hungry"); note that the child's words are not used
- The child may not be required to imitate the recast utterances, in which case it is a weakness of the method and quite likely inefficient

Cleave, P. L., Becker, S. D., Curran, M. K., Owen Van Home, A. J., & Fey, M. (2015). The efficacy of recasts in language intervention: A systematic review and meta-analysis. *American Journal of Speech-Language Pathology*, 24, 237–255.

Time-Delay. A parent-administered procedure to provide environmental stimuli to evoke verbal responses

from children; clinicians may use it as well; in implementing this procedure, teach the parents to:

- Establish joint attention—the parent and the child look at the same stimulus or activity; (e.g., the mother is blowing bubbles and the child is watching with interest)
- Wait for the child to make a request or a comment (e.g., the mother puts the wand close to her mouth but waits)
- Expand and reinforce the child's mand (e.g., the child says, "Blow bubbles" and the mother blows bubbles and expands, "I blow big bubbles")
- Prompt or model a response if the child does not mand or gives no response of any kind (e.g., the mother mands, "Tell me what you want" or "Say, 'blow bubbles'")
- Reinforce and expand the child's utterance (e.g., if the child said "blow bubbles," the mother says, "I blow bubbles for you" and blows plenty of them)
- Anytime there is no response, model the response, reinforce the child's imitation, and expand it

Hancock, T. B., & Kaiser, A. P. (2006). Enhanced milieu teaching. In R. J. McCauley & M. E. Fey (Eds.), *Treatment of language disorders in children* (pp. 203–236). Baltimore, MD: Paul H. Brookes.

Topic Initiation (Treatment for). The skill to start conversation with a new topic; a pragmatic (social) language skill, a treatment target; children with language disorders either fail to initiate topics or introduce inappropriate topics.

- Arrange a variety of stimuli that could trigger a new topic: objects, pictures, storybooks, topic cards (for children who can read), toys, structured play situations such as a kitchen and so forth
- Introduce one of the stimulus items or situations and draw the child's attention to it (e.g., a picture of a family setting up a tent in a park)
- Wait for the child to initiate conversation about the picture and the story

- Instruct the child to say something about the picture if the child does not initiate a topic
- Prompt by beginning the story when the child does not initiate, (“They are setting up a . . .”)
- Praise the child for saying anything related to the topic depicted
- Accept statements that are remotely connected to the topic at hand; gradually demand more relevant responses
- Do not interrupt the child or overly correct the forms of responses
- Ask the child to use the topic cards to initiate new topics
- Ask the child to think of new topics to talk about
- Prompt new topics
- Withdraw or fade such prompts, cues, cards, pictures, and other special stimuli to make topic initiation more spontaneous
- Train parents to use your techniques so they can continue intervention at home

Topic Maintenance (Treatment for). A pragmatic (social) language skill and treatment target; talking about a single general topic for extended duration; frequent and abrupt switching of conversational topics suggests lack of this skill.

- Target topic maintenance when training has moved to the conversational speech stage or sooner if the session structures allow it
- Let the child select topics of interest for talking
- Set a realistic duration for which you want the child to talk on a single topic; or set a target number of words to be produced on a topic
- Increase the duration or the number of target words in gradual steps
- Use such devices as *Tell me more. What about that? What happened next? Who said what? Where was it? When did that happen?* and so forth to stimulate more speech on the same topic
- Reinforce the child for maintaining the topic

- Stop the child when he or she abruptly switches the topic
- Move the child back to the target topic
- Train on a few topics and then probe with untrained topics to see if the skills have generalized
- Train on additional topic exemplars if the skills have not generalized

Turn-Taking (Treatment for). Appropriate exchange of speaker and listener roles during conversation; an advanced treatment target; interrupting a speaker and not responding to cues to talk are indicators of deficient turn-taking.

- Select turn-taking as a target when treatment has advanced to conversational speech or sooner if the child can handle it
- Baseline the number of interruptions and failure to take cues to talk
- Design a signal for the child to talk (e.g., such verbal cues as “Your turn” or nonverbal cues as a hand gesture to suggest *you speak*)
- Design a signal that says *do not interrupt* or *do not talk* because it is your (clinician’s) turn to talk (e.g., finger on your lips)
- Use such other discriminative stimuli as a real or toy microphone that you exchange with the child; the one holding the microphone talks and the other listens
- Reinforce the child for talking only when signaled or while holding the microphone
- Follow the same rule that you impose on the child (e.g., talk only when you hold the microphone)
- Teach the child to say, “It is your turn”
- Reinforce the child for yielding the floor
- Teach turn-taking until the child meets a performance criterion (e.g., no errors of turn-taking in two consecutive conversational exchanges)
- Fade the signals or other special discriminative stimuli used to prompt the child
- Probe without signals or special discriminative stimuli

- Train until a probe criterion is met (at least 90% accuracy in turn-taking while not receiving reinforcers)

Whole Language Approach. A philosophical approach, not a technique to teach language skills; advocates that in teaching language, it should not be broken down into components; believes that all aspects of literacy, including reading, writing, listening, and talking, should be simultaneously taught as an integrated whole; its use in teaching language to children with language disorders is questionable; there are better, experimentally supported, alternatives.

Language Disorders in Ethnoculturally Diverse Groups.

Treatment of language disorders in ethnoculturally varied children may generally use the same treatment procedures with suitable modifications to accommodate a particular child's and the family's unique cultural and language background; there is no evidence that contraindicates the use of such effective language treatment procedures as modeling, positive reinforcement, corrective feedback, shaping, and so forth; much of the multicultural literature on speech and language skills is concerned with appropriate diagnosis, based on the speech and language characteristics of the ethnocultural group the child belongs to; consult the cited sources and the two companion volumes, *Hegde's PocketGuide on Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, to make appropriate diagnosis of language disorders in children of varied backgrounds; see Ethnocultural Variables in Treatment for general guidelines; before starting treatment with a child of different cultural background, consider the following:

- Assure the parents that a diagnosis of language disorders was made on the basis of the child's and the family's language background and communication patterns and that it was not based on a comparison of the child's skills in relation to another ethnocultural group
- Discuss treatment recommendations with the parents so they understand that the need for treatment is based

on an appropriate diagnosis and will be designed to help the child meet his or her academic, personal, family, and social communication needs

- Discuss potential treatment targets with the parents and invite their suggestions, ask them to express their concerns, and offer to discuss the language treatment goals with the child's teachers
- Discuss the possibility of integrating language treatment with literacy treatment based on the child's classroom needs
- Select stimulus materials and treatment activities that are relevant to the child and the family
- Ask parents to supply stimulus materials from home or request suggestions on what kinds of ethnic stories, pictures, holiday celebration ideas and items, and other treatment materials may be better suited to the child's language treatment; use such materials in language treatment sessions
- Have a collection of ethnocultural story books, fairy tales, ethnic holiday celebration ideas, unique ways of celebrating birthdays or other kinds of family events; incorporate such resources into language treatment sessions
- Ask parents to periodically observe your treatment sessions and offer suggestions on treatment stimuli and treatment activities
- Make your display boards, activity boards, bulletin boards, and so forth culturally rich and varied; include different ethnic materials to represent varied cultural backgrounds

Treatment of Language Disorders in African American Children.

In treating language disorders of African American children, consider the following guidelines and the African American English characteristics; these characteristics, although different from mainstream English usage, are not indicative of language disorders; therefore, they are not targets of language treatment; acquiring mainstream English expressions may be a goal of some African American children and their families; in such cases, teaching patterns of mainstream English usage is acceptable.

- Make sure that the child and his or her family speak AAE; do not assume that all African Americans speak AAE at home
- Consult with the child's family members about their goals for language treatment; ask whether they want to have the mainstream English language patterns taught as well
- Consult with an African American speech-language pathologist in your area or with one who specializes in AAE characteristics; develop resources on AAE and culture for future clinical use
- Select language treatment targets that are consistent with AAE and the goals of the family, are useful to the child, help meet the child's educational and social demands, and will quickly improve the child's communicative skills
- Work with the teachers and other special educators to have them appreciate the child's language patterns and not make negative evaluations
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support
- Consider the following AAE characteristics as summarized by Roseberry-McKibbin (2014); select treatment targets that are consistent with these characteristics; see also [Treatment of Speech Sound Disorders in African American Children](#)

AAL Characteristic	Mainstream American English	Sample AAL Utterance
Noun possessives may be omitted.	That's the woman's car. It's John's pencil.	That <i>the woman</i> car. It <i>John</i> pencil.
Noun plurals may be omitted.	He has 2 boxes of apples. She gives me 5 cents.	He got 2 <i>box</i> of <i>apple</i> . She give me 5 <i>cent</i> .

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AAL Characteristic	Mainstream American English	Sample AAL Utterance
Third person singular may be omitted	She walks to school. The man works in his yard.	She <i>walk</i> to school. The man <i>work</i> in his yard.
Forms of <i>to be</i> (<i>is, are</i>) may be omitted.	She is a nice lady. They are going to a movie.	<i>She a</i> nice lady. <i>They going</i> to a movie.
Present tense <i>is</i> may be used regardless of person or number.	They are having fun. You are a smart man	<i>They is</i> having fun. <i>You is</i> a smart man.
Person or number may not agree with past and present forms.	You are playing ball. They are having a picnic.	You <i>is</i> playing ball. They <i>is</i> having a picnic.
Present tense forms of auxiliary <i>have</i> may be omitted.	I have been here for 2 hours.	I been here for 2 hours. He done it again.
Past tense endings may be omitted.	He lived in California. She cracked the nut.	He <i>live</i> in California. She <i>crack</i> the nut.
Past tense <i>was</i> may be used regardless of number and person.	They were shopping. You were helping me.	They <i>was</i> shopping. You <i>was</i> helping me.
Multiple negatives may be used to add emphasis to the negative meaning.	We don't have any more. I don't want any cake. I don't like Broccoli.	We don't have no more. I don't never want no cake. I don't never like Broccoli.

AAL Characteristic	Mainstream American English	Sample AAL Utterance
<i>None</i> may be substituted for <i>any</i> .	She doesn't want any.	She don't want <i>none</i> .
In perfective constructions, <i>been</i> may be used to indicate that an action took place in the past.	I had the mumps when I was 5.	I <i>been had</i> the mumps when I was 5. I <i>been known</i> her.
<i>Done</i> may be combined with a past tense form to indicate that an action was started and completed	He fixed the stove. She tried to paint it.	He <i>done fixed</i> the stove. She <i>done tried</i> to paint it.
The form <i>be</i> may be used as the main verb.	Today she is working. We are singing.	Today <i>she be</i> working. <i>We be</i> singing.
Distributive <i>be</i> may be used to indicate actions and events over time.	He is often cheerful. She's kind sometimes.	<i>He be</i> cheerful. <i>She be</i> kind.
A pronoun may be used to restate the subject.	My brother surprised me. My dog has fleas.	My brother, <i>he</i> surprise me. My dog, <i>he</i> got fleas.
<i>Them</i> may be substituted for <i>those</i> .	Those cars are antiques. Where'd you get those books?	<i>Them</i> cars, they be antique. Where you get <i>them</i> books?
Future tense <i>is</i> and <i>are</i> may be replaced by <i>gonna</i> .	She is going to help us. They are going to be there.	She <i>gonna</i> help us They <i>gonna</i> be there.

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AAL Characteristic	Mainstream American English	Sample AAL Utterance
<i>At</i> may be used at the end of <i>where</i> questions.	Where is the house? Where is the store?	Where is the house <i>at</i> ? Where is the store <i>at</i> ?
Additional auxiliaries may be used.	I might have done it.	<i>I might could have</i> done it.
<i>Does</i> may replace <i>do</i> .	She does funny things. It does make sense.	<i>She do</i> funny things. <i>It do</i> make sense.

Battle, D. E. (2002). *Communication disorders in multicultural children* (3rd ed.). Boston: Butterworth-Heinemann.

Hegde, M. N., & Pomaville, F. (2017). *Assessment of language disorders in children: Resources and protocols* (3rd ed.). San Diego, CA: Plural.

Kamhi, A. E., Pollock, K. E., & Harris, J. L. (1996). *Communication development and disorders in African American Children*. Baltimore, MD: Paul H. Brookes.

Roseberry-McKibbin, C. (2014). *Multicultural students with special language needs* (4th ed.). Oceanside, CA: Academic Communication Associates.

Treatment of Language Disorders in Children of Asian Cultures. Children of Asian cultures are a heterogeneous group; very few generalized statements can be made about these children; people of Asia speak a variety of languages, belonging to different language groups, and each language may have many dialects; languages of China, the Indian subcontinent, and South East Asia differ widely; much of the available information described under the Asian heading pertains to the languages of China; some of the characteristics listed may apply to other Asian children; use the guidelines offered in this section with caution while treating children of Asian background.

- Make sure that the child and his or her family do use a primary language that is other than English at home; some Asian children may acquire English as their first language although the parents speak a different language at home
- Consult with the child's family members about their goals for language treatment; ask whether they want to have the mainstream English language patterns taught as well
- Consult with a bilingual speech-language pathologist in your area or with one who specializes in Asian languages in general or the child's primary language in particular; develop resources on the different Asian languages spoken in your service area for future clinical use
- Select language treatment targets that are consistent with the child's primary language and the goals of the family, are useful to the child, help meet the child's educational and social demands, and will quickly improve the child's communicative skills
- Work with the teachers and other special educators to have them appreciate the child's English language patterns that are due to his or her primary Asian language and not make negative evaluations
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support
- Consider the following characteristics of Asian communication patterns Roseberry-McKibbin (2014) summarizes; select treatment targets based on these characteristics for a child whose primary language is an Asian language; note that not all characteristics may apply to all Asian languages; see also Treatment of Speech Sound Disorders in Bilingual Children

Asian Language Characteristics	Sample English Utterance
Omission of plurals	Here are two piece of toast. I got 5 finger on each hand.

continues

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Asian Language Characteristics	Sample English Utterance
Omission of copula	He going home now. They eating.
Omission of possessive	I have Phuong pencil. Mom food is cold.
Omission of past tense morpheme	We cook dinner yesterday. Last night she walk home.
Past tense double marking	He didn't went by himself
Double negative	They don't have no books
Subject-verb-object relationship differences/omissions	I messed up it. He like.
Singular present tense omission or addition	You goes inside. He go to the store.
Wrong ordering of interrogatives	You are going now?
Misuse or omission of prepositions	She is in home. He goes to school 8:00.
Misuse of pronouns	She husband is coming She said her wife is here.
Omission and/or overgeneralization of articles	Boy is sick. He went the home.
Incorrect use of comparatives	This book is gooder than that book.
Omission of conjunctions	You ____ I going to the beach.
Omission, lack of inflection on auxiliary "do"	She ____ not take it. He do not have enough.
Omission, lack of inflection on forms of "have"	She have no money. We ____ been the store.
Omission of articles	I see little cat.

- Cheng, L. L. (1991). *Assessing Asian language performance* (2nd ed.). Oceanside, CA: Academic Communication Associates.
- Hegde, M. N., & Pomaville, F. (2006). *Assessment of language disorders in children: Resources and protocols*. San Diego, CA: Plural.
- Roseberry-McKibbin, C. (2014). *Multicultural students with special needs* (4th ed.). Oceanside, CA: Academic Communication Associates.
- Shekar, S., & Hegde, M. N. (1995). Asian Indians: Their languages and culture. In L. L. Cheng (Ed.), *Integrating language and learning for inclusion—an Asian-Pacific focus* (pp. 125–145). San Diego, CA: Singular.
- Shekar, S., & Hegde, M. N. (1996). Cultural and linguistic diversity among Asian Indians. *Topics in Language Disorders*, 16(4), 54–64.

Treatment of Language Disorders in Hispanic Children.

Information on treating children whose primary language is Spanish is accumulating faster than information on other minority language groups in the United States; most guidelines offered here are relevant to all children whose primary language is other than English; adapt these guidelines to other groups by learning more about the primary language and communication patterns of your bilingual client in need of language treatment.

- Make sure that the child and his or her family do use a primary language that is other than English at home; some ethnoculturally diverse children may acquire English as their first language even though the parents speak a different language at home
- Consult with the child's family members about their goals for language treatment; ask whether they want to have the mainstream English language patterns taught as well
- Consult with a bilingual speech-language pathologist in your area or with one who specializes in the child's primary language characteristics; develop resources on the different languages spoken in your service area for future clinical use
- Select language treatment targets that are consistent with the child's primary language and the goals of the

family, are useful to the child, help meet the child’s educational and social demands, and will quickly improve the child’s communicative skills

- Work with the teachers and other special educators to have them appreciate the child’s English language patterns that are due to his or her primary language and not make negative evaluations
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support
- Consider the following characteristic of Spanish-influenced English as Roseberry-McKibbin (2014) summarizes; select treatment targets that are consistent with these characteristics; see also Treatment of Speech Sound Disorders in Bilingual Children

Spanish-Influenced Language Characteristics	Sample English Utterances
1. Adjective comes after the noun.	The house green.
2. <i>s</i> is often omitted in plurals and possessives.	The girl book is . . .
3. Past tense <i>-ed</i> is often omitted.	We walk yesterday.
4. Double negatives are required.	I don’t have no more.
5. Superiority is demonstrated by using <i>mas</i> .	This cake is more big.
6. The adverb often follows the verb.	He drives very fast his motorcycle.

Brice, A. E., & Brice, R. G. (2009). *Language development: Monolingual and bilingual acquisition*. Boston, MA: Allyn & Bacon.

Goldstein, B. A. (2004). *Bilingual language development & disorders in Spanish-English speakers*. Baltimore, MD: Paul H. Brookes.

Hegde, M. N., & Pomaville, F. (2017). *Assessment of language disorders in children: Resources and protocols*. San Diego, CA: Plural.

Roseberry-McKibbin, C. (2014). *Multicultural students with special needs* (4th ed.). Oceanside, CA: Academic Communication Associates.

Treatment of Language Disorders in Native American Children.

Children of Native Americans (American Indians) also speak a variety of languages belonging to different language families; however, it is likely that many children acquire English as their only language, as the American Indian languages are on the decline even within their own communities; follow the guidelines offered here with caution and try to develop information on the particular language and culture of the child being treated; if the child does speak one of the several Native American languages, he or she is then essentially a bilingual child with varying English proficiency.

- Make sure that the child and his or her family do use a Native American language at home and that English is a second language; many American Indian children do not learn their native language and English may be their only language
- Consult with the child's family members about their goals for language treatment; ask whether they want to have the mainstream English language patterns taught as well
- Consult with a speech-language pathologist who knows the child's Native American language or an expert in that language; develop resources on the different Native American languages spoken in your service area for future clinical use
- Select language treatment targets that are consistent with the child's Native American language and the goals of the family, are useful to the child, help meet the child's educational and social demands, and will quickly improve the child's communicative skills
- Work with the teachers and other special educators to have them appreciate the child's English language patterns that are due to his or her primary language and not make negative evaluations
- Refer the child to other specialists as needed (e.g., to an audiologist); educate the family about needed and available services, resources, and support

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- Consider the following characteristics of native American communication styles as Roseberry-McKibbin (2014) summarizes; select treatment targets that are consistent with these characteristics; note that the information is not specific to any one Native American language; you will have to get that language-specific information before you can plan an effective treatment program; see also Treatment of Speech Sound Disorders in Bilingual Children; among Native Americans:
 - Mutual respect is a high cultural value; avoiding eye contact and looking down is a standard method of showing respect
 - Children are especially taught not to maintain eye contact while talking to adults; maintaining eye contact during conversation with an adult is a sign of rudeness and defiance
 - Native American mothers, especially those in the Navajo population, may not talk much while caring for their infants
 - Children are taught to listen, observe, and learn thereby
 - In the judgment of parents, their children may have better auditory comprehension skills than their expressive language skills
 - Some parents may not encourage their children to speak their native language until their articulation is acceptable; therefore, children in the early years may be deprived of language learning opportunities
 - A long period of nonverbal communication (pointing and gesturing) may pass before children begin to use words
 - Talking too much or talking English may be viewed as imitating the white man
 - Native American etiquette requires that a speaker pause before answering a question; quick answers imply that the question did not require much thought
 - If unsure of an answer, children may not respond to a question

- Children may be reluctant to express their opinions until the adults indicate that they have earned their right to express their own opinions
- Public expression of strong feelings is generally discouraged
- Expression of grief in the presence of outsiders may be acceptable only during official mourning ceremonies

Robinson-Zanartu, C. (1996). Serving Native American children and families: Considering cultural variables. *Language, Speech, and Hearing Services in Schools*, 27(4), 373–384.

Roseberry-McKibbin, C. (2014). *Multicultural students with special needs* (4th ed.). Oceanside, CA: Academic Communication Associates.

Westby, C., & Vining, C. B. (2002). Living in harmony: Providing services to Native American children and families. In D. E. Battle (Ed.), *Communication disorders in multicultural populations* (3rd ed., pp. 135–138). Boston, MA: Butterworth-Heinemann.

Language Disorders in Infants, Toddlers, and Preschoolers.

Treatment of language (and speech) problems in young children requires a two-pronged approach: parent-implemented home treatment (language stimulation) program and the more formal treatment the clinician implements in the clinic; the clinician may conduct a few informally experimental treatment sessions to find out what works best for the child; during these sessions, the clinician may train parents in a home language stimulation program; frequent assessment is essential to evaluate the child's progress and to modify treatment plans; early intervention is essential to avoid or reduce the subsequent and serious negative consequences for the child's social development and academic achievement; treatment of infants, toddlers, and preschoolers should concentrate on all aspects of communication, including early speech development, nonverbal communication, physical and behavioral development; treatment outcome will depend on the presence of any intellectual and developmental disabilities, genetic syndromes, autism spectrum disorder, and hearing loss or other sensory loss.

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Ethnocultural Considerations

- Diagnose a language disorder in the context of the child's and the family's ethnocultural and linguistic background
- Recommend treatment that is consistent with the family's cultural and linguistic background
- Gauge the family members' sophistication in implementing a home language stimulation program; estimate the training needs
- Seek support and resources that the family may need to sustain a long-term intervention plan for the child
- Consider the additional guidelines offered under [Language Disorders in Children](#)

Treatment of Infants, Toddlers, and Preschoolers: General Guidelines

- Design an early intervention program that may include both treatment in a clinical facility and interventional activities at home conducted by parents, other caregivers, and other family members (see [Individual Family Service Plans](#))
- Base the family- and clinic-based intervention plan on an interdisciplinary assessment not only of the child, but also of the family's communication patterns and resources
- Do not recommend a home treatment program that you have not informally tested with the child during a few clinic-based treatment sessions
- Have the early home-based language stimulation program implemented in naturalistic communication and child care contexts
- Ask parents to repeat their attempts to stimulate language throughout the day; minimally, ask them to repeat stimulation several times during the day
- Describe specific skills they need to stimulate at any given period, as the target skills will change as the child learns the various elements of speech and language
- Design and implement a formal language treatment if the parents' work did not produce desired results in a matter of a few months

- Use the same procedures as you recommend the parents to increase the child's response to verbal and social stimuli and to teach specific words, phrases, and sentences
- Include parents or other caregivers in the formal treatment sessions to continuously refine their skills in implementing the home-based program
- Select treatment procedures described under Language Disorders in Children

Treatment of Infants, Toddlers, and Preschoolers

- Design a comprehensive interdisciplinary treatment for a young child; different specialists (e.g., audiologists, medical professionals, child psychologists) will address the varied needs of the young child; the speech-language pathologist takes greater responsibility for communication assessment and treatment and perhaps feeding and oral motor development as well.
- Consider the Child's Need for Aural Rehabilitation
 - Based on the audiological assessment results, help develop an aural rehabilitation program by working with an audiologist
 - Design the home-based language stimulation program that takes into consideration the hearing loss if the child has one
- Improve the Child's Response to Verbal and Nonverbal Stimuli
 - During the first 6 months, have the parents:
 - Present various kinds of environmental and human sounds, including the mother's voice throughout the day; have parents and others in the family approach the infant and make cooing and babbling sounds; make such sounds whenever the mother attends to the physical needs of the child (e.g., bathing, feeding)
 - "Talk" to the infant while taking care of the infant
 - Draw the baby's attention to one's own smiling face
 - Gently tickle and touch the baby as the baby looks at the face, turns in the direction of the mother's voice, smiles, or vocalizes

- Call the baby's name out frequently to teach the baby to respond to his or her name
- Speak to the baby in a soft, affectionate, and pleasant voice
- Hold infant at eye level and keep an expressive face while interacting with the infant
- Smile and laugh in general and exhibit contingent smiling and laughing (reaction to similar behaviors in the child)
- Avoid quiet play with the child; instead, play with speech and vocalizations; vary the tone of speech and vocalization; always give contingent response to child's reactions
- During 6 to 12 months of age, have the parents:
 - Utter the names of family members when they are around; reinforce with smiles, gentle touch, and appropriate vocalizations when the baby looks around or looks at the person whose name is called
 - Name objects that are present in the baby's environment (e.g., toys, food items, clothing) and reinforce the baby for looking at the named object or for trying to reaching it
 - Present different sounds associated with the baby's toys and reinforce the baby for looking at the toy when the sound is heard
 - Take note of child's response to "No" (e.g., cessation of activity; hesitancy; unpleasant facial expression) or to scolding (e.g., frowning, crying, unpleasant facial expression) to continuously assess the child's reactions and improvements to different kinds of verbal stimuli
 - Teach the baby to imitate various kinds of gestures (e.g., clapping when clapped, touching the body parts when modeled)
 - Draw attention to an object and reinforce the infant for joint attention
 - Keep written notes on the kinds and number of words the baby understands at home to informally

assess whether there is improvement due to such stimulation activities

- During 12 to 18 months of age, have the parents:
 - Reinforce the child for responding to familiar names and objects
 - Reinforce the baby's response to some names of absent objects and persons (e.g., when said "ball," looking at the place where a ball was kept, but is now absent; looking in the direction of a sibling's room when the sibling's name is uttered)
 - Model simple and everyday words for the child to imitate; reinforce vocal approximations or even appropriate nonverbal responses
 - Begin to model such constructions as possessor + possession (e.g., *Mommy's shoes*) for the child to comprehend (not produce)
 - Give simple commands and reinforce the child (e.g., "Clap your hand," "Pick up the car," "Show me your nose")
 - Keep track of the baby's progress
- During 18 to 24 months of age, have the parents:
 - Evoke the baby's response to absent objects more frequently; talk to the baby about siblings not present, objects not in the view; reinforce baby's vocal or nonvocal responses (e.g., imitating the name of the sibling, vocalizing noises associated with toys)
 - Model and reinforce productions of two-word combinations; model phrases constructed out of the words the child has learned to produce
 - Make simple requests or give commands (e.g., "Give me the pencil," "Find the ball") and reinforce correct responses
- During 24 to 36 months of age, have the parents:
 - Teach correct nonverbal response (comprehension) to three-word sentences and reinforce the actions
 - Teach the child to respond correctly to two-, three-, and four-element commands and reinforce correct nonverbal actions

- Make progressively more complex requests and reinforce correct responses
- Ask questions involving *what*, *who*, and *where* and reinforce correct responses
- Train the Parents in Promoting Verbal Communication
 - During the first 6 months, have the parents:
 - Reinforce any vowel-like sounds, sound vocalizations, and increased range of vocalizations produced spontaneously; touch, gentle tickle, smile when the baby is looking at the face, and pick up the infant
 - Reinforce marginal babbling (consonant-like sounds in babbling)
 - Model vocalizations if the infant's level of vocalizations is low; ask the parents to get closer to the infant, draw the infant's attention, and model babbling-like sounds and other kinds of vocalizations
 - Reinforce any kind of imitative or spontaneous vocalizations the infant makes
 - During 6 to 12 months of age, have the parents:
 - Model and reinforce consonant-vowel syllable productions like *bababa* or *mamama*)
 - Model and reinforce production of sounds produced in the front of the mouth (e.g., /p/, /t/, /d/)
 - Take turns in vocalizations (e.g., the baby and a caregiver may vocalize in an alternating fashion); reinforce baby's vocalizations
 - Vocalize additional and varied consonant-like sounds for the infant to imitate
 - Model adult vocalizations or gestures for the infant to imitate
 - Reinforce the child for differential responding in any form; the infant may produce specific sounds in specific situations (e.g., one vowel sound to suggest a desire for an object and another sound to suggest disapproval)
 - Model varied intonations as you vocalize for the infant to imitate
 - Reinforce any sign of intentional communication between 9 and 10 months of age (e.g., specific ges-

- turing or pointing; vocalizing while making eye contact)
- During 12 to 18 months of age, have the parents:
 - Continuously expand the range of sounds modeled for the child's imitation
 - Reinforce increased production of gestures and vocalizations to obtain objects, draw attention, and regulate the behavior of caregivers
 - Reinforce the approximations of the first words that should appear during this time; if not, frequently model simple everyday terms like *Mommy*, *Daddy*, names of the siblings and pets, child's food and clothing items
 - During 18 to 24 months of age, have the parents:
 - Model words in differentiated categories (nouns, action words, modifiers, or adjectives); reinforce the child's approximations
 - Model 10–15 selected functional names of objects while showing the objects; reinforce the child's approximations
 - Begin teaching two-word phrases, constructed out of the single words the child already produces; model and reinforce the correct productions or approximations
 - Model speech sounds for improved articulation while teaching simple words; target 50% intelligibility of the child's speech to the caregivers
 - Reinforce a sentence-like intonation, even if the child's production is an unintelligible strings of syllables
 - During 24 to 36 months of age, have the parents:
 - Model and reinforce the production of simple sentences
 - Model three- and four-word sentences for the child to imitate; model five-word sentences toward the end of this period
 - Model and reinforce the production of such questions as “What is that?” or “What you doing?”

- Continue to work on more precise production of speech sounds to increase intelligibility of speech (up to 75%)
- Continue to work on expanding the child's vocabulary by introducing new words for the child to imitate and produce spontaneously; frequently reinforce the production of new words (at 24 months, just under 300 and at 36 months, 900 to 1000 words are expected in a typical child)
- Reinforce any form of functional communication, including words, phrases, sentences, gestures, and other forms of nonverbal communicative behaviors while continuing to stimulate verbal expressions
- Withdraw reinforcement gradually for gestures and undifferentiated signs for which the child can substitute verbal responses
- Promote Interactive Play
 - Interactive or social play activities are a good means of promoting communication in naturalistic contexts; have the parents:
 - Play with the child, include a sibling or a peer, and teach cooperative play; have the playmates exchange toys or other materials first by requesting them; name objects involved in play; have the child describe the activities
 - Take part in the child's constructive activities (e.g., painting, block building, art or craft activities) and induce as much speech as possible
 - Arrange play situations and model the verbal responses that are appropriate for the different or alternating roles the child assumes; encourage the child to verbalize what he or she is supposed to do or say

Billeaud, F. P. (2003). *Communication disorders in infants and toddlers* (3rd ed.). St. Louis, MO: Butterworth-Heinemann.

Rossetti, L. M. (2001). *Communication intervention birth to three* (2nd ed.). Clifton Park, NY: Thomson Delmar Learning.

Paul, R., Norbury, C., & Gosse, C. (2017). *Language disorders from infancy through adolescence* (5th ed.). St. Louis, MO: C. V. Mosby.

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Language Disorders in Infants, Toddlers, and Preschoolers: Specific Programs.

Very few comprehensive language treatment programs specifically developed for the very young have been experimentally evaluated; the outline given in the previous entry is comprehensive, and uses the behavioral methods of modeling, shaping, and differential reinforcement that have been known to be effective in teaching speech-language skills to younger as well as older individuals; among the few procedures that have received research support, one or more of the following may be considered:

- Enhanced Milieu Teaching, described in a previous entry
- Focused Stimulation, described in a previous entry
- Recast (also part of the enhanced milieu teaching)
- It Takes Two to Talk—The Hanen Program for Parents

Language Disorders in Older Students and Adolescents.

Treatment of language disorders in older students and adolescents poses some special challenges; research information and clinical resources on language skills and disorders in older students and adolescents lag behind those that are available for younger children; unlike in children under 5 or 6 years of age, language skills in older students and adults change more gradually; disorders in their language may be more subtle; the clinician needs to treat a variety of higher-level semantic, abstract, and academic language skills in older students and adolescents; for details on language and literacy disorders in older students and adolescents, see the sources cited at the end of this entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*.

Treatment Objectives/General Guidelines

- Treatment of language disorders in older students and adults is multifaceted and includes the following problems or target skills:

- Deficiencies in basic language skills that may have persisted from early childhood
- Advanced social and technical discourse
- Critical and logical reasoning skills
- Mastery and expression of scientific, technical, literate, academic, logical, and discipline-specific terms
- Word retrieval, word definitions, word relations, and skills in the use of figurative language
- Advanced syntactic structures
- Advanced narrative and discourse skills
- Reading and writing skills
- Treatment is oriented toward academic demands and performance
- Treatment needs to be fully coordinated with the teacher's regular or special education programs for the student

Ethnocultural Considerations

- See Ethnocultural Variables in Treatment
- Pay special attention to the student's academic standing and social communication patterns, and to the family's communication patterns
- Check whether the student is bilingual; if so, obtain information from the teachers about the student's English proficiency to design appropriate treatment strategies

Treatment of Language Disorders

- Teach deficient basic skills (e.g., basic sentence structures, use of grammatical morphemes, pragmatic skills including turn-taking, topic initiation, topic maintenance, conversational repair skills)
- Use procedures described under Language Disorders in Children to remediate basic language deficiencies in older students and adolescents
- Teach advanced semantic skills
 - Teach comprehension and correct production of literate (academic, scholarly, learned) lexicon; make a list of such words as the following:

assume	suppose	infer	interpret	hypothesize	define
compare	contrast	criticize	evaluate	summarize	predict
explain	describe	conclude	confirm	support (a statement)	discriminate
imply	concede	presume	guess	reject (a statement)	allude
fact	opinion	evidence	belief	contradictory	logical
rational	irrational	implicit	explicit	contrary	affirm

- Have precise definition of academic terms selected for teaching; use the student's textbooks or other classroom materials; seek the teacher's help in selecting and defining terms to be used in intervention sessions
- Ask the student to define the terms on the list; prompt correct definitions and reinforce approximations to begin with
- Require progressively more precise and complete definitions of terms
- Ask the student to produce sentences with those terms
- Ask the student to write sentences that include the terms
- Ask the client to contrast the meaning of terms; shape correct responses, and reinforce approximations initially and require more precise comparisons as the student gains better mastery of the concepts:
 - Fact and opinion
 - Description and explanation
 - Inference and assumption
 - Suggestion and hypothesis
 - Belief and theory
 - Illusion and allusion
 - Since and because

- Further and farther
- Affect and effect
- Alternate and alternative
- Latter and later
- Thought and feeling
- Teach both comprehension and correct production of figurative language; make a list of common proverbs, metaphors, and idioms and ask the student what they mean; for example:
 - A stitch in time saves nine
 - A penny saved is a penny earned
 - Put the cart before the horse
 - Put all your eggs in one basket
 - Don't kill the goose that lays the golden eggs
 - He is cold as ice
 - She is fit as a fiddle
 - Time is money
 - They wanted to bury the hatchet
 - Off the wall
 - Off the record
 - She looks like a million bucks
 - Skeleton in the closet
- Prompt correct interpretations of the selected figurative language exemplars
- Contrast the literal and figurative meanings of the exemplars and ask the student to state the contrasts
- Ask the student to write sentences or short paragraphs in which figurative language exemplars are correctly used; give corrective feedback; show published writing samples for modeling
- Treat word retrieval problems in conversational speech:
 - Prompt the correct words at the earliest sign of false starts, pauses, revisions, repetitions, and beating around the bush—all signs of word retrieval problems
 - Teach the student to pause and think of the correct words instead of beating around the bush or getting stuck with false starts

- Teach the student to avoid general words but use instead specific words; prompt specific words and withdraw prompts
- Teach word relations:
 - Have the student define and contrast synonyms and antonyms
 - Ask the student to explain why the meanings are similar across synonyms and different across antonyms
 - Teach the student to write brief essays in which relational words are correctly used; give corrective feedback and ask the student to rewrite
- Teach the production of precise terms during conversation and narrative tasks:
 - Reduce the frequency of such expressions as “this,” “that,” “you know what I mean,” “this thing,” “that stuff”
 - Prompt the student for more precise words when the student uses just general terms
 - Withdraw prompts and reinforce the student for pausing and thinking of more precise words during conversation
 - Have the student write short essays for precise words; ask the student to avoid the general words in writing the essays; evaluate the student’s writing and give feedback
- Teach syntactic skills
 - Use the assessed and analyzed speech and language sample, narratives, writing samples, to target syntactic skills for teaching; generally, teach the production of low-frequency constructions the student has difficulty with:
 - Passive sentences (e.g., The examination *was thoughtfully constructed* to make sure that most students would pass it)
 - Modal auxiliary verb (e.g., They *should have* studied harder)
 - Perfect aspect (e.g., He *had been* spending money recklessly)

- Appositives (e.g., John Laughlin the *famous comedian* left the audience laughing hysterically)
- Elaborated subjects (e.g., *National parks such as the Yellowstone, Yosemite, Grand Canyon, and Bryce Canyon* are treasures of the American West)
- Post-modifications (e.g., The next person *to pull the handle* could win the jackpot)
- Prompt and teach the use of cohesion devices or connectives (e.g., such expressions as *therefore, as a result of, consequently, subsequently, because of*)
- Teach correct agreement (e.g., noun-verb agreement) in both connected speech and writing samples
- Teach the correct use of pronouns to avoid ambiguous referents; prompt and model correct pronoun usage
- Teach Pragmatic Skills
 - Teach the use of correct register, which is language appropriate to the context, situation, and the conversational partner; using the role-playing technique, teach the student to switch from one kind of register to the other:
 - You play the role of a peer and talk to the student
 - You play the role of a teacher and discuss a formal topic with the student
 - You play the role of a parent and talk to the student
 - Model, prompt, and reinforce correct form of speech that fits the role the student is asked to play; stop and give corrective feedback when the student uses incorrect registers (e.g., lay language while discussing a formal academic topic with the teacher, production of slang used among peers when talking to the parent)
 - Modify inappropriate gestures the student may exhibit during conversation
 - Have the student bring multiple topics to treatment sessions and ask the student to introduce one topic after another; reinforce the student for promptly introducing new topics
 - Ask the student to talk for extended durations on a selected topic of discussion; give such prompts as “Can you say more on that topic?” “Can you give me

more details?” and so forth; give feedback on the duration for which the student maintained conversation on a topic and target and reinforce progressively longer durations

- Teach the student to *modify statements* (a conversational repair strategy) when the listener fails to understand; play the role of a listener who does not understand what the student is saying:
 - Ask the student frequently to clarify his or her statements
 - Interrupt and request more information
 - Ask the student to say something differently
 - Ask the student to explain what he or she just did
 - Reinforce correct repair strategies
- Teach the student to *request clarification* (another conversational repair strategy) when a speaker's statement is not understood; play the role of a speaker who makes frequent unclear statements that force a request for clarification:
 - Make ambiguous statements (e.g., asking the student to point to the paragraph on a printed page when there are multiple paragraphs)
 - Speak too softly to be heard, especially at the end of certain sentences or just mumble something
 - Make other kinds of unclear or abrupt statements (e.g., “Tell me about your trip to Paris” when there is no hint of a trip to Paris)
 - Model different ways in which clarification may be requested
 - Reinforce varied and prompt requests for clarification
- Reduce the frequency of maze behaviors by stopping the student when he or she begins to exhibit false starts, hesitations, interjected extraneous statements, frequent revisions, repetitions, and other kinds of dysfluencies and repeated but unsuccessful attempts to express the same ideas; model more appropriate responses and reinforce them
- Reduce the frequency of irrelevant or extraneous comments by giving corrective feedback

- Teach narrative skills
 - Tell a story and ask the student to retell it
 - Ask the student to narrate a story or a personal experience
 - Prompt correct event sequencing, details of narration, correct representation of the characters, appropriate inference of feelings and thoughts of the characters; reinforce the student for progressively better narrative skills
- Teach discourse skills
 - Suggest an academic topic or a social issue and ask the student to give a speech on it (e.g., electoral college, civil rights, school discipline, cell phone technology, healthy school lunch program)
 - Ask the client to choose a topic of his or her interest and give a speech on it
 - Encourage the student to give detailed descriptions and coherent arguments, represent opposing views, be objective and state evidence, draw conclusions, and use technical or academic language
 - Model advanced sentences with abstract words when necessary
 - Record the speech and replay the recording to give feedback
- Teach reading skills
 - Ask the student to read aloud a printed passage that is at his or her grade level; follow the student's reading on a copy of the passage being read aloud; stop and give corrective feedback for
 - Misreading of words
 - Repetition of words or phrases, interjecting extraneous comments, and prolonging certain sounds
 - Teach a slower rate of reading if that promotes more fluent and smooth reading
 - Take note of the words the student does not know (hence struggles to read) and teach the meaning of those words
 - Teach syntactic structures that seem to give difficulty in reading

- Have the student learn to spell difficult words on which he or she stumbles while reading
- Ask questions about the read material to assess reading comprehension
- Ask the student to re-read and ask you questions about material not understood
- Give reading assignments and have the student write brief reports on the read material; give feedback to the student
- Teach writing skills
 - Dictate a passage for the student to write
 - Ask the student to spontaneously write something to reflect his or her experience
 - Ask the student to copy a brief printed passage
 - Ask the student to bring an extended writing assignment he or she may have submitted to the teacher and analyze the various writing samples for problems in writing and:
 - Give edited (written) feedback on spelling errors, poor formation of letters, general organization and neatness, and overall quality of handwriting
 - Give edited (written) feedback on errors in punctuation (the use of comma, semicolon, period, dash, quotation marks, and parenthetical constructions)
 - Discuss with the student the writing sample for its content; in case of the extended piece of writing (e.g., an essay submitted to the teacher) give feedback on the adequacy, depth, appropriateness, and details of information offered
 - Correct and give feedback on errors of syntax in the written samples; identify wrong or incomplete sentences; suggest varied syntactic constructions
 - Discuss with the student the appropriateness of the language to the topic of writing; correct errors in technical word usage, pro and con arguments elaborated, conclusions offered, and support for views advocated
 - Give feedback on overall organization and neatness of writing

Nippold, M. A. (2007). *Later language development: The school age and adolescent years* (3rd ed.). Austin, TX: Pro-Ed.

Nippold, M. A. (2014). *Language sampling with adolescents: Implications for intervention* (2nd ed.). San Diego, CA: Plural.

Paul, R. (2018). *Language disorders from infancy through adolescence* (5th ed.). St. Louis, MO: C. V. Mosby.

Language Impairment. Generally the same as language disorder; *disorder* is currently preferred to *impairment*, even in the case of what used to be called Specific Language Impairment; see Language Disorders in Children for treatment procedures.

Language-Learning Disorders. Generally the same as language disorders; links language disorder to a general learning disorder that negatively affects academic learning; often used in special educational contexts; see Language Disorders in Children for treatment procedures.

L

Language Stimulation by Parents. Activities parents implement at home to stimulate language in infants and toddlers; may be the only recommendation for a child; may supplement or parallel clinicians' treatment; supported by some evidence; see also Language Disorders in Infants and Toddlers; develop a parent-implemented language stimulation program:

- Assess the child and his or her family
- Assess the parents' education, sophistication, time commitment, and motivation to conduct regular activities at home
- Design a language stimulation program for the child
- Test the program in the clinic for a few sessions to make sure it works
- Have parents observe your sessions
- Train parents in the effective methods; model the methods frequently
- Have parents conduct a session or two in the clinic
- Give feedback and refine their skills
- Train them to keep records of therapy that you can evaluate

- Give parents simple, clear written instructions
- Give parents videorecorded samples of treatment techniques
- Periodically assess the child and the parents' sessions at home
- Suggest needed modifications and movement to higher levels of training
- Initiate formal treatment when your assessment indicate a need for it
- Consider using a parent-administered language stimulation program such as It Takes Two to Talk—The Hanen Program for Parents, Enhanced Milieu Teaching, Focused Stimulation, Responsive Education, or Conversational Recast

Laryngeal Cleft. A cleft between the larynx (cricoid cartilage) and the esophagus; caused by a failure of dorsal fusion of the cricoid lamina; may be accompanied by other congenital anomalies, including feeding and respiratory problems soon after birth; treatment is surgical closure of the cleft.

Laryngeal Cysts. Formation of small, fluid-filled sacs on the larynx, especially in the ventricle; often congenital; caused by accumulation of glandular secretion in submucosal; symptoms include swollen false (ventricular) folds; if enlarged, can result in hoarseness; treatment is surgical removal of the cyst with follow-up voice therapy if needed.

Laryngeal Hyperkeratosis. A thickening of the laryngeal mucosa resulting from an abnormal growth of the epithelium due to cigarette smoking, heavy alcohol use, environmental pollutants, and strained and tense speaking habits; usually occurs on the true vocal folds; may sometimes be premalignant.

- Modify the client behaviors that are causative (e.g., reduced use of alcohol and tobacco)
- Counsel the parents to reduce exposure to environmental factors that lead to laryngeal hyperkeratosis

Laryngeal Leukoplakia. Appearance of white patches on the laryngeal mucosa; voice may be hoarse; may be premalignant.

- Modify client behavior to reduce or eliminate smoking
- Do not offer voice therapy for hoarseness, as it is not effective

Laryngeal Stoma. An opening made into the trachea between the thyroid glands to allow for breathing in patients with laryngectomy.

Laryngeal Web. Growth of a thin membrane across portions of the vocal folds; may be congenital or induced by trauma later in life; negatively affects respiration; treatment is surgical removal of the web.

Laryngectomye. A person who has had a partial or total Laryngectomy.

L

Laryngectomy. Treatment of individuals who have undergone laryngeal surgery because of disease or trauma requires an understanding of the various types of laryngeal surgeries, their differential consequences for communication training, and communication options available to speak without larynx; speech-language pathologists help develop treatment and rehabilitation services for patients who are being considered for laryngeal surgery; the clinician is involved from the preoperative stage to postoperative rehabilitation, an important part of which is to teach a new mode of communication; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for surgical methods, reasons for surgeries, and potential etiological factors that cause diseases of the larynx; for assessment procedures, see the cited sources and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Preoperative Evaluation and Counseling

- Work as a member of the rehabilitation team
- In consultation with the surgeon, counsel the patient and the family about the effects of the planned medical treatment on communication
- Invite and answer all questions from the patient and the family members; give answers that are consistent with advice from other professionals on the team
- Obtain a sample of the patient's speech and writing; make an assessment of client's communication skills
- Describe various methods of speaking without a larynx; discuss communication options that may be preferable to the client; be consistent with the surgeon's preferences and recommendations
- Reassure the patient that he or she will talk again by using new techniques
- Have the patient meet and speak with a rehabilitated Laryngectomee who has mastered Alaryngeal Speech
- If possible, assist the patient in acquiring an intra-oral electrolarynx for use immediately following surgery
- Provide training in the use of the intra-oral electrolarynx

Postoperative Management

- Review the information provided during the preoperative counseling
- Discuss methods of alaryngeal speech (described later in this section)
- Demonstrate how electronic speech aids work
- If a tracheoesophageal puncture (TEP) has been done, measure the fistula, select and place an appropriate size and style voice prosthesis, and educate the patient on its care and maintenance
- Discuss the patient's rehabilitation plan; be cautious in making prognostic statements
- Give written information on rehabilitation plans and possibilities for the patient to read later
- Arrange a visit from a rehabilitated Laryngectomee to encourage the patient

Teaching New Methods of Communication

General Principles

- Select an appropriate method of communication judged to be efficient, practical, and preferred by the individual
- Teach the client to use the new method of communication
- Select a type of alaryngeal speech (described later in this section) for permanent communication; even if the patient chooses esophageal or TEP speech for long-term use, he or she may need a “backup” of an electronic device to the primary method
- Begin to teach the use of a neck placement electronic device only after the neck and throat areas recover from swelling and tenderness and the surgical suture lines heal
- Teach tracheoesophageal speech if the patient is surgically prepared for it and a voice prosthesis is in place
- Consider both individual and group therapy sessions
- Determine the frequency of treatment sessions based on the patient’s physical condition
- Consider daily sessions in the beginning if the patient’s physical stamina permits them
- Hold at least one weekly session
- Get family members involved in training sessions
- Let the patient’s performance and progress dictate the pace of therapy
- Ask the client to practice the new method of communication at home

Teach Alaryngeal Speech with Electronic Devices

- To assess the appropriateness of an electronic device, see the companion volume, *Hegde’s PocketGuide to Assessment in Speech-Language Pathology*
- Select an electronic larynx after discussing various models with the patient; transoral devices, held in hand, carry the sound into the mouth via a tube; transcervical (on the neck) types transmit the sound via skin into the mouth when held against the neck

- Recommend an adapter to convert a transcervical device to a transoral device; adapters may be affixed to the head of the device, with an attached tube that delivers the sound to the mouth
- Recommend an on-the-neck device only after the surgical wound is healed; if the neck is extensively scarred by surgery, on-the-neck type may not be used at all
- Recommend a transoral devices soon after the surgery, as the unit is not held against the neck
- Demonstrate first what the instrument sounds like and then how speech produced with it sounds like
- Experiment with the transcervical device to find the best position, usually under the jaw; let the head of the device make good contact with the skin without pressing it
- Manipulate the button for sound production and ask the patient to count aloud
- Ask the patient to clearly shape the words with the mouth; esophageal speech involves mild overarticulation of the speech sounds, particularly final consonant sounds
- Ask the patient not to exhale forcefully
- Teach the patient to handle the device
- Instruct the patient to coordinate sound and speech and to turn off the sound when not talking
- Reduce the patient's rate of speech to increase intelligibility
- Teach the patient to increase articulatory precision by practicing words that begin with voiceless consonants
- Shape progressively longer utterances
- Ask the person to practice phrasing with appropriate timing (voice on and voice off) of the electric larynx
- Teach the client to maintain eye contact with the listener

Teach Esophageal Speech

- Begin esophageal speech training soon after patient starts eating food orally

- Describe the anatomy and the physiology of esophageal speech production
- Describe esophageal sound production to the patient
- Use diagrams to explain esophageal speech
- Teach the client the production of esophageal sound
- Try various procedures and settle on the one most effective with the client
- Teach the patient to use the injection method of taking air into the esophagus
 - Ask the patient to press the tongue tip against the alveolar ridge to push the air back toward the esophagus without the tongue making contact with the pharyngeal wall (glossal press)
 - Ask the client to press the tongue tip against the alveolar ridge and to move the tongue back to make contact with the pharyngeal wall; thus push air back into the esophagus (glossopharyngeal press)
 - Ask the patient to keep the velopharyngeal port closed
 - Ask the client to inject the air in an audible manner, producing the sound called the “klunk”
- Teach the person to use the inhalation method of taking air into the esophagus if necessary
 - Teach the individual to synchronize the air intake through the stoma with air intake through the mouth into the upper esophagus
 - Ask the persons to relax the PE segment to create a negative pressure there to move air into the upper esophagus
- Ask the person to produce plosive consonants to stimulate esophageal sound
- Instruct the patient to say ta-ta-ta or sta-sta-sta
- Ask the patient to use easy injection of air and say a series of ta-ta-ta or sta-sta-sta
- Reinforce a likely emergence of esophageal sound
- Teach the patient to puff the cheeks out and move the air trapped in the mouth from one side to another; instruct the patient to move this trapped air quickly into the esophagus

- Ask the patient to produce words that typically trigger sound production: *church, stop, skate, scotch,* and *scratch*
- Use consonant-vowel (C-V) combinations initially, opening the mouth wide during the vowel production
- Move on to single syllable words
- Increase response complexity
- Ask the patient to slow down the rate of speech

Teach Tracheoesophageal Speech

- Select a Voice Prosthesis for a patient who has undergone Tracheoesophageal Fistulization/Puncture
- Insert the voice prosthesis into the fistula; make sure the fistula is properly healed; also make sure that there is no leakage of fluid around or through the prosthesis
- Ask the patient to inhale, occlude the stoma with a finger, and exhale
- Ask the client to produce sound as the air from the lungs enters the P-E Segment through the voice prostheses
- Have the patient practice sound production
- Shape the sound into speech
- Increase the length of utterances
- Give appropriate feedback
- Experiment with “hands free” speaking valves and train the patient to utilize them, if appropriate

Sapienza, C., & Ruddy, B. H. (2018). *Voice Disorders* (3rd ed.). San Diego, CA: Plural.

Stemple, J. C., Roy, N., & Klaben, B. K. (2014). *Clinical voice pathology* (5th ed.). San Diego, CA: Plural.

Laryngitis. Irritated and swollen vocal folds; causes include vocally abusive behaviors and infection; see Voice Disorders for additional information.

Laryngitis, Chronic. Irritated and swollen vocal folds of long history; Hoarseness is the primary result; lowered vocal pitch and vocal tiredness also may result; may lead to vocal nodules or polyps.

- Impose vocal rest without whispering
- Reduce vocally abusive behaviors; see [Treatment of Vocally Abusive Behaviors](#) under [Voice Disorders](#)

Laryngitis, Traumatic. Irritated and swollen vocal folds; result of such vocally abusive behaviors as shouting, screaming, and loud cheering; hoarseness is the primary result.

- Do not recommend voice treatment for such temporary laryngitis as that following enthusiastic participation in ball games; natural period rest (one night's sleep) may be adequate
- Reduce vocally abusive behaviors if they persist; see [Treatment of Vocally Abusive Behaviors](#) under [Voice Disorders](#)

Laryngoplasty. Surgical treatment to improve phonation in people with vocal cord paralysis or weakness; involves medial displacement of vocal cords with the help of implant materials to promote better approximation.

L

Left Neglect. To treat reduced awareness of stimuli in the left visual field in individuals who have sustained right brain injury, see [Right Hemisphere Disorders](#).

Lesson Plan. A brief treatment plan which describes short-term goals and procedures; in case of student clinicians, approved by the clinical supervisor; in preparing lesson plans:

- Use [Operational Definitions](#) in writing treatment goals
- Give clear and brief description of procedures to be used

Levodopa. A neurotransmitter; various kinds of movement disorders result from its deficiency; drugs containing this neurotransmitter may be prescribed for patients to control their movement disorders (e.g., those with Parkinson's disease).

Lipreading. Understanding speech by watching the mouth of the speaker; gaining cues from the movement of the lips and the tongue; a skill that may supplement lim-

ited comprehension of speech through residual hearing; see also [Speechreading](#).

Literacy and Literacy Skills in Children. Because they are language-based, speech-language pathologists may teach literacy skills in the context of speech and language treatment and thus promote academic success of students; if practical, literacy and speech-language interventions may be integrated; see the cited sources at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for details on literacy problems; for details on assessment of literacy skills, see the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Literacy Intervention: Main Objectives

- To teach the emergent literacy, reading, and writing skills in children, ideally integrated with the early speech and language intervention
- To identify literacy skills that may need more intensive and dedicated treatment or instruction
- To help design a home literacy program based on the assessment results; to offer suggestions to parents on stimulating and encouraging literacy skills in their children

Literacy Intervention: General Guidelines

- Design a program to enrich the literacy resources available to the child at home (e.g., books, supplies, a work area, facilities such as a suitable chair and desk to read and write)
- Consider the possibility of increasing the parental behaviors that will better promote literacy skills at home (e.g., encouraging the parents to read and write in the child's presence, reading stories aloud to them, teaching writing skills)

Ethnocultural Considerations

- Pay special attention to the home environment and literacy resources available to minority families
- Find out ways of improving the support systems the parents may need (e.g., access to a public library;

libraries and bookstores that offer story reading sessions for children; organizations that may supply free books for children)

- Select intervention materials that are at the child's grade and are culturally appropriate for the child's background

Intervention for Literacy Skills

- Modify the child's home literacy environment to increase behavior support for literacy
 - Depending on the results of a home literacy environment assessment, design and implement a program to support literacy skills at home; consider the following:
 - Encourage the parents to supply the writing materials to the child (e.g., crayons, pencils, pens, paper, eraser, a child's desk and chair)
 - Encourage the parents to borrow children's story books from the town or community libraries; introduce them to used book markets where they can find children's books at low cost
 - Explore community support for acquiring such materials in case of families with limited income
 - Ask the parents and older siblings to read and write at home to provide literacy models to the child
 - Ask parents and older siblings to read stories to the child in an interactive manner; ask the reader to let the child see the words and the lines being read; move the finger along the line; let the reader ask questions and invite comments from the child
 - Ask the parents to encourage the child to draw, copy, and color
 - Suggest to the parents that they may teach the child to write the letters of the alphabet
 - Suggest educational toys (e.g., magnet letters, map puzzles, counting beads) the parents might consider buying for gifts for special occasions (e.g., birthdays, Christmas)
 - Suggest incentive programs they can establish for learning literacy skills (e.g., small gifts and inex-

- pensive treats for listening to stories, tracing letters, drawing, printing the letters of the alphabet)
- Teach the emergent literacy skills in preschool children
 - Teach print awareness to preschool children
 - For each skill you teach, reinforce immediately the child's correct responses; correct errors immediately so the child does not practice mistakes
 - Model correct verbal responses and give print examples when appropriate
 - Prompt correct responses to prevent frequent errors
 - Use manual guidance in teaching the child to print letters or words
 - Have the child print the letters of the alphabet; in teaching the production of single words as a part of early language intervention, encourage the child to print the first letter of words that you use as stimuli
 - Read a short story aloud to the child as a form of reinforcement for good work on other treatment tasks, including regular speech and language tasks; position yourself such that the child can see the printed material you read aloud
 - Point to the words and letters as you read the story aloud; reinforce the child for looking at the printed material
 - Teach the child to tell the meaning of commonly encountered logos and signs (e.g., a stop sign or signs in supermarkets)
 - Display a few children's books the child can have easy access to; reinforce if the child picks up a book and looks into it; offer to read a story from a book the child picks up
 - Teach the child to count numbers; teach this skill in the context of speech or language treatment
 - Teach the child to recite the alphabet, nursery rhymes, days of the week, or names of the months
 - Teach the child to point to single words and sentences in printed passages
 - Hand a book to the child and ask the child to show the front and the back of the book

- Hand a book upside down to the child; if the child fails to turn it around for correct orientation, guide the movement manually
- Ask the child to point to the beginning and the end of a story in a simple story book; teach if necessary
- Name the letters of the alphabet and ask the child to point to each on a printed sheet (e.g., “Can you find the letter *s*? The letter *b*?”)
- Teach reading and writing at the early elementary level
 - Teach letter identification skills
 - Ask the child to name all the letters of the alphabet in a systematic manner (A through Z); prompt when the child makes a mistake
 - Ask the child to name a few letters presented in a random manner; prompt when necessary
 - Ask the child to make the kind of sound a letter represents (while pointing to different printed letters, ask, “How does this sound?”); model sounds the child missed; reinforce correct imitative or spontaneous responses
 - Teach early reading skills
 - Show simple printed words and ask the child to sound them out
 - Have the child read a passage from the child’s grade level; correct misreading of words; prompt when the child hesitates, and draw attention to omitted words
 - Ask the child to read slightly faster; reinforce small increments in faster but fluent reading; measure the effects of this procedure by counting the number of words read per minute
 - Have the child read a short passage silently and ask questions about the material; reiterate what the child missed; ask the child to re-read to gain better comprehension
 - Teach elementary writing skills
 - Have the child draw simple drawings; give corrective feedback, ask the child to redraw
 - Ask the child to trace the letters; prompt when necessary

- Ask the child to print all the letters of the alphabet; prompt correct sequencing
- Have the child spontaneously write a few sentences; give online feedback to correct such errors as spelling mistakes
- Teach reading and writing at the more advanced level
 - Teach more advanced reading skills
 - Ask the child to read aloud a longer and more complex piece of printed material, selected from the child's grade level; prompt to minimize errors
 - Ask questions about the read material; correct errors in comprehension; have the child re-read the material; ask questions and reinforce correct comprehension
 - Ask the child to distinguish between literary genres (e.g., an essay, a poem, a novel, nonfiction); describe the differences to the child and have the child restate the differences
 - Teach more advanced writing skills
 - Have the child write a brief essay on a topic of his or her choice; it may be a personal narrative (e.g., a vacation experience, a book recently read); evaluate the writing for word usage, spelling, letter formation, capitalization, punctuation, organization, logical sequence, sentence variety, grammaticality, and other features of good writing; give written feedback; discuss orally the strengths and weaknesses; give suggestions on how to overcome the weaknesses
 - Repeat the writing exercise until the child can demonstrate writing skills that are appropriate for his or her grade level
 - Teach abstract language skills
 - Ask the child to define selected terms that you find in the child's textbook; prompt correct definitions; reinforce progressively more accurate definitions
 - Ask the child to give examples of concepts described in his or her grade level books; help the child refine the examples or correct any errors

- Ask the child the meaning of selected proverbs, idioms, and slang; model correct responses or prompt them; teach progressively more complex and abstract expressions
- Ask the child to define selected antonyms and synonyms sampled from the child's book or graded for appropriateness for the child's grade; prompt when the child hesitates; have the child restate the meanings of words he or she missed
- Ask the child to describe how he or she would plan for a camping trip; prompt details and comprehensive descriptions
- Tell a story and ask the child to retell it; correct errors; prompt details and correct sequencing of story elements; have the child tell multiple stories
- Ask the child about the moral of the story he or she tells; correct any errors
- Ask the child to talk on a topic or social issue (homelessness, earthquakes or floods, school lunch program)
- Consider whether to teach phonological awareness or not
 - Teach literacy skills directly, instead of wasting time on presumed underlying or prerequisite skills; in spite of its popularity, there is no significant evidence that phonological awareness intervention causes improvements in literacy skills; some evidence suggests that direct literacy intervention causes changes in skills subsumed under phonological awareness.
 - If you decide to teach it, see Phonological Awareness for procedures.

Catts, H. W., & Kamhi, A. (2012). *Language and reading disabilities* (3rd ed.). Boston, MA: Allyn & Bacon.

Hegde, M. N., & Pomaville, F. (2017). *Assessment of communication disorders in children: Resources and protocols* (3rd ed.). San Diego, CA: Plural.

Pence, K. L. (2007). *Assessment in emergent literacy*. San Diego, CA: Plural.

Whitehurst, C. G., & Lonigan, C. J. (1998). Child development and emergent literacy. *Child Development*, 69(3), 848–872.

Lombard Effect. Increased vocal intensity in noisy environments or under auditory masking with white noise; typically reflexive, but can be brought under the individual's control by instruction and reinforcement.

Loudness. To increase vocal loudness in individuals who speak with excessively soft voice, see under [Voice Disorders](#).

M

Maintenance Strategy. Use a variety of strategies to promote the production of treated communicative skills in natural environments and have them sustained over time; plan these strategies from the beginning of treatment; they require the extension of treatment to natural settings and training the client's significant others to help evoke and reinforce the target skills; all aspects of treatment, including stimulus variables, response characteristics, and response consequences, should be manipulated to achieve maintenance.

Stimulus Manipulations

- In teaching speech and language skills, especially to children, select common, functional, child-specific stimulus items, preferably objects; let the child bring stimuli from home (e.g., a girl could bring her toys to serve as stimuli in speech or language training)
- Even for adult clients (e.g., those with aphasia or dysarthria), select stimuli that are functional in their environment
- Select stimuli that are ethnoculturally appropriate for clients of all ages; in the case of children, let the parents suggest culturally relevant stimuli that the child is familiar with
- Select colorful, unambiguous, and realistic pictures
- Select simple and common verbal stimuli that are used to evoke the target responses
- Vary the audience; have family members and other persons participate as conversational partners in treatment sessions
- Vary the physical settings; conduct informal treatment outside the clinic room, in cafeterias, campus walks, library, bookstore, home, and other natural settings

Response Considerations

- Select client-specific and functional responses for treatment targets
- Select ethnoculturally relevant and appropriate treatment targets
- Select target behaviors that are likely to be produced at home

- Select target behaviors that can easily be expanded into more complex communicative behaviors
- Train multiple exemplars of each target skill and at each level of response complexity
- Take training to complex levels of target skills: always end treatment with sufficient training at the conversational level

Contingency Manipulation

- Use intermittent reinforcement schedules in the latter stages of training
- Use conditioned reinforcers (tokens with backup reinforcers)
- Delay reinforcement in the latter stages of training
- Let the family members and others watch treatment sessions so they can better understand the treatment targets and teaching methods
- Train significant others in evoking and prompting the target behaviors at home and in other nonclinical settings
- Train significant others in reinforcing the production of target behaviors at home and in other nonclinical settings
- Reinforce generalized responses; have parents and others reinforce generalized productions at home
- Teach Reinforcement Priming to the client (e.g., teach the client to draw attention to his or her production of target behaviors at home so the ignoring parents can pay attention and reinforce the client)
- Hold informal Training Sessions in Natural Environments
- Teach Self-Control (Self-Monitoring) Procedures (e.g., counting one's target behaviors)
- Give treatment for a sufficient duration
- Provide Follow-Up and arrange for Booster Treatment

Hegde, M. N. (1998). *Treatment procedure in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Management of Behavioral Contingencies. A clinician's or a family member's skill in arranging effective

stimuli for target communication skills, requiring the production of specified skills, and in promptly and effectively providing differential feedback for the correct and incorrect productions; inherent to all behavioral interventions; controlled evidence supports the use of behavioral contingencies in the treatment of communicative disorders.

- Provide effective stimuli for target behaviors; use pictures, objects, enacted events, instructions, demonstrations, models, prompts, manual guidance, visual and tactile cues, and other stimuli for the target behavior
- Specify the response form; demonstrate what the client is expected to produce
- Reinforce promptly, clearly, naturally, and as frequently as needed
- Positively reinforce imitated or evoked target behaviors with Verbal Praise, Tokens that are exchanged for backup reinforcers, Informational Feedback, Biofeedback, and High Probability Behaviors
- Use Corrective Feedback, Response Cost, Time-Out, and Extinction to reduce undesirable behaviors
- Use Differential Reinforcement to teach desirable behaviors that replace undesirable behaviors

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Mand-Model. Use this naturalistic language treatment method when working with children with language disorders; it is a child language intervention method which uses components of the Incidental Teaching Method; uses typical adult-child interactions in a play-oriented setting to teach functional communication skills; for procedures, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs; well researched and supported by experimental evidence.

Mands. A class of verbal behaviors that are triggered by a state of motivation; includes requests, commands, demands, and many kinds of questions; need to create a state of motivation to teach mands; often reinforced with primary reinforcers.

- Create a state of motivation:
 - Arrange treatment around lunch or breakfast time so food may be used as a reinforcer (hunger is the state of motivation)
 - Hold food in front of the child until the child asks for it
 - Place attractive toys on a high shelf to encourage the child to request it; ask a question (e.g., “do you want that car?”) and if there is no response, model the request for the child (e.g., “Say, I want that car”)
 - Offer a food item the child does not like to encourage the child to verbally refuse it; model the response if the child fails to respond verbally (e.g., “Say, I don’t like it”)
 - Eat something the child is fond of without offering it (the child should request it; model the request if necessary)
 - Give a tightly closed jar with candy in it (the child should ask you to open it; model the request if necessary)
- Reinforce promptly with the displayed or held-back item; remove promptly an aversive item presented when the child makes an appropriate response

Manual Communication. Use these methods of non-verbal communication that may include sign language, finger spelling, gestures, and other forms of non-oral communication to children and adults with severe oral language difficulties; when taught to young nonverbal children, manual modes of communication may promote verbal communication; for general procedures, see [Augmentative and Alternative Communication](#).

Manual Guidance. Physical guidance provided to shape a response; the [Phonetic Placement Method](#) is similar to manual guidance; needed when the client cannot imitate a response; used in teaching practically all types of communicative skills.

- In teaching speech sound productions, use your fingers to shape articulators
- Take the client’s hand and make it touch the target picture while training comprehension of words

- Use tongue depressors to move the tongue to desired positions
- Apply slight digital pressure to the laryngeal area to lower a client's pitch
- Apply slight pressure on the chin of a child who does not readily open the mouth
- Fade manual guidance to promote the production of target responses without it

Manual Pointing. A method of Augmentative and Alternative Communication in which the client points to a correct message among the several displayed on a screen or on a board; may or may not use a pointing device.

Masking Noise. Delivery of noise through headphones to reduce or eliminate auditory sensitivity of pure tones or speech during auditory assessment; normally induces the Lombard Effect, an increased vocal intensity; masking also tends to induce stutter-free speech in people who stutter, although not a recommended method because of only short-term gains in unnatural sounding fluency; used in the treatment of Voice Disorders.

Matching. A scientific method of selecting participants to experiments, including treatment efficacy research studies that use the group experimental design strategy; a method in which subjects of similar characteristics are placed in the experimental and control groups used to evaluate treatment effects; an alternative to random selection and assignment of participants; part of the Group Design Strategy.

- Find pairs of subjects with the same or similar relevant characteristics (age, gender, severity of the disorder, socioeconomic status)
- Assign one of the pair to the experimental group and the other to the control group
- Match groups on the basis of group means if pair-wise matching is not possible (the two groups with the same average IQ, for instance)

Hegde, M. N. (2003). *Clinical research in communication disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Maximal Contrast Method. A speech sound (articulation) training method in which word pairs that contrast the most are used to train target phonemes; contrasted with Minimal Pair Contrast Method, in which word pairs differ by one phoneme (e.g., *pat*, *bat*); in maximal contrast pair, the words may differ by several features or phonemic contrasts; for procedural details, see Speech Sound Disorders: Specific Techniques or Programs.

Mechanical Corrective Feedback. A method to reduce incorrect responses in treatment; also known as Biofeedback; feedback is presented soon after an incorrect response is made; includes such feedback as provided on a computer monitor for incorrect responses (e.g., undesirable vocal pitch or intensity) and electromyographic feedback on muscle tension; helpful in treating Voice Disorders and neuromuscular disorders associated with Dysarthrias.

Melodic Intonation Therapy. An aphasia treatment program that uses musical intonation, continuous voicing, and rhythmic tapping to teach verbal expressions to patients with severe nonfluent aphasia with good auditory comprehension; see Aphasia: Treatment of Aphasia: Specific Technique or Programs for procedures.

Memory Impairments. Impairments in remembering, recalling, or acting on the basis of remote or recent experiences; impairment in learning or retaining current events or recently experienced events; typically described in such unhelpful mentalistic or mechanistic terms as storage and retrieval and a variety of presumed mental or neurological patterns; nonetheless, almost always it is the presence or absence of actions and behaviors that lead to such presumptions, theoretical speculations, and analogical reasoning; found in many persons with a variety of neurological and psychiatric disorders; speech-language pathologists are unlikely to treat memory problems found in individuals with no communication disorders (psychiatrists and psychologists may); therefore, treatment of memory problems concern speech-language pathologists only when they work

with individuals who have any form of Dementia, including Alzheimer's Disease, Traumatic Brain Injury in Adults, and to some extent, individuals who have Aphasia; Attention Disorders also are relevant to memory management; see those main entries for procedures to manage problems.

Memory Log Books. A memory aid for patients with memory impairments; consists of written material that helps sustain skills or prompt actions; recommended for patients who have retained at least a basic level of reading and writing skills; not useful for patients who are confused, have left-sided neglect, or have severe uncorrected visual defect.

- Design a simple log book that will contain only essential information
- Design a colorful cover for the book so it is easy to locate
- Designate a regular place where book will always be kept; let the health care staff or family members know the place
- Select functional information to be included in the log book; consult with the patient, staff, and the family members in making this selection
- Include the patient's biographical information (name, age, address, phone number, family members' names); the current month, year, and the name of the hospital; names of main health care workers and individual clinicians treating the patient
- Train the patient to take the book to all appointments, including treatment sessions
- Include in the book pictures and names of therapists and family members; train the client to frequently consult this information to facilitate memory for their names and faces
- Designate a single health care worker who will help maintain, update, and modify the book as found appropriate
- Divide the book into easily identifiable sections, preferably of different color; organize information in the sections for easy consultation (e.g., a section on treatment session appointments, a section on daily activities, a

section on medications and their schedules, a section on recreational activities)

- Train the client to consolidate all written forms of memory aids into the log book and not have multiple and odd pieces of information strewn around
- Teach the patient to write down information and then to periodically review the written information to act on it

Mendelsohn Maneuver. A swallowing maneuver that helps elevate the larynx more and for longer duration, resulting in an increased width and duration of cricopharyngeal opening; see Dysphagia to teach this maneuver to individuals with swallowing problems.

Mental Retardation. Dated term; see Intellectual and Developmental Disabilities.

Metronome-Paced Speech. Consider using this method to slow down the rate of speech in which the client is asked to pace a syllable or a word to each beat of a metronome; may be used in the treatment of stuttering, cluttering, and certain forms of dysarthria; see Dysarthrias and Stuttering, Specific Techniques or Programs under Fluency Disorders.

Minimal Pair Contrast Method. Use this method of treating speech sound disorders in children and speech production problems in adults with Dysarthrias; the method is based on the assumption that it is necessary to contrast the semantic differences between a child's (misarticulated) production and the correct adult production in words; it involves the use of word pairs that differ only in one phoneme (hence the name, *minimal contrast*); for procedural details, see Speech Sound Disorders: Specific Techniques or Programs; see also Maximal Contrast Method.

Mixed Dysarthrias. To treat this type of motor speech disorder that is a combination of two or more pure dysarthrias, see Treatment of Dysarthria: Specific Types under Dysarthrias.

Mode of Response. Manner or method of a response; includes imitation, spontaneous production, conversational speech, and oral reading; useful in sequencing treatment targets:

- Teach a target behavior first in the imitative mode if necessary; model the target response
- Fade the modeling and teach a target behavior in evoked mode
- Teach the target behavior in conversational mode
- Teach the target behaviors in oral reading mode if judged useful

Modeled Trials. Structured opportunities to imitate a response when the clinician models it; trials are separated by brief time interval; response accuracy scored for each trial; faded when imitation becomes more consistent; applicable in the treatment of almost all communication disorders.

- Place stimulus item in front of the client; show an object, or demonstrate an action
- Ask the predetermined question (e.g., “What is this?”)
- Immediately model correct response (e.g., “Johnny, say . . .”)
- Wait a few seconds for the client to respond
- Consequence the response if it is a modeled training trial
- Do not consequence the response if it is a modeled baseline trial
- Record the response on the recording sheet
- Remove stimulus item
- Wait 2–3 seconds to signify end of trial

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Modeling. Clinician’s production of a target behavior for the client to imitate; needed when the clinician cannot evoke a response; used frequently in treating communicative disorders; there is much experimental evidence to support its use in treatment; see Modeled Trials for procedural details.

Modification of Treatment Procedures. See Treatment of Communicative Disorders: Procedural Modifications.

Monterey Fluency Program. To treat stuttering with this behavioral program, see Stuttering: Specific Techniques or Programs under Fluency Disorders.

Moto-Kinesthetic Method. An articulation treatment method developed by Young and Stinchfield-Hawk; is similar to Phonetic Placement Method; emphasizes awareness of kinesthetic movement involved in articulation.

- Consider using the technique in the initial stages of treatment
- Manipulate the client's articulators with your fingers (Manual Guidance)
- Provide visual stimulation of the movements with the help of a mirror

Motor Speech Disorders. To treat a group of speech disorders associated with impaired motor control of speech muscles or motor programming of speech movements, see Cerebral Palsy, Dysarthrias, and Apraxia of Speech.

Multi-Infarct Dementia. To clinically manage this type of vascular dementia, see Dementia and Vascular Dementia.

Multiple Baseline Designs. A set of single-subject designs in which the effects of treatment are demonstrated by showing that untreated baselines did not change and that only the treated baselines did; practical designs to demonstrate treatment effects; has been extensively used in researching behavioral treatment procedures both in speech-language pathology and behavioral science; useful in integrating treatment research with service delivery; has three variations: across behaviors, settings, and subjects.

Multiple Baseline Across Behaviors Design.

A single subject design in which several behaviors are sequentially taught to show that a behavior changed only when brought under treatment and untreated behaviors

remained unchanged; helps rule out extraneous variables leading to the conclusion that the treatment was effective.

- Select three or more target behaviors (e.g., three or more phonemes, grammatic morphemes)
- Establish baselines on all selected target behaviors on discrete trials and in conversational speech
- Teach the first behavior to a Training Criterion
- Repeat baselines on the remaining untreated behaviors
- Teach the next behavior and repeat the baselines on the remaining untreated behaviors
- Continue to alternate baselines and treatment until all the behaviors are trained
- Expect the untreated behaviors not to change; if changed, note that the experimental control is weakened and it is difficult to claim treatment effectiveness
- If all behaviors changed only when brought under treatment, conclude that treatment was effective and that no other factor is responsible for the changes

Multiple Baseline Across Settings Design. A single-subject design in which a behavior is sequentially taught in different settings to show that the behavior changed only in a treated setting and hence the treatment was effective.

- Base rate a target behavior in three or more settings (e.g., clinic, home, school, office)
- Teach the behavior in one setting (e.g., fluency in the school clinician's office)
- Repeat the base rates in the remaining untreated settings
- Teach the behavior in another setting (e.g., fluency in the classroom)
- Continue to alternate base rates and teaching in different settings until the behavior is trained in all settings
- Conclude that the treatment was effective only if the repeated base rates show that the target behavior changed in a setting only when treatment was offered in that setting

Multiple Baselines Across Subjects Design. A single-subject research design in which several subjects

are treated sequentially to show that only treated subjects changed and hence treatment was effective.

- Select a target behavior that needs to be taught to three or more clients
- Base rate the target behaviors in all subjects
- Treat one of the subjects
- Repeat the base rates on the untreated subjects
- Treat the second subject
- Repeat the base rates on untreated subjects
- Alternate treatment and base rates until all the clients are trained
- Conclude that the treatment was effective only if the repeated base rates show that a client showed positive changes only when treated and that until the treatment was offered, no one changed

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Multiple Phoneme Approach. To use this behaviorally based method of teaching children with multiple misarticulations, see [Speech Sound Disorders: Specific Techniques or Programs](#).

Multiple Sclerosis (MS). To treat communication disorders associated with this neurological disease, see [Dysarthrias](#).

Mutational Falsetto. To treat the undesirable continuation of prepubertal, high-pitched voice after attaining puberty:

- Have medical confirmation of laryngeal maturation
- Establish a lower pitched voice; use techniques described under [Voice Disorders; Treatment of Disorders of Loudness and Pitch](#).

Mutism. Treatment of mutism requires a knowledge of various clinical conditions that lead to, or are associated with, lack of oral speech; treatment of mutism may be a part of treating various kinds of neurological diseases that cause dysarthria and dementia that may terminate in

mutism; mutism may also be a voice disorder; it may be classified as a psychiatric disorder in some cases; see the companion volume, *Hegde's PocketGuide to Communication Disorders*, for etiology and symptomatology of mutism; see *Psychiatric Communication Disorders* for treating a type of mutism not associated with any neurophysiological problem:

- See *Aphasia, Apraxia of Speech, Dysarthrias, and Traumatic Brain Injury in Adults* for treatment of associated mutism
- Mutism is sometimes transient in patients with aphasia and traumatic brain injury; no treatment may be needed for such cases
- Oral language treatment may be contraindicated in cases of severe apraxia of speech and aphasia, especially after a period of competent but still unsuccessful oral communication training
- *Augmentative and Alternative Communication* choices may be considered for patients who are no longer candidates for oral communication

Duffy, J. R. (2013). *Motor speech disorders: Substrates, differential diagnosis, and management* (3rd ed.). St. Louis, MO: Elsevier Mosby.

McInnes, A., & Manassis, K. (2005). When silence is not golden: An integrated approach to selective mutism. *Seminars in Speech and Language*, 26(3), 201–210.

Toppelberg, C. O., Tabors, P., Coggins, A., Lum, K., & Burger, C. (2005). Differential diagnosis of selective mutism in bilingual children. *Journal of the American Academy of Child and Adolescent Psychiatry*, 44(6), 592–595.

Wintgens, A. (2005). Selective mutism in children. *Child Language Teaching and Therapy*, 21(2), 214–216.

N

Narrative Skills. Treatment of narrative skills is an important element in language treatment in children, adolescents, and adults; to treat language skill in describing events in a sequential, chronologically correct, and logically consistent manner, see [Language Disorders in Children](#); [Treatment of Language Disorders: Specific Techniques or Programs](#); [Narrative Skills Training](#).

Nasal Assimilation. A phonological pattern in which nasal consonants are substituted for oral consonants (e.g., /n/ or /d/); see [Speech Sound Disorders](#) for treatment approaches.

Nasal Emission. Audible escape of air through the nose during speech; often found in children with cleft palate; see [Cleft Palate](#) and [Velopharyngeal Dysfunction](#) for treatment procedures.

Nasogastric Tube (NG). A method of feeding patients with dysphagia by inserting a tube through the nose and into the stomach and introducing solid and liquid food through the tube; see [Dysphagia](#).

N

Natural Settings. Nonclinical settings where clients communicate for the most part; communication in such settings is always a final treatment target; in the case of infants and toddlers, treatment may be implemented in such settings; extending treatment to such settings is essential to promote response maintenance.

Natural-Sounding Fluency. A stuttering treatment target when such techniques as [Delayed Auditory Feedback](#), [Metronome-Paced Speech](#), and [Rate Reduction](#) (syllable prolongation) are used; see [Stuttering; Treatment of Stuttering: Specific Techniques or Programs under Fluency Disorders](#) for additional information; to reinstate natural-sounding speech:

- Fade explicit management of airflow
- Fade the use of a metronome
- Fade the use of delayed auditory feedback

- Increase the rate of speech to near-normal levels
- Teach variations in intonation
- Teach normal rhythm of speech

Neck Brace. A brace around the neck used to stabilize the weakened neck muscles; often used in treating clients with dysarthrias.

Negative Reinforcers. Aversive events that are removed, reduced, postponed, or prevented; responses that accomplish these tend to increase in frequency; less useful than positive reinforcers in teaching communicative skills; avoidance behaviors seen in people who stutter, for example, may be negatively reinforced; prevention of avoidance is effective in eliminating negative reinforcement.

Neglect. To treat neglect of visual stimuli, often described as left-side neglect, see Right Hemisphere Disorders with which the problem is most often associated.

Neologism. To reduce the frequency of new but meaningless words patients with aphasia create, see Aphasia.

Neural Anastomosis. Connecting a branch of an undamaged nerve to a damaged nerve; a surgical treatment for certain dysarthric clients; a branch of the intact cranial nerve XII may be connected to the damaged cranial nerve VII to restore function and appearance.

Nonexclusion Time-Out. Use this procedure of response-contingent arrangement of a brief duration of time in which all interaction is terminated to reduce a variety of undesirable behaviors, including stuttering; in this procedure, the client is not removed from the clinical or educational setting; one of the Direct Methods of Response Reduction.

- Give response contingent signal to start time-out (e.g., saying “Stop” as soon as a dysfluency occurs); do not let the client talk during time-out
- Turn your face away from the client
- Stay motionless for 5 seconds
- Turn toward the client, and continue the interaction

Nonfluent Aphasia. To treat this type of aphasia, see [Aphasia](#); [Treatment of Aphasia: Specific Types](#).

Non-Iconic Symbols. Geometric shapes that do not look like what they suggest are used to teach a form of non-verbal communication; the meaning of such shapes need to be established by training; more difficult to learn than [Iconic Symbols](#), but more flexible; plastic chips or various shapes are an example; used in teaching [Augmentative and Alternative Communication-Aided](#).

Nonpenetrating (Closed-Head) Injury. To treat communication disorders associated with this type of head injury in which the skull may or may not be fractured or lacerated and the [Meninges](#) remain intact, see [Traumatic Brain Injury in Adults](#).

Non-SLIP (Non-Speech Language Initiation Program). Use this nonspeech communication program that includes the Premack-type, color-coded plastic shapes, each associated with a word; developed and researched by Joseph Carrier, Jr.; the client learns to communicate by arranging them in sequence to form sentences; also used to promote oral language acquisition in initially minimally verbal children; used in teaching [Augmentative and Alternative Communication, Aided](#).

Nonverbal Communication. Modes of communication that do not involve spoken speech; use of gestures, signs, symbols, printed material, electronic display, communication boards, and so forth to communicate; also includes such fully developed non-oral languages as [American Sign Language](#); see [Augmentative and Alternative Communication](#).

Nonverbal Corrective Feedback. A method used to reduce incorrect responses in treatment by giving immediate feedback for them; includes various forms of gestures, hand signals, and facial expressions that suggest to the client that the response was wrong (e.g., the stereotypic sad

face); a form of Corrective Feedback; often paired with Verbal Corrective Feedback.

Nonverbal Oral Apraxia. To treat difficulty in executing oral movements that are typically involved in speech, see Apraxia and Apraxia of Speech, with which it may be associated.

Normal Prosody. Treatment of prosodic features is essential to restore normal or socially acceptable rhythm, stress, intonation (pitch variation), intensity, transition between words and phrases, correct phrasing and pausing at appropriate junctures, and acceptable rate of speech; patients with apraxia of speech, cluttering and stuttering, dysarthria, foreign accent reduction, hearing impairment, voice disorders, and so forth may need treatment for prosodic problems; prosodic impairments are a side effect of treatment in which the speech rate is slowed down to either improve intelligibility or reduce stuttering.

- Select a particular aspect of prosody for treatment (e.g., pitch variations or increased speech rate)
- Model the target behavior (e.g., a higher pitch)
- Demonstrate the target on a computer screen, if possible
- Audio-record the model and play it
- Ask the client to match the live or recorded model (imitate the higher pitch)
- Shape the target behavior in successive and progressively more complex steps
- Reinforce any movement in the direction of the model
- Set a higher level of response (e.g., sentences) when the target (a certain pitch or intensity) is achieved at a lower level (e.g., phrases)
- Give maximum feedback, including auditory and visual feedback
- Teach other prosodic features in the same manner

Hargrove, P. M., & McGarr, N. S. (1994). *Prosody management of communication disorders*. San Diego, CA: Singular.

Normative Strategy. An approach to selecting treatment target behaviors for clients, especially children, based

on age-based norms and developmental sequences; often used in selecting target speech sounds and language structures for children; some clinicians question its relevance and assumptions; contrasted with Client-Specific Strategy.

- Assess the communicative behaviors of the child to determine potential treatment targets
- Select behaviors the child should already have acquired based on the age-based norms
- Teach the selected behaviors in the normative sequence in which they are acquired
- Note that this is by no means the only approach to target behavior selection and sequencing
- Do not hesitate to experiment with different sequences that may not conform to the normative sequence

O

Objectivity. Agreement among different observers who observe or measure the same event in the same manner; important in treatment research so that different observers agree that a treatment had an effect; helps replicate treatment procedures by other clinicians.

Obturator. A structure that helps close an opening; a prosthetic device that helps close a cleft of the hard palate; the device has a plate that covers the cleft.

Omission. This type of speech sound disorder is treated commonly by teaching the missing sounds; see Speech Sound Disorders.

Omission Training. Use this behavioral method of reinforcing a person for not exhibiting a certain undesirable behavior to indirectly decrease that undesirable behavior and to increase the reinforced desirable behavior; the same as Differential Reinforcement of Other Behavior.

Open-Head Injury. To treat communication disorders associated with this type of head injury, also called Penetrating Head Injury, see Traumatic Brain Injury in Adults.

O

Operant. A behavior that is affected by its consequences; most voluntary behaviors, including communicative behaviors, are operants; behaviors that can be increased or decreased by reinforcing and punishing consequences, respectively.

Operant Aggression. Controlling aggressive behavior directed against the source of an aversive stimulus may be essential with certain kinds of clients and while using specific kinds of behavioral treatment procedures; a potential undesirable side effect of punishment; contrasted with Elicited Aggression.

- A child may say “I hate you” when the clinician says “No” for an incorrect response
- A child may fight the clinician’s attempt to take a token away in a response cost procedure

- To reduce operant aggression in treatment, use punishment procedures sparingly and use more positive reinforcement
- Use Differential Reinforcement to indirectly control undesirable behaviors by increasing desirable behaviors

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Operant Conditioning. Skinnerian conditioning; method of selecting and strengthening behaviors of an individual by arranging reinforcing consequences; roughly the process and procedures by which most behaviors and skills are taught in treatment sessions.

Operational Definitions. Definition of variables in measurable terms; in treatment, it is essential to define target behaviors in measurable terms so the progress or the lack of it can be objectively documented:

- Specify the topographic aspects of the target behavior (e.g., production of /s/ in word-initial positions, phrases, sentences)
- Specify the mode in which the response will be measured (e.g., reading, conversational speech)
- Specify the stimuli and settings (e.g., when shown pictures, in the clinic, at home)
- Specify the accuracy criterion (e.g., 90% correct)

Oral Apraxia. Deficits in making movements unrelated to oral speech; see Apraxia and Apraxia of Speech.

Oral Language. A form of communication based on articulated speech and language; the most common form of communication in most societies; most disorders of communication treated are the disorders of oral language.

Oral Phase. To treat swallowing disorders associated with this phase, see Dysphagia.

Oral Preparatory Phase of Swallow. To treat swallowing disorders associated with this phase, see Dysphagia.

Orofacial Examination. An element of assessment; an examination of the structures of the face and mouth to detect their overall integrity and any deviations that may be present; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for procedures.

Overarticulation. Consider teaching exaggerated articulatory movements that may improve speech intelligibility when intelligibility is impaired; may be a treatment target in individuals with Dysarthrias.

Overcorrection. A procedure used to reduce behaviors by requiring the person to eliminate the effects of his or her misbehavior (Restitution) and practice its counterpart, a desirable behavior (Positive Practice); both described under Imposition of Work.

P

Pacing. A speech treatment procedure in which the rate of speech is reduced with rhythmic external stimulus to cue the production of syllables or words; used in the treatment of Aphasia and Dysarthrias.

Pacing Board. A wooden board that has a series of colored slots that are separated by ridges; used in reducing the speech rate of clients with motor speech disorders; the speaker touches one slot for each word spoken; see Dysarthrias.

Paired-Stimuli Approach. An articulation treatment method; uses correct production of sounds in a Key Word to teach correct production of the same sounds misarticulated in other words; for the procedures, see Speech Sound Disorders: Specific Techniques or Programs.

Palatal Lift Prosthesis. Constructed by a prosthodontist in consultation with a speech-language pathologist, this device helps achieve velopharyngeal closure to reduce hypernasality in clients with flaccid dysarthria; anchored to the teeth, the device has a plate that covers the hard palate; the rear end is custom-shaped to fit the patient's oropharynx; it pushes the soft palate up and back to make contact with the posterior pharyngeal wall.

P

Palate Reshaping Prosthesis. An intra-oral device that lowers the palatal arch by artificially increasing its bulk; may be designed with teeth to replace the missing teeth of the patient; helps the tongue with limited vertical movement to make contact with the hard palate to chew food.

Palatoplasty. Surgical repair of the clefts in the palate; see Cleft Palate.

Palilalia. Reducing this speech disorder in which a word, a phrase, or a sentence is repeated with increasing speed and decreasing intelligibility is a treatment target in some neurogenic communication disorders; often a symptom of Parkinson's Disease.

Palliative Treatment. Medical treatment that may reduce the intensity of some symptoms, control pain, and increase the patient comfort level, but may not cure the disease.

Pantomime. A method of communication in which the speaker acts out a message by gestures and bodily movements; a target communication skill for some nonverbal or minimally verbal individuals who can use gestures and bodily movements; unlike in other gestural systems, uses whole- as well as part-body movements; often more concrete and easier to understand than other gestures; used in teaching Augmentative Communication–Unaided.

Papillomas. Wart-like growths on the larynx; thought to be of viral origin; may be life-threatening if they block the airway; may be a recurring condition; treatment is laser surgery, which also needs to be repeated; may need voice therapy to make the best possible use of the compromised larynx.

- Teach the client to achieve appropriate pitch and loudness; use techniques described under Voice Disorders: Treatment of Disorders of Loudness and Pitch.
- Teach proper respiratory control; treat any other voice symptom with Specific Normal Voice Facilitation Techniques (described under Voice Disorders)

Paradoxical Effects of Punishment. Unexpected increase in response rates when a known response reduction procedure (e.g., time-out or verbal “No”) is used; potential side effect of punishment.

- Always watch for undesirable side effects when using response reduction (punishment) procedures
- Terminate the response reduction procedure when paradoxical effects are evident
- Always reinforce desirable target behaviors and minimize the use of response reduction procedures
- Consider one of the Differential Reinforcement procedures in which an undesirable response is indirectly reduced and by reinforcing alternative desirable behaviors

Paraphasias. Treatment of unintended word or sound substitutions that result in fluent but somewhat meaningless speech is a treatment target in patients with Aphasia.

Paraplegia. Paralysis of both legs, often due to spinal cord injury or disease; see Cerebral Palsy.

Parent Training. Preparing parents (or other family members, or other caregivers) to conduct informal treatment at home; to conduct maintenance activities to sustain treatment gains at home and other natural settings; see Language Stimulation by Parents; Maintenance Strategy; Peer Training.

Parkinson's Disease. To treat communication disorders associated with this progressive neurological syndrome, see Dementia.

Partial Modeling. Use this procedure to withdraw modeling of complete sentences in gradual steps; a method of Fading to promote more spontaneous production of clinically established skills.

- Initially model complete sentences for the client to imitate (e.g., “The book is on the table”)
- Drop the last word when it is time to fade modeling (e.g., “The book is on the . . .”)
- Drop additional words, one word at a time, on subsequent trials (“The book is on . . .”; “The book is . . .”; “The book . . .”; etc.)

P-E Segment. Pharyngeal-esophageal segment; a part of the pharynx and the esophagus; muscle fibers from the cricopharyngeus, esophagus, and inferior constrictor blend at this site to create a sphincter that can reduce the cross-sectional area of the esophagus.

Peer Modeling. Use this naturalistic child language intervention method in which peers are trained to model the target skills for the child client; see Language Disorders

in Children; Treatment of Language Disorders: Specific Techniques or Programs; briefly:

- Select a peer who agrees to help and is acceptable to the client
- Have the peer observe your treatment sessions
- Describe the target behaviors, modeling and imitation sequence, and reinforcement procedures
- Let the peer model and reinforce the child's productions in your presence
- Refine the peer's skills in modeling the target behaviors
- Ask the peer to submit recorded language samples that document appropriate modeling outside the clinic
- Periodically assess the results of peer modeling and provide additional training to the peer

Peer Training. Training peers of clients to evoke and reinforce target behaviors in natural settings is an important treatment as well as maintenance strategy; it has been well researched with adequate experimental support; see also Maintenance Strategy.

- Ask the peers to initially observe your treatment sessions
- Describe the target skills the client is being taught
- Let the peers count the occurrence of the skill along with you
- Give them feedback on their counting
- Train the peers to prompt, evoke, model, and reinforce the target communication skill
- Have peers conduct a session in your presence
- Give peers feedback and refine their skills
- Give peers simple, clear written instructions
- Give peers a sample of videorecordings of treatment procedures
- Ask peers to monitor the target skills in natural settings
- Ask peers to audio-record a monitoring session outside the clinic or submit data recorded on paper
- Review submitted data
- Assess the client who is taught by the peers on an ongoing basis
- Initiate clinical treatment if peer training is not effective or their training cannot be improved

Goldstein, H. (2002). Communicative interaction for children with autism: A review of treatment efficacy. *Journal of Autism and Developmental Disabilities*, 35(5), 373–396.

Goldstein, H., & Wickstrom, S. (1986). Peer intervention effects on communicative interaction among handicapped and nonhandicapped preschoolers. *Journal of Applied Behavior Analysis*, 19, 209–214.

Penetrating (Open-Head) Injury. To treat communication disorders associated with brain injury where the skull is perforated or fractured and the Meninges are torn or lacerated, see Traumatic Brain Injury in Adults.

Perceptual Training. The same as auditory comprehension training of speech sounds or language; in the context of treatment for speech and language in children, perceptual training may be unnecessary and therefore poor use of intervention time; comprehension training has little or no effect on production, whereas production almost always involves comprehension; see Speech Sound Disorders on some approaches to teach speech sound comprehension or auditory discrimination, see Aphasia for some approaches to teach language comprehension.

Pervasive Developmental Disorders. Currently referred to as Autism Spectrum Disorder.

P

Pharyngeal Flap Operation. A surgical procedure designed to reduce hypernasality in persons with repaired cleft or in those with weak or paralyzed soft palate (as in flaccid dysarthria), and to improve swallow in some people with swallowing disorders; to improve velopharyngeal closure, the surgeon:

- Cuts a flap of tissue from the posterior pharyngeal wall
- Brings the flap down or raises it up (depending on how the flap is cut)
- Attaches the flap to the velum to provide extra muscular mass that helps achieve velopharyngeal closure
- Leaves an opening on either side of the flap to allow breathing, nasal drainage, and production of nasal sounds

Pharyngeal Phase of Swallowing. A normal swallow stage in which the food is propelled through the pharynx and into the pharyngeal-esophageal (P-E) segment; to treat disorders associated with this phase, see Dysphagia.

Pharyngoplasty. A surgical procedure designed to reduce hypernasality in persons with repaired cleft or in those with weak or paralyzed soft palate (as in flaccid dysarthria); in this procedure, the surgeon:

- Injects Teflon or other substance (e.g., Dacron wool or silicone gel bag) into the posterior pharyngeal wall
- Creates a bulge through such injection in the pharyngeal wall to help close the velopharyngeal port

Phonatory Disorders. To treat disorders of phonation due to laryngeal structural problems or habitual patterns, see Voice Disorders.

Phonetic Derivation. Use this Shaping procedure (progressive approximation) to teach correct speech sound production to clients who do not imitate the clinician's productions; in using this procedure:

- Break the target sound production into its simpler components (e.g., teaching the production of /m/ by first having the client put the lips together, a simplified component of the total response)
- Teach the next component that will move the sequence in the right direction (e.g., add humming to the closed-lip posture)
- Teach other response components to achieve the total response (e.g., opening the mouth while humming through the nose, resulting in *ma*)
- Have the client practice the integrated response (e.g., *mommy*)

Phonetic Placement Method. Use this articulation treatment method of showing the placement of articulators to produce different speech sounds; the method is essential when the client cannot imitate the modeled sound production; it uses instruction, physical guidance, and visual

feedback on how target sounds are produced; often used as a component of a comprehensive articulation treatment program.

- Describe how the target sound is produced
- Demonstrate how the sound is produced
- Show the placement of articulators
- Give maximum visual feedback; use a mirror and a drawing of articulatory placements; use palatograms and breath indicators
- Show the differences between correct and incorrect productions of the same sound
- Help position the tongue of the client with tongue blades
- Use your fingers to manipulate and correctly position the client's articulators
- Let the client feel the presence and absence of laryngeal vibrations
- Reinforce correct responses

Phonological Awareness. Many clinicians recommend phonological awareness skill training to children with speech disorders; such skills as rhyming, alliteration, phoneme isolation, sound blending, syllable identification, sound segmentation, and invented spelling are thought to be prerequisites for effectively treating speech sound disorders and to prevent reading disorders; the assumption is based on correlations between phonological awareness and speech and literacy problems; although popular and gaining momentum, the assumption that phonological awareness training is essential to treat speech and literacy problems remains experimentally unsupported; there is evidence contradicting the assumption; it is more effective to directly teach speech production and literacy skills than to spend time on presumed prerequisites; see the cited sources and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for a description of phonological awareness skills; if still preferred, teach the following skills:

- Rhyming. Present selected words to the child and ask the child to produce words that sound alike or rhyme

with each word you present; present rhyming and non-rhyming words, each printed on a card, and ask the child to sort them out

- Alliteration. Ask the child to name words that begin or end with certain sounds
- Phoneme isolation. Present printed words or present words orally and ask the child to tell whether a specific sound occurs in the beginning, end, or middle of a word
- Sound Blending. Orally present words but pause in between the word productions and ask the child to say the word so produced
- Syllable Identification. Orally present words of different lengths (different number of syllables), and ask the child to say how many syllables are in each word you present
- Sound Segmentation. Orally present words and ask the child say how many phonemes are in each of the words presented
- Invented Spellings. Orally present words and ask the child to spell each word phonetically

Catts, H. W., & Kamhi, A. E. (1999). *Language and reading disabilities*. Boston, MA: Allyn & Bacon.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Phonological Disorders. To treat multiple errors of articulation that form phonological rule-based patterns, see Speech Sound Disorders and Phonological Patterns; the treatment target is to eliminate phonological patterns.

Phonological Patterns. Multiple and undesirable ways in which children simplify adult production of speech sounds; become *error patterns* when they persist beyond certain age limits; see Speech Sound Disorders, Treatment of Speech Sound Disorders: Specific Techniques or Programs; by teaching specific sounds within a pattern, eliminate the following patterns:

Assimilation Patterns. Productions of dissimilar phonemes that sound more alike; eliminate the following:

- *Alveolar assimilation*: Substitution of an alveolar sound by an non-alveolar sound (e.g., substitution of /d/ for /p/)
- *Devoicing*: Substitution of a voiceless final sound for a voiced (e.g., /k/ for /g/ in final positions)
- *Devoicing of final consonants*: Substitution of a voiceless final consonant for a voiced (e.g., /t/ for /d/)
- *Labial assimilation*: Substitution of a labial sound for a nonlabial (e.g., /b/ for /d/)
- *Nasal assimilation*: Substitution of a nasal consonant for a nonnasal (e.g., /n/ for /d/)
- *Postvocalic voicing*: Substitution of a voiceless sound for a voiced sound that follows a vowel (e.g., /t/ for /d/)
- *Prevocalic voicing*: Substitution of a voiced sound for voiceless sound preceding a vowel (e.g., /b/ for /p/ in prevocalic positions)
- *Reduplication*: Repetition of a syllable, resulting in substitution of one for another (e.g., wawa for *water*)
- *Velar assimilation*: Substitution of a velar consonant for a nonvelar (e.g., /g/ for /d/)

Deletion Patterns. One or more consonants or a syllable in a word is deleted or omitted; teach the missing phonemes to eliminate the patterns:

- *Cluster reduction*: One or more consonants are deleted in a cluster of consonants (e.g., bu for *blue*, sike for *strike*)
- *Initial consonant deletion*: Omission of an initial consonant of a syllable (e.g., og for *dog*, ink for *sink*)
- *Final consonant deletion*: Omission of a final consonant (e.g., ha for *hat*, goo for *good*)
- *Unstressed syllable deletion*: Omission of a syllable (e.g., medo for *tomato*, nana for *banana*)

Substitution Patterns. Substitution of one class of sounds for another; eliminate such patterns by teaching specific phonemes:

- *Deaffrication*: Substitution of a fricative for an affricate (e.g., /t/, /s/, or /k/ for /tʃ/; /d/ or /z/ for /dʒ/)

- *Denasalization*: Substitution of an oral consonant for a nasal consonant (e.g., /d/ for /n/)
- *Gliding*: Substitution of a glide for a liquid (e.g., /w/ for /r/)
- *Stopping*: Substitution of a stop for a fricative or an affricate (e.g., /p/ for /f/; /p/ for /v/; /t/ for /s/)
- *Velar fronting*: Substitution of an alveolar for a velar (e.g., /t/ for /k/; /d/ for /g/; /n/ for /ŋ/)

Phrases (Word Combinations). Teaching language productions that contain two or more words yet grammatically incomplete, is an intermediate treatment target for language impaired children; see [Language Disorders in Children](#) for details.

- Teach a few [First Words](#)
- Create two-word phrases out of words the child already has learned (e.g., such nouns and adjectives as *big man* or *small box*)
- Teach them with a variety of procedures described under [Language Disorders in Children](#)

Physical Prompts. Use these effective techniques that include visual signs or gestures that you give before a response is produced to increase its probability of occurrence (e.g., showing a lifted tongue tip just before the child attempts to produce a /t/; showing lip closure to prompt the production of a bilabial sound).

Physical Setting Generalization. A functional treatment target that seeks to establish production of trained responses in a setting not used in training; an important clinical goal; measured on a [Probe](#); typically not reinforced.

- Select stimuli for treatment targets from the client's home
- Use common stimuli found in nonclinical settings
- Give training in varied physical settings such as outside the treatment room, outside the clinic building, and in other places where target behaviors may be practiced in conversational speech

Physical Stimulus Generalization. A functional treatment target that seeks to establish production of trained responses in the presence of untrained stimuli because of their similarity to trained stimuli; an important treatment goal; typically measured on a Probe; usually not reinforced.

- Use varied stimuli in training
- Use stimuli from the client's home
- Prefer objects to pictures
- Use multiple exemplars to train each target behavior
- Probe frequently with the help of untrained stimuli to evaluate physical stimulus generalization
- Provide additional training until the Probe Criterion is met

Pic Symbols. A set of symbols drawn in white on a black background; used in teaching a form of Augmentative and Alternative Communication.

Picture Communication Symbols. A large collection of pictures that represent words, phrases, sentences, social exchanges widely used in Augmentative and Alternative Communication; most symbols are transparent (meaning readily apparent).

Picture Exchange Communication System (PECS).

A nonverbal communication system in which the individual picks a picture and hands it to a caregiver or therapist who then gives what the picture implies or depicts; the response is spontaneous because the clinician does not give any prompts; has been used in teaching communication to children with autism; children who cannot point to pictures to indicate what they want may nonetheless pick and hand a picture to someone to achieve the same effect; a transitional system eventually leading to verbal communication training; supported by experimental evidence.

Bondy, A. S., & Frost, L. A. (2001). The picture exchange communication system. *Behavior Modification*, 25, 725–744.

Picsyms. A set of symbols containing line drawings that can be used to teach non-oral expression of nouns, verbs,

prepositions, and so forth; each symbol also is associated with a printed English word; an open-ended system to which clinicians can add their own drawings; used in teaching Augmentative Communication–Aided.

Pictographic Symbols. Pictorial representation of objects and events; easier to learn than abstract symbols; used in teaching Augmentative Communication–Aided.

Polyps. Protruding, soft, fluid-filled growths on the inner margin of the vocal folds; result of vocal abuse, often from a single abusive episode; typically unilateral; may be *sessile* (broad-based) or *pedunculated* (the mass of the polyp is connected to the vocal fold by a stalk-like structure); associated with hoarseness and breathiness; surgically removed; see Voice Disorders: Treatment of Vocally Abusive Behaviors.

- Identify the vocally abusive behaviors
- Reduce vocally abusive behaviors
- Teach appropriate vocal behaviors (e.g., gentle onset of phonation, soft speech)

Population. A large, defined group with certain characteristics identified for the purposes of a study; part of the Group Design Strategy of research; a representative Sample is randomly drawn from the population.

- Identify a large group of persons with defined characteristics (e.g., persons who stutter; people who have aphasia with additional defined characteristics relative to age, gender, severity, and so forth)
- Randomly draw a sample of subjects needed for the study
- Assign them randomly to an experimental group and a control group in a treatment research study

Positive Practice. Required and unreinforced practice of a desirable behavior following Restitution for an undesirable behavior; a Direct Methods of Response Reduction; a part of Imposition of Work.

Positive Reinforcers. Events that, when presented immediately after a response is made, increase the

future probability of that response; an effective method to select and increase the frequency of target communicative behaviors; extensively researched with a variety of clinical populations; commonly used in communication training.

- Select potential reinforcers after consulting with the client, the family, or both
- Present potential reinforcer immediately after the correct response is made
- Use a Continuous Reinforcement schedule in the beginning and an Intermittent Reinforcement schedule subsequently
- Prefer Conditioned Generalized Reinforcers (e.g., Tokens) to Primary Reinforcers
- Use verbal praise (even when you use other kinds)
- Use a different event when the one selected does not increase the response rate
- Call an event a reinforcer only when it increases a response rate

Post-Reinforcement Pause. A period of no responding after receiving a reinforcer; markedly evident in Fixed Interval Schedule of reinforcement.

Posttests. Measures of behaviors established after completing an experimental or routine teaching program; compared with Pretests; in a group design study, post-test measures help rule out the influence of extraneous variables.

Postural Strategies. Techniques of manipulating body positions, especially head and neck positions to prevent aspiration in patients with dysphagia; see Dysphagia.

Pragmatic Language Skills. Aspects of appropriate social language produced in naturalistic communicative contexts; targets of language intervention; include such skills as Conversational Repair; Eye Contact; Narrative Skills, Topic Initiation, Topic Maintenance; and Turn-

Taking (all described under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs).

Pragmatics. The study of social language production and the rules that apply to it.

Prephonation Airflow. A target behavior for people who stutter and those who exhibit hard glottal attacks; includes a slight exhalation before initiating phonation; for procedures, see Fluency Disorders and Voice Disorders.

Premack-Type Symbols. Plastic shapes or tokens designed by David Premack to teach communication to chimpanzees; Non-iconic symbols that may be used to teach communicative skills to individuals with limited oral communication skills; used in teaching Augmentative Communication–Aided.

Pretests. Measures of behaviors established before starting an experimental or routine teaching program; compared with Posttests; in a group design study, posttest measures help rule out the influence of extraneous variables.

Pretraumatic Memory Loss. To treat loss of memory for events *preceding* brain injury, see Traumatic Brain Injury in Adults.

Primary Progressive Aphasia. To treat this atypical form of aphasia that is due to progressive neurodegenerative disease terminating in a type of frontotemporal dementia, see Dementia for clinical management.

Primary Reinforcers. Reinforcers whose effects do not depend on past learning; often fulfill biological needs; contrasted with Conditioned, Secondary, or Social Reinforcers; also known as unconditioned reinforcers.

- Use primary reinforcers with infants, toddlers, and other children who do not respond well to Social Reinforcers
- Use with children who have intellectual disabilities, minimal verbal skills, and those who are autistic
- Use with persons who have brain injury in the initial stages of treatment
- Combine them with social reinforcers
- Withdraw primary reinforcers gradually and maintain the skills with social reinforcers

Principles (of Treatment). Empirical rules from which treatment procedures are derived; procedures may be modified (e.g., a response consequence found to be ineffective may be modified), but experimentally supported principles do not change (e.g., the principle of positive reinforcement remains valid).

Probe Criterion. A rule that specifies when to terminate training at a given topographic level of training or on a specified target behavior.

- A 90% correct Intermixed Probe response rate at each topographic level of training may suggest that the training may be moved to the next level (e.g., from the word to the phrase level)
- A 90% correct Pure Probe response rate for a behavior at the conversational level may suggest that the behavior is sufficiently trained and that the training may move on to another target behavior.

Probe Procedure. Procedure to assess generalized production of target behaviors; see Intermixed Probes and Pure Probes for procedures.

Probe Recording Sheet. A prepared sheet for recording probe response rates.

- Design and use a probe recording sheet similar to the following; modify as found necessary

Name of Client	Treatment Target
Clinician	Date
Probe Recording Sheet	
Stimulus Items	Responses: + (Correct), - (Incorrect), 0 (No response)
1.	
2.	
3.	
4.	

Probes. Procedures used to assess generalized production of clinically established responses; administered every time a few exemplars are trained to assess generalized productions; may be Intermixed Probes, Pure Probes, or Conversational Probes.

Procedures of Treatment. Technical operations the clinician performs to effect changes in client behaviors; actions of clinicians; contrasted with Treatment Targets; in describing treatment procedures:

- Specify the target communication skills
- Specify what you ought to do to achieve those target skills
- Describe the stimulus conditions you need to arrange
- Specify the kinds of positive reinforcement and corrective feedback you should give to the client under the differing conditions of correct, incorrect, and lack of responses
- Clarify how you measure the skills during treatment to document progress
- Describe how you plan to promote generalized productions and maintenance over time and across situations
- Specify the follow-up and booster treatment procedures

Production Training. Treatment designed to teach a client to produce a specified speech or oral language target; contrasted with Auditory Discrimination Training or Perceptual Training; emphasis is on what the client ought to say rather than just listen or respond nonverbally.

- Model the target skills and ask the client to imitate your productions
- Fade modeling when imitation is established
- Evoke the target skills by appropriate questions and other devices

Prognosis. A statement about the future course of a disorder under specified conditions, which typically include treatment or no treatment; good prognosis implies that the patient will recover from the clinical condition (with or without treatment, usually specified).

Programmed Learning. A method of teaching skills in a systematic manner with immediate positive and corrective feedback using operant conditioning principles; used in the treatment of various disorders of communication, especially language, articulation, and fluency disorders.

Program of Treatment. An overall description of target behaviors, treatment variables, measurement procedures, generalization measures, maintenance strategies, follow-up, and so forth.

P

Progressive Assimilation. A phonological pattern in which a sound takes on the properties of a preceding sound; see Phonological Patterns and Speech Sound Disorders.

Prolonged Speech. To use this method of slowing down the speech of people who stutter, see Treatment of Stuttering: Specific Techniques or Programs under Fluency Disorders.

Prompts. Special stimuli that increase the probability of a response; prompts may be verbal or nonverbal.

- Prompt promptly, as the client hesitates (e.g., in treating naming in a client with aphasia: “What is this?” “The word starts with a /t/”)

- Prompt more frequently in the beginning to reduce errors
- Prefer a subtle or short prompt to ones that are loud or long (e.g., in treating a person who stutters to speak slowly: “Slower” instead of “Speak at a slower rate”)
- Prefer a gesture to a verbal prompt (e.g., in treating a person who stutters to speak slowly: make a hand gesture to suggest a slower rate)
- Use Partial Modeling as a prompt
- Fade prompts as the responses become more consistent

Prosthesis. A device fashioned for individual clients and fitted to compensate for deficient or deformed structure to improve its function; see Cleft Palate for some examples.

Pseudobulbar Palsy. Paralysis of the muscles of mastication, articulation, and swallowing caused by bilateral brain damage; so called because the symptoms mimic those caused by brainstem damage; associated with Dysarthria, especially the spastic variety.

Pseudodementia. To understand the clinical management of apparent dementia-like symptoms in people who are depressed, see Psychiatric Problems Associated with Communication Disorders.

Pseudo Supraglottic Swallow. A procedure to protect the airway during swallowing; used with patients who have dysphagia.

- Ask the patient to inhale and hold the breath
- Swallow
- Cough

Psychiatric Problems Associated with Communication Disorders. Treatment of several forms of communication disorders needs a basic knowledge of psychiatric and neurological problems that may also be part of the clinical picture or may be entirely psychiatric in nature; while treating communication disorders in patients with dementia and neurodegenerative diseases, speech-language pathologists may encounter specific psychiatric symptoms

that need to be dealt with; in other cases, the clinician may encounter a communication disorder that is entirely psychiatric in nature; stuttering that is due to malingering or faking, certain forms of mutism, a type of aphonia, and a type of dementia may all be entirely a psychiatric problem; some of these psychiatric disorders may not have a neurological basis; in yet another set of patients, communication disorders may be a part of a major psychiatric disorder; patients with schizophrenia are an example; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for details on psychiatric problems associated with communication disorders; to assess these problems, see the cited sources and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*; consider the following in clinically managing psychiatric disorders while treating associated communication disorders:

Treatment of Psychiatric Disorders: ***General Guidelines***

- With appropriate specialization, speech-language pathologists may treat communication disorders that may be described as psychiatric in nature; just because a person seems to fake stuttering, the clinician need not make a referral to a psychiatrist or a psychologist who may lack the expertise of the speech-language pathologist in either diagnosing the typical stuttering, differentiating it from malingering, and treating either or both of them
- In most cases, there may be no need to refer the client to a psychiatrist or psychologist after treating a psychiatric communication disorder; in some cases, the clinician may judge that it is necessary to make that referral to additional, independent, and significant emotional problems (e.g., patients who remain extremely depressed or highly anxious)
- Some individuals may spontaneously recover from their psychiatrically based communication disorders; others need treatment
- Persons who receive both behavioral therapy to change their maladaptive behaviors and treatment for their

communication disorders are more likely to sustain improvement than those whose communication disorder is changed but their overall behavior pattern remains maladaptive; a certain number of patients with psychiatric speech or voice disorders may need to change the way they react to stress

- In some cases, a psychological or behavioral explanation, found during an analysis of the problem, may help diminish the symptoms; in other cases, the symptoms may diminish without any analysis or explanation
- Most persons with psychiatric communication disorders believe that their disorder is organically based; the clinician should handle this belief in a serious and respectful way, not in a dismissing manner; the topic may be discussed at various stages of treatment and the person may be more inclined to hear the explanation when the treatment has produced positive changes
- In some cases, improvement may be so rapid that treatment may be initiated in the diagnostic session and completed in the next few sessions
- When a neurologic disorder is also present along with a psychiatric communication disorder, it is essential to make it clear to the person the different bases of their different symptoms
- A medical report that clears any organic basis for the disorder to be treated will be of help in establishing credibility of behavioral management
- A careful case history, taken with a view to establish the conditions of onset, may be the initial strategy in treating the client; lack of a physical cause for the disorder may be established during the interview when the medical reports also are reviewed; the attention can then be turned to behavioral history, stressful situations the client has been facing, the client's typical reactions to stress, possible consequences of the disorder, and so forth

Abulia. Treatment of abulia may be a responsibility of a speech-language pathologist who encounters clients with an extreme lack of motivation to do anything; in such individuals, the clinician may have to treat a total lack of

drive to talk, which is a form of mutism called *akinetetic mutism*; even though it may be considered a psychiatric disorder, it has a neurological basis; treatment of akinetic mutism involves several considerations:

- Defer treatment if the person is just emerging from a mute state in the process of recovering from strokes and trauma; when the physiological state of the patient is stabilized, motivation to talk may also improve; address this type of mutism in the context of language treatment; use strong reinforcers to increase motivation
- An alert and attentive individual who does not talk may be treated like a functionally aphonic individual; digital manipulation of the larynx to reduce tension, identifying vocalization that may be associated with coughing, and then reinforcing vocalizations may all be effective
- A person who communicates with whispers may also be prompted to vocalize by laryngeal manipulation and by identifying “accidental” vocalizations that are shaped into more sustained vocalizations and normal speech production

Aphonia—Functional. Treatment of aphonia without a neurological basis, also called *functional aphonia* or *psychogenic aphonia*, requires that a medical examination rule out a physical (organic) basis for the disorder (e.g., bilateral vocal fold paralysis with a wide-open glottis); treat this as a psychiatric or behavioral disorder; note that in treating such disorders, taking a detailed case history and interviewing the individual are also opportunities to explore treatment ideas and strategies as well:

- Use the interview and explore the case history to identify possible stress factors in the client’s life that surrounded the onset of aphonia
 - Were there traumatic or stressful situations surrounding the onset of aphonia?
 - Were there stress-related difficulties in the past?
 - Is there significant tension in the laryngeal structures?
 - If so, the problem is more likely a reaction to stress, which may be maintained by other factors

- Discuss the relation between stress, laryngeal tension, and aphonia to set the stage for treatment manipulations
- During the interview, explore indirectly the consequences that followed the loss of voice:
 - Was there a reassignment of job responsibilities that reduced the patient's work-related stress? Did a student escape from stressful oral presentations because of aphonia? If so, aphonia in both cases may be stress related and it is negatively reinforced because the disorder led to a reduction in stress
 - How have the listeners reacted to the client's whispering? Perhaps the client may have been, and still is being, complimented for effective whispering and gesturing; if so, the client's aphonia is likely maintained by positive reinforcement
 - Discuss the possible connection between stress, aphonia, and reduced work-related stress or academic stress that followed the onset of the problem; this may help induce insight and hint at the direction of treatment
- Ask the person to cough, laugh (crack some jokes), and clear the throat
 - Quickly point out that you heard the voice; do not make the person feel that he or she was "caught"; instead, use such observations to reassure the patient that there is no physical problem that is preventing voice production and to induce hope that improvement is possible
 - Make humorous statements to induce laughter, and then draw the patient's attention to fleeting voice
 - Repeat the procedure with throat clearing or coughing
 - Digitally manipulate the laryngeal area (laryngeal massage), to reduce tension and to induce voice production
 - Lower the larynx by gently kneading or massaging the laryngeal muscles, because a high larynx is a tensed larynx; if pain is felt, reassure the patient that it will pass and that the manipulation is essential to relieve tension; to do this, encircle the hyoid bone

with the thumb and middle finger; make circular and up-and-down motions with gentle pressure

- Ask the patient to clear the throat, cough, or grunt
- Ask the patient to prolong vocalization associated with throat clearing, coughing, or grunting
- Reinforce positively even the slightest improvement in phonation or its duration
- Fade out the digital manipulation of massage; shape the brief or faint phonation the patient has achieved up to this point into stronger phonation of progressively longer duration
- Continue treatment with words, phrases, and sentences produced with phonation; reinforce voice in oral reading as well
- Use the yawn-sigh approach described under [Voice Disorders](#) to reduce tension and increase the likelihood of normal voice production

Delusions and Hallucinations. The need to treat delusional speech associated with other types of communication disorders may vary across individuals; most speech-language pathologists will not be involved in treating *delusional disorder*, which is a specific psychiatric problem treated by psychiatrists and psychologists; in various ways, speech-language pathologists may address coexisting delusions, hallucinations, or both in their clients who have neurodegenerative diseases and dementia; clinical management of such delusions and hallucinations may be a part of an overall interventional strategy, often implemented by a team of specialists including psychologists:

- General management strategies for delusional speech and expression of hallucinations include the following:
 - Find out if the psychologist or a psychiatrist is handling delusions, hallucinations, confusion, and other psychiatric disorders; if so, find out their strategies; try to include them in your treatment approaches (e.g., ignoring them)
 - Do not positively reinforce utterances that contain reference to delusions and hallucinations; ignore

them completely (extinction); such utterances increase in frequency when people pay attention to them or ask the person to elaborate on them

- Use time-out if extinction does not work; as soon as the person begins to express undesirable speech, cease all interaction for 5 seconds; turn your face away from the person; re-establish eye contact and ask a question that will prompt a different response
- Tell the individual that you need to work on other aspects of speech and that you would rather not hear “that kind of speech,” referring to speech that is full of delusional thoughts and hallucinatory experiences; do not keep repeating this instruction, which may actually reinforce the patient; once stated, ignore the patient’s inappropriate speech
- Educate staff and family caregivers not to reinforce speech that contains reference to delusions and hallucinations
- Educate staff to pay attention to speech that is more reality oriented and free from reference to delusions and hallucinations
- Consider such strategies for patients with Alzheimer’s disease, frontotemporal dementia, Parkinson’s disease, Huntington’s disease, AIDS dementia complex, and patients with other kinds of dementia who may have speech that refers to delusions and hallucinations

Depression. Treatment of individuals with mild and severe forms of depression may require different strategies; many with moderate to severe depression may be taking prescription drugs to control it; those with mild depression may or may not be treated with drugs; speech-language pathologists do not treat depression, but may have to modify their communication treatment procedures to maximize their effects on individuals with depression; some general suggestions include the following:

- Conduct a few trial treatment sessions with depressed individuals to evaluate the usefulness of intervention;

depending on the results, defer treatment to those who are severely depressed until psychiatric treatment improves their mood and responsiveness

- Use the most colorful and realistic stimulus materials that may cheer up the person; arrange activity-based treatment sessions in which the person's favorite activities, hobbies, and interests are included; physical activities may improve the speech response rate
- Find functional reinforcers; most may not respond to verbal praise; evaluate if listening to music or engaging in some favorite activity may interest the patient; experiment with different kinds of reinforcers to find the one that seems to be effective
- Give more time to respond when you ask questions or show pictures; frequently prompt the correct responses
- Give frequent but short breaks during treatment
- Find a suitable time to conduct communication treatment; ask the caretakers if the patient is more alert and responsive at certain times during the day than at certain other times

Euphoria and Mania. Treatment of an exaggerated feeling of well-being (euphoria) and an extreme degree of hyperactivity (constantly on the go) combined with an unrealistic sense of well-being (mania) is left to the psychiatrist or psychologist; to supplement or support such treatment, speech-language pathologists may do the following during communication training:

- Ignore excessive jocularity and statements of exaggerated self-esteem (in patients who have frontotemporal dementia); euphoric-like false sense of superiority (in patients with Huntington's disease), or outright euphoria (in patients with dementia caused by repeated head injury, vascular dementia, and Creutzfeldt-Jakob disease)
- Suspend communication treatment when the patient is extremely euphoric or in a state of mania, who are always jumping from one activity to another, without finishing any (e.g., patients with AIDS dementia complex)

- Suspend treatment when the patient is in an extremely depressed state, as euphoria or mania often alternate with depression

Factitious Disorders. Treatment of communication disorders that are factitious in nature may be an occasional clinical challenge faced by speech-language pathologists; factitious disorders—those that are simply invented by the patient—typically take the form of physical diseases that require hospitalization and sometimes serious and painful treatment; occasionally, they may take the form of stuttering or a voice disorder; there is very little research on treating factitious communication disorders; some general suggestions include the following:

- Note that a differential diagnosis of a factitious disorder is half the treatment effort; consult the cited sources in *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for specific signs that help make a diagnosis; among others, document:
 - A desire to play the role of a sick person
 - No evidence of gain, actually evidence of pain, inconvenience, or extraordinary expenses
 - Mysterious and frequently treated illnesses in the past; self-inflicted injuries; extensive and frequent travels to distant places to consult many specialists; a history of many ineffective treatments
 - A tone of dramatic flare in reporting the symptoms
- Discuss the results of your analysis with the client; there are differences between the typical stuttering and the factitious stuttering or typical aphonia or factitious aphonia; point them out to the client
- Offer trial treatment; for instance, for a person suspected of factitious stuttering, conduct a few treatment sessions involving such devices as Delayed Auditory Feedback (DAF) or masking noise; use such other procedures as shadowing, choral reading, metronome speech, and extremely slow speech; if there are no beneficial effects—a likely outcome if the disorder is factitious—discuss with the patient that the problem

may not be the kind of stuttering you would typically treat

- It is not evident from the literature whether factitious communication disorders can be eliminated with confrontation; factitious disorders of other kinds have been difficult to treat

Malingering. Treatment of malingering can be challenging because a symptom complex is being faked (feigned) by a client to gain a certain advantage; the clinician has to scrutinize not only the symptoms and the conditions of their onset, but also the motivation of the client for exhibiting the symptoms; the conscious and well-planned symptoms may resemble a variety of disorders; speech-language pathologists occasionally see malingered stuttering, voice disorders (especially mutism or aphonia), or cognitive dysfunction (especially faked memory loss) that is disproportionate with the extent of traumatic brain injury, movement disorders with no neurological basis, and hearing loss; more common in medical and psychiatric practice; see the companion volume, *Hegde's Pocket Guide to Communication Disorders*, for details; to treat malingered stuttering or voice disorders:

- Treat malingered aphonia as you would functional aphonia, described previously
- Treat malingered stuttering by taking the following steps:
 - Scrutinize the total symptom complex carefully; though the person consciously faking a disorder has studied it, he or she is still not an expert in it; there will be discrepancies between actual stuttering and feigned stuttering
 - Offer experimental treatment with such devices as DAF and masking noise; offer trial therapy with shadowing, choral reading, extremely slow speech induced by instruction and modeling; these methods reduce the typical stuttering; if they do not, discuss this and the discrepant features of his or her stuttering

- Continue those procedures that show positive results until fluency is stabilized
- Counsel the client about stressful life events; reported stressful events include a history of marital problems, traumatic divorce, various kinds of tragedies in personal or family life, death of loved ones, failure in meeting employment responsibilities, feelings of frustration and anger because of chronic unemployment or sudden loss of employment, unjust accusation of wrongdoing, religious differences with family members, and so forth; raise the possibility that such traumatic events may cause fluency problems (not necessarily implying that the client is feigning the disorder); there is some clinical evidence that an understanding of the relationship between stressful and conflicting experiences and stuttering (or other disorder of communication) may help resolve it
- Discuss any features of the client that are at odds with the known characteristics of the disorder, only to impress the client that the stress, not some other organic or genetic factor, is responsible for stuttering
- Discuss any potential gain that may have accrued to the client; while not suggesting that the client is faking the disorder to obtain the benefits, point out the relation between events that may include a discussion of positive reinforcement (financial or other kinds of gain) or negative reinforcement (avoidance of difficult situations) that may be involved in the genesis of the disorder; do not force the connection between the gains and the disorder, but suggest the idea for the patient to think about
- Continue treatment methods that have worked in the initial stages, as the client who feigns stuttering needs a face-saving strategy; to give a feeling that the improvement is due to treatment, not to the disclosure of the underlying problems, treatment needs to be continued for at least a few sessions

Mutism—Selective. Treatment of selective mutism—a psychiatric variety of mutism—requires a differential diagnosis from other kinds of mutism; paralysis of the vocal folds or an advanced stage of neurodegenerative disease may be associated with mutism; rule out physical causes for mutism before you treat it as selective and functional (psychiatric in nature); see the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* to make the differential diagnosis; once the diagnosis of selective mutism is established, use the following suggestions to treat it:

- Find out from the case history and interview the situations in which the client has spoken because selective mutism is situation specific
- Get an audio-recorded sample of speech from the situation in which the client speaks; ask the client to reproduce speech like he or she could as evident on the recording; reassure the client that phonation is not lost
- Observe limited verbalizations that may occur during treatment; they may be characterized by monotonous or altered voice; tell the client that this would be the starting point of regaining voice
- Ask the client to produce the monotonous or altered voice on repeated trials
- Reinforce such voice or phonated speech; reassure the client that there is no physical barrier to voice production and that he or she can shape normal voice and speech out of what he or she can produce
- Shape and systematically reinforce progressively more normal sounding voice and speech production
- Use additional suggestions or strategies described under functional aphonia

Pseudodementia. Treatment of dementia may be complicated by the presence of depression which often gives the impression of cognitive impairment that may be confused with actual dementia; on the other hand, a misdiagnosis of dementia may be made in elderly individuals who are depressed for various other reasons;

pseudodementia, which has dementia-like symptoms in depressed persons, needs a careful assessment; see sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, for assessment to be completed before initiating treatment; there is very little research on treating pseudodementia; consider the following general suggestions:

- Do not offer communication treatment to individuals diagnosed with pseudodementia; let the psychiatrist or psychologist treat depression first
- Reassess the client for cognitive or memory impairments when the patient is no longer depressed
- Treat the communication deficits if they still exist, an unlikely event, with procedures described under Dementia
- Reassess the client because persistent symptoms of dementia when the individual is no longer depressed suggests that the initial diagnosis of a pure pseudodementia was wrong; the person perhaps had dementia and depression
- Use suggestions offered under Depression within this main entry to treat if there are still lingering signs of depression but it is evident that the person needs treatment for dementia as well
- Take note of more rapid progression of cognitive impairment than is found in generally slowly progressing dementia associated with neurodegenerative diseases
- Assess visual memory tasks on which the depressed individuals with symptoms of dementia may do better than those with dementia without depression
- Take note of a history of psychiatric problems in the past
- Work closely with the clinical psychologists or psychiatrists in the diagnosis of depression, dementia, and associated communication problems

Schizophrenia. Speech-language pathologists typically do not treat aberrant speech patterns associated with

schizophrenia; it is thought that psychiatric treatment of schizophrenia, when successful, will reduce or eliminate schizophrenic speech and language; nonetheless, modification of schizophrenic speech and language characteristics may be a significant part of the overall treatment plan; if successful, reduced frequency of schizophrenic speech will make the person less abnormal and socially more appropriately interactive; behavioral methods of reducing schizophrenic speech and language are known to be effective; to reduce these abnormal speech and language patterns, generally use the strategy that was outlined under Delusions and Hallucinations in this entry; in managing schizophrenic speech and language:

- Work with behavioral analysts to develop an overall behavior modification program; reducing speech and language deviations may be effective only when all therapists and all health care workers take the same steps in modifying both the verbal and nonverbal aberrant behaviors
- Do not try to reason with the person that his or her language does not make sense, because such attempts are ineffective
- Do not reinforce aberrant speech and language expressions; ignore them completely
- Redirect the conversation to something concrete, specific, and present in the situation when you hear aberrant speech
- Reinforce appropriate speech and language behaviors
- Have the person describe pictures, narrate personal experiences, or retell stories; differentially reinforce only the appropriate behaviors
- Design and implement a pragmatic language intervention program; ignore frequent and abrupt changes in the topic of discussion or conversation; redirect the person to the topic under discussion
- Stop the client from speaking on topics or ideas that are remotely or loosely related to the point of discussion
- Ignore irrelevant answers to questions and irrelevant comments; model relevant answers and comments and reinforce the client for producing them

- Consider using such behavior reduction procedures as time-out for meaningless schizophrenic speech
- Teach appropriate stress patterns and other prosodic features by modeling and reinforcement
- Reinforce eye contact; frequently draw attention to your face

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Pull-Out Therapy Model. A special education service delivery model in which children are taken out of the classroom for special services, including speech-language services.

Punisher. A stimulus or a consequence that, when delivered soon after a response is made, is likely to reduce that response; a response-reducing consequence; the most common form used in treating communicative disorders is corrective feedback when a wrong response is made; other

procedures include response cost and time-out (pause and talk); all supported by experimental evidence.

Punishment. Procedures of reducing undesirable behaviors by response-contingent presentation or withdrawal of stimuli; includes Direct Methods of Response Reduction and Indirect Methods of Response Reduction.

- Minimize the use of response reduction procedures
- Simplify the target response and shape it to avoid or reduce the use of punishers
- Let the positive:corrective ratio be in favor of the positive (more reinforcers than corrective feedback)
- Prefer indirect methods of response reduction in which you replace undesirable behaviors with desirable behaviors that you positively reinforce
- Change the treatment procedures when the client's correct responses do not increase
- Watch for potential undesirable Side Effects of Punishment
- Use Time-Out and Response Cost because they are especially effective in reducing various disorders of communication and other undesirable behaviors children might exhibit during treatment sessions

Pure Agraphia. To treat writing problems with no reading or other associated language problems due to recent brain injury, see Agraphia and Aphasia, in which such problems are typically treated.

Pure Alexia. To treat reading disorders without writing problems associated with brain injury, see Alexia and Aphasia, in which such problems are typically treated.

Pure Probes. Procedures to assess generalized production with only untrained stimulus items; to be administered when the client has met the intermixed probe criterion, preferably toward the end of treatment: contrasted with Intermixed Probes, in which trained and untrained items are alternated.

- Prepare a Probe Recording Sheet on which you have at least 10 untrained exemplars (untrained words, phrases, or sentences that contain the target sound or language feature)
- Present each exemplar on discrete trials
- Provide no reinforcement or corrective feedback
- Calculate the percent correct probe response rate
- Give additional training when an adopted probe criterion is not met (e.g., 90% accuracy)
- Move on to the next level of training or to new target behaviors when the criterion is met

Pushing Approach. A voice therapy procedure to promote better vocal fold approximations in clients who have weakened or paralyzed folds; for procedures, see Voice Disorders: Specific Normal Voice Facilitating Techniques.

Q

Question. Interrogative verbal form that is often a request; treatment targets for clients with language disorders; elaborate situations may have to be set up or described to teach questions; a few examples:

- Describe the scene such as this one to the client: You meet a new student in the school cafeteria. You don't know the person's name.
 - Ask the child, "What do you ask?"
 - If no response, model the question: "Say, 'What is your name?'"
 - Reinforce the correct imitation; repeat the trials a few times
 - Fade the modeling and just ask the question ("What do you ask?")
 - Reinforce the correct response ("What is your name?")
- You may show pictures that the child is not familiar with to teach "What is this?" in the same manner.
- You may show photographs of unfamiliar persons to teach "Who is this?" with the same procedure.
- Use similar strategies to teach other forms of questions (*which, when, where, whose, why, and how*).

R

Random Assignment. A method of assigning participants selected for a study to either the experimental or the control group without the experimenter bias; used in treatment research; part of the Group Design Strategy.

- Select participants randomly
- Assign a number to each subject
- Assign every even-numbered subject to one group and every odd-numbered subject to the other group

Random Selection. A method of selecting participants to evaluate the effects of new treatments; theoretically, each potential participant has an equal chance of being selected for the study, hence no experimenter bias in participant selection; need a large number of potential participants for the method to work; part of the Group Design Strategy.

- Identify a large number of potential participants
- Assign a number to each person
- Select the required number of participants randomly (e.g., every second, every fourth, every tenth person)

Rate Reduction. See Stuttering, Cluttering (described under Fluency Disorders) and Dysarthrias to reduce the speech rate to achieve fluency or speech intelligibility.

Ratio Strain. Reduction in response rate due to a sudden thinning of reinforcement as when the clinician shifts from continuous reinforcement to an FR10 (every tenth response is reinforced) .

- Start with continuous reinforcement for target responses
- Move to an FR2 or FR3 (every second or third response is reinforced)
- Increase the ratio of reinforcement gradually

Rebuses. Pictures of objects and persons used in teaching Augmentative Communication–Aided; different from ordinary pictures in that words and grammatic morphemes are combined with rebuses.

Recast. To use this child language intervention procedure, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Recurrent Laryngeal Nerve Resection. A surgical treatment procedure for adductor Spasmodic Dysphonia; the recurrent laryngeal nerve is unilaterally resected to paralyze one of the folds to prevent hyperadduction; voice therapy may be needed following this operation.

Regulated Breathing. To use this stuttering treatment target, see Stuttering, Treatment; Treatment of Stuttering: Specific Techniques or Programs under Fluency Disorders.

Reinforce. To strengthen and increase behaviors by arranging immediate consequences for them; an important action clinicians perform in teaching target responses to children as well as adults; see Reinforcers.

Reinforcement. A method of selecting and strengthening behaviors of individuals by arranging consequences under specific stimulus conditions; widely used in the treatment of communicative disorders; see Reinforcers.

Reinforcement Priming. Seeking reinforcers for one's own behaviors; useful strategy for the client to learn in getting parents, teachers, peers, and others to notice the production of clinically established behaviors in natural settings and thus get reinforced; part of Maintenance Strategy.

- Teach others to reinforce the client for the production of target behaviors established in the clinic
- Teach the client to draw attention to his or her desirable communicative behaviors from others; for instance:
 - Teach the child who stuttered but has mastered fluency skills to make such statements as “Mom, did you notice my smooth speech?” “Dad, did you notice that I didn’t stutter once?” and so forth
 - Teach the child who has mastered specific language features to make such statements as “Mrs. Lyndner [the teacher], did you notice that I said *boys are* and not *boys is*?” “Dad, did you hear my long sentence?” and so forth
- Verify that others are indeed reinforcing the client when attention is drawn to the production of target behaviors

Reinforcement Withdrawal. Taking away reinforcers to decrease a response; one of the Direct Methods of Response Reduction; includes Response Cost and Time-Out; effective in reducing various kinds of general undesirable, noncooperative behaviors as well as specific disorders of communication, including stuttering and specific speech sound misarticulations.

Reinforcers. Events that follow behaviors and thereby increase the future probability of those behaviors; widely used in treating communicative disorders.

- Select either the Primary, Secondary, Conditioned Generalized, Informative Feedback, or High Probability Behaviors to reinforce clinical targets
- Reinforce initially on a Continuous Reinforcement schedule
- Switch to an Intermittent Reinforcement schedule later
- Always use verbal (conditioned) reinforcers in conjunction with other types
- Teach the client's significant others to reinforce the skills you establish

Reinke's Edema. Also known as polypoid degeneration; sausage-shaped, grayish-pink, or red projection-like growth on vocal fold margins associated with severe and persistent hoarseness.

- Treatment is surgical removal of the growth; performed on one fold at a time; if normal voice is restored, the second operation may be avoided
- Voice treatment involves modifying smoking and drinking behaviors and any residual voice problems

Reliability. Consistency with which the same event is repeatedly measured; important in clinical work and clinical as well as nonclinical research; includes inter- and intra-observer reliability; for reliability associated with standardized test measure, see the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*.

Interobserver Reliability. The extent to which two (or more) observers agree in measuring an event.

- Measure a behavior of interest with its location identified for a unit-by-unit analysis (count not only the behaviors being measured, but also their locations in transcribed speech samples)
- Have another trained observer measure in the same manner (unit-by-unit analysis)
- Score the total number of locations for which both of you agreed for an Agreement count (A)
- Count the total number of locations for which only one of you, not both of you, scored the behavior (stuttering, pitch breaks, articulatory error) for a Disagreement count (D)
- Calculate the unit-by-unit Agreement Index by using the following formula: $A/(A+D) \times 100$

Intra-Observer Reliability. The extent to which the same observer repeatedly measures the same event consistently.

- Measure the behavior of interest using the unit-by-unit method
- Measure again by the same method
- Calculate the Agreement Index using the same formula as given under Interobserver Reliability.

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Replication. Repeated experimental evaluation of the same treatment procedure to show that it works with different clients, in different settings, and for different clinicians; important in treatment efficacy research; includes direct replication and systematic replication; both designed to show treatment Generality; one of the Treatment Selection Criteria.

Direct Replication. The same investigator repeats the same treatment experiment in the same setting but with different participants who have the same characteristics as the original participants.

- Initially, an investigator shows that a treatment works with some participants
- Selects different participants who share the same personal (age, gender, health) and clinical characteristics (severity, age of onset) as the original participants
- Repeats the treatment experiment
- Analyzes the results to evaluate generality of the treatment method

Systematic Replication. The same or different investigators repeat a treatment experiment in different settings, with participants who have different characteristics than the original participants; may even include individuals with totally different diagnoses.

- Initially, an investigator shows that a treatment is effective with a sample of clients
- The same or a different investigator repeats the treatment research with another sample, with different personal (gender, age, health) and clinical characteristics (severity, age of onset), and in a different setting than the original
- The investigator analyzes the results to evaluate the broader generality of the treatment method

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Response Class. A group of responses created by the same or similar contingencies; functionally, but not necessarily structurally, similar responses; good treatment targets because teaching a few responses (exemplars) within a class can cause generalized production of other responses within the same class; exemplars that belong to different classes are discriminated (not generalized).

Response Complexity. Different topographic levels of a target behavior; structural complexity of communicative behaviors typically create a sequence of treatment.

- Teach words before phrases
- Teach phrases before sentences
- Teach sentences before conversational speech

- Do not assume, however, that simpler behaviors must always be taught before teaching more complex behaviors (e.g., phrases may be taught without teaching individual words in the phrases)
- Experiment with different sequences (generally, if a more complex behavior can be taught first, the more simple behaviors may be produced without training or maybe taught quite easily)

Response Cost. A direct response reduction strategy in which the production of each response scheduled for reduction results in the loss of a reinforcer; in the Earn and Lose variety, clients earn a token for every correct response and lose one for every incorrect response; in the Lose-Only variety, the client who receives unearned tokens at the beginning of a session loses one for every incorrect response; an effective procedure to reduce speech sound disorders and stuttering; also effective in reducing various behavior problems children exhibit during any kind of treatment session.

Earn and Lose

- Let the child select a backup reinforcer for the session (typically a small gift)
- Give a token for correct responses (to be exchanged for a backup reinforcer)
- Take a token away each time the client produces an incorrect response
- Exchange the tokens the client still possesses for a backup reinforcer at the end of the session

Lose-Only

- Let the child select a backup reinforcer for the session (typically a small gift)
- Give a certain number of tokens at the beginning of a session
- Take a token away each time the client produces an incorrect response
- Exchange the tokens the client still possesses for backup reinforcers at the end of the session

Response Generalization. Production of unreinforced (new, untrained) responses that are similar to trained responses; a goal of treatment; typically achieved through various strategies designed to promote Generalization because stimulus generalization in language training also involves response generalization.

Response Mode Generalization. Production of unreinforced responses in a mode not involved in training; an important treatment target.

- Train skills in a certain mode (e.g., fluency in oral reading)
- Probe to assess generalized production (fluency in conversational speech)
- If there is no generalization, train the skills in that mode (fluency in conversational speech)

Response Recording Sheet. A prepared sheet for recording correct, incorrect, and no responses given in treatment sessions.

- Design and use a response recording sheet similar to the following; modify as necessary

Name of Client	Treatment Target
Clinician	Date
Response Recording Sheet	
Stimulus Items	Responses: + (Correct), - (Incorrect), 0 (No response)
1.	
2.	
3.	
4.	

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Response Reduction Strategies. To use one of several methods of reducing undesirable behaviors during treatment, see Direct Methods of Response Reduction and Indirect Methods of Response Reduction.

Response Substitution. Increase in an undesirable behavior or the emergence of a new undesirable behavior when another behavior is reduced; exemplified by increased frequency of wiggling in the chair when a child's disruptive hand movements are reduced by a response reduction method.

- Apply a response reduction strategy to the newly emerged undesirable behavior
- Apply such strategies sequentially if new undesirable behaviors keep emerging
- Emergence of new undesirable behaviors are likely to stop when a few new responses are reduced

Response Unit. A training target in the Paired Stimuli Approach to treating speech sound disorders (described under Speech Sound Disorders: Specific Techniques or Programs); the client is asked to produce a key word and a target word as a single response unit (e.g., *this-bus*); the client earns one reinforcer only by correctly producing the target sound in both of the words.

Restitution. An element of overcorrection in which the person eliminates the effects of his or her misbehavior and then improves the situation; described under Imposition of Work.

- Ask the child who disorganizes your stimulus materials to reorganize them
- Next, ask the child to organize the toys on the floor when the disorganized toys were not the child's making

Retrograde Amnesia. To treat memory loss for events that preceded brain injury, also known as pretraumatic memory loss, see Traumatic Brain Injury in Adults.

Reversed Imitation. Clinician's imitation of a child's utterance during indirect language stimulation; it is not clear to what extent this strategy works; in the more typical operant Imitation, it is the client who imitates and the clinician who models; this strategy works.

Right Hemisphere Disorders. A constellation of symptoms due to brain injury sustained in the right cerebral hemisphere and characterized by various perceptual, attentional, emotional, and communicative deficits; varying degrees of functional involvement depending on the site, nature, and extent of damage; see the cited sources and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for etiology and symptomatology of right hemisphere disorders.

Treatment: General Considerations

- Use all treatment suggestions with caution as treatment research is limited, and for several procedures, nonexistent; recommendations are based on clinical experience of several clinicians; collect data during treatment and keep abreast of treatment research
- Counsel the family about communication treatment soon after the onset
- Target specific skills for teaching, instead of presumed underlying, often mental or cognitive processes; when underlying processes are treated, skills do not improve.
- Begin treatment as soon as it is practical
- Expect to spend significant treatment effort on re-establishing pragmatic language skills and overall communicative effectiveness instead of grammatical accuracy
- Select the client-specific treatment targets that:
 - Will result in the most improvement in family, social, and vocational communication
 - Help build other, more advanced communication skills
 - Help focus on communicative treatment targets (e.g., attentional deficits may have to be treated before other language skills)

- The clients can imitate
- Develop stimulus materials that:
 - Range from simple to progressively more complex and from fewer to greater number of elements
 - Are clear, unambiguous, and relatively concrete
 - Are familiar, meaningful to the client, and attractive
- Establish baselines of target behaviors
- Provide extensive and intensive practice
- Be aware that there is no controlled clinical evidence to support the use of computerized cognitive rehabilitation programs
- Structure treatment sessions initially and loosen them as the client becomes more proficient in producing the target responses
- Use instructions, modeling, and prompts in all stages of treatment
- Fade the special antecedents used in early stages of treatment
- Shape the target behaviors
- Give prompt and effective feedback
- Work with the family members to promote generalization and maintenance

Treatment Targets and Procedures

Treat Lack of Awareness of Problems

- Give immediate verbal feedback on communication problems, left neglect, and other kinds of errors you notice in all treatment sessions
- Give visual feedback on errors whenever possible (e.g., point to the neglected left side of a printed page the client reads aloud)
- Audio-record and replay the speech to the client and discuss the errors
- Teach the client Self-Control (Self-Monitoring) skills

Treat Impaired Attention

- Do not target attention as an independent skill; there is no compelling evidence to treat such skills as the following:
 - Letter cancellation (crossing out a target letter printed among other letters on the same page)

- Pressing a certain key on the computer key board when certain dots appear or disappear against a background of dots
- Treat attention during communication training; improving attention to treatment stimuli and treatment tasks, including talking on a specific topic without getting distracted, is more functional than isolated attending tasks:
 - Make attending to stimuli presented during communication training a treatment target
 - Reinforce both the communication target behaviors and attending behavior
 - Specify why you are reinforcing what action (e.g., “Mrs. Bobcat, thanks for looking at the picture!” “That was a nice description of the picture!”)
- Use the Shaping procedure to establish sustained attending behaviors
- Reinforce specific attending behaviors during communication training:
 - Reinforce the individual for paying attention to the stimulus material
 - Reinforce maintenance of eye contact
 - Reinforce extended talk on a given topic (concentration on a topic of conversation); stop the person when the topic diverges inappropriately
 - Reinforce the client for not watching extraneous stimuli when asked to talk about a particular part of the picture presented
 - Reinforce appropriate responses to questions you ask and commands you give; reinforcement of such responses also reinforces attention
- Structure the initial treatment sessions:
 - Limit the number of stimuli and the number of response choices in the initial stages of treatment
 - Conduct treatment in a quiet and distraction-free room
 - Conduct treatment in more naturalistic settings after the attending behaviors improve
- Prompt and give attention-getting cues before presenting the treatment stimuli, before giving instruc-

tions, or before asking the client to perform an action:

- Say, “Look at me” before modeling a response
- Say, “Look at the picture before you say something about it”
- Say, “Talk only about this part, not any other part of this picture” and point out the target stimulus part
- Say, “Get ready, here comes the next picture” before you present the next stimulus picture
- Say, “Please listen, I am going to tell you something”
- Say, “Be careful now! I want you to do something different!” and then describe the new action required of the client
- Touch the client before presenting a treatment stimulus to draw attention
- Give a nonverbal cue to draw attention (e.g., point to your face or to the stimulus picture to keep the client focused)
- Use only effective stimuli:
 - Vary the treatment stimuli
 - Drop unattractive stimuli that do not help sustain the client’s interest or attention
 - Use clear, colorful, and forceful stimuli
- Structure the treatment session to suit the individual
 - Watch for signs of fatigue
 - Give frequent, brief breaks in the initial phase of treatment
 - Reduce the number and duration of the breaks gradually as the client’s attention and performance improve
- Introduce gradually some distracting stimuli while still reinforcing attention to treatment tasks

Treat Left Visual Neglect

- To treat left neglect:
 - Force attention to the neglected side of the visual field by a variety of methods with several kinds of cues and prompts

- Reinforce the client for paying attention to the neglected side
- Reinforce attention to printed stimuli on the left side of the page by several specific techniques:
 - Teach the patient to keep a finger on the left margin while reading and track back to it before beginning a new line
 - Color the entire left-side margins to draw attention
 - Draw a colored line through the margin
 - Place a few colored dots on the left margin
 - Point to the left margin with a pencil or a ruler as the client reaches the end of the right margin
 - Tell the client to “Look to the left” when he or she reaches the end of the right margin
 - Fade all such special stimuli used to draw attention
 - Design reading materials with large print and progressively smaller print and ask the client to read them aloud
 - Impress on the client that what is read may not make sense because of missed material on the left side of the page
 - Reinforce the person for reading aloud the words on the left side
 - Interrupt reading each time the client misses words on the left side

Control Impulsive Behaviors

- Teach the client to wait and withhold responses so the responses eventually given are more likely to be correct
- Give nonverbal signals to delay responses soon after asking a question:
 - Instruct the client to respond only when you give a hand signal to respond
 - Give a hand gesture that suggests *wait*
 - Give a gesture that suggests *think*
- Give verbal signals to delay responses soon after asking a question:
 - Say, “Please wait for a few seconds before you answer”

- Say, “Please wait until I am finished”
- Say, “Please respond when I say *now*”

Treat Pragmatic Language Skills

- Teach the client to initiate conversation; have the client discuss several topics in which you teach specific skills by modeling them, ask the person to practice them, and reinforce the client for practicing them; for instance:
 - Teach the client to introduce the topic explicitly
 - Teach the client to give background information on narratives and stories
 - Teach the client to periodically restate the topic of discussion
 - Prompt the client to maintain focus on the main topic
 - Teach the client to ask such questions as “Do you follow me?” or “Do you understand?”
 - Give corrective feedback
- Teach the client to request clarification when messages are not understood; model and reinforce a variety of requests, for instance:
 - “Please repeat that”
 - “I do not understand”
 - “I am not sure what you mean”
- Teach the client to maintain eye contact during conversation
 - Frequently tell the client to look at your face
 - Give nonverbal signals to look at your face or eyes
 - Reinforce and offer corrective feedback as found appropriate
- Teach the client to maintain topics of conversation for reasonable lengths
 - Set progressively longer target durations for reinforcement (e.g., two minutes on the same topic, progressively increased to an appropriate length)
 - Start with simple topics and increase the complexity of topics progressively
 - Record and show video recordings of conversations to point out the problems and to reinforce improved performance

- Stop the individual at the earliest sign of departure from the topic
- Teach the person to pay attention to listeners on all sides
 - Impress on the person the need to look at people on the left side in group interactions
 - Arrange small-group interactions, preferably with family members
 - Frequently prompt the individual to look at persons on the left side; reinforce correct responses
 - Teach the client's conversational partners on the left side to frequently draw attention to themselves as the client speaks and ignore them
 - Teach the conversational partners to first draw the client's attention before speaking (e.g., "Look at me, I am going to say something to you")
- Teach the person to take turns in conversation
 - Stop the person for inappropriate turn-taking (e.g., interrupting you)
 - Instruct the person to speak when you say, "It is your turn"
 - Teach the person to say, "It is your turn" and yield the floor to you
- Teach appropriate prosodic features
 - Model and reinforce appropriate speech rate and rate variations
 - Model and reinforce normal patterns of intonation
 - Model and reinforce progressively louder speech if it is too soft
 - Model and reinforce inappropriate stress patterns within utterances
 - Take note of hypermelodic speech
 - Ask the patient to imitate an intonational pattern of a modeled sentence; take note of deviations in intonational patterns, including variations in stress and pitch
- Teach the client socially appropriate speech
 - Stop the client when he or she makes irrelevant, tangential, or inappropriate comments or jokes
 - Discourage excessive talking

- Consider the PACE program (Promoting Aphasics' Communicative Effectiveness; described under Aphasia; Treatment of Aphasia: Specific Techniques or Programs) for teaching social communication

Treat Impaired Reasoning Skills

- Teach the individual planning activities
 - Ask the person to describe how he or she might plan a vacation
 - Ask the person how he or she might solve a billing problem with the credit card company or utilities
 - Reinforce when the client describes the steps in a logical manner
 - Stop and question the client when the steps described are unreasonable, illogical, or potentially unproductive
- Pose different kinds of problems one might encounter in real life and ask the client to solve them
 - Ask, "How do you buy an airplane ticket?"
 - Ask, "How do you prepare a peanut butter sandwich?"
 - Reinforce and provide corrective feedback as necessary

Treat Impaired Inference

- Teach correct or reasonable inference by a variety of strategies:
 - Tell stories and ask questions to have the client describe information not explicitly stated, but implied in the stories
 - Show pictures (such as those by Norman Rockwell) and ask the client to infer what is going on and what had just gone on
 - Show such line drawings as that of a dog on a beach, shaking its body, and ask what the dog had been doing
 - Show pictures that show sad or happy faces and ask the client why they might be so
 - Prompt correct inferences and give corrective feedback on incorrect ones

Treat Impaired Recognition of Absurdities

- Show pictures that depict logical and absurd events and ask the client to separate them; for instance:
 - Show the picture of a cat chasing a rat and the picture of a rat chasing a cat
 - Ask the client to point out the one that is correct
 - Ask the client to point out the one that does not make sense (absurd)
- Present verbal or written statements that are logical or absurd and ask the client to separate them; for instance:
 - Present printed absurd and sensible statements and ask the client to separate them (e.g., “Yesterday, I was so hungry when I was coming home from work that I went to a park and sat there” versus “Yesterday, I was so hungry when I was coming home from work that I stopped by a restaurant to eat before reaching home”)
- Present a series of absurd statements without their logical counterparts and ask the person to explain why a statement is absurd
- Show pictures that depict absurd situations and ask the individual to explain why they are absurd

Treat Impaired Comprehension of Proverbs, Metaphors, or Idioms

- Select proverbs, metaphors, or idioms that are appropriate for the client’s level of education and premorbid language skills
 - For each abstract statement (proverb, metaphor, or idiom), write a literal interpretation and the abstract meaning it implies
 - Ask the client to select statements that give literal meanings
 - Ask the client to select statements that give figurative (abstract) meanings
 - Prompt correct responses
 - Reinforce and give corrective feedback
 - Tell a few proverbs and ask the person to state the abstract meanings

- Ask the person to state proverbs and explain their abstract meanings
- Start with more common figurative or abstract expressions and move on to more complex expressions

Treat Comprehension of Humor

- Use cartoons to teach correct comprehension of humor
 - Show cartoons and their captions separately
 - Ask the client to study the cartoons and then match them with their captions
 - Ask the client to say what is humorous about the cartoon
- Ask the client to read a few jokes whose body of the joke and the punch lines are separated
 - Ask the client to match the joke with its punch line
 - Ask the client why a joke is funny

Teach Functional Reading and Writing Skills

- Note that premorbid reading skills may be re-established in the context of treating left neglect
- Give feedback on all kinds of reading errors, not just left neglect
- Treat writing skills only if the client needs them
- Ask the client to write a paragraph and give feedback as the client writes (e.g., repetition of letters, poor letter formation, confused spacing, crowding of the letters on the right side, failure to give the left margin)
- Dictate a brief passage and give feedback on errors

Teach Compensatory Strategies

- Teach the use of functional memory aids; teach the client to
 - Make lists of things to do
 - Write down appointments
 - Write a checklist of things to do before leaving the house
 - Keep related things together in one place and the same place (e.g., the house key and the wallet)
- Teach Self-Control (Self-Monitoring) skills, including the generation of memory aids

- Teach the client to stop and self-correct when errors are made
- Make a few socially appropriate and inappropriate statements and ask the client to evaluate them
- Have the client evaluate social appropriateness of his or her own productions
- Teach the client to rephrase inappropriate comments to make them more appropriate

Blake, M. L. (2018). *The right hemisphere and disorders of cognition and communication: Theory and clinical practice*. San Diego, CA: Plural.

Hegde, M. N. (2018). *A coursebook on aphasia and other neurogenic language disorders* (4th ed.). San Diego, CA: Plural.



S

Sample. A smaller number of individuals selected from a larger population for a research study; participants included in a treatment research study; some may receive treatment (the experimental group) and others may not (control group).

Satiation. Temporary termination of a drive or need because it has been satisfied; a potential problem with Primary Reinforcers (e.g., food); such reinforcers lose their effect as the client experiences satiation; to control this problem in treatment sessions:

- Give only small amounts of food or drink to reinforce individual responses
- Let the client accumulate food that he or she can eat later
- Arrange treatment sessions, especially with infants and toddlers who need primary reinforcers, around breakfast or lunchtime

Schedules of Reinforcement. Different patterns of reinforcement that generate different patterns of responses; include Continuous Reinforcement schedules and Intermittent Reinforcement schedules.

- Use continuous reinforcement schedule in the beginning of treatment
- Shift to an intermittent schedule as learning becomes more stable

Scripts. Written descriptions of frequently occurring, sequenced events, episodes, or personal experiences; used in teaching advanced language skills, including narrative skills; a description of *baking cookies* or *running a lemonade stand* is a script; it has a beginning and an end, actions people take, or roles people play; for procedures, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Script Therapy. To use Scripts in teaching language skills to children with language disorders, see Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Secondary Reinforcers. Conditioned reinforcers whose effects depend on past learning; appropriate reinforcers for all kinds of verbal responses except for certain kinds of mands that request food and drink; include Social Reinforcers, Conditioned Generalized Reinforcers, Informative Feedback, and High Probability Behaviors.

Self-Control (Self-Monitoring). A behavior that monitors and modifies other behaviors of the same person; a necessary skill in maintaining clinically learned communicative behaviors; see Maintenance Strategy; teach such skills as the following to promote self-control or self-monitoring of newly learned skills:

- Teach clients to discriminate their own incorrect and undesirable responses; for instance:
 - Model a child's misarticulated sound productions and ask the child to listen to them
 - Demonstrate a stuttering person's associated motor behaviors
 - Model a child's phrases or sentences that do not contain required grammatic morphemes (e.g., "Two cup," "Boy walking")
 - Play back a recorded sample of client's speech to point out undesirable voice quality
- Teach clients to contrast their desirable and undesirable responses; for instance:
 - Contrast easy and hard glottal attacks of a voice client or a fluency client by modeling the two kinds of phonatory initiations
 - Contrast incorrect and correct productions of phonemes by modeling them
 - Contrast correct and incorrect sentence forms (e.g., "Two cup," "Two cups")
 - Contrast the desirable lower pitch with the client's excessively high pitch by modeling the two pitch levels or by audio-recording the client's voice that might catch both the levels
- Teach clients to measure their behaviors reliably; for instance:

- Ask a client to measure his or her dysfluencies in select- ed situations outside the clinic and submit the data
- Ask a client to measure the frequency of his or her vocal- ly abusive behaviors for three days and submit the data
- Ask a client to measure his or her misarticulations in certain specific situations and submit the data
- Have clients audio-record a home speech sample and play it as you give feedback on the target behaviors to be measured
- Let clients measure their behaviors along with you in treatment sessions; note that children, too, can learn to make a tally mark to measure correct productions of sounds, grammatic morphemes, voice qualities, and so forth; for instance:
 - In the treatment sessions, ask the client to chart his or her own correct and incorrect productions of target phonemes along with you
 - Ask a fluency client in therapy to chart his or her own dysfluencies, hard and abrupt voice onset, excessively fast speech, easy onset, appropriate rate
 - Ask an apraxic client to chart correct and incorrect articulatory movements
 - Ask a child in language therapy to chart sentences with and without the target grammatic features
 - Ask a voice client in therapy to chart his or her desir- able vocal qualities
- Give clients frequent feedback on their measurement to improve their skill; for instance:
 - Give all clients feedback when they fail to record their correct and incorrect responses; point out the charac- teristics of those responses, model them if necessary, and ask them to resume their charting
 - Throughout the session give them informative feed- back (“You and I agreed on three of the five instances”; and “This time, you and I agreed on four of the five instances”)
 - Monitor the progress and make sure the clients’ skill in measuring their behaviors improves over the sessions
- Teach clients to monitor their newly learned skills by measuring them outside the clinic; for instance:

- Ask the fluency client to keep a record of his or her use of slower rate or gentle phonatory onset; minimally, ask the client to gain a clear impression of the frequency of their use
- Ask a voice client with excessively soft voice to keep a record of the number of social situations in which he or she spoke with adequate loudness
- Ask a client with aphasia to keep a record of the number of times he successfully recalled his wife's name in a given day
- Teach clients to generate signals and prompts for their own actions in the natural environment; for instance:
 - Ask a client with aphasia to write down the names of family members and frequently consult the list before engaging in conversational speech
 - Ask a client to wear electronic devices that remind him or her of appointment, to slow down the speech, to speak more softly or loudly, and so forth
- Teach clients to pause after they produce a wrong response both in the clinic and outside the clinic; for instance:
 - Teach a client who stutters to pause briefly after each dysfluency and say the same word fluently without your help
 - Teach a voice client to pause soon after a hard glottal attack is produced and start again with a softer attack without your help
 - Teach a child in articulation therapy to stop as soon as he or she produces a target sound incorrectly without your help
 - Teach a client in apraxia therapy to stop as soon as a wrong articulatory gesture is initiated and then initiate the correct gesture without your help
- Teach clients to correct their own mistakes or nontarget responses
 - Initially, give corrective feedback for every incorrect response
 - Tell the client that you will on occasion not give correct feedback and the client should catch himself or herself making a wrong response and immediately correct the mistake without help

- Withhold corrective feedback on certain occasions
- If the client does not immediately self-correct, give corrective feedback; remind the client that he or she should self-correct without your help
- Continue the procedure until the client reliably self-corrects; reduce the frequency of corrective feedback further while maintaining self-corrections
- Teach clients to anticipate problems and take corrective actions
 - Teach a person who stutters to practice saying “Hello” with appropriate airflow when the telephone starts ringing
 - Teach a person with aphasia to practice names of people who are expected to visit shortly
 - Teach a voice client with vocal abuse to reduce talking before a scheduled and unavoidable speech to be given to a group
 - Teach another voice client to drink more water to ward off vocal problems related to vocal dryness
- In group therapy, teach clients to correct other participants’ errors
 - In a group therapy involving children who have received treatment for stuttering, teach each child to monitor other children’s speech rates and to give appropriate feedback
 - In group therapy involving individuals with aphasia, teach each to remind the other to practice specified skills (e.g., self-cueing words with the first phoneme of the word)
 - In language group therapy, teach each child to monitor the use of a specific grammatic feature in other children and give appropriate feedback

S

Sequence of Treatment. Movement within a treatment program from the beginning to the end; a description of steps involved in implementing a treatment plan; progression of treatment from a simple to a more complex level; see Treatment in Communicative Disorders: General Sequence.

Service Delivery Models. Different arrangements of providing assessment and treatment services to children with communication disorders include the Collaborative Model, the Consultant Model, the Language-Based Classroom Model, and the Pull-Out Therapy Model.

Shaping. A method of teaching nonexistent responses that are not even imitated; also known as successive approximation; supported by experimental evidence; highly useful in teaching a variety of skills.

- Select a terminal target response (e.g., the production of /m/ in word-initial positions)
- Identify an initial component of that target response the client can imitate (e.g., putting the two lips together)
- Identify intermediate responses (e.g., humming or other kinds of vocalizations, opening the mouth as humming is continued)
- Teach the initial response by modeling and immediate positive reinforcement (e.g., putting the lips together on several trials)
- In successive stages, teach the intermediate responses (e.g., adding humming when the lips are closed; opening the mouth when the humming is continued; adding other sounds to form words)
- Continue until the terminal response is taught

Side Effects of Punishment. Undesirable effects of punishment procedures; include emotional reactions, aggressive reactions, unexpected increase in the punished response, increase in a different response than the one punished, and so forth; to be eliminated or minimized by prudent use of response reduction methods.

- Always use positive reinforcers for desirable behaviors
- Keep reinforcement:punishment ratio in favor of reinforcement
- Shape a difficult response to avoid using response reduction methods
- Consistently apply response reduction methods to all undesirable behaviors

- Remove or reduce reinforcement for undesirable behaviors
- Never associate response reduction methods with positive reinforcement
- Use Differential Reinforcement techniques to increase desirable behaviors that replace undesirable behaviors

Significant Others. People who typically interact with a client on a daily basis; include family members, teachers, friends, colleagues, baby sitters, and health care workers; important in helping the client learn and maintain communicative behaviors; recipients of training within a Maintenance Strategy.

Single-Subject Experimental Design Strategy.

Methods of demonstrating treatment effects by showing contrasts between conditions of no treatment, treatment, withdrawal of treatment, and other control procedures; typically, all subjects selected receive treatment because the method does not use a control group of individuals who are denied treatment; includes, among others, ABA Design, ABAB Design, and Multiple Baseline Designs; contrasted with Group Design Strategy, in which the participants in a control group are denied treatment.

Social Reinforcers. A variety of conditioned reinforcers frequently used in treatment sessions; include verbal praise, attention, touch, eye contact, and facial expressions; resistant to satiation effect; may not work with nonverbal clients.

- With children who are nonverbal, profoundly retarded, and very young (infants and toddlers), pair social reinforcers with primary reinforcers
- Eventually, fade the primary reinforcers and maintain the responses on social reinforcers only
- Do not use to reinforce requests for food and drink, as the natural reinforcers for such mands are food and drink

Spasmodic Dysphonia. Treatment of this voice disorder, characterized in most cases by severe overadduction of

vocal folds and strained or choked-off voice quality (in the adductor type) or excessively breathy and irregular voice (in the abductor type), has been both difficult and controversial, partly because its etiology is poorly understood; traditional voice therapy techniques have been unsuccessful in producing lasting treatment effects; some clinicians combine voice therapy with medical management in a team effort; see the cited sources and *Hegde's PocketGuide to Communication Disorders* for etiology and symptomatology of spasmodic dysphonia; see the cited sources and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for assessment procedures; current medical management includes:

- For the abductor type, injection of Botox (botulinum toxin A) into one or both of the vocal folds to induce paralysis of the folds; some data suggest that unilateral injection may be better than the bilateral, although generally the outcomes have been unsatisfactory
 - Initially, the voice is weak and breathy, which improves with voice therapy
 - Re-injection may be needed, as the effects last only a few months
- For the adductor type, surgical, medical, and behavioral approaches are available, although the results are mixed
 - Recurrent laryngeal (laryngeal adductor) nerve sectioning; a surgical procedure in which the recurrent laryngeal nerve is resectioned, selectively sectioned, or crushed; the effects last only a few months; in some cases, the symptoms worsen; the laryngeal nerve also may regenerate, resulting in the symptoms' return; several other surgical procedures have been tried with varying degrees of success
 - Botulinum toxin type A injection has been used to paralyze one or both vocal folds to eliminate vocal spasms and other symptoms of adductor spasmodic dysphonia; relapse is common; need to be combined with voice therapy; procedures to raise the pitch, head position changes, and digital manipulation of the larynx have all been clinically tried and found to be helpful to varying degrees

- Voice clinicians may try several more or less traditional approaches, including inverse phonation (phonation in inhalation), slightly elevated pitch, whispered speech, breathy voice, singing and humming that are shaped into speech; relaxed phonation, relaxation of the laryngeal muscles and the general body; expect only moderate effects
- A newer, behavioral approach is *conversational training therapy* designed to increase awareness of one's own voice and techniques to produce efficient voice in conversational narratives from the very first session; needs more research

Sapienza, C., & Rudy, B. F. (2018). *Voice disorders* (3rd ed.). San Diego, CA: Plural.

Stemple, J. C., Roy, N., & Klaben, B. K. (2014). *Clinical voice pathology* (5th ed.). San Diego, CA: Plural.

Spastic Dysarthria. To treat this type of motor speech disorder, see [Treatment of Dysarthria](#); [Dysarthria: Specific Types: Spastic Dysarthria](#).

Specific Language Impairment (SLI). Language disorders in children who do not have any other significant neural, behavioral, or intellectual disabilities; impairment is limited to language; in 2017, an international panel of experts renamed SLI *Developmental Language Disorder* (DLD); because both the terms—SLI and DLD—pose theoretical and clinical problems, a more descriptive and theoretically neutral term, *Language Disorders in Children*, is preferred in the three *PocketGuides*; for treatment procedures, see [Language Disorders in Children](#).

Bishop, D. V. M., Snowling, M. J., Thompson, P. A., Greenhaig, T., & the CATALISE-2 Consortium (2017). Phase 2 of CATALISE: A multinational and multidisciplinary Delphi consensus study of problems with language development: Terminology. *Journal of Child Psychology and Psychiatry*, 58, 1068–1080.

Specific Normal Voice Facilitating Techniques. A collection of voice therapy procedures used to promote normal voice productions; see [Voice Disorders](#); [Specific Normal Voice Facilitating Techniques](#).

Speechreading. Understanding speech by watching the face of the speaker as well as using the residual hearing; while Lipreading involves watching mostly the lips and tongue, speech reading includes all available visual and auditory cues.

Speech Sound Disorders. Treatment of speech sounds in children is a major responsibility of speech-language pathologists; children who misarticulate their speech sounds constitute a significant workload of school clinicians; difficulty in producing a few sounds with no pattern or derivable rule is often treated as an articulation disorder; multiple errors that can be grouped on some principle or characteristics and thus form patterns are typically treated as phonological disorders; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for the epidemiology, ethnocultural factors, symptomatology, etiology, and theories of Speech Sound Disorders.

Treatment of Speech Sound Disorders: General Guidelines. Consider the issues and approaches in treating speech sound disorders:

- *Phonetic vs. phonemic approach.* A pseudo issue because all speech sound disorders are remediated by teaching production (phonetic) of sounds in error; there is no phonemic or linguistic treatment procedure for any disorder of communication.
- *Treatment targets.* Treatment targets may be defined either as the correct production of phonemes in error or the remediation or elimination of phonological pattern; note also that either way, individual sounds need to be taught; there is no other way to eliminate a phonological pattern; correct production of all phonemes in conversational speech produced in natural settings is the final treatment target
- *Sequencing the targets sounds.* Although all sounds need to be taught, misarticulated multiple sounds cannot be taught all at once; need to select certain sounds for initial training, certain others for subsequent

training, and the remaining sounds for training during the final stages of therapy; consider the following varied and some contradictory recommendations:

- Select sounds that are *functional* for the child for immediate training; sounds that are important from the standpoint of social and academic performance (e.g., even though /r/ may not be an initial target for all 4-year-old children who misarticulate that sound, it may be for a child named Robert Roberson, who refers to himself as *Wobert Woberson*; in this case, /r/ is *functional* for the child); least controversial
- Select misarticulated sounds that when corrected, will produce the greatest improvement in intelligibility; select the errors that have the greatest negative effect for initial correction; also not controversial
- Select for immediate training sounds the child imitates correctly (stimulable sounds) when modeled; contrarily, select the sounds that are most difficult for the child, the sounds that are not even imitated because when taught they have the most beneficial generalization effect; two opposing views
- Select *high frequency* sounds for immediate training; correct production of those will improve intelligibility faster and more noticeably than correcting sounds that occur less frequently; see [Frequency of Occurrence of English Consonants](#); contrarily, select sounds of low frequency that tend to be more difficult sounds; correction of such difficult sounds make the rest of the teaching easier and may produce better generalization
- Teach first sounds that are normally *acquired earlier*, because they may be easier to teach; there is very little experimental evidence to suggest that sounds acquired earlier need to be taught earlier or that the sounds need to be taught in their normative sequence; contrarily, select sounds that are *normally acquired later*, not earlier, because when corrected, they produce the greatest change in the child's sound system; there is some evidence to support this

- Select for immediate training sounds that the child produces at some low-level on assessment and baseline trials because they may be taught more easily; contrarily, select sounds the child does not produce at all because they tend to be more complex, and when taught they produce greater degree of generalization
- Select for elimination phonological patterns that are less frequent in the child's speech, less complex, and more common across children, because the sounds in those patterns may be easier to teach first; contrarily, target first the elimination of phonological patterns that affect the greatest number of sounds, patterns that are idiosyncratic, patterns whose occurrence is 100% because such an approach yields better and quicker speech intelligibility
- Evidence is emerging that better generalization and intelligibility is achieved by addressing the more difficult sounds and patterns first
- Regardless of the strategy used, always select sounds that are appropriate for the child's ethnocultural and linguistic background; select sounds that are in the child's dialect; select sounds that are not in the child's dialect only when the child or the parents demand their teaching
- Select academically useful words to teach specific sounds; consider the academic demands made on the child; consult with teachers in selecting words, phrases, and sentences
- *Sequencing the Response Topography.* Use the following guidelines in sequencing response topography (levels of response complexity at which the sounds are trained); consider Specific Techniques or Programs described in a later section for somewhat varied recommendations:
 - Teach the target sounds at the highest level of response complexity that the individual can initially handle with success
 - Do not begin teaching at a lower level if the individual can handle training at a higher level (e.g., do not

train at the syllable level if in the initial treatment trials the individual can learn to imitate the modeled productions at the word level)

- Begin training a sound minimally at the word level; if practical at the phrase level; or even at the sentence level; expect in most cases to begin training at the word level
- Drop to syllable level only if the child fails to imitate the modeled production of sounds in words on several trials
- Drop to the isolated phoneme level only if the child fails to imitate the modeled production of sounds in syllables on several trials
- To begin with, experiment with a higher level; you can always drop to a lower level if necessary
- Move up to a higher level of complexity when training is completed at a lower level (e.g., move on to training the sound in phrases when it has been trained in words)
- End training at the level of conversational speech produced in more naturalistic settings

Ethnocultural Considerations in Treating Speech Sound Disorders

- Note that variables related to ethnic, cultural, and linguistic background of individuals significantly affect the assessment, diagnosis, and treatment of articulation disorders; see the cited sources and the two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, for ethnocultural factors and their assessment in individuals with varied ethnocultural background
- Do not recommend treatment for an individual who speaks a different dialect of English, as all dialects of a language are its accepted forms with their own cultural heritage; note that this is the official position of the American Speech-Language-Hearing Association on social dialects

- See Treatment of Speech Sound Disorders in African American Children in this section to design an appropriate treatment program for such children
- See Treatment of Speech Sound Disorders in Bilingual Children in this section to design an appropriate treatment program for such children
- Recommend treatment only if there is a speech sound disorder within the dialectal framework of an individual's speech
- Offer treatment to teach the sound patterns of mainstream English only if the individual's errors are not due to the influence of a primary language, an accepted social dialect, or African American English
- Offer treatment to change someone's otherwise acceptable dialect (such as African American English or Spanish-influenced English) only if the individual, the family, or both seek it

A Comprehensive Treatment Program for Speech Sound Disorders

- Select the target speech sounds for modification:
 - Use previously described criteria for selecting treatment targets (functional targets, sounds that greatly improve intelligibility, sounds that when taught cause significant generalization)
 - Sequence the target sounds for training them in the early, middle, and final treatment phases using the previously described guidelines
 - Group the target sounds—especially in the case of multiple misarticulations—according to phonological patterns (e.g., elimination of the final consonant deletion pattern)
- Write measurable objectives for each target sound; include in each objective statement:
 - The response to be taught (e.g., The production of /s/ . . .)
 - The complexity level at which the response will be taught initially (e.g., The production of /s/ in words . . .)

- The position of the target sound in words (e.g., The production of /s/ in word-initial positions . . . ; and similar statements for other word positions)
- The performance criterion to judge success (e.g., The production of /s/ in word-initial positions *at 90% accuracy in a set of 10 training words* . . .)
- The response setting (e.g., The production of /s/ in word-initial positions *at 90% accuracy in a set of 10 training words* . . .); note that for initial training, only one setting—the clinical setting—may be specified, and as training advances, new objectives may be written to include additional settings
- The number of speech samples or sessions in which the production of sounds is measured (e.g., The production of /s/ in word-initial positions at 90% accuracy in a set of 10 training words *measured across three consecutive speech samples or treatment sessions*)
- Write different measurable target objectives for different levels of response complexity
 - At the word level (e.g., The production of /l/ in word-final positions at 90% accuracy in a set of *10 training words* measured across three consecutive speech samples recorded in the clinic)
 - At the phrase level (e.g., The production of /t/ in word-medial positions at 90% accuracy in a set of *10 two-word training phrases* measured across three consecutive speech samples recorded in the clinic)
 - At the sentence level (e.g., The production of /d/ in word-initial positions at 90% accuracy in a set of *10 training sentences* measured across three consecutive speech samples recorded in the clinic)
 - At the conversational speech level (e.g., The production of /k/ in word-medial positions at 90% accuracy in three consecutive, spontaneous conversational speech samples recorded in the clinic)
- Write different measurable target objectives for different settings
 - For the clinical setting (e.g., The production of /k/ in word-medial positions at 90% accuracy in three

consecutive, spontaneous conversational speech samples produced in the clinic); write similar objectives for other word positions and for phrases and words

- For the setting just outside the clinic, but within the clinic premises (e.g., The production of /z/ in 10 training phrases at 90% accuracy produced outside the clinic but within the general clinical setting); write similar objectives for other word positions and for words and sentences
- For the individual's home setting (e.g., The production of /k/ in conversational speech at 90% accuracy in three consecutive samples evoked by parents in their home setting); write similar objectives for other sound targets; note that when the parents do the home training and measurement, the response mode is typically conversational speech; parents are rarely asked to pay attention to position of sounds in words, although they can monitor sound productions in words
- For the individual's classroom (e.g., The production of /t/ in conversational speech at 90% accuracy in three consecutive samples evoked by the teacher in the child's classroom setting); write similar objectives for other sound targets; note that when the teachers monitor production of target speech sounds, the response mode is typically conversational speech; teachers are rarely asked to pay attention to position of sounds in words, although they can monitor sound production in words
- For the individual's school setting other than the classroom (e.g., The production of /t/ in conversational speech at 90% accuracy in three consecutive samples evoked by the teacher in the school's dining hall); write similar objectives for all target phonemes to be monitored in all academic settings (e.g., library, playground)
- For the individual's varied settings (e.g., *The production of /b/ in conversational speech at 90% accuracy in three consecutive samples evoked by*

family members or others in such natural settings as restaurants and shopping centers); write similar objectives for all target sounds; note that when the production of a target sound is monitored in natural contexts and settings, the response mode is usually conversational speech

- Prepare stimulus materials to be used in therapy sessions
 - Prepare 20 words, phrases, and sentences for each target sound in each word position (i.e., 20 words *each* for initial, medial, and final word positions; 60 target words for each sound to be trained); select picturable words to the extent possible
 - Select pictures, drawings, objects, or a combination of these to evoke each target word, phrase, or sentence; preferably, select colorful and realistic pictures from popular magazines
 - Test the stimulus materials with the child to make sure that the child is familiar with them; show the materials to parents to get their feedback; replace any ethnoculturally inappropriate stimulus material (including picture, objects, words, and phrases) with those that are appropriate, familiar to the child, and approved by the parents
- Design a treatment response recording sheet of the following kind; modify as necessary with the required features of space available to list the target behaviors and score the child's responses as correct, incorrect, or absent (no response):

Treatment Response Recording Sheet	
Name:	Treatment target:
Clinician:	Date: Session #:
Target Behavior: Correct production of /s/ in word-initial positions	Responses (+ = correct; - = incorrect; 0 = no response; m = modeled; e = evoked)
1. soup	+ m
2. sun	- m
3. soap	+ m
4. seed	0 m
. . . 20.	

Note that in the beginning, all trials will be modeled; subsequently, when modeling is dropped, evoked trials are administered.

- Write training and probe criteria that help make clinical decisions throughout training:
 - *Imitative accuracy criterion:* When the child gives five consecutively correct imitated responses, modeling will be discontinued
 - *Modeling reinstatement criterion:* When the child gives three incorrect responses on evoked (without modeling) trials, modeling will be reinstated and then withdrawn after five correct imitated responses as before; reinstated and withdrawn in this manner
 - *Shaping criterion:* When the child fails to correctly imitate the target sound in the first word selected for training on 10 consecutive trials, a different word will be selected for training; if the child fails to imitate the sound in the second word in 10 trials, the training of the sound will be initiated in syllables or at the level of isolated sounds; when the child gives five correct responses at the phoneme level, the training will shift to syllable level; five correct

responses at the syllable level will move training to the word level

- *Training criterion:* A 90% accurate production of a sound at each level of response complexity (i.e., 90% accuracy in sound production at the word, phrase, and sentence levels evoked on a block of 10 trials; and 90% accuracy in conversational speech with at least 20 productions of the target sound)
- *Probe criterion:* A 90% accuracy in the production of sounds in untrained words presented on a series of Probe trials with at least 10 untrained words
- *Probe timing and frequency criterion:* A probe will be administered when a child meets the training criterion on 4 to 5 stimulus items or exemplars (e.g., when the child's correct response rate on 4 words—*soup*, *soap*, *sun*, and *seed*—reaches 90% accuracy on a block of 10 evoked training trials, a probe will be conducted with at least 10 untrained words interspersed with trained words; the same criterion for all word positions, phrases, and sentences)
- *Reinstatement of training criterion:* Treatment on the same sound in the same word position will be reinstated if the probe criterion is not met (similar criterion for phrases and sentences); after training the sound in four new words, a probe will be conducted again; training and probes will be alternated until the probe criterion of 90% accuracy in untrained words is achieved
- Establish Baselines of target sounds in words, phrases, sentences, and conversational speech
 - Measure the correct production of target sounds in conversational speech; use the initial sample recorded during assessment; repeat the procedure to obtain another sample in the first intervention session; if the initial assessment sample and the first intervention-session sample show different percentages of errors for the same sound, repeat the sample

- Measure the production of target sounds in words with the discrete trial procedure; administer one set of trials with modeling and one without:
 1. Place a picture or an object in front of the child
 2. Ask a relevant question (e.g., “What is this?”)
 3. Model the correct response (e.g., “Say *soup*”); emphasize the target sound
 4. Wait a few seconds for the child to imitate
 5. Move the picture toward you and record the sound production as correct, incorrect, or no response
 6. Re-present the same picture; ask the same question; but do not model the response; record the response as before
 7. Administer all target words once with modeling (modeled trial) and once without (evoked trial)
 8. Calculate percent correct imitated and evoked response rate for each sound and each word position
- Teach the baserated speech sound production in words, phrases, and sentences using the discrete trial procedure
 - Initially, train the target sound at the word (or syllable) level with Modeling:
 1. Describe the target sound and how it is produced; demonstrate its production; and point out its Phonetic Placement; if necessary, use Manual Guidance to move articulators to their correct production; see Sound-Evoking Techniques for English Consonants that describe specific instructions and manipulations that facilitate the production of specific sounds
 2. Place a picture or an object in front of the child
 3. Ask a relevant question (e.g., “What is this?”)
 4. Model the correct response (e.g., “Say *sun*”); emphasize the target sound
 5. Wait a few seconds for the child to imitate

6. Positively reinforce the child by verbal praise, a token, or a primary reinforcer as soon as he or she correctly imitates the sound
 7. Give corrective feedback if the child does not imitate correctly (say, "That is not correct; you said _____; I want you to say _____"); use Time-out or Response Cost to control incorrect sound productions and uncooperative or interfering behaviors
 8. Move the picture toward you and record the sound production as correct, incorrect, or no response
 9. Re-present the same picture; repeat the instructions on correct production; show the phonetic placement for its production; ask the same question; model the response; record the response as before
 10. Continue this training with modeling until the child gives five consecutively correct imitated responses; discontinue modeling
 11. If the child fails to imitate the sound in a word in 10 trials, use shaping by dropping to the syllable or isolated phoneme production level; when the child correctly imitates the phoneme on 5 trials, move up to the syllable level; with 5 correct imitative responses at the syllable level, move up to the word level; with 5 correct imitations at the word level, discontinue modeling
- Administer evoked trials
 - Omit modeling on evoked trials:
 1. Present the same picture, ask the same question, but do not model
 2. Reinforce the correct responses or give corrective feedback for incorrect responses; repeat instructions, with demonstrations of phonetic placement if necessary
 3. Continue this training without modeling until the child gives nine correct responses in a block

- of 10 trials; when this happens, consider the particular word tentatively trained
4. Select another word with the same target sound and begin training with modeling; discontinue and reinstate modeling as before; drop to phoneme or syllable level if necessary; move up to the word level; when the child gives nine correct evoked responses in a word in a block of 10 trials, consider this word tentatively trained; follow this procedure until the child is trained on at least four words, perhaps up to six words
- Probe for generalized production of the trained sound:
 - Conduct a probe to assess generalized production of trained sounds in untrained contexts
 - Conduct an Intermixed Probe initially, with at least 10 untrained words intermixed with trained words; alternate trained and untrained words; reuse trained words; note that untrained words are also called the probe words; correct production of the sound in probe words indicates that the child will produce the sound in new and untrained words
 - Prepare an Intermixed Probe Recording Sheet as shown here and record the probe responses
 - Do not model correct productions because all probe trials are evoked only
 - Reinforce or provide corrective feedback for sound productions only in already trained words; do not reinforce or provide corrective feedback for correct or incorrect productions in untrained (probe) words; see also Probe Procedure

Intermixed Probe Response Recording Sheet	
Name:	Treatment target:
Clinician:	Date: Session #:
Target Behavior: Correct production of /s/ in word-initial positions	Responses (+ = correct; - = incorrect; 0 = no response)
1. soup (trained)	
2. superman (untrained)	
3. sun (trained)	
4. sunflower (untrained)	
5. soap (trained)	
6. soda (untrained)	
7. seed (trained)	
8. seal (untrained)	
Note: Repeat the trained words until at least 10 probe words are administered.	

- Calculate the percent correct probe response rate based only on the number of probe words, ignoring the trained words in this calculation (e.g., If the child's production of the target sound was correct in 5 of the 10 probe words, the correct probe response rate is 50%)
- Alternate between probe and training
 - Provide additional training with new training words if the child's probe response rate is below 90%; after training four or six more words, administer another intermixed probe; if the probe criterion is not met, train more words and probe again until the child meets the probe criterion
 - Administer a Pure Probe when the intermixed probe criterion is met; present only the untrained words; calculate the percent correct pure probe response

rate; if it is below 90% correct, provide a few additional training trials on all the trained words until the pure probe criterion is met

- Switch training to another target sound or the same sound in a different word position; follow the same procedure as outlined to complete training the sound at the word initial, medial, and final positions; train other sounds at the word level with the same procedure
- Shift training to higher levels of response complexity
 - Construct two- or three-word phrases that contain a word in which the sound is produced correctly (i.e., when a sound meets either the intermixed or pure probe criterion for a given word-position)
 - Administer the phrases the same way as the words; model initially; withdraw modeling when at least five correct imitations are produced; reinstate modeling if incorrect responses return; withdraw modeling again
 - Administer an intermixed probe after training four to six phrases to the same training criterion as 90% correct on a block of 10 trials; provide additional training on new phrases if the intermixed probe criterion is not met; administer a probe when the intermixed probe criterion is met; when the pure probe criterion is met, shift training to sentence levels in which the word with correct production of the target sound is included
 - Train sentences in the same manner as words and phrases; after presenting the stimulus item, ask a question that would evoke a sentence (e.g., ask, “What is the boy doing?” to evoke “He is having some soup”); initially, model the sentences and withdraw modeling; probe generalized production of sounds in conversation in the same way as described for words and phrases
 - Shift training to conversational speech level when the pure probe criterion is met for sentences; use pictures and other stimulus materials to engage

the child in conversational speech and positively reinforce correct productions on an Intermittent Reinforcement Schedule; prompt the correct productions; give corrective feedback

- Conduct periodic pure conversational probes in which you engage the child in conversation without reinforcement or corrective feedback; continue training until the child meets the 90% correct production of all target sounds in conversational speech with little or no prompting and reinforcement
- Integrate components of various programs of articulation treatment described under Treatment of Speech Sound Disorders: Specific Techniques or Programs into this comprehensive treatment program; for instance:
 - Use aspects of the Contrast Approach; in using the Minimal Pair Contrast Method, write minimal pair contrast words for training each sound; for instance, in teaching a child to produce final consonants (elimination of final consonantal deletion), write such word pairs as *bow-boat*, *bee-beet*, *toe-toad*, *pie-pine*; to teach the correct production of a single phoneme such as /b/ for which the child substitutes /p/, write such pairs as *bye-pie*, *bat-pat*, *beat-peat*, *bike-pike*; instead of placing a single stimulus item as previously described, place a pair of stimulus items in front of the child; model both the words and ask the child to imitate; withdraw modeling and prove evoked trials; move through response complexity levels as described earlier
 - Use aspects of the Paired-Stimuli Approach; instead of presenting only the target word in which the child misarticulates the sound (as described earlier), present both the target word and a Key Word in which the sound is correctly produced; ask the child to produce the key word first and then the target word; see Paired-Stimuli Approach for details; when the individual meets the training criterion for words, move on to probes and then to training the words in sentences

- Use aspects of the Multiple Phoneme Approach if the child exhibits multiple errors; instead of training one sound at a time as described, train multiple phonemes
- Use aspects of the Phonological Pattern Approach if the individual exhibits multiple misarticulations with identified patterns; note that the phonological pattern approach does not have a unique treatment *procedure*; to eliminate a pattern, individual phonemes still need to be taught; this teaching involves instructions, modeling, phonetic placement, manual guidance, positive reinforcement, corrective feedback, and so forth; in using the phonological pattern approach, identify patterns or patterns in misarticulations; select patterns for elimination; teach the phonemes as described in the comprehensive program; when you teach a few phonemes within a pattern, probe the other phonemes within the pattern to see if they are now produced on the basis of generalization; if they are, move on to the other patterns or to higher levels of response complexity; if they are not produced, train additional sounds within the pattern
- Implement a maintenance program
 - Always train the correct production of sounds in sentences and in naturalistic conversational speech during the final stages of treatment
 - Conduct informal training sessions in Extraclinical Settings; monitor the child's correct speech sound production outside the therapy room but within the clinic, outside the clinic but in the surrounding areas, and in such natural settings as the classroom, the school dining hall, and the library
 - Train family members, teachers, and peers in reinforcing the correct production of sounds in natural settings; ask the parents or other caregivers to observe your treatment session; educate them about the correct production of target sounds; ask them to pay attention to the correct method of providing positive reinforcement and corrective feedback; train them to prompt and promptly reinforce the correct

productions at home; train them to hold brief training sessions at home; ask parents to audio-record the session and review the sample to give them feedback

- Teach the child Self-Control (Self-Monitoring) techniques to self-manage the correct and incorrect productions; in the treatment sessions, ask the child to judge the accuracy of his or her productions; give feedback on this judgment; ask the child to mark on a sheet his or her own responses as either correct or incorrect; teach the child to pause briefly as soon as an error is produced and then try to produce is correctly
- Follow up the child periodically and assess the correct production of phonemes in conversational speech; provide booster treatment when the correct production in conversational speech falls below 90% accuracy

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Hegde, M. N., & Peña-Brooks, A. (2006). *Treatment protocols for articulation disorders*. San Diego, CA: Plural.

Treatment of Speech Sound Disorders in African American Children.

African American English is a product of unique historical and cultural forces; it is a recognized form of English; therefore, note that:

- African American English (AAE) has its own phonologic, syntactic, semantic, and pragmatic rules and conventions
- The phoneme inventory of children speaking AAE will consist of the same phonemes as in Mainstream American English (MAE); only some phonemes will be used differently, substituted for other phonemes, or omitted in certain contexts
- A majority of phonemes are used in the same way in both AAE and MAE
- The clinician needs to know which AAE phonemic usages that differ from those of MAE are indeed characteristics of AAE; see *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for pho-

nological patterns that are accepted in AAE and hence are not to be treated as disorders.

- A treatable articulation disorder for a child who speaks AAE is a disorder in the context of AAE, not in the context of MAE
- Note that not all African Americans speak AAE; family communication patterns and the parents' preferences will dictate the target phonological patterns
- Plan a speech sound treatment program for an African American child, with the following guidelines:
 - Select the misarticulated speech sounds of AAE (even if they vary from those of MAE English) as treatment goals for a speaker of AAE
 - Select for treatment the following misarticulated phonemes for children in the age range of 33 to 36 months and older, because I. Stockman's research has shown that typically learning AA children produce these accurately: /m/, /n/, /p/, /b/, /t/, /d/, /k/, /g/, /f/, /s/, /h/, /w/, /y/, /l/, and /r/
 - Accept and reinforce sound patterns that are accepted in AAE (e.g., a child's production of *baf-tub* for *bathtub* is acceptable and reinforceable in treatment)
 - Treat first the phoneme usages that are the same in AAE and MAE (many actually are)
 - Treat next the unique phoneme usages of AAE that the child does not use correctly; the goal is to teach what is acceptable in AAE, even if it deviates from MAE
 - Change AAE dialectal patterns to MAE patterns only if the individual, the family, or both demand it; in this case, treatment of MAE sound patterns is *elective*
- Select functional speech sounds, child-specific sounds, and sounds that when treated will rapidly improve the child's speech intelligibility
- Select stimulus items for treatment from the child's home environment; consult with parents about pictures, objects, toys, and other materials for their appropriateness and child familiarity

- Consult the family members about appropriate reinforcers for the child; ask the child about his or her preferences; but as always, determine that a consequence is a reinforcer only after data show that correct productions have increased during treatment
- Assume that standard treatment procedures (e.g., modeling, phonetic placement, positive reinforcement involving especially verbal praise, a token system backed up with a variety of culturally appropriate reinforcers) will be effective with African American children; in all treatment sessions, record the response rates systematically to support or correct this assumption

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Roseberry-McKibbin, C. (2014). *Multicultural students with special language needs* (4th ed.). Oceanside, CA: Academic Communication Associates.

Stockman, I. (2008). Toward validation of a minimal competence phonetic core for African American children. *Journal of Speech-Language-Hearing Research*, 51, 1244–1262.

Treatment of Speech Sound Disorders in Bilingual Children.

Bilingual children are a large and varied group, whose primary language may be one of many varieties of Spanish, an Asian language, a Native American language, or an East European language, including Russian. Therefore, it is not possible to list the characteristics of all the various primary languages that influence American English. Follow the guidelines specified below to develop appropriate speech sound treatment programs for bilingual children:

- Analyze errors in the primary language assessment data; select these as the treatment targets; note that such treatment requires the working knowledge of the

child's primary language; in the absence of such knowledge, refer the child to a speech-language pathologist (SLP) who has the knowledge

- Analyze errors in mainstream American English (MAE) that are *not* due to the influence of the primary language; these errors too may be treatment targets; note that a monolingual English-speaking SLP can offer this treatment, assuming that an assessment was made with the help of a bilingual clinician
- Analyze English sound productions that vary from those in MAE but are due to the influence of the child's primary language; these are not normally the targets of speech sound modification (e.g., final consonant devoicing, as in *dose* for *doze*, is not a treatment target because that is a feature of Spanish)
- Select functional speech sounds, child-specific sounds, and sounds that when treated will rapidly improve the child's speech intelligibility
- Select stimulus items for treatment from the child's home environment; consult with parents about pictures, objects, toys, and other materials for their appropriateness and child familiarity
- Consult the family members about appropriate reinforcers for the child; ask the child about his or her preferences; but, as always, determine that a consequence is a reinforcer only after data show that correct productions have increased during treatment
- Assume that standard treatment procedures (e.g., modeling, phonetic placement, positive reinforcement involving especially verbal praise, a token system backed up with a variety of culturally appropriate reinforcers) will be effective with bilingual children; in all treatment sessions, record the response rates systematically to support or correct this assumption
- Modify treatment techniques to suit the individual child; keep good records of such modifications to evaluate whether they were needed because of individual differences or ethnocultural variations
- Refer the child to a bilingual clinician who knows the child's primary language

- See *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for the unique characteristics of Spanish language that influence English learning; these characteristics are not a basis to offer speech sound intervention
- To treat speech sound disorders in bilingual Spanish-English speaking children, follow these guidelines:
 - Correct the errors in phonemes that are common to the child's primary language and the secondary MAE on a priority basis
 - Teach the speech sounds that are misarticulated in Spanish if this is feasible and the treatment plan is to modify a speech sound disorder in Spanish
 - Correct MAE phoneme misarticulations if they are not due to an influence of the primary language (this is a true speech sound disorder in English in a bilingual child)
 - Correct MAE sounds whose productions are affected by the primary language only if the individual or the family request such treatment because of the advantage MAE offers in educational, social, and occupational settings; most likely such a request will be made
- To treat speech sound disorders in other bilingual children (such as those whose primary language is one of many Asian languages):
 - Use the general guidelines already specified for working with bilingual children
 - Determine the phonological characteristics of the specific Asian American child because Asian languages belong to different language families and hence are highly varied in their phonological and other characteristics
 - Use the guidelines offered in the previous section on Spanish-English bilingual children to select and correct speech sounds

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Peña-Brooks, A., & Hegde, M. N. (2007). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Roseberry-McKibbin, C. (2014). *Multicultural students with special language needs* (5th ed.). Oceanside, CA: Academic Communication Associates.

Sound-Evoking Techniques for English Consonants.

There is no systematic research on the relative effectiveness of sound-evoking procedures; they are based on clinical experience of many clinicians; therefore, use them with caution and modify or abandon procedures that do not produce favorable results; generously reinforce any slight improvement in the individual's target sound production or a movement in the right direction:

- Evocation of /p/, a bilabial, voiceless, stop-plosive:
 - Model the sound production several times; draw attention to the lip closure, building up of air pressure in the mouth, and air explosion as the sound is produced
 - With a piece of paper, show the plosive characteristic that can move the paper held in front of the mouth
 - Manually guide the two lips to the required articulatory posture; ask the child to forcefully expel the air from the mouth
 - Shape the /p/ from the explosive airstream
- Evocation of /b/, a bilabial, voiced stop-plosive:
 - Use the same procedures as those described for /p/ except that, as the child forcefully releases the air, ask him or her to add voice or to turn on the voice box; let the child feel the vocal fold vibrations as you produce the sound; ask the child to produce those vibrations as the air is released from the mouth; shape the /b/ out of these manipulations
- Evocation of /t/, a lingua-alveolar, voiceless stop-plosive:
 - Model the sound production several times, drawing attention to the placement of the tongue tip against the alveolar ridge; with the help of a moving feather,

draw attention to the air that escapes through the mouth as the sound is produced

- Ask the child to place the tongue tip firmly against the alveolar ridge; if necessary, place a piece of food on the alveolar ridge with a Q-tip to teach the exact tongue tip placement; hold the breath briefly, lower the tip slightly, and release the air as the tongue tip is lowered
- Shape /t/ with such manipulations
- Evocation of /d/, a lingua-alveolar, voiced stop-plosive:
 - Use the same procedures as those described for /t/; in addition, teach the child to add voicing by helping the child feel the vocal fold vibrations; ask the child to turn on the voice box
- Evocation of /k/, a lingua-velar, voiceless stop-plosive:
 - Model the sound production several times, demonstrate the tongue tip placement behind the front teeth and the raised back portion of the tongue that makes firm contact with the soft palate; build up air pressure in the mouth and suddenly lower the back of the mouth to release the air in a plosive manner
 - Shape the sound with this manipulation
- Evocation of /g/, a lingua-velar, voiced stop-plosive:
 - Use the same procedure as described for /k/ and ask the child to turn on the voice; let the child feel the vocal fold vibrations
- Evocation of /f/, a labiodental, voiceless fricative:
 - Model the production of the sound several times; draw attention to the lower lip position under the front upper teeth; manually guide the lips and the teeth to the right placement
 - Ask the child to blow air through the teeth and the lip when they are still in contact with each other; ask the child to turn off the voice box
 - Use a feather or a piece of paper to demonstrate the air flow out of the mouth
 - Shape the sound out of such manipulations
- Evocation of /v/, a labiodental, voiced fricative:
 - Use the same procedure as described for /f/ and ask the child to add voice or turn on the voice box; let the child feel the vocal fold vibrations

- Evocation of /θ/, linguadental voiceless fricative:
 - Model the sound production several times; draw attention to the slightly protruded tongue tip between the upper and lower front teeth and the air being blown over the tongue and through the constriction between the tongue tip and the teeth
 - Instruct the child to position the tongue as demonstrated; if necessary, ask the child to stick the tongue out of the mouth, then draw it slowly in until it is correctly positioned; use manual guidance with a tongue depressor
 - Ask the child to blow the air over the tongue and through the constriction between the tongue tip and the teeth; shape the sound through these manipulations
 - Alternatively, shape the sound while the child produces /f/; ask the child to push the tongue tip forward to come in contact with the teeth as the /f/ is being produced
 - Alternatively, shape the sound while the child produces /s/; ask the child to move the tongue tip to position as the /s/ is being produced
- Evocation of /ð/, a linguadental voiced fricative:
 - Use the same procedure as described for /θ/ and ask the child to turn on the voice box; let the child feel the vocal fold vibrations
- Evocation of /s/, a lingua-alveolar voiceless fricative:
 - Model the sound production several times; draw attention to either the tongue-tip-up position or the tongue-tip-down position with which this sound can be produced
 - Teach the child to place the tip of the tongue behind the upper teeth and groove the tongue-midline
 - Teach the child to bring the upper and lower teeth together
 - Teach the child to blow the air out along the groove of the tongue
 - Alternatively, shape an /s/ out of /θ/ by asking the child to draw the tongue inward until the sound approximates the /s/

- Evocation of /z/, a lingua-alveolar voiced fricative:
 - Use the procedures described for /s/ and ask the child to add voice; let the individual feel the vocal fold vibrations
 - Alternatively, shape /z/ out of /ð/ or from /θ/ to which the child adds voice
- Evocation of /ʃ/, lingua-palatal voiceless fricative:
 - Model the sound production several times; draw attention to the articulatory position
 - Shape /ʃ/ out of /s/; while the child is producing /s/, ask the child to pucker the lips and to move the tongue back until /ʃ/ results
 - Ask the child to produce “shh” (the “be quiet” sound) and shape /ʃ/ out of this production
- Evocation of /ʒ/, a lingua-palatal voiced fricative:
 - Use the procedures described for /ʃ/ and ask the child to add voice
 - Alternatively, shape /ʒ/ out of /z/
- Evocation of /tʃ/, alveo-palatal voiceless affricate:
 - Model the sound production several times; draw attention to the articulatory positions
 - Shape /tʃ/ out of an initial /t/ the child is asked to hold and then explode into a /ʃ/, which may result in /tʃ/
 - Alternatively, have the child place the tip of the tongue right behind the front teeth; then ask the child to move the tip slightly back; ask the child to make the sneezing sound (*choo!*) while puckering the lips
 - Alternatively, have the child say phrases in which the first word ends with /t/ and the second word begins with /ʃ/ (e.g., *that ship*); bring the production of the two sounds together to achieve /tʃ/
 - Shape the sound with these manipulations
- Evocation of /dʒ/, alveo-palatal, voiced affricate:
 - Use procedures described for /tʃ/ and ask the child to add voice
 - Alternatively, have the child produce phrases in which the first word ends with /t/ and the second

- word starts with /j/ (e.g., *meet you*); bring the /t/ and the /j/ sounds closer to achieve /dʒ/
- Evocation of /m/, a bilabial, voiced nasal:
 - Model the sound production several times; draw attention to the articulatory positions; emphasize the nasal resonance; let the child feel the nasal vibrations
 - Instruct the child to hum continuously and, while doing this, open the mouth; this might result in *ma*, from which you can shape /m/
 - Alternatively, teach the child to breathe in deeply through the nose, close the mouth, and let the air come out of the nose while saying “Ah”; this might result in /m/
 - Shape /m/ out of these manipulations
 - Evocation of /n/, a lingua-alveolar, voiced nasal:
 - Model the sound production several times; draw attention to the articulatory positions; emphasize the nasal resonance; let the child feel the nasal vibrations
 - Shape /n/ from /d/; teach the child to let the air out through the nose while producing /d/
 - Alternatively, ask the child to breathe out through the nose with voice added while keeping the tongue tip positioned for /n/
 - Shape /n/ with such manipulations
 - Evocation of /ŋ/, a lingua-velar, voiced nasal:
 - Model the sound production several times; draw attention to the articulatory positions; emphasize the nasal resonance; let the child feel the nasal vibrations
 - Shape /ŋ/ out of /g/ by teaching the child to produce /g/ while keeping mouth closed to direct the air through the nose
 - Alternatively, ask the child to produce a prolonged /i/ and, while doing this, ask the child to raise the back of the tongue to make a firm contact with the roof of the mouth
 - Shape /ŋ/ out of these manipulations

- Evocation of /j/, a lingua-palatal voiced glide:
 - Model the sound production several times; draw attention to the articulatory positions
 - Shape /j/ with a prolonged /i/; teach the individual to prolong the /i/ and quickly produce /u/, resulting in /iju/; teach the individual to shorten or unvoice the /i/, resulting in /j/
 - Shape /j/ with /ɜ:/; ask the child to produce /ɜ:/ in quick succession, resulting in /j/
 - Shape /ɜ:/ with these manipulations
- Evocation of /w/, a bilabial, voiced glide:
 - Model the sound production several times; draw attention to the articulatory positions
 - Shape /w/ with /u/; ask the child to produce a prolonged /u/ and then quickly add the schwa, resulting in /uwa/; teach the child to shorten the /u/ or turn the voice off on it, resulting in /wa/
 - Alternatively, teach the child to raise the back of the tongue without touching the roof of the mouth, round the lips and bring them closer, and breathe out with voicing
 - Shape /w/ with such manipulations
- Evocation of /r/, an alveo-palatal voiced glide:
 - Model the sound production several times; draw attention to the articulatory positions
 - Ask the child to place the tongue tip slightly behind the upper front teeth; ask the child then to “curl the tongue backward” without making contact with the roof of the mouth; round the lips slightly, and breathe out with voice on; this should result in /r/
 - Alternatively, shape /r/ from /d/; from the /d/ position, ask the child to lower the tongue tip; retract the tongue, round the lips slightly and breathe out with voice on; this should result in /r/
 - Alternatively, ask the child to growl like a tiger and shape /r/ from that growl
 - Shape /r/ with such manipulations
- Evocation of /l/, an alveolar voiced lateral:
 - Model the sound production several times; draw attention to the articulatory positions

- With a tongue depressor, lift the tip of the tongue and position it for /l/; ask the child to breathe out and turn on the voice, resulting in /l/
- Alternatively, ask the child to produce a prolonged /a/ and, while producing this, ask the child to raise the tongue tip to the alveolar ridge, resulting in /l/
- shape /l/ by such manipulations

Speech Sound Disorders: Specific Treatment Techniques or Programs.

There are many packaged programs to treat speech sound disorders and only a few are highlighted here; the core of all programs is a set of behavioral techniques that include picture stimuli, instructions on sound production, modeling, shaping, manual guidance of the articulators, visual cues (position, transition, and movement of the articulators), immediate reinforcement for approximations and correct productions, and immediate corrective feedback for errors. In essence, programs that appear different from each other have at their core behavioral contingencies of stimulus-response-consequence; programs that describe themselves as phonological or linguistic or cognitive achieve their goals only by teaching individual sound productions through behavioral methods; neither phonology nor linguistics nor cognitive science offers a treatment procedure.

Concurrent Treatment Approach. A behavioral treatment procedure that avoids the traditional simple-to-complex target skill hierarchies and randomly varies the simple and complex levels of teaching speech sounds; in essence, all levels of response complexity are concurrently targeted for teaching; supported by experimental evidence; researched by S. Skelton and associates.

- Establish the baselines of target speech sounds
- Develop stimuli at different levels of training; the simplest level of training is imitation of sound in single syllables moving through multisyllable, words, phrases, and sentences, and finally evoked conversational speech—the most complex level of training;

consult the cited sources for the specific levels because they may be as many as 29

- Establish the target sound in CV and VC syllables to meet the criterion of correct responses on 8 out of 10 blocked trials
- Randomly intermix the simple and complex levels of training tasks and train all exemplar types in a randomized order, presenting them at least once (except for the conversational speech segment, which should be presented at least four times); present the randomized levels on as many passes as possible in each session
- Use the behavioral procedures of modeling, shaping, positive reinforcement, token reinforcement with backup reinforcers, and corrective feedback
- Extend the same to multiple sound training, each sound first established in the CV and VC trials and then intermixing simple and complex training tasks
- Consider using the method in small groups (especially in public schools), as this too has been shown to be effective
- Consider using the method with childhood apraxia of speech, as the method has been found to be effective with this problem as well

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Skelton, S. L. (2004). Concurrent task sequencing in single phoneme phonologic treatment and generalization. *Journal of Communication Disorders*, 37, 131–155.

Contrast Approaches. Several approaches often used in remediating phonological patterns; all use contrasting pairs of words that contain minimal, maximal, or multiple differences between the target sounds and those contrasted; the actual training of sounds may involve behavioral contingencies; researched by multiple investigators.

Minimal Pair Contrast Method. Teaches word pairs that have minimal phonemic contrast (e.g., *bat-pat*,

which differ in only one feature—voicing-nonvoicing of the initial sound); difference in one or just a few phonemic contrasts is considered *minimal*.

- Analyze the individual's misarticulations
- Write minimal contrast word pairs; for instance, to remediate deletion of final consonants, write such pairs as *boat-bow*, *bee-bead*, and *tee-teeth*; to remediate fronting, write such pairs as *can-tan*, *key-tea*, and *gate-date*; select 8 to 10 word pairs for a phonemic contrast ((e.g., to eliminate final consonant deletion, select such word pairs as *bow-boat*, *bee-beet*, *toe-toad*, and *pie-pine*)
- Obtain pictures for words in selected pairs
- Place the pictures representing the word pairs in front of the child
- Model a given word pair (e.g., *bow-boat*) and ask the child to imitate; reinforce correctly imitated productions or approximations
- Provide extensive trials on imitative production of the target and contrast words
- Ask the individual to spontaneously name the picture pairs
- Ask the individual to name the pictures and then sort them into separate piles
- Alternatively, ask the individual to say the target word as you pick the correct picture (the individual says *boat* and you pick up the picture of *boat*; if the individual says *bow*, you pick up the picture of *bow* and then correct the individual)
- Ask the individual to match two pictures by first picking a picture from several displayed and then selecting its minimal pair match
- Arrange controlled play activities in which the target sounds and words are practiced in conversational speech (e.g., a toy *soap* used to pretend wash hands, pots, dolls, and so forth while talking about the activity)
- Move on to more naturalistic conversational speech and reinforce correct productions of target speech sounds

Maximal Pair Contrast Method. Uses word pairs that have multiple (maximal) phonemic contrasts or maximal opposition.

- Select word pairs that contrast maximally; for instance, select such word pairs as *chain-main*; *can-man*; *gear-fear* (the initial phoneme in the first word of each pair is the target of treatment; the initial phoneme in the second word in each pair is the phoneme with maximal opposition because it differs from the treatment target on more than one feature)
- Use the general procedure outlined for Minimal Pair Contrast Method

Multiple Contrast (Opposition) Approach. The procedure uses minimal word pairs, but the pairs are created to treat all or most of the errors simultaneously, something not done in other contrast approaches; use this method to treat children who substitute a single sound for several sounds (e.g., a child who substitutes /t/ for /s/, /k/, /tʃ/ and /tr/ will substitute a single word such as *tip* for multiple words—*sip*, *Kip*, *chip*, and *trip*); in using this contrast method, simultaneously treat all error contrasts, instead of treating one contrast at a time (as in minimal pairs contrast).

- Create multiple sets of minimal pairs, but include all error types in each set (e.g., to treat the previously mentioned single sound substitution for multiple sounds, create such pairs as *tip-sip*, *tip-Kip*, *tip-chip*, *tip-trip*); note that all errors are contrasted in this set
- Create additional sets to achieve the same multiple contrasts (e.g., *tease-ease*, *tease-keys*, *tease-cheese*, *tease-trees*)
- Teach the contrasts simultaneously, using the behavioral methods of modeling, positive reinforcement, and corrective feedback

Cycles Approach. A phonological pattern approach designed to treat children with multiple misarticulations and highly unintelligible speech; approach consists of treatment cycles which vary between 5 weeks

and 16 weeks; includes auditory stimulation and production practices; developed by B. Hodson and E. Paden.

- Assess the individual's phonological performance with 50 spontaneous naming responses and continuous speech samples; may use Hodson's *Assessment of Phonological Patterns—Revised*
- Arrange a hierarchy of stimulable phonological patterns that occur in at least 40% of the relevant contexts
- Treat the most stimulable pattern first, then the next most stimulate pattern, and so on
- Target only one phonological pattern in any single session
- Treat each phoneme within a target pattern for about 60 minutes per cycle (one 60-minute, two 30-minute, or three 20-minute sessions) before moving to other phonemes within the pattern or to other patterns
- Review the prior week's production practice word cards (see below); skip this step if introducing a new pattern for treatment
- Begin treatment with auditory bombardment:
 - Ask the individual to listen attentively for about 2 minutes as you produce 12 words with the target sound and sentences containing those words
 - Do not ask the individual to produce the sounds
 - Periodically contrast the correct and the incorrect production of the target sound
- Use five production-practice word cards: Ask the individual to first say a target word and then draw, color, or paste the picture of the word on 5 × 8 index cards; write the word on the card
- Begin production practice:
 - Ask the individual to name about five target pictures (five words per sound)
 - Model the target word; use auditory, tactual, and visual cues
 - Engage the individual in conversation
 - Use a game format

- Probe for stimulability of next session's target sounds
- Repeat the amplified auditory bombardment; present the same 12 words as before
- Ask the family members or teachers to read the same 12-word list to the individual; ask the individual to name the five picture cards used in production practice during the week
- Recycle a pattern that persists in conversational speech

Hodson, B., & Paden, E. (2004). *Targeting intelligible speech: A phonological approach to remediation* (2nd ed.). Austin, TX: Pro-Ed.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Multiple Phoneme Approach. A method in which all speech sound errors are treated in all sessions; appropriate for children with six or more errors; explicitly based on behavioral procedures for teaching sound production in conversational speech without auditory discrimination training; consists of establishment, transfer, and maintenance phases; each phase has several steps; highly structured and carefully sequenced; developed and researched by R. McCabe and D. Bradley.

- Obtain conversational speech sample of about 150 words
- Mark each word that contains at least one error
- Calculate percentage of words spoken correctly (Whole Word Accuracy: WWA)
- Use WWA measure to supplement single-word articulation tests

Phase I, Step 1. Establishment. Goal: Production of consonants in response to a printed letter or phonetic symbol representing it.

- Show an upper- or lowercase letter and ask, "Do you know what sound this letter makes?" (visual cue only)

- Ask the individual to produce the sound in isolation on five successive trials (record the correct responses)
- Record the error if that is what is observed and move to the next step
- Give verbal instructions along with auditory and tactile stimuli; use any other effective procedure (auditory, visual, and phonetic-placement); continue until four out of five attempts are correct; move to the next step
- Show the letter and model the sound for the child to imitate (auditory and visual stimuli only); seek five consecutively correct responses; then, move to the next step
- Present only the letter (visual stimulus only); ask the individual to make the sound; seek five consecutively correct responses
- (Skip *visual only step* for children under age 5)
- In the first session or two, include sounds produced correctly to give experience of success; omit these sounds in subsequent sessions
- Reinforce correct responses (verbal praise, tokens)

Phase 1, Step 2. Holding Procedure. Designed to maintain the correct production of sounds produced in isolation when they are not yet advanced to syllable or word levels; other sounds are moved to these higher levels.

- Evoke one correct response by showing the letter once and asking the individual to produce the sound (visual stimulus only)

Phase II. Transfer. Goal: Production of all target sounds in conversational speech; simultaneous training of five or more sounds; sounds may be at different levels.

Phase II, Step 1: Syllable. Used only when the individual fails to produce the sound correctly in 6 out of 10 probe words (5 words with the sound in

the initial position and 5 words with the sound in the final position)

- Provide one auditory-visual model or one visual-only stimulus
- Ask the individual to produce the sound with a variety of vowels
- Ask the individual to produce the sound in both initial and final positions
- Seek five productions for each stimulus presentation
- Use a criterion of 80% correct over two sessions or 90% correct in one session

Phase II, Step 2: Word. Goal: Accurate production of target sounds in 25 to 30 varied words to be later included in sentences (nouns, verbs, modifiers, and prepositions).

- Present printed words or picture stimuli
- Ask the individual to produce the word
- Accept erred production of nontarget phonemes
- Move training to the sentence level when the sound in a given position (e.g., initial position) is produced with 80% accuracy over two sessions or 90% accuracy in one session
- Continue training at the word level when the sound in a given position (e.g., final position) does not meet the criterion
- Consider using another approach, such as the minimal contrast therapy or phonological pattern approach, to eliminate the final consonant deletion pattern

Phase II, Step 3: Phrase and Sentence. Goal: Correct production of all sounds in words; self-monitoring.

- Construct phrases and sentences (imperatives, declaratives, and interrogatives) with words already trained, adding new words as needed
- Present Rebuses, Blissymbolics, or pictures for nonreaders
- Model phrases and sentences

- Ask the individual to imitate
- Analyze the phonetic contexts in which errors occur and have the individual practice the production in these contexts
- Have the individual practice words in which sounds are produced incorrectly as well as those that precede or follow such words
- Vary stress, rhythm, timing, and accent patterns
- Seek 80% accuracy over two sessions or 90% accuracy in one session, calculating accuracy with target sounds only

Phase II. Step 4: Reading and Storytelling. Goal: Accurate production of target sounds in connected utterances containing four to six words.

- Select reading materials that are easy for the child
- For nonreaders, select comic books, picture books, and sequence cards
- Tell a story and ask the child to retell it
- Seek whole word accuracy and 80% correct production over two sessions or 90% in one session

Phase II. Step 5: Conversation. Goal: Accurate production of all sounds used in conversational speech.

- Begin to monitor conversational speech when even one or two sounds reach this level
- Encourage discussions, descriptions, comments, questions, state facts, identify cause-effect relations, talk about emotions and desires; do not just answer questions
- When multiple sounds need to be monitored, group sounds on the basis of manner or place of articulation; monitor sounds in one group for 3 to 5 minutes; then, monitor sounds in another group, and so on
- Count every spoken word as a response and calculate the whole word accuracy level
- Note the context in which certain sounds are misarticulated and use these contexts for additional practice

- Seek 80% correct production of all words over two sessions or 90% in one session for children 6 years and older; seek 69% criterion for younger children

Phase III: Maintenance. Goal: Maintenance of 90% whole word accuracy in conversational speech produced in various speaking situations without treatment or external monitoring.

- Have the individual return to the clinic; assess and monitor sound productions
- Visit classrooms
- Maintain telephone contact with the individual and the family
- Obtain report from others
- Have others monitor accuracy in various speaking situations
- Monitor for 3 months

McCabe, R., & Bradley, D. (1975). Systematic multiple phonemic approach to articulation therapy. *Acta Symbolica*, 6, 1–18.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Paired-Stimuli Approach. A method that uses a Key Word in which a target sound appears only once in either initial or final position and is correctly produced 9 out of 10 times to teach the production of sounds in other contexts; explicitly uses operant reinforcement contingencies; uses pictures to evoke the target words; highly structured and carefully sequenced; a single speech sound is the target at any one time; developed and researched by J. Irwin and A. Weston.

Word Level. Consult Weston and Irwin (1971/1975) for assigned key words, questions to be asked, and expected answers.

- Select the target phonemes for the individual
- Find four key words; two containing the target sound in the initial position and two containing it in the final position

- Create key words, if they are absent, by teaching them
- Select at least 10 training words in which the target sound is misarticulated and the sound appears only once in the same position as in the key word
- Select pictures as stimuli to evoke the word productions
- Place the first key word (picture) with sound in the initial position in the center and arrange the 10 training words (pictures) around it
- Point to the key word (picture) and ask the individual, “Say this”
- Reinforce the likely correct production
- Ask the individual to name one of the 10 target words
- Ask the child to name the key word again
- Ask the child to name another target word; alternate the key word and a training word in this manner
- Reinforce the individual by giving a token for the correct production of the target sound in both the key and the training words; ignore misarticulations of other sounds
- Complete a *training string* by pairing each of the 10 target words with the key word
- Include three training strings in each session that lasts about 30 minutes
- Adhere to a training criterion of 8 correct out of 10 productions of the training words in two successive training strings without reinforcement
- Arrange the second key word with the same sound in the final position and pair it with 10 training words
- Move to the next stage of training; ask the child to say the third key word with the target sound in the initial position and a training word as a Response Unit with only a brief pause between the two (e.g., “said-salad”; *s* is the target; *said* is the key word and *salad* is the target word)

- Reinforce only if the sounds in both the words are correctly produced
- Adhere to a training criterion of 8 out of 10 correct *response units* over two successive training strings
- Ask the child to say the fourth key word with the target sound in the final position and a training word as a *response unit* with only a brief pause between the two
- Reinforce the correct productions in *response units* as before

Sentence Level

- Pair the first key word with its 10 training words; ask a question designed to evoke a response in the sentence form (e.g., “What do you see?” “I see a cat” with *see* as the key word for /s/).
- Reinforce with a token on a fixed ratio 3 (FR3) schedule of reinforcement
- Complete a training string of 10 questions
- Adhere to the training criterion of 8 out of 10 correct sentences over two training strings
- Ask two questions alternately (e.g., “What do you see?” for key word *see* and “That’s what?” for key word *that’s*) as you present the first and the second key words and their 10 training words
- Reinforce with a token for three correct sentences (FR3)
- Adhere to the training criterion of 8 out of 10 correct sentences over two training strings
- Ask four questions (e.g., “What is this?” “What do you see?” “That’s what?” “What did you say that was?”) as you present the first and the fourth key words and their 10 training words
- Adhere to the criterion of 8 out of 10 correct sentences over two successive training strings

Conversational Level

- Engage the child in conversation
- Stop the conversation (a) when the child correctly produces a target sound in four words or (b) when

the child incorrectly produces a target sound in any word; model the correct production; ask the child to repeat it

- Require subsequently the correct production of a target sound in seven words; probe when the child can do this
- Require the correct production of a target sound in 10 and 13 words In subsequent stages; probe when the child can do this
- Give verbal praise and visual feedback of scoring only when all productions are correct
- Take a conversational speech samples for all probes,; no feedback of any sort during probes
- Terminate training on a given sound when the child gives 15 consecutively correct productions of a target sound in conversation held on two successive treatment sessions separated by at least one day

Irwin, J. V., & Weston, A. J. (1971/1975). *Paired Stimuli Kit*. Milwaukee, WI: Fox Point.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Phonological Pattern Approach. Not a treatment method, but a procedure to analyze speech sound errors and target phonological error patterns for elimination by teaching the correct production of sounds by behavioral methods; based on the assumption that teaching only a few individual phonemes within a pattern will be sufficient to realize generalized production of other sounds within the pattern; an assumption with mixed evidence; untrained phonemes may or may not be produced; researched by multiple investigators; includes multiple programs with varying degrees of empirical research, some with negligible controlled research base.

- Obtain a conversational speech sample that reflects a variety of words and linguistic contexts in which all sounds are produced; may use one of the several available protocols of phonological analysis

- Identify the Phonological Patterns that account for error patterns
- Select patterns for elimination through teaching specific sounds or groups of sounds; for instance:
 - Identify all phonemes that are misarticulated within a pattern (e.g., all final consonants a child omits—the final consonant deletion pattern)
 - Teach only a few final consonants
 - Use any effective procedure to teach the consonants
 - Use the comprehensive treatment approach described at the beginning of this main entry with modeling, shaping, manual guidance (phonetic placement), positive reinforcement, and corrective feedback
 - Alternatively, use the paired-stimuli approach, contrast approach, or any other technique supported by controlled evidence
 - Teach sounds in syllables, words, phrases, and sentences
- When a few final consonant productions meet the training criterion, probe to see if other, *untreated sounds* within the pattern are produced correctly without training, based on generalization (e.g., other untreated final consonants that are omitted)
- If there is no generalized production of untreated phonemes, then treat them as well
- If untreated phonemes are produced on the basis of generalization, select another phonological pattern for elimination; teach selected phonemes within the pattern; administer probes
- Schedule maintenance activities as appropriate

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Programmed Conditioning for Articulation. A treatment program that uses behavioral principles and programmed learning concepts; developed by R. Baker and B. Ryan.

- Criterion of Performance: Ten correct responses in a row.

*Establishment Phase: Training Sequence**Sound in Isolation*

- Teach the sound in isolation with Continuous Reinforcement (crf)

Nonsense Syllable Level

- Teach the sound in initial, final, and medial positions of nonsense syllables in that order (crf)

Word Level

- Teach the sound in word-initial, final, and medial positions (50% reinforcement; rf)

Phrase Level

- Teach the sound in word-initial, final, and medial positions produced in two- or three-word phrases (50% rf)

Sentence Level

- Teach the sound in word-initial, final, and medial positions produced in four- to six-word sentences (50% rf)

Contextual Reading Level

(Go to the next level if the individual is a nonreader)

- Teach the sound in orally read sentences (crf)

Story Narration Level

- Teach the sound in story retelling (after silently reading a story) (crf)

Picture Description Level

- Teach the sound in sentences and phrases produced to describe a story (crf)

Conversational Speech Level

- Teach the sound in conversational speech (crf)
- Teach the sound in conversational speech (10% rf)

Administer the criterion test

Move to the Transfer Phase and begin training on new sounds

*Transfer Phase: Training Sequence**Home Training*

- Teach the sound in words, phrases, and sentences (crf)

- Teach the sound in oral reading or picture description (crf)
- Teach the sound in conversation (crf)

Clinician Trains in Different Settings

- Conversation outside the clinic room door, down the hall, outside the clinic building or in another room, in playground or cafeteria, outside the classroom away from the school or clinic, in that order (crf)

Training in Classroom

- Conversation with the clinician in classroom, with the clinician and teacher, in small-group activity, in large-group activity, *show and tell* in front of the class (crf)

Administer the transfer criterion test

Maintenance Phase: Training Sequence

- Conversation during weekly meetings for the first 4 weeks (crf)
- Conversation during one monthly meeting (crf)
- Dismiss the individual

Baker, R. D., & Ryan, B. P. (1971). *Programmed conditioning for articulation*. Monterey, CA: Monterey Learning Systems.

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Traditional Approach. An early treatment approach developed for the most part by Van Riper, who included several techniques from various sources; sounds are trained in isolation, in syllables, in words, and in sentences; used or researched by multiple investigators; aspects of this program are now part of many existing programs; it uses the following procedures that are described under the head of various programs:

- Perceptual Training (Ear Training). Teach auditory discrimination of speech sounds by asking the child to repeatedly listen to the correct and incorrect productions of the sounds the child misarticulates; ask the child to raise the right hand when the sound is correct and the left hand when it is not

- **Production Training: Establishment.** Model correct productions of target sounds in isolation, in syllables, and in words; and ask the child to imitate; use Phonetic Placement and Shaping
- **Production training: Stabilization.** Continue to train the sound in isolation as well as in nonsense syllables; in words, in phrases, in sentences, and finally in conversational speech; vary the sentence lengths and complexity; reinforce correct productions in oral reading
- **Production training: Transfer (generalization).** Give speech assignments the child completes at home; get parental reports; teach self-monitoring, have the child practice correct speech sound productions in varied settings

Peña-Brooks, A., & Hegde, M. N. (2015). *Assessment and treatment of speech sound disorders in children* (3rd ed.). Austin, TX: Pro-Ed.

Van Riper, C., & Emerick, L. (1984). *Speech correction: An introduction to speech pathology and audiology* (7th ed.). Englewood Cliffs, NJ: Prentice-Hall.

Stimulation Versus Treatment. In language intervention, stimulation tends to be less directed, more naturalistic, without specific language targets, and without a requirement that the child imitate modeled responses; stimulation is often implemented by family members and other caretakers; to the contrary, treatment or teaching tends to be more clinician directed, less naturalistic, with specific language structures as targets, with the requirement that the child imitate modeled responses; imitated or spontaneous responses are reinforced or provided with corrective feedback; vague stimulation procedures with no specific target skills and stimulus-response-consequence arrangements are not as effective as direct treatment or teaching implemented by speech-language pathologists with explicit behavioral contingencies.

Stimulus Generalization. Production of a newly learned response to stimuli not used in training; an important goal of intervention; to promote stimulus generalization.

- Use a variety of stimuli during treatment
- Use common stimuli
- Ask clients to bring objects, toys, books, and so forth from home to use as stimuli
- Train multiple exemplars
- Periodically Probe the production of target behaviors to assess their generalized productions

Story Grammar. The structure of narratives which may be treatment targets for children with language disorders; described under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Stridency. A voice disorder characterized by an unpleasant, shrill, and metallic-sounding voice; caused by excessive pharyngeal constriction and an elevated larynx.

- Use the Chewing Method
- Model and contrast good vocal production with a strident production
- Lower the pitch; use Glottal Fry
- Teach relaxation
- Teach oral openness
- Use Yawn-Sigh method

Stridor. A harsh and shrill (high-pitched) sound during breathing; often found in persons with laryngeal obstruction; associated with various laryngeal pathologies, including congenital laryngeal cyst, congenital laryngeal papilloma, congenital subglottic stenosis, and laryngocele; treatment is medical.

S

Subcortical Aphasia. Aphasia presumably due to damage to subcortical structures, especially to basal ganglia and surrounding areas; somewhat controversial because some experts suspect that subcortical aphasia involves cortical damage as well; characterized by generally fluent speech, intact repetition, and articulation problems; see Aphasia.

Super-Supraglottic Swallow. Teach this swallowing maneuver to help close the airway entrance before and during the swallow; see [Dysphagia](#) for details; to implement this maneuver, ask the patient to:

- Inhale and hold the breath tightly by bearing down
- Swallow while holding the breath and bearing down

Supraglottic Swallow Maneuver. Teach this procedure to reduce or control aspiration while modifying swallowing behavior during the oral phase of the swallow; see [Dysphagia](#) for details; ask the patient to:

- Inhale and hold the breath
- Place food in the mouth
- Tilt the head back and swallow
- Cough after each swallow to clear any residual food from the pharynx

Swallow Reflex. A series of reflexive actions needed to complete the swallow act; includes the reflexive elevation of the soft palate, closure of the airway, peristalsis (constriction of the pharyngeal constrictors), relaxation of the cricopharyngeal muscle to passage of food into the esophagus; often delayed in patients with dysphagia; may be triggered by stimulating the base of the anterior faucial arch.

Syndromes Associated with Communication Disorders. A constellation of signs and symptoms that are associated with a morbid process; see the companion volume, *Hegde's PocketGuide to Communication Disorders*, for various syndromes associated with communication disorders; see the cited sources and the two companion volumes, *Hegde's PocketGuide to Communication Disorders* and *Hegde's PocketGuide to Assessment in Speech-Language Pathology*, for additional information.

General Guidelines in Treating Individuals with Genetic Syndromes:

- An understanding of genetic syndromes, their etiologic factors, and varied phenotypic expressions is essential to diagnose and treat communication disorders

- There are no special treatment procedures to improve the communication skills of individuals with genetic disorders; their speech, language, voice, and fluency problems are treated with the same procedures described under the main entries
- Give guarded prognosis for improvement with treatment; the extent of improvement may depend on the severity of symptoms, medical prognosis for stabilized, improving, or deteriorating condition of the individual
- Work with a team of specialists, including genetic counselors, medical specialists, psychologists, special educators, social workers, occupational therapists, and other health care professionals who may be involved in the treatment and rehabilitation of individuals with genetic syndromes
- To treat speech production problems, see [Speech Sound Disorders](#) and [Cerebral Palsy](#)
- To treat language disorders, see [Language Disorders in Children](#), [Apraxia](#), and [Dysarthria](#) if relevant
- To treat dysfluent speech, see [Fluency Disorders](#)
- To treat deviant vocal characteristics, see [Voice Disorders](#)
- Be skeptical of generalized recommendations because of the unique needs of each individual
- Consider each client as a unique individual; the same syndrome may express itself differently or with different severity across individuals
- Make adjustments in treating communication disorders depending on the medical needs of the child; e.g., prosthetic or surgical needs if the child has cleft palate; hearing aids or group auditory trainers if the individual has a hearing loss; orthopedic rehabilitation if the individual has a need for it; and so forth
- Train family members to promote generalization and maintenance
- Integrate academic and literacy goals with speech and language treatment; see [Language Disorders in Children](#) and [Literacy and Literacy Skills](#)

Jones, K. L. (2005). *Smith's recognizable patterns of human malformations* (6th ed.). Philadelphia, PA: W.B. Saunders.

National Institutes of Health. (n.d.). <https://rarediseases.info.nih.gov/diseases>; key word search with the name of a syndrome is productive.

Zajac, D. J., & Vallino, L. D. (2017). *Evaluation and management of cleft lip and palate*. San Diego, CA: Plural.

T

Teflon or Collagen Injection. A medical treatment procedure for clients with paralyzed vocal folds; the material is injected into the middle third of the fold, to increase the bulk and the chances of adduction.

Terminal Response. The final response targeted in Shaping.

Time-Out (TO). Time-out from positive reinforcement; also described as pause-and-talk as used in Stuttering treatment described under Fluency Disorders; effective in reducing a variety of undesirable behaviors; direct response reduction procedure in which one of the following three contingencies is placed on a behavior to be reduced: (1) a brief period of no reinforcement (nonexclusion TO); (2) exclusion of the person from the stream of activity (exclusion TO), but not from the current environment; or (3) removal of the person from the current environment and placing the person in an isolated place for a brief period (isolation TO); exclusion TO is of limited use, if any, and isolation TO is not to be used at all in communication treatment sessions; they are used in mental hospitals to control extremely violent and dangerous behaviors.

Nonexclusion TO

- Recommended for reducing behaviors during communication treatment:
 - Begin TO as soon as the client produces an undesirable response
 - Ask the client not to respond for the specified duration
 - Limit TO to no more than 10 seconds
 - Do not interact with the client during TO; avoid eye contact
 - At the end of the TO duration, smile, re-establish eye contact, and resume interaction

Tokens. Conditioned generalized reinforcers; objects that are earned during treatment and exchanged later for backup reinforcers; effective in teaching a variety of communication skills.

- Always have backup reinforcers the child can exchange the tokens for
- Let the child choose a backup reinforcer at the beginning of each session
- Let the child understand the ratio of tokens to a backup reinforcer (e.g., say, “You can get this toy car at the end of the session if you have ten tokens left with you”)
- Set a low ratio in the beginning and gradually raise the number of tokens needed to receive the backup reinforcer (e.g., a token for every correct response in the beginning, one for every third response, fifth response)

Tongue Thrust. A deviant swallow in which the tongue is pushed forward against the central incisors; treatment considerations include the following:

- Experimental treatment research on either tongue thrust modification or speech error modification is limited
- Some clinicians strongly believe that tongue position and lip closure should be modified before treating the speech sound errors (if present)
- Correct productions of /z/ and /s/—typically distorted—are primary articulatory targets
- Additional target sounds may include /ʃ/, /tʃ/, /dʒ/, /ʒ/, /t/, /d/, /l/, and /n/; these may be produced with a more anterior tongue position than normal; treatment is concerned with more normal tongue position in producing these sounds
- See Speech Sound Disorders for treatment procedures

Topic Initiation. A pragmatic language skill in initiating conversation on a topic; a frequent language intervention target; procedures described under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Topic Maintenance. A pragmatic language skill in maintaining conversation for socially acceptable time periods; a frequent language intervention target; procedures described under Language Disorders in Children; Treatment of Language Disorders: Specific Techniques or Programs.

Topographic Sequence of Treatment. Sequencing treatment based on response complexity; traditionally, simpler targets were taught in the initial stages; more recent research on the treatment of speech sound and language disorders suggest that teaching first the more complex, difficult, low-frequency skills, and the skills not produced at all during assessment may produce better and quicker generalization; informal experimentation may be the best strategy.

- Begin treatment not necessarily with the simplest responses but at a relatively complex level
- In treating speech sound disorders, begin teaching a phoneme at the word or even phrase level; drop to the syllable level or isolated phoneme level only if the child cannot handle the more complex level; as the client becomes proficient in producing the sound at the successful initial level, shift training to the next higher level; consider moving sooner rather than later to the conversational speech
- In language therapy, begin teaching grammatic morpheme in phrases or even sentences (*two cups* or *I see two cups*); shift training to the higher levels, also sooner rather than later
- In fluency therapy, train such skills as gentle onset, prolonged speech, and airflow management initially in phrases and sentences and subsequently in conversational speech
- In voice therapy, begin to modify voice quality at the sentence level and if unsuccessful drop to single word or vowel productions (e.g., /a/) to improve voice quality; subsequently, shift training to conversational speech

Topography. Description of natural and physical properties of an object or event; topographic aspects of skills refer to their physical form or shape, including how complex they are, and how they appear, sound, and feel; words, phrases, and sentences are different topographic responses targeted in treatment.

Total Communication. The simultaneous use of multiple modes of expression to enhance communication;

includes speech, gestures, informal and formal (e.g., American Sign Language) signs, and facial expressions.

Tracheoesophageal Fistulization/Puncture (TEF/TEP). A surgical procedure that helps laryngectomy patients to produce laryngeal speech with the help of a voice prosthesis; the tracheal wall is punctured to create a small tunnel into the esophagus; the puncture acts as a shunt to allow air into the esophagus through a Voice Prosthesis inserted into the opening; air goes up through the P-E Segment and results in the production of sound; see Laryngectomy for additional information.

Stemple, J. C., Roy, N., & Klaben, B. K. (2014). *Clinical voice pathology* (5th ed.). San Diego, CA: Plural.

Traditional Orthography. Written natural language; a normal form of communication; a method of nonvocal communication for individuals with limited oral communication skills; used in teaching Augmentative and Alternative Communication-Aided.

Training Criterion. A rule that specifies when an exemplar or a target skill has met a specified performance level; a 90% correct response rate is an often accepted training criterion.

- Specify a training criterion in measurable terms (e.g., 9 out of 10 correct responses)
- Continue training until that criterion is met
- Probe when the training criterion is met
- If the probe criterion (90% correct in untrained contexts) is not met, resume training
- When the probe criterion is met on a given target, select another target for training

Training in Natural Environments. Part of extra-clinical training strategy used to promote maintenance.

- Initially, hold training in varied settings in and around the clinic
- Next, hold informal training sessions in nonclinical settings

- Train parents to hold training sessions at home
- Take the client to such natural settings as shopping centers and restaurants
- Let the training in natural settings be less conspicuous, involving mostly conversational speech
- Prompt the target responses in a subtle manner
- Deliver reinforcers and corrective feedback in a subtle manner

Transcortical Motor Aphasia. To treat this type of nonfluent aphasia, see [Aphasia; Treatment of Aphasia: Specific Types](#); in most respects, treatment is similar to that of [Broca's Aphasia](#).

Transcortical Sensory Aphasia. To treat this type of fluent aphasia, see [Aphasia; Treatment of Aphasia: Specific Types](#); for the most part, treatment is similar to that of [Wernicke's Aphasia](#).

Traumatic Brain Injury (TBI) in Adults. Treatment of communication disorders in individuals who have sustained traumatic brain injury follows the more urgent medical management; the individuals may have either [Penetrating \(Open-Head\) Injury](#) or [Nonpenetrating \(Closed-Head\) Injury](#); also known as craniocerebral trauma; immediate concern is medical; long-term concern is rehabilitation.

Treatment of Traumatic Brain Injury: General Principles

- Serve as a member of rehabilitation teams that include different professionals
- Work with the family and the rehabilitation staff from the beginning; counsel the family members on the effects of TBI and the initially limited goals for functional communication
- Plan for long-term treatment and rehabilitation, especially in the case of more severe injury
- Use direct behavioral treatment procedures, as they are known to be effective; these include simplifying

the tasks with shaping procedure, measurable, practical skill targeting instead of indirect underlying and presumed cognitive or linguistic process training; and immediate positive reinforcement for skill management

- Select effective reinforcers; consider using tangible or primary reinforcers, as some patients with TBI may not respond well to verbal praise
- Schedule different kinds of therapeutic activities at different stages of recovery from TBI (acute, postacute, outpatient, and long-term)
- Plan on communication training gaining momentum as the patient recovers from the initial effects of TBI
- Select client-specific functional treatment goals that help improve immediate communication, orientation to the environment, memory for events and persons, and those that help reduce confusion
- Revise treatment targets as the patient's condition improves (or deteriorates); select goals that are appropriate and practical for the current physical condition of the client
- Let the client's family members participate in treatment target selection; have them rate the importance of potential communication skills and skill hierarchies
- Emphasize functional communication effectiveness instead of grammatical accuracy; accept gestures, words, phrases, or grammatically incorrect expressions if they effectively communicate
- Integrate such cognitive skills as orientation, memory, and attention into communication training instead of concentrating on them in an isolated manner (e.g., reinforce increasingly longer durations of attention while training such communication skills as maintaining eye contact or topic maintenance) instead of paying attention to printed dots or squares
- Include behavioral self-management exercises in your treatment program
- Select treatment activities that are real-life activities (e.g., in improving memory skills, use pictures of

family members instead of irrelevant pictures or geometric shapes)

- Begin with simple activities and move through a sequence of more complex activities; but always experiment to see if the client can handle more complex skills
- Hold brief and frequent treatment sessions in the initial stages of rehabilitation
- Increase the duration of sessions as the client's general physical condition improves; hold longer sessions less frequently
- Consider teaching compensatory strategies whenever necessary
- Structure treatment sessions to eliminate distraction, especially in the initial stages of recovery; loosen the structure gradually to better resemble everyday living conditions
- Sequence treatment tasks to suit the individual's success in handling them; don't rigidly follow the commonly used simple-to-complex strategy
- Work closely with the members of the interdisciplinary team so that an integrated plan of rehabilitation is implemented
- Design and implement a community reentry program; consider the educational, social, and occupational environment that the client will enter after discharge from the hospital and rehabilitation facilities
- Note the similarities in treatment goals and procedures for patients with TBI and those with Right Hemisphere Disorders
- Note that there is little research on ethnocultural variables that affect treatment; consequently, consider general guidelines given under Ethnocultural Variables in Treatment; generally, select treatment stimuli and activities that are ethnoculturally appropriate for the client; invite family members' suggestions on treatment stimuli and activities that will be consistent with the client's education, interests, hobbies, and cultural background

Traumatic Brain Injury, Treatment Procedures

Treatment During the Initial Stage

- Simplify the patient's daily activities and routines
- Decrease variability in activities and stimulation; let the patient experience only a few structured activities to begin with
- Induce consistency in staff care and imposed activities and events
- Improve the client's orientation and attention to surroundings
 - Arrange familiar cues by pasting familiar pictures, posters, and objects
 - Play favorite music
 - Post written signs and lists about the daily routines; train the patient to consult the signs and lists; ask the patient to read them aloud; ask the patient to describe scheduled activities and their timings
 - Ask questions about time, place, and people (e.g., "Where are you now?" "What time is it?" "Who am I?" "Who is she?") and prompt correct responses
 - Frequently model any response you expect from the client; reinforce correct imitative responses
 - Increase the patient's attention to the surroundings by drawing the patient's attention to events and persons around him or her, including health care staff; by encouraging the patient to talk about his or her surroundings and people and giving corrective feedback and positive reinforcement
 - Simplify all demands so that the client experiences successes
 - Use tangible reinforcers, as the patients with TBI may not initially respond to verbal praise (use such tangible reinforcers as sweets, music, touching, massage)
 - Keep the treatment sessions brief
 - Prompt and assist the client to engage in self-care activities (dressing, eating)

- Reduce gradually the amount of physical help offered
- Have the client participate in group treatment sessions as soon as it is practical
- Place behavioral contingencies on appropriate behaviors
- Shape desired targets
- Teach the use of Augmentative and Alternative Communication in the initial stage of recovery
 - Organize a simple communication board that helps communicate basic needs
 - Always pair gestures and other nonverbal means of communication with verbal expressions as you speak to the person; train the caretakers to do the same
 - Fade the nonverbal device as the person's physical condition improves and is more inclined to use verbal communication
 - Assess the potential need for long-term use of an augmentative or alternative mode of communication
- Do not overstimulate
- Use colored objects and pictures in treatment
- Start with strong cues and fade later
- Use graphs and charts to show the patient relationships between objects
- Relate the information to experiences that have occurred in the patient's life
- Teach the patient to respond with *yes* or *no*
- Introduce familiar sounds from the patient's home (e.g., the bark of the family dog)
- Use familiar odors to reorient patient to previously identifiable smells
- Gain the patient's attention before talking to him or her (e.g., "Listen, carefully, now"; "I want to say something to you"); educate the clinical staff to do the same
- Give introduction to new topic (e.g., "I am now going to tell you about . . ."); do not introduce topics abruptly

- Assess comprehension of spoken speech frequently by asking the patient to restate what was just said or summarize the main points of discussion, prompt correct responses to minimize errors
- Reinforce nonverbal communication or signs of attention (e.g., eye contact, smiling, nodding); still, assess comprehension to make sure the patient understands
- Withhold attention from irrelevant, inappropriate, and tangential responses; without responding, ask a simple question that might evoke a more relevant response; reinforce it; use the nonexclusion Time-Out to decrease more serious undesirable behaviors
- Reduce complexity and rate of speech if necessary
- Use statements instead of questions when initially communicating with the patient
- Prompt, gesture, and use verbal instructions to help the patient comprehend
- Allow the patient time to listen to instructions
- Use sentence completion tasks for patients with initiation or inhibition difficulties
- Promptly reinforce correct responses and given corrective feedback for incorrect or inappropriate responses

Treatment During the Intermediate Stage

- Establish more complex routines
 - Increase the number of daily routines and activities
 - Periodically assess to ensure that the increase is gradual and that the patient's progress is acceptable
- Teach the patient to request information
 - Ask questions about time, space, or persons (e.g., "What is today's date?")
 - Model a request for information if the patient cannot answer them (e.g., "I am not sure, can you tell me, please?")
- Improve the patient's memory skills
 - Make a list of scheduled activities the patient is expected to participate in at different times of the

day; post it in a prominent place in the patient's room

- Post signs and symbols to remind the patient of activities or appointments
- Teach the patient to consult the lists, look at signs and symbols periodically; ask the patient to read them, state their meaning
- Ask the patient to write a list of activities after consulting the posted list
- Ask the patient to silently read the list and then describe the activities to be attended to
- Use pictures of family members to re-establish the names; show the pictures frequently and ask the patient to name them
- Repeat the names of health care staff throughout the daily routines
- Ask the patient to write down the names of family members and those of health care staff
- Repeat treatment trials
 - Note that repetition is the key to rehearsing the lost skills
 - Repeat the successful trials more than unsuccessful trials
 - Arrange repeated trials with needed rest pauses in between
- Improve selective attention and comprehension by asking the patient to
 - Match pictures to sentences
 - Follow spoken instructions
 - Retell a message to another person
 - Answer simple questions
- Work closely with health care workers; teach them to
 - Recognize the client's problems
 - Respond promptly to positive changes in communication skills
- Invite family members and health care workers to the treatment session; teach them to
 - Prompt the client when there is hesitation
 - Model appropriate behaviors

- Reinforce the behaviors naturally and sustain those behaviors
- Increase awareness of deficits
 - Use simple explanations to describe the problem to the patient
 - Give contingent feedback on problem behaviors
 - Use group therapy to allow the patient to see that others have similar problems
- Reduce irrelevant, inappropriate, and tangential responses
 - Do not reinforce such responses with attention or verbal response
 - Change the topic immediately; reinforce appropriate responses
 - Use the nonexclusion time-out (5 seconds of no interaction) to decrease such responses
- Treat dysarthric speech
 - Postpone treatment for mild dysarthria if speech intelligibility is acceptable and occupational or social demands are minimal
 - Assess what problematic aspects of the complex speech production the patient needs treatment for
 - Offer treatment to improve respiratory support for speech for postural modifications, reduce hyperadduction by teaching gentle voice onset, improving laryngeal adduction by pulling and pushing exercises, improving vocal qualities, reducing hypernasality, and improving articulatory precision, especially that of consonants
 - See [Dysarthrias](#) for details
- Begin to diminish special stimuli and reminders as performance improves toward the end of the intermediate stage or anytime
 - Post reminders and schedules in less conspicuous places
 - Print reminders and schedules in smaller prints
 - Reduce the frequency of verbal reminders
 - Reinstate the original form of reminders when such fading results in skill deterioration; fade again after a while

- Begin to teach self-control (self-monitoring) skills
 - Have the patient pause soon after a wrong or irrelevant response is given
 - Prompt self-correction
 - Reinforce the client for self-correction
 - Ask the client to count the correct and incorrect responses along with you
 - Give feedback on such self-assessment
 - Delay corrective feedback to promote self-correction
 - See [Self-Control \(Self-Monitoring\)](#) for additional suggestions
- Continue to evaluate the need for compensatory skills
 - Continue the use of initial compensatory strategies
 - Add additional strategies as found appropriate (e.g., the use of hand-held electronic device)

Treatment During the Late Stage

- Train more complex activities that enhance independence
 - Teach the client to write detailed instructions to self
 - Teach the client to use hand-held electronic devices to keep track of appointments and activities
 - Teach the client to plan the next day's activities, write them down, and consult the list periodically throughout the day
- Teach conversational narrative skills
 - Initially tell a brief and simple story and ask the client to retell it
 - Prompt details or chronology of events
 - Reinforce for well-sequenced narration
 - Tell progressively more complex stories the patient will retell
 - Reduce the frequency of your prompts
 - Ask the patient to tell a story from his or her experience
 - Ask the patient to describe how he or she would fix a sandwich, make a grocery list, pay utility bills

- Prompt correct responses in correct sequence of steps; reinforce the patient
- Integrate such pragmatic skills as topic maintenance and topic initiation into narrative skill teaching
 - Promote topic maintenance by prompting the patient to “Say more,” “Give details,” “What happened next?” “What about this?” or “What about that?” and so forth
 - Prompt the client to initiate conversation on new topics; fade the prompts
 - Draw attention to your face by saying, “Look at me” or by giving a nonverbal cue to have the client maintain eye contact
 - See specific techniques under Language Disorders in Children to teach such additional conversational skills as turn-taking and conversational repair strategies
- Integrate work-related communication into your treatment activities
 - Have the patient or a family member supply words, phrases, and sentences that are work related
 - Include activities that are specific to work (e.g., drawing in the case of an artist, typing on a computer key board, writing assignments, making brief oral presentations on technical topics)
- Further diminish special stimuli (posters, verbal reminders, written instructions) that control behaviors
- Continue to use shaping, modeling, prompting, and manual guidance to enhance correct responses and to reduce the probability of errors
- Continue to treat dysarthria
- Continue to teach Self-Control (Self-Monitoring) skills as before
- Teach compensatory strategies if necessary, by teaching the patient to
 - Break down tasks into smaller, more manageable components; teach the patients to write down steps involved in performing an action (e.g., steps in preparing a breakfast)

- Request information relative to time, date, and so forth
- Request others to modify their speech (e.g., teach the client to request others to speak slowly or to repeat)
- Rehearse important information (e.g., teach the client to self-talk about how to perform such activities as fixing lunch or changing light bulbs)
- Write down instructions, appointments, important information, and so forth
- Ask for written instructions from people
- Use active instead of passive cues (an alarm instead of a reminder in a diary that may not be consulted)
- Use electronic devices (digital watches that display time, day, and date; auditory signals for appointments); use data bank watches that store messages and appointments; use electronic pill boxes that remind the patient to take medications; use recorders (e.g., tablets or cell phones) to record lectures, instructions, and discourse they can listen to later; use hand-held electronic spell checkers; use notebook computers for more complex information management
- Limit distractions or modify environment by finding quiet places to study or rearrange work environments
- Keep possessions at specific and constant places to improve access
- Teach the patient to self-cue activities, names, places, and appointments
- Teach organizational strategies by teaching the patient to
 - Separate relevant from irrelevant material
 - Summarize, highlight, and take notes
 - Self-monitor errors and self-correct them
- Ask patients to copy symbols, letters, and words that commonly occur in their surroundings (e.g., signs that say “No drinking, smoking, and eating”)

- Develop a core vocabulary that the patient is likely to use every day
- Teach the patient to recognize letters, syllables, words, phrases, and sentences
- Hold group treatment sessions; note, however, that there is no strong empirical evidence to support this, although it is a common practice; structure these group interactions to
 - Promote pragmatic communication skills (discourse, topic maintenance, and topic initiation)
 - Increase general socialization
 - Increase socially appropriate verbalization
 - Encourage self-evaluation of strengths and limitations
- Promote community reentry; note that rehabilitation should end with successful community reentry in which steps are taken to ensure a smooth transition from the rehabilitation setting to home, school, work, and the larger social situations
 - Prepare the patient for reentry; in the final stages of treatment, emphasize self-help skills and independent living skills; stimulate the patient's interest in academic, social, occupational, and household activities
 - Educate family members, teachers, and supervisors about the current status of the patient; let them appreciate the patient's strengths and limitations; let them understand the patient's compensatory strategies and the continued support needed
 - Modify the communication styles of family members and significant others to suit the remaining and perhaps permanent deficits (e.g., teach them to speak slowly, repeat often, speak in simpler sentences)
 - Modify the number and nature of demands people make (e.g., the teacher may give reduced amount of work or simplified work; work supervisor may give extra time to complete a task)

- Teach family members and others to recognize reasons for oppositional behaviors (e.g., oppositional behaviors may diminish if the demands are modified or tasks are simplified)

Blake, M. L. (2018). *The right hemisphere and disorders of cognition and communication*. San Diego, CA: Plural.

Hegde, M. N. (2018). *A coursebook on aphasia and other neurogenic language disorders* (4th ed.). San Diego, CA: Plural.

High, W. M., Sander, A. M., Struchen, M. A., & Hart, K. A. (2006). *Rehabilitation for traumatic brain injury*. New York, NY: Oxford University Press.

Hollowell, B. (2017). *Aphasia and other acquired neurogenic language disorders*. San Diego, CA: Plural.

Traumatic Brain Injury (TBI) in Children. Treatment of communication disorders in children who have sustained cerebral injury due to external force requires a few special considerations because, unlike in adults, still developing skills are impaired; compared with TBI in adults, injury to an immature brain will have different consequences to academic and social development of children; generally, the treatment procedures described under Traumatic Brain Injury in Adults are applicable with suitable modifications to accommodate a child's special needs and demands; in addition, because the child's language acquisition may be impaired, the procedures described under Language Disorders in Children also are generally applicable; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for epidemiology, etiology, symptomatology, and neurological and behavioral symptoms associated with traumatic brain injury; see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* to make a thorough assessment of communication and related disorders before developing an intervention plan; a few special considerations in treating children with TBI include the following:

- Design a treatment program that is based on the child's social, family, and academic communication needs

- Develop a family-centered management plan; work closely with family members in selecting the most functional and effective communication targets for the child
- Offer counseling to family members who may be overwhelmed with the multiple problems associated with TBI in children, including aggressive and antisocial behaviors
- Discuss the child's needs with the child's teachers, educational psychologists, and special educators including reading specialists
- Analyze the child's curricula to select treatment targets that will help the child meet his or her academic needs; include academic terms in language training
- Integrate literacy skills into communication training; see [Literacy and Literacy Skills](#)
- Work with the teacher to help the child with classroom communication and general behavior; suggest to the teacher that it might be helpful to
 - Simplify the academic tasks for the child
 - Shape difficult tasks
 - Use simpler language spoken at slower rate
 - Limit distractions in the classroom
 - Keep the classroom situation organized with little variation
 - Use gestures and signs along with verbal expressions
 - Repeat instructions, give written instructions
 - Ask the child to repeat the instructions
 - Ask the child to take adequate notes
 - Encourage the child to request help and promptly reinforce such attempts
 - Accept any mode of expression initially, but expect more refined verbal communication eventually
- Keep the teacher and other educators serving the child informed of your treatment targets, general procedures, and outcome
- Develop and implement a comprehensive communication intervention program
 - Retrain to restore lost skills
 - Treat [Memory Impairments](#)
 - Treat [Dysarthria](#), which tends to be associated with TBI

- Treat language disorders; see Language Disorders in Children
- Train conversational skills involving the child's peers in the treatment sessions (e.g., Topic Initiation, Topic Maintenance, Turn-Taking)
- Implement a habilitative program to teach skills not yet acquired
 - Assess speech and language skills not yet mastered before the injury
 - Teach grammatical morphemes, sentence types and varieties, and social communication skills the child had not mastered
- Consider Augmentative and Alternative Communication options for the child whose verbal skills do not meet his or her communication needs
 - Consider aids (Picture Exchange Communication System, drawing, hand-held devices, tablets, smart phones)
 - Consider unaided forms (gestures, formal or informal signs) for the child
- Consider environmental modifications and accommodations the child might need
 - Optimize the classroom for the child's learning; reduce noise, request the teacher to give more time to complete the test, give frequent rest breaks, arrange for a note taker, and so forth
 - Arrange for a temporary reduction in the number and complexity of assigned home work, give tests in different formats (e.g., multiple choice questions versus essays)
- Use behavioral management techniques in all aspects of teaching, as well as in
 - Controlling undesirable behaviors by reinforcing alternative desirable behaviors; use Differential Reinforcement techniques
 - Prefer extinction, time-out, and response cost to other direct response reduction procedures
- Develop a home treatment program and train parents in its implementation; train family members to support the

child's communicative attempts by prompts and positive reinforcement

- Implement an intervention program if the child has Dysphagia
- Select relevant recommendations from “Promote community reentry” under Treatment During the Late Stage (previous entry)

Allison, B., Byom, L. J., & Turkstra, L. S. (2017). Traumatic brain injury in children and adults. In A. Johnson & B. Jacobson (Eds.), *Medical speech-language pathology* (3rd ed.). New York, NY: Thieme.

Togher, L., McDonald, S., Tate, R., Power, E., & Rietdijk, R. (2013). Training communication partners of people with severe traumatic brain injury improves everyday conversations: A multicenter single blind clinical trial. *Journal of Rehabilitation Medicine*, 45, 637–645.

Ylvisaker, M. E., Turkstra, L., Coelho, C., Yorkston, K., Kennedy, M., Sohlberg, M. M., & Avery, J. (2007). Behavioural interventions for children and adults with behaviour disorders after TBI: A systematic review of the evidence. *Brain Injury*, 21, 769–805.

Treatment. Application of a variable that can induce changes; use of any effective procedure in teaching new communicative skills; behaviorally, management of contingent relations between antecedents, responses, and consequences; conceptually, a rearrangement of communicative relationships between a speaker and his or her listener.

Treatment of Communicative Disorders: Procedures That Apply Across Disorders. Common procedures used in treating most if not all disorders of communication; modified to suit the individual client, his or her specific problems, the specific target behaviors, and in light of the performance data.

- Assess the client
 - Determine the diagnosis
 - Describe the strengths and limitations of the client
 - Describe the client's current level of communicative performance
- Evaluate the client's family constellation
 - Assess the family communication patterns

- Describe the family support and resources
- Describe the social, educational, or occupational demands made on the client
- Select functional, client-specific target behaviors
 - Select behaviors that, when treated, will have the greatest effects on the client's communication in social situations
 - Select both short- and long-term targets
 - Define the dismissal criterion
- Establish the pretreatment or baseline measures of target behaviors
 - Select stimuli for evoking the target behaviors
 - Repeat the measures to establish reliability
 - Use the Baseline Evoked Trials and Baseline Modeled Trials
 - Take an extended conversational speech sample
 - Obtain home sample if possible
- Design a flexible therapeutic environment
 - Use the degree of control and structure that is necessary
 - Gradually, loosen the structure to make the treatment environment more like the client's everyday environment
- Write a treatment program; specify
 - the target behaviors
 - treatment procedures
 - Reinforcers or Correct Feedback procedures
 - Criteria for Making Clinical Decisions (moving from one level of treatment to another)
 - Probe procedure
 - a Maintenance Strategy
 - the Follow-Up procedure
 - Booster Treatment
- Implement the treatment program
 - Use objects, pictures, demonstrated actions, and so forth to evoke the target behaviors
 - Give instructions, demonstrations, explanations
 - Model the target responses
 - Prompt the target responses
 - Use manual guidance to assist the client in producing the target responses

- Shape the responses
- Fade the special stimuli including pictures, objects, modeling, prompts, and manual guidance
- Give prompt, positive feedback to the client; use natural reinforcers; if you used tangible reinforcers, fade them; decrease the amount of feedback given
- Give prompt, corrective feedback to the client; say “No” or “Wrong”; use other procedures as found appropriate (time-out, response cost)
- Do not always start treatment at the lowest level (e.g., simplest phonemes, isolated phonemes, words), unless shaping is needed; start treatment at a level of complexity the individual can handle (more complex phonemes, words instead of isolated phonemes; more complex sentences instead of the simplest forms); drop down a level only if the performance is not sustained at a higher level
- Probe for generalized production as often as necessary
- Shift treatment, in progressive steps, to more complex levels as the client meets the probe criterion
- Always train the target behaviors in conversational speech with natural consequences
- Implement the maintenance program
 - Train family members, teachers, friends, and professional caregivers in supporting the client’s communicative behaviors
 - Teach them to evoke the target behaviors and reinforce the client naturally
 - Shift training to nonclinical settings
 - Invite other persons into the treatment sessions
 - Have family members conduct informal treatment sessions at home
 - Have teachers reinforce the target skills you teach in conversation with the child and integrate those skills into their literacy and other academic work
 - Teach the client to self-monitor his or her errors and target behaviors
 - Teach the client to count his or her relevant behaviors
 - Teach the client to self-correct mistakes
 - Teach the client to cue himself or herself

- Teach the client to pause soon after an error response is produced
- Dismiss the client when responses are reliably produced in natural settings
- Follow up the client
 - Set up a schedule for follow-up
 - Follow up a client for a duration necessary to document maintenance
 - Take conversational speech samples during follow-up sessions
 - Measure the production of relevant communicative skills
 - Recommend booster treatment if the skills have deteriorated
- Arrange for booster treatment
 - Give the same or better treatment
 - Probe the response rates
 - Schedule the next follow-up if necessary

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Treatment of Communicative Disorders: A General Sequence that Applies Across Disorders.

It is important to design treatment programs that include step-wise progression in target skills mastered; the sequence may be based on response topography, response modes, multiple targets, training and maintenance, and response consequences.

- Simplify and sequence the target behaviors topographically; never start treatment at a lower level if the client can perform at the higher level (e.g., if the child can imitate correct production of sounds in words, do not start treatment at the syllable level); experiment to see if the client can manage at a higher level; always start treatment at the highest level possible; topographic levels of treatment include:
 - Syllables or words
 - Phrases
 - Sentences that are imitated or otherwise controlled
 - Sentences that are more spontaneously produced

- Sentences that are fully spontaneously produced
- Conversational speech
- Narratives and discourse
- Sequence the response modes; the two most common modes, often both necessary include:
 - Imitated; begin treatment with imitation as the initial response mode if necessary; it often is necessary
 - Evoked; fade modeling and move to evoked response level
- Sequence the multiple targets; most clients need treatment for multiple targets that need to be properly sequenced for effective and efficient training time; most commonly:
 - Teach the most useful behaviors earlier than the less useful ones
 - Teach relatively more complex behaviors before you teach the most complex behaviors; experiment with different levels of complexity
 - Teach first those behaviors that are building blocks for other behaviors
 - When one target behavior reaches the probe criterion, select another behavior or shift training to a more complex level of the behavior under training
- Sequence training and maintenance strategies; typically, behaviors need to be established to some extent before maintenance strategies may be implemented:
 - Initially establish the behavior under structured clinical situations
 - Loosen the structure gradually and make treatment conditions more similar to natural conditions
 - Shift treatment to more natural conditions in and around the clinic
 - Shift training to natural conditions away from the clinic
 - Shift training to home situations, but do this as soon as possible (do not wait until the last stage of training)
- Sequence response consequences or feedback variations; how much and what kind of feedback is offered also create sequences in treatment:
 - Give more frequent and consistent feedback in the beginning; perhaps a fixed ratio 1 (FR1; every correct

response is reinforced and every incorrect response is given corrective feedback)

- Reduce the amount of feedback as the learning stabilizes (shift the ratio gradually to FR2, FR4, FR6, and so forth)
- Give tangible reinforcers if necessary and only in the beginning
- Shift to social and more natural reinforcers (verbal praise, agreement, simile)
- Train others to give natural feedback in naturalistic settings

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Treatment of Communicative Disorders: Procedural Modifications. It is critical to make prompt changes in treatment procedures because of their ineffectiveness or less than optimum effectiveness; modifications may be made in antecedents, responses, and consequences; treatment procedures, not principles, are modified; all modifications are based on the client's performance data.

- Modification of antecedents
 - Change stimuli that are ineffective in evoking the target responses
 - Shift from pictures to objects
 - Shift from line drawings to photographs
 - Shift from abstract to concrete stimuli
 - Shift from pictorial representation to enacted stimuli
 - Discard clinical stimuli in favor of stimuli from the client's home
 - Model if evoking is not effective
 - Prompt if evoking is not effective
 - Provide manual guidance (physical assistance to execute a response) if the evoking techniques are not effective
 - Give instructions and repeat them
 - Ask effective, common questions to evoke the responses
 - Rephrase ineffective questions

- Modification of responses
 - Simplify the response if a more complex topographic feature is ineffective (too difficult)
 - If the target is not produced in sentences, shift downward in progressive steps
 - Abandon training on a behavior that is too difficult for the client in favor of one that is easier; use the baseline data for guidance
 - Move quickly to a higher level of complexity if the level being trained seems too easy to the client
 - Abandon training on a behavior that is not imitated in favor of the one that is
 - Return to abandoned behaviors at a later date; shape them in small steps
- Modification of consequences
 - Use the operational definition of consequences; to be called reinforcers, events should increase the behaviors on which they are made contingent; punishers or correctives should decrease the behaviors on which they are made contingent
 - Change consequating events that do not increase behaviors
 - Change consequating events that do not decrease behaviors
 - Use primary reinforcers if social consequences do not reinforce
 - Shift back to social reinforcers after the behaviors are established
 - Use tokens backed up by a variety of reinforcers or time-out if other forms fail
 - Use one of the Differential Reinforcement procedures to simultaneously increase a desirable behavior and decrease an undesirable behavior
 - Use biofeedback if other forms fail

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Treatment Evaluation. Clinicians should have a knowledge of procedures by which the immediate and long-term

effects of treatment procedures are established through controlled experimental analysis; such a knowledge is essential to select treatment procedures that will work; see [Treatment Selection Criteria](#); treatment evaluation involves:

- Using group or single-subject experimental designs in which a new treatment (or hitherto untested treatment) is compared with no treatment—an essential feature of treatment evaluation
- Selecting participants for the experimental evaluation of the new treatment procedure; in group designs, participants are selected randomly and then assigned randomly to the experimental group; in single-subject designs, available individuals, preferably homogeneous individuals, are selected
- Comparing the participants' skills under treatment versus no treatment to show that the treatment is better than no treatment
- Ruling out extraneous variables to show that it was the treatment, and not some other factor (e.g., maturation, teacher's work, parents' actions), that was responsible for the documented positive changes
- Replicating the study by the original investigator as well as other investigators to show that the technique is effective (or not effective) when applied in different settings, by different clinicians, and in treating different clients

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Treatment Evoked Trials. Treatment conducted with evoked trials is a common feature in speech-language pathology; such evoked trials are structured and temporally separated opportunities for the client to produce a target response in the absence of clinician's modeling; useful in establishing target behaviors, especially with clients who perform better under a highly structured treatment session.

- Place the stimulus item in front of the client or demonstrate an action
- Ask the relevant predetermined question
- Wait a few seconds for the client to respond
- If the response is correct, reinforce the client

- If the response is incorrect, give corrective feedback
- Record the response on the recording sheet
- Remove the stimulus item or move it slightly toward yourself
- Wait 2–3 seconds to signify the end of the trial
- Begin the next trial
- Calculate the percent correct response rate

Treatment Modeled Trials. Treatment almost always starts with modeling of the target skill for the client to imitate; modeled trials are structured and temporally separated opportunities for the client to imitate a clinician-modeled response; when the client imitates correctly on a certain number of trials (e.g., five consecutively correct), modeled trials are faded into evoked trials.

- Place a stimulus item in front of the client or demonstrate an action
- Ask the predetermined question
- Immediately model the correct response
- If the response is correct, reinforce the client
- If the response is incorrect, give corrective feedback
- Wait a few seconds for the client to respond
- Record the response on the recording sheet
- Remove the stimulus item
- Wait 2–3 seconds to signify the end of the trial
- Calculate the percent correct response rate

Treatment Selection Criteria. General guidelines on selecting treatment procedures; select procedures according to the following criteria.

Importance of Critical Selection of Treatment

- Many treatment procedures in communicative disorders remain experimentally untested to show that treatment is better than no treatment
- It is the ethical responsibility of clinicians to use techniques that have been shown to be effective in controlled experimental research
- It is necessary for clinicians to have a good knowledge of treatment research designs, without which they cannot evaluate treatment procedures offered to them

- Many specialists vigorously advocate and offer “new and revolutionary” treatments without controlled experimental data and many clinicians enthusiastically accept them
- Rejection of treatment procedures based solely on opinions, speculation, questionable and speculative theories, bandwagon, or popularity is not just good clinical practice, it is a clinician’s ethical responsibility
- Techniques that appear logical, appealing, likable, and so forth may not necessarily be effective
- Widespread use of a technique is not an assurance of its effectiveness
- Certain unpopular techniques (e.g., time-out and response cost in the treatment of stuttering) are known to be effective, as shown by controlled research
- Speech-language pathology has been highly vulnerable to faddish trends in treatment
- Getting on a bandwagon quickly is not the best way of staying current in the discipline
- Some of the techniques you have been using with conviction and enthusiasm for years may never have been tested and may in fact be useless
- Useless or ineffective treatments are not benign; they are harmful because they prolong the negative consequences of a disorder for the client and discourage people from seeking more effective treatments

Evaluating the Levels of Evidence to Select or Reject Treatment Procedures.

Note that the criteria are hierarchically arranged based on the levels of evidence; a higher criterion is more stringent, more difficult to meet, although more preferable than a lower criterion; a treatment technique is accepted or rejected based on the level of evidence on which it is advocated:

- Level 1. Expert Advocacy. Some techniques are simply advocated by experts; no evidence of any kind is offered; may sound logical and appealing, and the advocate may be influential or well-known; reject all such procedures

- Level 2. Unreplicated, uncontrolled case studies. Some techniques may have been clinically tried with a few clients and a case study published; the study has not been repeated (replicated); no control groups or single-subject controlled conditions were used; clients have improved under treatment (see Improvement); no assurance that treatment was effective because extraneous variables have not been ruled out (see Effectiveness); because at least improvement with the technique was documented, you may select this type of treatment and use it with caution, provided that no technique with better evidence is available to treat that particular disorder
- Level 3. Replicated, uncontrolled case studies. A technique has been applied more than once and multiple case studies have been published; no control groups or controlled conditions yet; clients have improved under treatment; such a technique may be used, although no one can claim that the treatment is effective
- Level 4. Unreplicated, controlled experimental studies. The technique has been experimentally evaluated with a control group or a single-subject experimental design with control conditions; treatment was shown to be effective by ruling out extraneous variables; but the generality and applicability of the technique is unknown because it has not been applied in different settings by different clinicians (replication); clinician can use a technique that has been experimentally verified; not just improvement, but effectiveness is claimed for the technique
- Level 5. Replicated, controlled experimental studies. A technique has been shown to be effective in experimental research and then shown to have generality by repeating its application in different settings, by different clinicians, in treating a variety of clients; therefore, this technique is the most desirable; this is the kind of technique the clinician would want to select

Hegde, M. N. (2003). *Clinical research in communicative disorders: Principles and strategies* (3rd ed.). Austin, TX: Pro-Ed.

Treatment Targets. Skills or behaviors selected for teaching can make a difference between successful versus unsuccessful treatment; select treatment targets that will ensure success in treatment and improve the client's social communication:

- Select treatment targets after a thorough assessment
- Select functional targets that are useful to the client; consult with the significant others in the client's life (teachers, family members, colleagues) to select target skills that will be effective in family, social, educational, or employment contexts
- Select targets that are verbally and culturally appropriate to the client; select targets only after understanding the client's and the family's ethnocultural background
- Select skills that can make an immediate and socially significant difference in the communicative skills of the client; keep in perspective the more advanced academic or occupational communication skills that may be integrated with treatment in different stages
- Select behaviors that serve as building blocks for more complex functional skills

Treatment Variables. Technical operations performed by the clinician to create, increase, or decrease behaviors; these include:

- Antecedents or stimuli used in treatment, including modeling, instructions, demonstrations, manual guidance, pictures, objects, recreated events, storytelling (by the clinician), topics of conversation, and so forth
- Consequences or feedback the clinician gives, including verbal praise, tokens, tangible reinforcers, opportunities to indulge in various activities, privileges offered by parents, and so forth

T

Treatment Versus Stimulation. Whether to actively and directly treat skills or to offer indirect stimulation programs can be a dilemma in some contexts, especially in case of infants and toddlers; see [Stimulation Versus Treatment](#); see also [Language Disorders in Infants and Toddlers](#); in most cases, direct treatment, if possible at all, is preferred

to stimulation—a vague and generally inefficient approach; inefficient because lack of specific target behaviors and antecedent and consequent control.

Trials. Measurable and sequentially repeatable opportunities to produce a response; may be more or less structured; include Baseline Evoked Trials, Baseline Modeled Trials, Treatment Evoked Trials, and Treatment Modeled Trials.

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Turn-Taking. To teach the role of a listener as well as talker during conversation, see Language Disorders in Children.

U

Unconditioned Reinforcers. Reinforcers whose effects do not depend on past learning or conditioning (e.g., food items); the same as Primary Reinforcers; see also Conditioned Reinforcers.

Unconditioned Response. A response given to unconditioned stimulus; typically a response without a conditioning or learning history (e.g., salivary response to food in the mouth); see Conditioned Response.

Unconditioned Stimulus. A stimulus that elicits a response without the benefit of conditioning or learning (e.g., food in the mouth that automatically elicits a salivary response); see Conditioned Stimulus.

Unilateral Upper Motor Neuron Dysarthria. To treat this type of motor speech disorder, see Dysarthrias and Treatment of Dysarthria: Specific Types.

v

Validity. May refer to assessment instruments or treatment procedures; validity of assessment instruments (standardized tests) is the degree to which a measuring instrument measures what it purports to measure (see *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for types of assessment test validity; in treatment evaluation and research, valid procedures indeed produce the attributed effects of those procedures; validity is typically considered a matter of several kinds of generality in treatment research; treatment procedures with multiple kinds of validity are preferred to those with no or limited kinds of validity:

- Inferential (statistical) generality: Extending the conclusions of a group treatment research study to the population from which the study sample was randomly drawn; there virtually are no studies that randomly draw participants from the general population
- Logical generality: Extending the conclusions of a single-subject design study to the clients who are similar to those (and only to those) who participated in that study
- Generality across individuals: Experimental demonstration that a method is effective when applied to other individuals with the same disorder
- Generality across settings: Experimental demonstration that a method is effective when applied in different clinical or educational settings
- Generality across clinicians: Experimental demonstration that a method is effective when applied by different clinicians
- Generality across response classes: Experimental demonstration that a method is effective across different kinds of skills or across disorders

Hegde, M. N. (1998). *Treatment procedures in communicative disorders* (3rd ed.). Austin, TX: Pro-Ed.

Variable Interval Schedule (VI). An intermittent reinforcement schedule in which the time duration between reinforcers is varied around an average; not as applicable as the Fixed Interval or Variable Ratio Schedules in the treatment of communicative disorders; difficult to use in routine

clinical work; efficient with electronic behavioral programming equipment.

Variable Ratio Schedule (VR). An intermittent reinforcement schedule in which the number of responses needed to earn a reinforcer is varied around an average; more effective in generating response rates that last; useful in a maintenance strategy.

- Vary the number of responses required for reinforcement from one occasion to the other
- Initially, reinforce on a smaller ratio schedule (e.g., a VR_4) and increase the ratio gradually (e.g., VR_6 , VR_8 , VR_{12})

Vascular Dementia. Treatment of vascular dementia follows the same general principles and procedures described under Dementia; the clinician needs to manage communication problems associated with three common types of vascular dementia.

- Large vessel occlusions. To treat this type of dementia:
 - Target cognitive impairments, including confusion, in the early stage of the disease
 - Treat inconsistent memory impairments that may range from total difficulty to recall an event and sudden recall of the same event
- Lacunar State. To treat dementia due to lacunar states:
 - Treat Dysarthrias
 - Treat Dysphagia
 - Treat other communication problems characteristic of Dementia, especially in the late stage of the disease
- Binswanger's Disease. To treat dementia associated with this disease:
 - Target symptoms that are similar to those found in patients with lacunar states
 - Target symptoms of Dementia in the later stage of the disease

Hegde, M. N. (2018). *A coursebook on aphasia and other neurogenic language disorders* (4th ed.). San Diego, CA: Plural.

Jagust, W. (2001). Understanding vascular dementia. *Lancet*, 358, 2097–2098.

Velopharyngeal Dysfunction (VPD). Communication-based treatment is not appropriate for certain kinds of VPDs; when appropriate, the treatment may help control resonance disorders; communication and other kinds of clinical management, including prosthetic and surgical treatment, may be considered for children with clefts, those with dysarthria, and many clients who present voice and resonance disorders; disorders that need to be treated include hypernasality, hyponasality, denasality, cul-de-sac nasality, nasal air emission, weak consonant productions, and compensatory articulation errors; see the sources cited at the end of this main entry and the companion volume *Hegde's PocketGuide to Communication Disorders* for etiologic and symptomatic details; see also *Hegde's PocketGuide to Assessment in Speech-Language Pathology* for diagnostic procedures.

Treatment for VPD: General Guidelines

- Treatment for clients with velopharyngeal dysfunction, especially in children with palatal clefts, may target correct placement of articulators for producing different speech sounds or reduction of hypernasality and audible nasal emission
- Clients who are not candidates for resonance treatment may be candidates for teaching correct placement of articulators
- If abnormal, inadequate, or insufficient mechanism is the sole cause of poor velopharyngeal closure, resulting in significant speech and resonance disorders, speech treatment is unwarranted, as it is likely to be ineffective; it is also unethical to offer speech treatment when the neuromuscular mechanism is incapable of closure; the client should be referred to other kinds of intervention (e.g., surgical or prosthetic)
- When the client has received other kinds of intervention (e.g., surgical or prosthetic), speech-language pathologist may offer speech treatment to maximize the benefits, to teach effective speech production following such other interventions; clients need help in using the new structures or prosthetic devices in place; speech treatment can accomplish this goal

- Compensatory speech treatment may be justifiable and offered to individuals with physiologically inadequate mechanism only if no other intervention is possible; in such cases, the clinician may try to teach compensatory skills that might minimize the resonance disorders caused by a defective velopharyngeal mechanism that cannot be corrected
- When hypernasality and nasal air emission are mild, and it is therefore difficult to judge the candidacy for speech treatment, it is acceptable to offer trial speech treatment with a clear understanding given to the clients and the families that the treatment is purely exploratory or diagnostic in nature and that it may have to be discontinued if no progress is evident
- Trial treatment also may be offered to clients who show minimal hypernasality and audible nasal emission which is evident only when fatigued, and the problems are inconsistent (i.e., sometimes there seems to be good closure); it is likely that in such cases, treatment will be successfully concluded
- Any time trial therapy is offered, the clinician should make frequent evaluations of the effect to stop treatment when continuation appears unethical
- Treatment for articulatory placement may be offered anytime to any client, even if there is significant velopharyngeal insufficiency, causing prominent hypernasality and audible nasal emission; clients who are waiting for surgery or have not yet been evaluated for it may still be scheduled for articulatory placement therapy but not for resonance therapy; velopharyngeal insufficiency may lead to compensatory misarticulations but will not prevent learning of correct articulatory placements

Treatment Strategies for VFD

- Treatment of hypernasality, audible nasal emission, or both includes a variety of strategies; these are offered when there is expectation of improvement and the problems are mild
- Note that many techniques and programs designed to treat speech problems resulting from velopharyngeal

closure deficiencies have not been experimentally evaluated; many of them cannot be recommended

- Nonspeech exercises that include blowing, sucking, whistling, cheek puffing, swallowing, and playing wind instruments are offered as treatment to improve velopharyngeal closure; such techniques are ineffective and are not recommended
- Such other tactics as increased mouth opening, light articulatory contacts, and decreased rate of speech have been used with questionable outcomes
- Potentially useful approaches include the following:
 - Temporary speech prostheses: Have a speech bulb prepared for the client and evaluate its use as a temporary means of closing the velopharyngeal gap; this requires a nasopharyngoscopy examination to properly construct, modify, and fit the prosthetic bulb; evidence on the effectiveness of speech bulbs in reducing hypernasality is weak, and therefore be cautious about its recommendation
 - Mechanical feedback techniques: Biofeedback with the help of a nasometer and similar instruments is judged practical, not necessarily because of some strong evidence in its favor, but because of its availability; some instruments give feedback on both the oral and nasal resonance, and this feedback might help shape a more appropriate resonance pattern; nasal endoscopic feedback gives the client a chance to observe the closure during continuous speech production; select clients who achieve velopharyngeal closure during blowing and production of pressure consonants for this procedure, as they are likely to benefit the most

Kummer, A. W. (2014). *Cleft palate and craniofacial anomalies* (3rd ed.). Clifton Park, NY: Thomson Delmar Learning.

Peterson-Falzone, S. J., Hardin-Jones, M. A., & Karnell, M. P. (2001). *Cleft palate speech* (3rd ed.). St. Louis, MO: Mosby.

Zajac, D. J., & Vallino, Linda, D. (2017). *Evaluation and management of cleft lip and palate*. San Diego, CA: Plural.

Ventricular Dysphonia. In treating this type of voice disorder characterized by low pitch, monotone, decreased loudness, harshness, and arrhythmic voicing, modify the way the voice is produced; limit the participation of ventricular folds in voice production.

- Teach the client to take a prolonged inhalation through the open mouth and sustained exhalation without phonation
- Teach the client Inhalation Phonation (this is usually true fold phonation)
- Ask the client to produce inhalation phonation–exhalation phonation on the same breath
- Ask the client to produce a matching exhalation phonation
- Have the client practice exhalation phonation
- Teach the client to vary the pitch
- Fade inhalation–exhalation; stabilize normal phonation in conversational speech

Verbal Apraxia. An older term for Apraxia of Speech.

Verbal Corrective Feedback. A preferred method of reducing incorrect responses in treatment; most frequently used in treating virtually all disorders of communication; it is feedback presented soon after an incorrect response is made; a consequence known to decrease undesirable responses.

- Give corrective feedback immediately after a wrong response is made or even when it is imminent
- Give feedback in an objective tone
- Say, “No,” “Wrong,” or “Not correct”
- Frequently model the correct response and shape responses to reduce the chances of incorrect responses
- Combine them with Nonverbal Corrective Feedback.

Verbal Praise. A common method of strengthening correct responses given during treatment; a type of positive reinforcement in which the client is praised for giving correct responses or for imitating modeled responses; a

response consequence known to increase the frequency of target behaviors.

- Praise the client promptly for producing or imitating a correct response
- Use such phrases as “Good Job!,” “I like that!,” “That was correct!,” “Excellent!,” and so forth; select the phrases and words that are appropriate for the client’s age, education, and cultural background
- Deliver verbal praise in a natural manner, with appropriate emotion and facial expression
- Keep an accurate record of response frequency to make sure that the verbal consequences are indeed functioning as reinforcers (i.e., the response rate is increasing)
- Use other forms of reinforcers if verbal praise does not work (e.g., tokens or primary reinforcement)

Verbal Prompt. Use this effective technique to evoke an unsure response; they are similar to *hints* of everyday life; commonly used in the treatment of communication disorders; see [Prompts](#).

Verbal Stimulus Generalization. Production of unreinforced verbal responses when untrained verbal stimuli are presented; measured on a [Probe](#).

- Present the same physical stimulus as used in training (e.g., the picture of *two books* used in teaching an exemplar of the plural morpheme)
- Ask a question other than the one used in training to evoke the response (e.g., if you asked, “What do you see?” on training trials, ask, “What are these?” on these probe trials)
- Do not model and do not provide any response consequences
- Record the response
- Present at least 10 trials, each involving a different picture and probe question (different from the training question)

Visual Alerting Systems. A variety of assistive listening devices for the deaf; generate visual signals (light) that

alert the person to incoming phone calls, door bells, and smoke alarms; see [Aural Rehabilitation](#).

Vocal Emphasis. Use this effective treatment technique in which the target feature within a word, phrase, or sentence is highlighted in some manner while modeling it.

- In speech sound production training, emphasize the target phoneme with extra stress, increased loudness, or a slight prolongation
- In language treatment, emphasize the modeled grammatic morphemes or other targets with increased stress, loudness, or slight prolongation of the vowels

Vocal Fold Paralysis. Treatment of voice disorders associated with vocal fold paralysis requires a determination of whether the paralysis is bilateral or unilateral; respiratory survival and feeding are the main concerns when the paralysis is bilateral; unilateral vocal fold paralysis is more common, often due to trauma or accidental cutting of the recurrent laryngeal nerve, and the resulting aphonia or dysphonia may be treated; Teflon or collagen may be injected into the paralyzed fold to make it bulge and help approximate; collagen is currently preferred over Teflon; although appropriate, voice therapy in the case of unilateral vocal fold paralysis may only produce temporary effects, and the goals may be limited to having the client produce functional voice.

- Model and reinforce a higher-pitched voice that might improve the voice
- Use the [Half-Swallow Boom](#)
- Use [Digital Manipulation of the Larynx](#)
- Try different [Head Positioning](#) maneuvers

Vocal Fry. Popcorn popping or bubbling type of voice occurring toward the lower end of the pitch range; may be slightly hoarse; a normal characteristic that may be abnormal if exhibited too frequently.

Vocal Hyperfunction. Treatment of vocally abusive behaviors that cause nodules, polyps, and associated voice disorders is the bulk of routine voice therapy; *hyperfunc-*

tion is speaking with excessive muscular effort and force which must be reduced during treatment.

- Assess the disorder to find out the specific kinds of vocally abusive behaviors the client exhibits
- Reduce the vocally abusive behaviors
- Experiment with different Specific Normal Voice Facilitating Techniques (described under Voice Disorders) to promote normal or vastly improved voice
- Use those techniques to teach the client the more relaxed and normal voice production

Vocal Nodules. Treatment of voice disorders resulting from benign lesions of the vocal folds targets a reduction of such symptoms as hoarseness, harshness, periodic aphonia, frequent throat clearing, hard glottal attacks, vocal tension, and a dry vocal tract; voice therapy designed to reduce vocally abusive behaviors is preferred over surgical intervention; see Treatment of Vocally Abusive Behaviors under Voice Disorders; essentially:

- Prescribe mandatory vocal rest if surgery has been performed to remove the nodules
- Reduce Vocally Abusive Behaviors
- Increase the breath support for speech
- Reduce vocal intensity
- Use Specific Normal Voice Facilitation Techniques (described under Voice Disorders) to teach the client to produce healthy voice

Vocally Abusive Behaviors. A variety of behaviors that negatively affect the laryngeal mechanism and result in voice disorders; intervention described under Voice Disorders; Treatment of Vocally Abusive Behaviors; generally, reduce the frequency of the following actions:

- Excessive talking, singing, or humming
- Abusive singing habits (e.g., singing at inappropriate pitch levels, with poor laryngeal positions, with poor general posture, trying to maintain singing levels that exceed the singer's capabilities)
- Constant and excessive religious chanting
- Excessively loud talking or singing

- Excessive whispering
- Yelling, cheering, and screaming
- Excessive and chronic coughing and throat clearing
- Constant or excessive talking or singing during episodes of allergy, upper respiratory infection, and during menstruation
- Hazardous occupational use of voice (teaching, preaching, sports coaching, aerobic instruction, cheering, pep club activities, and political campaigning)
- Smoking and constant exposure to smoke (secondhand smoke)
- Excessive alcohol intake
- Constant and excessive grunting during exercises
- Habitual name shouting (calling) from a distance
- Constant talking in noisy environments (e.g., bars, sports arenas, construction sites)
- Speaking with hard glottal attack
- Singing or talking at the upper or lower end of the pitch range
- Excessive crying or laughing
- Inadequate fluid intake
- Playing with excessive vocal activity (making loud mechanical and animal noises during pretend play)
- Frequent pushing or lifting heavy objects
- Forceful effort during bowel movements

Voice Disorders. Treatment of various disorders involves addressing faulty, abnormal, or inappropriate phonation, loudness, pitch, and resonance; reduction of vocally abusive behaviors is a significant part of treating voice disorders in children and adults; speech-language pathologists treating voice disorders need to work with medical professionals, especially laryngologists who medically treat conditions associated with some of the voice disorders; see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Communication Disorders*, for various kinds of voice disorders and their classification, their epidemiology, etiology, symptomatology, and associated physical diseases or abnormalities; to make both an instrumental and clinical assessment of voice disorders,

see the sources cited at the end of this main entry and the companion volume, *Hegde's PocketGuide to Assessment in Speech-Language Pathology*. Generally, treatment targets include:

- Disorders of loudness. Socially inappropriate voice that is too loud or soft.
- Disorders of pitch. Voice characterized by inappropriate pitch; speech at the low end of one's pitch range, which requires too much effort and force; or speech at the high end of the range, which causes fatigue.
- Disorders of resonance. Voice characterized by inappropriate resonance, including *Hypernasality* and *Hyponasality*.
- Disorders of vocal quality. Voice that is characterized by varying degrees of breathiness, hoarseness, harshness, and pitch and loudness deviations.

Treatment of Voice Disorders: ***General Principles***

- Consider the goal of voice therapy as normal sounding voice with little or no effort and tension
- Maintain a cooperative working relationship with a laryngologist
- Have a good knowledge of laryngeal anatomy and physiology
- Understand the surgeries of larynx and related structures, medication to treat various voice disorders or underlying diseases, and the effects of all medical and surgical procedures on, and interactions with, voice treatment methods
- Have a medical evaluation done before starting voice therapy
- Have periodic medical examinations during voice therapy to assess the effects of voice therapy on the organic problems associated with the voice disorder
- Combine techniques designed to reduce vocally abusive behaviors with those that facilitate efficient and normal voice production
- Individualize the facilitating techniques, because what works with one client may not work with another

- Reduce vocally abusive behaviors first and teach healthier (adaptive) vocal behaviors next
- Work closely with parents of children and spouses or other family members of adults to help sustain healthy vocal behaviors at home
- Establish baselines of vocally abusive behaviors and the frequency of abnormal voice productions in and outside the clinic

Treatment of Vocal Loudness Disorders

Treatment of Excessively Soft Voice. Shape progressively louder voice in a client with too soft a voice:

- Make sure that the excessively soft voice is not due to such physical problems as hearing loss and vocal fold pathology
- Baseline vocal loudness with an audio-record speech sample to give feedback to the client about his or her excessively soft voice
- Instruct, demonstrate, and model voice at different loudness (e.g., role-play whispered voice, a quiet voice, normal conversational voice, loud and louder voice)
- Start with words and phrases and move on to conversational speech; reinforce speech produced at progressively higher intensity until the voice is produced with desired loudness
- Experiment with pitch changes to see if a slight increase in the pitch level might produce a slightly louder voice; if so, teach the client to speak at that level
- Ask the person to take deeper breaths to see if increased subglottic air pressure would help produce louder voice; if so, ask the client to practice speaking with such increased air pressure
- Use the auditory masking on an audiometer to induce the Lombard effect (increased vocal intensity under auditory masking); fade the noise gradually and let the client practice louder speech
- Shape a progressively louder voice by reinforcing it voice until an acceptable level is reached and stabilized

- Use such biofeedback instruments as the Visi-Pitch, and any available computer programs for voice therapy to shape the desired vocal pitch
- Use the Changing Criterion method to shape up the desired vocal loudness
- Use the Pushing Approach (described later in this section under Specific Normal Voice Facilitating Techniques) only if instruction, modeling, and biofeedback fail

Treatment of Excessively Loud Voice. Shape progressively softer voice in a client who speaks too loudly:

- Use essentially the same procedure as outlined to increase vocal loudness, except for setting a softer voice as the target
- If shaping is started at a very soft and quiet voice with modeling and imitation, model and reinforce progressively louder voice until the vocal loudness is judged acceptable (well below the baseline)
- If started with a louder voice (closer to the baseline), reinforce progressively softer voice until the loudness is judged acceptable
- Use the Changing Criterion method to shape down the vocal loudness

Treatment of Vocal Pitch Disorders

Treatment to Raise the Baseline Pitch

- Baseline the client's vocal pitch; let the client listen to his or her pitch
- Instruct the client on pitch, its variations, acceptable range, and the target pitch for the person
- Model different levels of pitch for the client
- Experiment with the client to see if he or she can produce a desirable pitch even if briefly
- Audio-record the client's desirable pitch and use it as a model for self-imitation
- Provide also a live model or a mechanical model on such computerized instruments as Visi-Pitch
- Use such an instrument to give immediate feedback during training trials or durations; supplement

mechanical feedback with verbal praise and other reinforcers

- Begin treatment with single words, preferably those that begin with vowels; have the client produce them with the desirable pitch
- Increase the response complexity by moving to words, phrases, sentences, and conversational speech
- Use the Changing Criterion method to shape up the pitch
- Use oral reading to stabilize the desired pitch

Treatment to Lower the Baseline Pitch

- Use the same procedures used for raising the pitch except for setting a lower pitch as the target
- Lower the pitch in carefully graded steps, if necessary
- Use the Changing Criterion method to shape down the pitch

Treatment of Resonance Disorders

General Guidelines

- Rule out the presence of cleft palate or congenital palatopharyngeal incompetence as the source of resonance problems; do not offer behavioral voice therapy for such cases unless the organic problems are eliminated or significantly improved by surgical or prosthetic means and clients are now good candidates for voice therapy; see Velopharyngeal Dysfunction for additional information on treating resonance disorders
- Work with a prosthodontist who might fabricate and fit a prosthetic device that may temporarily help close the velopharyngeal port (e.g., a speech bulb)
- Assess speech before and after surgical and prosthetic treatment
- Use biofeedback instruments to monitor and reinforce appropriate oral and nasal resonance

Treatment of Hypernasality

- Assess Hypernasality and treat it only when there is adequate or at least marginal velopharyngeal adequacy and the disorder needs behavioral management

- Use treatment procedures described under Hypernasality; see also Velopharyngeal Dysfunction

Treatment of Hyponasality

- Assess Hyponasality and treat it only when it is clear that too broad a pharyngeal flap or too big an obturator bulb is not the source of reduced nasality; be aware that very few persons exhibit hyponasality with no physical basis
- Use treatment procedures described under Hyponasality; see also Velopharyngeal Dysfunction

Treatment of Vocal Quality Disorders. The typical individual who seeks voice therapy has an abnormal voice quality that is often a result of vocally abusive behaviors; voice therapy in these cases is aimed at reducing such maladaptive behaviors; the therapy is entirely behavioral.

Treatment of Vocally Abusive Behaviors

- Educate the client and the family about the harmful results of vocally abusive behaviors the client exhibits; use charts and diagrams to show vocal nodules and how they affect voice
- Impress on the client and the family members the relationship between abusive behaviors and the resulting damage to the vocal folds and the consequent voice disorders
- Identify all the vocally abusive behaviors of the individual through case history and interview; identify the conditions under which these behaviors are most likely (e.g., a child may be more likely to indulge in noisy vocal behaviors while playing with certain aggressive toys; an adult might be prone to shout and scream at the sports stadium; discuss these with the client, the parents, spouses, or other family members)
- Make a hierarchical list of vocally abusive behaviors to reduce them on a priority basis
- Ask the client to measure (base rate) his or her prioritized vocally abusive behaviors for a few days and graph their frequency on a daily basis to establish

the baselines of vocally abusive behaviors in natural settings; give them a prepared recording sheet

- Ask the parents of young children to count and graph vocally abusive behaviors on a daily basis; perhaps a spouse of an adult client can help with this
- Target for initial elimination the most frequent and/or most damaging vocally abusive behaviors; go down the hierarchy as the more troublesome behaviors are reduced in frequency; use procedures described later in this section
- Design and implement a program to reduce the vocally abusive behaviors
 - Use the Changing Criterion method to shape down the frequency of vocally abusive behaviors; set an initial criterion that is just below the baserated number so it is easily met; examples:
 - For an adult with a baserate of 10 shouts/screams a minute inside a sports stadium, set an initial criterion on no more than 7 such abusive episodes; subsequently, set progressively lower criteria of 5, 3, and zero episodes; have an independent observer, a family member or a friend, count the episodes in the stadium
 - For a woman whose average baserate on-the-phone talking is 50 minutes, reduce the duration in 5-minute decrements; have her or a family member log the number and duration of calls
 - For a child who typically plays noisily with toys for 30 minutes on the average, have a parent model and reinforce quiet play initially for 5 minutes and increase the duration of quiet play in small durational increments (8 minutes of quiet play, increased to 12 minutes, 15 minutes, and so forth)
 - Similarly decrease the frequency of visits to noisy bars, frequency of throat clearing, excessive conversation with friends, hard glottal attacks, and so forth
 - Ask the client, parents, a spouse, teacher, or a friend to help establish the reliability of measures

of vocally abusive behaviors in the natural environment; give them a prepared recording sheet to document the occurrence of such behaviors before, during, and at the conclusion of treatment.

- Eliminate a condition under which a vocally abusive behavior is highly likely (e.g., teach the child to play with less aggressive themes and toys; gradually reduce the frequency of play involving toys that promote vocal abuse; in the case of an adult, encourage the client to avoid noisy and smoke-filled bars where the client tends to talk loudly and excessively)
- Implement a token system for reinforcing the child for having met the criterion of reduced frequency, duration, or both (of a specific vocal abuse); in the case of adolescents and adults, have a family member reinforce the vocally healthy behaviors by offering a treat of some kind that is likely a reinforcer
- In the treatment sessions, teach healthier vocal behaviors by teaching the client to:
 - Initiate speech sounds softly
 - Speak with optimum pitch
 - Speak at an appropriate loudness
 - Reduce the frequency of coughing or throat clearing
 - Breathe through the nose
 - Use an easy, relaxed breathing pattern when speaking
 - Speak with relaxed speech muscles
 - Open mouth more widely during talking
 - Teach whole-body relaxation
- Work with parents, teachers, friends, and others to monitor the production of vocally more appropriate behaviors (e.g., gentle vocal onset, soft speech)
- Make periodic assessment of voice if and when the medical or surgical treatment is repeated
- Measure the effects of the program to document the desirable changes in the voice disorder being treated
- Refer the client for periodic laryngological examinations as an ongoing assessment of effects of

treatment on any structural deviations that may be present (e.g., nodules)

- Follow up the client and arrange for booster treatment

Boone, D. R., McFarlane, S. C., Von Berg, S. L., & Zraick, R. L. (2014). *The voice and voice therapy* (9th ed.). Boston, MA: Allyn & Bacon.

Sapienza, S., & Ruddy, B. F. (2018). *Voice disorders*. San Diego, CA: Plural.

Stemple, J. C., Roy, N., & Klaben, B. C. (2014). *Clinical voice pathology* (5th ed.). San Diego, CA: Plural.

Specific Normal Voice Facilitating Techniques.

A collection of procedures used in voice therapy; most of them described by Boone, McFarlane, Von Berg, and Zraick (2014) and Sapienza and Rudy (2018); most are based on clinical experience; little or no controlled experimental evidence to demonstrate their effectiveness and efficacy.

Auditory Feedback. Providing feedback of client's own voice to help improve its quality, pitch, or intensity; amplified real-time feedback may be the most effective in modifying speech quality.

- Use an instrument such as the Kay Facilitator (Kay Elemetrics), which provides real-time amplified feedback of client's voice through head phones; also provides delayed auditory feedback, loop feedback, metronomic pacing, and masking noise
- Ask the client to listen carefully to his or her voice as the feedback is provided; teach the client to use a different voice facilitation approach (e.g., chanting or glottal fry)
- Ask the client to pay attention to improvement in voice quality when a facilitating approach is used; contrast this with the typical deviant voice of the patient

Chant-Talk Method. A manner of speaking in which words are spoken in a connected manner, with even stress, prolongation of sounds, soft glottal attack, and continuously with the absence of stress for individual

words; recommended for clients with hyperfunctional voice including Hard Glottal Attacks.

- Ask the patient to reduce the effort required to speak
- Play a recording of a chant and model the method
- Ask the patient to imitate the audio-recorded production by using a chant
- Ask the patient to read aloud by alternating the chant voice with the regular one
- Ask the patient to read for 20 seconds at a time
- Playback oral reading samples of the patient's voice and ask him or her to differentiate the chanted voice from the normal one
- Fade the chant and maintain normal conversational voice without the chant

Chewing Technique. Requires individuals to imagine that they are chewing food while voicing; recommended for reducing vocal hyperfunction, improving voice quality, reducing vocal stress, and increasing oral resonance.

- Describe and justify the procedure to the client
- Let the client face a mirror along with you
- Ask the client to pretend that he or she is chewing some food
- Teach exaggerated open-mouth chewing motions
- Pretend to move the food from one side of the mouth to the other
- Ask the client to phonate various sounds softly by constantly moving the tongue around in chewing motions
- Ask the client to say words while chewing
- Ask the client to chew and count
- Ask the client to chew and produce connected speech
- Fade chewing movements

Digital Manipulation of the Larynx. Physical manipulation of the larynx to promote desirable voice quality; may be used to reduce vocal pitch and decrease laryngeal tension.

- Use digital pressure to lower the pitch
 - Ask the patient to prolong a vowel

- Apply slight finger pressure to the thyroid cartilage as the vowel is prolonged (the pitch will drop)
- Fade the digital pressure and let the client practice the lower pitch
- Lower the larynx to reduce tension
 - Apply a slight downward pressure with the middle finger and the thumb just above the thyroid notch
 - Ask the client to prolong vowels with the larynx in the lowered position
 - Fade the downward pressure on the larynx

Glottal Attack Changes. Modifying either hard glottal attacks typically associated with vocal abuse or inducing slightly harder glottal attacks which may be beneficial to individuals with unilateral vocal fold paralysis; thus, the method includes two opposite manipulations applied to opposite problems.

- Reduce hard glottal attacks
 - Educate the client about the harmful effects of hard glottal attacks
 - Model and demonstrate hard and soft glottal attacks; audio-record the client's hard glottal attacks and let the client listen to them to understand their characteristics
 - Begin training on soft initiation of voice with monosyllabic words that contain the /h/ in the initial position; when the client learns to initiate the /h/ words softly, train with words containing initial unvoiced consonants and then with words containing vowels
 - Shape gentle onset with whisper as the initial response; ask the client to whisper the initial vowel and increase the intensity gradually until a soft phonation is heard
 - Use also the yawn-sigh method to reduce hard glottal attacks
 - Use also the chant to reduce hard glottal attacks
 - Use simultaneous chewing and chanting to reduce hard glottal attacks

- Increase the response complexity gradually until the client produces conversational speech with soft glottal contacts
- Teach harder glottal attacks
 - Educate the client about hard and soft forms of voice initiation
 - Demonstrate and model harder glottal attacks by producing such words as *pop*, *peep*, *bob*, and *beet* in a sudden plosive manner; ask the patient to imitate them and reinforce imitated or approximated productions
 - Prove visual feedback of soft and harder glottal attacks on an oscillograph or Visi-Pitch
 - Use the pushing approach briefly to induce harder glottal attacks
 - Increase the response complexity gradually to induce better approximation of vocal folds in conversational speech

Glottal Fry. A normal voice characteristic useful in treating voice disorders associated with vocal nodules, polyps, fold thickening, functional dysphonia, ventricular phonation, and spasmodic dysphonia; produced with relaxed vocal folds, minimal vocal fold tension, and minimal airflow.

- Educate the client about relaxed vocal folds being better able to approximate in spite of the nodules and other masses on them
- Teach the client to say /i/ softly after a slight exhalation
- Ask the client to “Open the throat” and make the sound deep, with a slow series of pops
- Ask the client to produce the same tone on inhalation to see whether the fry is better produced on inhalation or exhalation because individuals differ; alternate between inhaled and exhaled phonations
- Ask the client to say words like *on* and *off* and *in* and *out* while producing the fry; audio-record the client’s productions that will serve as a model

- Teach phrase or short sentence productions in the fry mode; ask the client to say, “Easy does it,” “Squeeze the peach,” or “See the eagle” and point out the improved voice quality to the client
- Ask the patient to practice speech in the fry mode at home
- Have the vocal fold pathology monitored by a laryngologist; as the voice improves, the health of the folds is likely to improve as well

Head Positioning. Manipulation of head positions to promote better voice quality; recommended especially for clients with neurological disorders, including dysarthria; may be used with clients who have hyperfunctional voice.

- Give instructions, model different head positions, demonstrate their effects on voice, and justify the procedure to the client
- Experiment with different head positions to find the one that promotes better voice (e.g., head rotated toward left or right; neck flexed downward with the face looking down); ask the client to produce some prolonged vowels (e.g., /i/, /ɪ/, /o/, or /u/) as different head positions are tried to identify the best position
- Ask the client to hold the head position that helps produce the best voice
- Ask the client to produce words, phrases, and sentences
- Gradually fade the unusual head position into a more normal position

Inhalation Phonation. A technique of voice therapy designed to evoke true vocal fold vibrations in clients who are aphonic or those who exhibit ventricular phonation.

- Raise your shoulders, inhale, and phonate a high-pitched hum
- Raise your shoulders, inhale, phonate the high-pitched hum and lower the shoulders; exhale and produce the same sound; repeat this

- Teach the client to produce inhalation phonation
- Teach the client to produce inhalation and exhalation phonation with corresponding shoulder movements
- Demonstrate the movement from the high-pitched voice to the exhaled low-pitched voice
- Reinforce the client's attempts to bring the pitch down
- Fade the shoulder movements
- Have the client practice single words until a normal sounding voice is stabilized
- Move on to more complex responses (phrases and sentences)

Laryngeal Massage. A technique of voice therapy in which the laryngeal area is gently massaged and manipulated to improve vocal quality; recommended for clients with functional voice disorders with no organic component characterized by a larynx that is positioned high during speech with some evidence of neck tension; some clinicians first try yawn-sigh with these clients and use laryngeal massage if the first attempt fails.

- Ask the client to prolong vowels as you massage the larynx through the steps specified
- Encircling the hyoid bone with the thumb and the middle finger, make a circular motion with your fingers while applying light pressure
- Repeat the procedure, this time with the fingers working posteriorly from the thyroid notch
- Repeat the procedure at the posterior borders of the thyroid cartilage
- Gently move the larynx in downward and lateral directions by placing the fingers over the superior borders of the thyroid cartilage
- Reinforce any improvement in the client's voice

Masking. A technique of voice therapy to treat clients with functional aphonia and those with poor voice quality because of inadequate auditory monitoring of one's own voice; masking noise introduced through headphones.

- Use a standard audiometer to introduce masking noise; do so without any explanation
- Ask the client to read orally; turn the masking on and off for brief periods
- Audio-record the client's reading to document possible changes in voice quality or the emergence of voice in the whispering aphonic patient
- Playback the recorded sample to demonstrate improved voice quality or voiced productions by an aphonic client; contrast voice with and without masking
- Ask the client to match his or her improved voice or emergence of phonation without masking
- Have the client read aloud under masking and, as the voice improves or phonation emerges, abruptly end masking; repeat this process until the client can sustain the gains

Nasal/Glide Stimulation. A voice treatment technique in which nasal glide consonants are used to promote better voice quality in clients with functional dysphonia, spasmodic dysphonia, and voice problems associated with vocal fold thickening, nodules, and polyps; words loaded with nasal glides are produced with least effort.

- Prepare a list of words with nasal consonants in them (e.g., *man, moon, many, morning, many men, moon man, morning singing*)
- Prepare also a list of words that are repeated with /a/ inserted between repetitions (e.g., *man a man a man* or *wing a wing a wing*; use this list as an alternative)
- Prepare a list of words with glide consonants /r/ and /l/ combined with nasal sounds (e.g., *marrow, married women, only lonely memory, Laura ran around*)
- Prepare a list of words with /l/ and /r/ that are repeated with /a/ inserted between each repetition (*lee a lee a lee* or *rah a rah a rah*)
- Have the client practice the selected words; reinforce improved voice quality

- Gradually move on to words, phrases, and sentences containing other consonants to improve voice in meaningful, spontaneous speech

Open-Mouth Approach. Oral openness during speech to increase oral resonance, reduce speaking effort, induce more relaxed speech, and promote appropriate loudness, pitch, and quality of voice.

- Give feedback on lack of mouth opening during speech; use a mirror if necessary
- Model greater and reduced oral openness; use a puppet (greater mouth opening); contrast that with the speech of a ventriloquist (minimum mouth opening)
- Ask the client to imitate the two ways of speaking that you model
- Ask the client to tilt the head down and speak
- Teach the client to self-monitor oral openness in natural settings
- Let the client practice speech with oral openness and reinforce for doing so

Pushing Approach. A voice therapy technique to promote better approximation of vocal folds; appropriate for increasing vocal loudness.

- Instruct and demonstrate pushing
- Ask the client to push down on the arm of the chair or push up by trying to lift the chair by gripping the bottom of the seat while seated
- Ask the client to phonate and push simultaneously
- Reinforce the louder voice that typically results
- Increase the length of utterances with the louder voice
- Fade pushing

Relaxation Training. A method to teach deep muscle relaxation with or without the help of biofeedback (e.g., electromyographic feedback); recommended for clients with excessive tension, anxiety, and stress; may be appropriate for some voice clients because of their excessive muscle tension.

- Use biofeedback instruments

- If no instruments are used, give instructions to contract and relax muscles
- Teach the client to discriminate between tensed and relaxed muscles by alternately asking him or her to tense and relax selected muscles (e.g., shoulder, neck, jaw muscles)
- Select facial, neck, and head muscles for relaxation training; ask the client to relax one set of muscles and tense them to appreciate the difference
- Manipulate head positions to induce relaxation
- Ask the client to imagine speaking situations that induce greater tension and immediately let the client relax the speech muscles
- Use relaxing head movements (positions) if necessary
- Use other appropriate voice therapy techniques in combination with relaxation
- Stabilize a relaxed speaking posture and improved voice quality

Respiration Training. Teaching clients to manage inhalation-exhalation cycles optimally for the purpose of phonation and sustained vocalization; recommended for clients with functional voice disorders who do not seem to use their breath stream properly in voice production.

- Explain the relation between breathing and speaking and between airflow and vocal fold vibrations
- Teach the client to inhale more quickly, more deeply than usual but exhale more slowly and in a controlled manner; to extend exhalation, ask the client to count to 5 slowly and then to 10 slowly
- Ask the client to prolong vowels to teach controlled and prolonged exhalation that would better support speech; in progressive steps, teach the client to prolong a vowel for about 20 seconds
- Teach the client to inhale quickly between utterances
- Teach good posture, which promotes normal airflow management

Tongue Position Modification. Manipulating tongue position in the oral cavity to affect changes in voice

quality and resonance; tongue typically positioned too far back results in cul-de-sac resonance; tongue typically carried too far forward creates “thin voice,” giving the baby-talk effect.

- Teach clients to carry tongue in its neutral position
- Modify the excessively backward tongue position
- Modify the excessively forward tongue position
- Instruct, model, demonstrate, and reinforce correct tongue positions

Vocal Rest. A voice therapy technique that requires little or no talking, typically for 4–7 days; vocal rest may be complete or partial.

- Recommend mandatory vocal rest for clients who have undergone any form of laryngeal surgery; this helps promote normal healing of the surgical wounds
- Recommend vocal rest as initial treatment for clients who have such types of laryngeal lesions as vocal fold hemorrhage and mucosal tear to let the healing process begin
- Recommend partial (modified) vocal rest for clients who have a severe cold (and resulting laryngeal inflammation), vocal nodules, and vocal fold edema; note that partial vocal rest means talking only when absolutely essential and with appropriate vocal habits
- Instruct the client either to totally avoid or markedly reduce
 - Speaking
 - Shouting or screaming
 - Singing or humming
 - Whispering
 - Coughing or throat clearing
 - Laughing or crying
 - Lifting or pushing heavy objects
- Have a family member monitor these activities
- Teach the client to keep a record of such activities
- Teach the client to self-monitor

Yawn-Sigh Method. A voice therapy technique for clients with hyperfunctional voice; uses the relaxing

effects of the inspiratory yawn followed by an expiratory sigh and phonation.

- Instruct and demonstrate the relaxing effects of prolonged inspiration involved in a yawn and the relaxed phonation that results with a sigh
- Ask the client to yawn and then exhale slowly while phonating lightly
- Ask the client to say words that start with /h/ after each yawn
- Teach the client to produce a gentle, voiced sigh while exhaling
- Teach the client to produce an easy, prolonged, open-mouthed exhalation after each yawn
- Ask the client to skip the yawn and teach the client to inhale normally and exhale a prolonged sigh with the open mouth
- Ask the patient to say “hah” after beginning each sigh
- Ask the patient to say additional words all beginning with the glottal /h/
- Ask the patient to blend in an easy, relaxed, phonation during the middle of a sigh
- Fade the sigh and move on to words, phrases, and sentences

Boone, D. R., McFarlane, S. C., Von Berg, S. L., & Zraick, R. L. (2014). *The voice and voice therapy* (9th ed.). Boston, MA: Allyn & Bacon.

Sapienza, S., & Ruddy, B. F. (2018). *Voice disorders*. San Diego, CA: Plural.

Voice Prosthesis. A small (1.8 to 3.6 cm) silicone device that has a valve at the back end and an opening at the front end; inserted into the tracheoesophageal puncture in patients who have undergone laryngectomy; allows air into the esophagus, which vibrates; the sound is shaped into speech; see [Laryngectomy](#).

Voluntary Stuttering. A treatment target in the fluent stuttering approach of Van Riper; for procedures, see [Treatment of Stuttering: Specific Techniques or Programs](#) under the main entry, [Fluency Disorders](#).

W

Wernicke's Aphasia. To treat this type of fluent aphasia, see [Aphasia](#); [Treatment of Aphasia: Specific Types](#).

Whole Language Approach. An approach to language and literacy intervention that requires the teaching of all aspects of language (speaking, reading, writing) simultaneously; lacking in experimental support and now highly questioned; there are procedures with better experimental support and hence not recommended; see [Language Disorders in Children](#); [Treatment of Language Disorders: Specific Techniques or Programs](#).

Whole Word Accuracy (WWA). A criterion measure used in a multiple-phoneme approach of articulation treatment; the entire word is judged for accuracy (as against judging the accuracy of only the target phoneme).

Wh-Questions. Questions that begin with *wh*; interrogative statements that begin with *what*, *when*, *where*, and *who*; intervention targets for language impaired children; see [Language Disorders in Children](#) for teaching strategies.

Wilson's Disease. To treat motor speech disorders associated with this and other neurological disorders, see [Dysarthrias](#).

Wireless Systems. [Assistive Listening Devices](#) that transmit messages from a speaker to a listener without wire connections; include FM auditory trainers and infrared systems; see under [Aural Rehabilitation](#).

Word Fluency. To teach clients the skill of producing words that start with a sound or words that belong to a particular category (e.g., animals or vegetables), see [Aphasia](#).

Word Retrieval Problems. To treat difficulties in recalling names and other kinds of words often associated with recent brain injury, see [Aphasia](#), in which they are a dominant problem; see also [Dementia](#), [Traumatic Brain Injury in Adults](#), and [Right Hemisphere Disorders](#).

Additional Titles by M.N. Hegde

- *Hegde's PocketGuide to Assessment in Speech-Language Pathology, Fourth Edition* (© 2018)
- *Hegde's PocketGuide to Communication Disorders, Second Edition* (© 2018)
- *A Coursebook on Aphasia and Other Neurogenic Language Disorders, Fourth Edition* (© 2018)
- *A Coursebook on Scientific and Professional Writing for Speech-Language Pathology, Fifth Edition* (© 2018)
- *Assessment of Communication Disorders in Adults: Resources and Protocols, Second Edition* (© 2017)
- *Assessment of Communication Disorders in Children: Resources and Protocols, Third Edition* (© 2017)

